

# Mensuration

2- D: Under 2 dimensions we will study

- Triangles
- Quadrilateral
- Polygons
- Circle

3- D: Under 3 dimensions we will study

- Cube
- Cuboids
- Box
- Cylinder
- Prism
- Cone
- Pyramid
- Frustum
- Sphere
- Hemisphere
- Tetrahedral

**Note:** For 2D figures, area and perimeter are calculated and for 3D figures, volume and surface area are calculated.

## TWO DIMENSIONAL FIGURES

### TRIANGLE

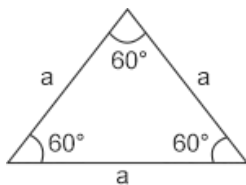
For a triangle with height, h, and base, b;

$$\text{Area} = \frac{1}{2} \times b \times h$$

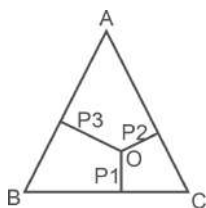
Perimeter = Sum of all sides

**Special cases:**

**1) Equilateral Triangle** - All sides are equal and each angle is  $60^\circ$ .



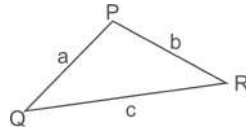
- Area =  $\frac{\sqrt{3}}{4} a^2$
- Height =  $\frac{\sqrt{3}}{2} a$
- Perimeter =  $3a$
- Inradius(r) =  $\frac{a}{2\sqrt{3}}$
- Circumradius(R) =  $\frac{a}{\sqrt{3}}$



If P1, P2 and P3 are perpendicular to a side from a point inside the triangle, then

- Height of triangle =  $P1 + P2 + P3$

**2) Scalene Triangle** - All sides will be unequal lengths.



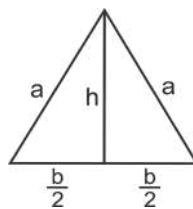
Area =  $\sqrt{s(s-a)(s-b)(s-c)}$ ,  
This formula is called Heron's formula.

Where, Semi-perimeter,

$$s = \frac{a+b+c}{2}$$

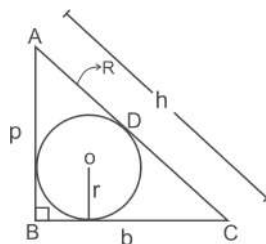
- Perimeter =  $a + b + c$

**3) Isosceles Triangle** :- two sides and two angles are equal. Altitude bisects the base.



- Area =  $\frac{b}{4} \sqrt{4a^2 - b^2}$
- Height =  $\sqrt{a^2 - \left(\frac{b}{2}\right)^2}$   
 $= \frac{1}{2} \sqrt{4a^2 - b^2}$
- Perimeter =  $a + a + b$   
 $= 2a + b$

**4) Right angled Triangle** :- One of the angles is  $90^\circ$ .



Here, p = perpendicular, b = base and h = hypotenuse

- Area =  $\frac{1}{2} \times b \times p$
- Perimeter =  $p + b + h$
- Pythagoras Theorem:  
 $h^2 = p^2 + b^2$
- Inradius(r) =  $\frac{p+b-h}{2}$
- Circumradius(R) =  $\frac{h}{2}$

**Note:** Common Pythagoras triplets:

(1, 1,  $\sqrt{2}$ ); (1, 2,  $\sqrt{5}$ ); (3, 4, 5); (5, 12, 13);  
(7, 24, 25); (20, 21, 29); (9, 40, 41);  
(8, 15, 17) (12, 35, 37) (60, 11, 61)  
(65, 72, 97) (96, 110, 146).

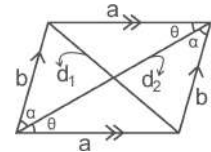
Try remembering them.

## QUADRILATERAL

A figure enclosed by four sides is called a quadrilateral. A quadrilateral has four angles and sum of these angles is equal to  $360^\circ$ .

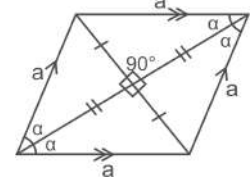
**Special Cases:**

**1) Parallelogram** - It is a quadrilateral with opposite sides parallel and equal.



- Area = base x height
- Perimeter =  $2(a + b)$
- $d_1^2 + d_2^2 = 2(a^2 + b^2)$
- In a parallelogram, opposite sides are equal, opposite angles are equal and diagonals bisect each other.

**2) Rhombus** - It is a parallelogram with all four sides equal. The opposite angles in a rhombus are equal.

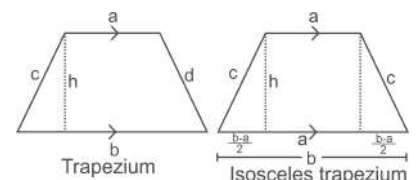


Here, a = side;  $d_1$  and  $d_2$  are diagonals.

- Area =  $\frac{1}{2} \times d_1 \times d_2$
- Side (a) =  $\frac{1}{2} \sqrt{d_1^2 + d_2^2}$
- Perimeter =  $4a$
- $4a^2 = d_1^2 + d_2^2$

Diagonals bisect each other at right angles.

**3) Trapezium** - It is a quadrilateral with one pair of opposite sides parallel.



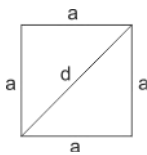
Here, a and b are parallel sides and h is the height or perpendicular distance between a and b.

- Area =  $\frac{1}{2} \times \text{height} \times (\text{sum of parallel sides})$   
 $= \frac{1}{2} \times h \times (a + b)$

**4) Rectangle**– It is a parallelogram with equal opposite sides and each angle is  $90^\circ$

- Area = Length  $\times$  Breadth =  $L \times B$
- Perimeter =  $2(L + B)$
- Diagonal ( $d$ ) =  $\sqrt{L^2 + B^2}$
- Area of the walls of a rectangular room  
 $= 2 \times (L + B) \times H$

**5) Square** – It is a parallelogram with all four sides equal and each angle is equal to  $90^\circ$ .



- Area =  $(side)^2 = a^2 = \frac{1}{2}d^2$
- Perimeter =  $4 \times side = 4a$
- Diagonal ( $d$ ) =  $a\sqrt{2}$

**NOTE: Important points about Quadrilaterals**

1. The diagonals of a parallelogram bisect each other.
2. Diagonal of a parallelogram divides it into two triangles of equal area.
3. The diagonals of a rectangle are of equal lengths and bisect each other.
4. The diagonals of a square are equal and bisect each other at right angles.
5. A rhombus has unequal diagonals and they bisect each other at right angles.
6. A parallelogram and a rectangle have equal areas if they are on the same base and between the same parallel lines.

### REGULAR POLYGON

In a regular polygon all sides and all interior angles are equal. A polygon is called a pentagon, hexagon, heptagon, octagon, nonagon and a decagon as they have 5, 6, 7, 8, 9, 10 sides, respectively.

If each side of a regular polygon of 'n' sides is equal to 'a' then:

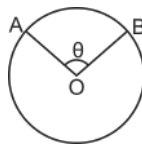
- Area of regular pentagon  
 $= \frac{6.9}{4} \times a^2$
- Area of regular hexagon  
 $= 6 \times \frac{\sqrt{3}}{4} a^2$
- Each exterior angle =  $\frac{360^\circ}{n}$
- Each interior angle

=  $180^\circ$  – Exterior angle

- Number of diagonals  
 $= \frac{n(n-3)}{2}$
- Sum of all interior angles  
 $= (n-2) \times 180$
- Sum of all exterior angle =  $360^\circ$

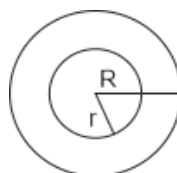
### CIRCLE

It is a plane figure enclosed by a line on which every point is equally distant from a fixed point (centre) inside the circle.



- Area =  $\pi r^2$
- Circumference (perimeter)  
 $= 2\pi r$
- Diameter =  $2r$
- Length of Arc (AB)  
 $= 2\pi r \times \frac{\theta}{360^\circ}$
- Area of sector AOB  
 $= \pi r^2 \times \frac{\theta}{360^\circ}$
- Length of Arc (AB) = *angle*  $\times$  *radius*, (where angle in radian)

### CIRCULAR RING



Here, R = radius of bigger ring,  
 r = radius of smaller ring

- Area =  $\pi(R^2 - r^2)$
- Difference in circumference of both the rings =  $2\pi R - 2\pi r$

### Short Cut methods/Tricks

1. If the length and breadth of a rectangle are increased by a% and b%, the area of the rectangle will be increased by

$$(a + b + \frac{ab}{100})\%$$

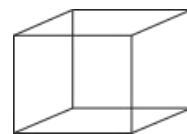
2. If any of the two sides of the rectangle is decreased then use '-ve' values for that side.
3. All the sides of any two dimensional figure changed by a%, then its area will change by  $(2a + \frac{a^2}{100})\%$   
 Whenever there is a decrease, use negative value for 'a'
4. If all the sides of any two dimensional

figure has changed (increased or decreased) by a% then its perimeter also changes by a%. In the case of a circle such changes take place because of the change in radius (or diameter).

5. If the area of a square is 'a' square unit. Then the area of the circle formed with the same perimeter is given by square units.
6. Area of a square inscribed in a circle of radius 'r' is equal to  $2r^2$ .
7. The area of the largest triangle inscribed in a semicircle of radius r is equal to  $r^2$ .
8. If a pathway of width x is made inside or outside a rectangular plot of length L and breadth B, then area of the pathway is  
 (i)  $2x(L + B + 2x)$ , if path is made outside the plot  
 (ii)  $2x(L + B - 2x)$ , if a path is made inside the plot.
9. If two paths, each of width x are made parallel to length (L) and breadth (B) of the rectangular plot in the middle of the plot crossing each other, then,  
 Area of the path =  $x(L + B - x)$

### THREE DIMENSIONAL FIGURES

#### CUBE



All sides are equal. It has six faces and 12 edges.

- Volume =  $a^3$
- Total surface area =  $6a^2$
- Diagonal =  $a\sqrt{3}$
- Sum of all edges =  $12a$

Here, a = length of the side

#### CUBOID

A rectangular body having 3D rectangular shape, is called a cuboid.



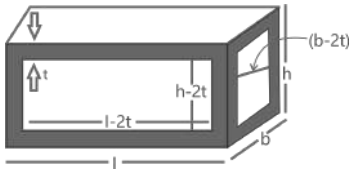
- Volume =  $l \times b \times h$
- Total surface area =  
 $2(lb + bh + lh)$
- Diagonal =  $\sqrt{l^2 + b^2 + h^2}$

#### BOX

A box has its shape like a cube or cuboid. The amount that a box can hold or contain is called the capacity of the box.



Capacity means internal volume.



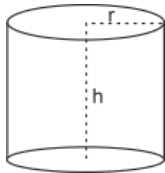
- Surface area of an open box  
 $= 2(\text{length} + \text{breadth}) \times \text{height} + \text{length} \times \text{breadth}$   
 $= 2(l + b) \times h + l \times b$
- Capacity of box  
 $= (l - 2t)(b - 2t)(h - 2t);$   
 where,  $t$  = thickness of box
- Volume of the material of the box = External volume – Internal volume (or capacity)  
 $= lbh - (l - 2t)(b - 2t)(h - 2t)$

### ROOM

A rectangular room has four walls (surfaces) and opposite walls have equal area.

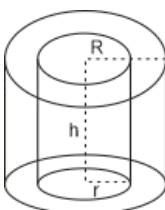
- Total Area of walls  
 $= 2(l + b) \times h$
- Total volume of the room  
 $= l \times b \times h$
- Area of floor or roof =  $l \times b$

### CYLINDER



- Volume of cylinder = area of base  $\times$  height =  $\pi r^2 h$
- Curved surface area = Perimeter of base  $\times$  height =  $2\pi r h$
- Total surface Area = curved surface area + area of both the circles  
 $= 2\pi r h + 2\pi r^2 = 2\pi(r + h)r$
- When the rectangular sheet is folded along its length, then the length becomes the circumference of the base of the cylinder and breadth becomes the height of the cylinder.

### HOLLOW CYLINDER



- Volume of hollow cylinder  
 $= \pi(R^2 - r^2)h$
- Curved surface area  
 $= 2\pi(R + r)h$
- Total surface area  
 $= 2\pi(R + r)h + 2\pi(R^2 - r^2)$   
 $= 2\pi(R + r)\{h + R - r\}$

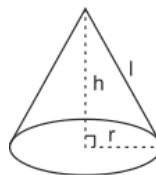
Where,  $R$  = External radius of cylinder,  $r$  = internal radius of cylinder,  $h$  = height

### PRISM

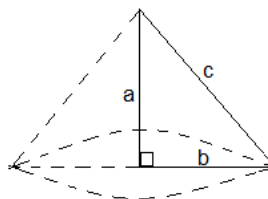
- Volume of prism  
 $= \text{area of base} \times \text{height}$
- Lateral surface area = Perimeter of base  $\times$  height
- Total surface Area = Lateral surface area + area of base and top surface

### CONE

A solid and round body with a round base and pointed peak.



- Volume =  $\frac{1}{3} \times \text{base area} \times \text{height} = \frac{1}{3} \pi r^2 h$
- Slant height ( $l$ ) =  $\sqrt{r^2 + h^2}$
- Curved surface area =  $\pi r l$   
 $= \pi r \sqrt{r^2 + h^2}$
- Total surface area =  $\pi r l + \pi r^2$   
 $= \pi r(l + r)$
- Cone formed by rotating right angled triangle about its height:



- Volume of cone so formed  
 $= \frac{1}{3} \pi b^2 a$
- Similarly, Cone formed by rotating right angled triangle about its base:  
 Volume of cone so formed  
 $= \frac{1}{3} \pi a^2 b$

- Similarly, Cone formed by rotating right angled triangle about its hypotenuse:  
 Volume of cone so formed  
 $= \frac{1}{3} \pi r^2 c$ , (where  $r$  is the altitude on hypotenuse and  $r = \frac{a \times b}{c}$ )

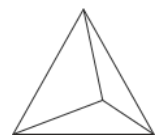
**Note:** If the base is not round, it will be called a pyramid. A pyramid can have various shapes of the base example: square, rectangular, triangular etc.

### PYRAMID

- Pyramid means a structure with regular polygon as its base and sloping sides that meet in a point at the top.
- In Pyramid, with  $n$  sided regular polygon at its base, total number of vertices =  $n + 1$
- Volume =  $\frac{1}{3} \times \text{base area} \times \text{height}$
- Slant height ( $l$ ) =  $\sqrt{r^2 + h^2}$
- Lateral surface area  
 $= \frac{\text{Perimeter} \times \text{slant height}}{2}$
- Total Surface area = Lateral surface area + Area of base

### TETRAHEDRON

It is a 3D figure made by joining four equilateral triangles.



- Volume ( $V$ )  
 $= \frac{1}{3} \times \text{base area} \times \text{height}$   
 $\Rightarrow V = \frac{1}{3} \times \frac{\sqrt{3}}{4}$   
 $a^2 \times \left( \frac{\sqrt{6}}{12} a + \frac{\sqrt{6}}{4} a \right)$

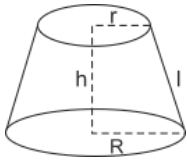
$$\Rightarrow V = \frac{\sqrt{2}}{12} a^3 \text{ (Remember this formula)}$$

- Lateral surface area =  $3 \times \frac{\sqrt{3}}{4} a^2$
- Total surface area  
 $= 4 \times \frac{\sqrt{3}}{4} a^2$
- Height of Tetrahedron =  $\frac{\sqrt{6}}{3} a$

### FRUSTUM OF CONE

If a cone is cut by a plane parallel to its base, so as to divide the cone into two

parts: upper part and lower part, then the lower part is called frustum.



- Slant height ( $l$ )  

$$= \sqrt{h^2 + (R - r)^2}$$
- Curved Surface Area  

$$= \pi(R + r)l$$
- Total surface area  

$$= \pi(R + r)l + \pi R^2 + \pi r^2$$

$$= \pi\{(R + r)l + R^2 + r^2\}$$
- Volume  $= \frac{1}{3}\pi h(r^2 + R^2 + rR)$

### SPHERE

- Volume of sphere  $= \frac{4}{3}\pi r^3$
- Curved Surface area  

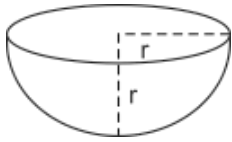
$$= \text{Total surface area} = 4\pi r^2$$

### HOLLOW SPHERE OR SPHERICAL SHELL:

- Volume of hollow sphere  

$$= \frac{4}{3}\pi(R^3 - r^3)$$
  - Internal surface area  $= 4\pi r^2$
  - External surface area  $= 4\pi R^2$
- Here  $R$  = external radius and  $r$  = internal radius

### HEMISPHERE



- Volume of the hemisphere  

$$= \frac{2}{3}\pi r^3$$
  - Total surface area  $= 3\pi r^2$
  - Curved surface area  $= 2\pi r^2$
- Where,  $r$  = radius

### IMP. UNIT CONVERSION :

- $1m^3 = 1000$  litres and
- 1 litre =  $1000\text{ cm}^3$
- 1 meter = 10 decimeter  
 $= 100\text{ cm} = 1000\text{ millimeter}$
- 1 meter =  $10^{-1}$  decameter  
 $= 10^{-2}$  hectometer  
 $= 10^{-3}$  kilometer

## Variety Questions

**Q.1.** A large regular polygon is made by combining 12 regular hexagons around a common center. What is the ratio of area of the combined shape to two hexagon?  
 SSC CGL 20/09/2025 (Shift 2)  
 (a) 4 : 1 (b) 6 : 1 (c) 3 : 1 (d) 5 : 1

**Q.2.** A right circular cone having height of 15 cm is cut by two parallel planes at heights 5 cm and 10 cm from the base. What is the ratio of the volumes of the three parts (from top to bottom) ?  
 SSC CGL 19/09/2025 (Shift 3)  
 (a) 8 : 19 : 27 (b) 1 : 6 : 20  
 (c) 8 : 19 : 64 (d) 1 : 7 : 19

**Q.3.** A right circular cone is cut parallel to its base, forming a frustum. The height of the original cone is 20 cm, and the cut is made 5 cm from the top. What is the ratio of the volume of the frustum to the volume of the entire cone?  
 SSC CGL 18/09/2025 (Shift 1)  
 (a) 63 : 64 (b) 62 : 63 (c) 16 : 19 (d) 14 : 15

**Q.4.** In a trapezoid PQRS with PQ parallel to SR, the diagonals PR and QS intersect at T. What is the ratio of the area of  $\Delta PQT$  to the area of  $\Delta SRT$  ?  
 SSC CGL 17/09/2025 (Shift 2)  
 (a) The ratio of PQ to SR squared.  
 (b) The ratio of PQ to SR.  
 (c) The ratio of the perimeter of  $\Delta PQT$  to the perimeter of  $\Delta SRT$ .  
 (d) The ratio of the area of  $\Delta PQT$  to the area of  $\Delta SRT$ .

**Q.5.** A chord in a circle with radius 10 cm subtends an angle of  $90^\circ$  at the center. What is the area of the minor segment ?  
 Graduate Level 26/07/2025 (Shift - 3)  
 (a)  $(25\pi - 50)$  sq. cm  
 (b)  $(100\pi - 50)$  sq. cm  
 (c)  $(50\pi - 100)$  sq. cm  
 (d)  $(25\pi - 100)$  sq. cm

**Q.6.** A solid hemisphere is drilled with a cylindrical with radius  $\frac{R}{4}$  hole through its center. If the height of the removed cylinder equals the hemisphere's radius  $\frac{R}{2}$ , find the ratio of the volume removed to the volume remaining.  
 Graduate Level 25/07/2025 (Shift - 3)  
 (a) 5 : 1 (b) 5 : 4 (c) 3 : 4 (d) 3 : 5

**Q.7.** There is a slide in a park. One of its side walls, shaped like a triangle, is painted with a message: "RESPECT NATURE, PRESERVE GREENERY." If the sides of the wall are 7 m, 8 m, and 9 m, what is the area painted in color ?

Higher Secondary 02/08/2025 (Shift - 2)

- (a)  $12\sqrt{5}\text{ m}^2$  (b)  $15\sqrt{5}\text{ m}^2$   
 (c)  $12\sqrt{3}\text{ m}^2$  (d)  $16\sqrt{3}\text{ m}^2$

**Q.8.** A field roller is in the shape of a cylinder whose diameter is 2.8 metres and length is 1.5 metres. If the roller rolls at a speed of 12 revolutions per minute, then find the maximum area (in  $\text{m}^2$ ) that it can roll in 1 hour (60 minutes).  
 Higher Secondary 29/07/2025 (Shift - 2)

- (a) 9,504  $\text{m}^2$  (b) 8,640  $\text{m}^2$   
 (c) 9,000  $\text{m}^2$  (d) 10,250  $\text{m}^2$

**Q.9.** A semi-circular window has a diameter of 2.8 meters. What is the total length of the frame required for this window?  
 Matriculation Level 01/08/2025 (Shift - 2)

- (a) 6.8 m (b) 7.2 m (c) 8.6 m (d) 9.0 m

**Q.10.** A cube is a special type of rectangular parallelepiped where: Which of the following correctly describes why a cube is considered a special type of rectangular parallelepiped?  
 Matriculation Level 26/07/2025 (Shift - 1)

- (a) All edges (sides) are of equal length.  
 (b) The length is greater than the breadth, which equals the height.  
 (c) The length and breadth are equal, but greater than the height.  
 (d) The length, breadth, and height are all different.

**Q.11.** At what angle is the arc length half the perimeter of the circle?  
 (a)  $120^\circ$  (b)  $180^\circ$  (c)  $90^\circ$  (d)  $270^\circ$

**Q.12.** What is the central angle of a sector of a circle whose area and perimeter are, respectively, equal to 209  $\text{cm}^2$  and 63 cm if it is given that its radius is a non-negative integer (roundup to one decimal place) ?  
 (a)  $55.5^\circ$  (b)  $53.5^\circ$  (c)  $49.5^\circ$  (d)  $51.5^\circ$

**Q.13.** The area of a square is  $16x^2 + 40x + 25$  square units. Find the perimeter of the square.  
 (a)  $3(10x + 8)$  (b)  $3(8x + 10)$   
 (c)  $2(10x + 8)$  (d)  $2(8x + 10)$

**Q.14.** A right triangle with sides 3 cm, 4 cm and 5 cm is rotated about the sides of 3 cm to form a cone. The volume of the cone so formed is:  
 (a)  $20\pi\text{ cm}^3$  (b)  $16\pi\text{ cm}^3$   
 (c)  $25\pi\text{ cm}^3$  (d)  $28\pi\text{ cm}^3$

**Q.15.** Find the area (in  $\text{cm}^2$ ) of the sector whose perimeter is  $\frac{64}{3}$  cm and central angle is  $60^\circ$ . (Use  $\pi = \frac{22}{7}$ )  
 (a)  $\frac{47}{3}$  (b)  $\frac{77}{3}$  (c)  $\frac{68}{3}$  (d)  $\frac{85}{3}$



**Q.16.** A cuboid with dimensions 20cm, 12cm, and 10 cm is cut into 8 identical pieces by 3 cuts. What will be the total surface area of all the pieces?

- (a) 2240 cm<sup>2</sup> (b) 1220 cm<sup>2</sup>  
(c) 3360 cm<sup>2</sup> (d) 1680 cm<sup>2</sup>

**Q.17.** A rectangle with the largest area is cut out of a circle with radius 5 units. What is the area of the remaining portion?

- (a)  $25(2\pi - 3)$  (b)  $25(2\pi - 5)$   
(c)  $25(\pi - 3)$  (d)  $25(\pi - 2)$

**Q.18.** A cylinder has a curved surface area equal to 350% of the curved surface area of another cylinder. If their radii are in the ratio 3 : 1, then the volume of the smaller cylinder is approximately \_\_\_\_% of the larger cylinder (the smaller cylinder has its radius and height smaller when compared to the larger).

- (a) 10.16 (b) 8.86 (c) 7.28 (d) 9.52

**Q.19.** The inner and outer radii of a hemispherical wooden bowl are 6 cm and 8 cm, respectively. Its entire surface has to be polished and the cost of polishing  $\pi$  cm<sup>2</sup> is Rs.50. How much will it cost to polish the bowl?

- (a) Rs.11,600 (b) Rs.11,400  
(c) Rs.10,000 (d) Rs.12,000

**Q.20.** Which of the following statements is INCORRECT about the circle?

- (a) The area of a sector =  $\frac{1}{2}(\text{arc length} \times R)$   
(b) The radius of an incircle of an equilateral triangle of side 's' is  $\frac{s}{2\sqrt{3}}$   
(c) The length of an arc =  $\frac{\text{central angle made by the arc}}{360^\circ} \times \pi r^2$ .  
(d) Circles are congruent if they have same radii while similar circles may have different radii.

**Q.21.** The average of the areas of 2 similar triangles is 706.5 m<sup>2</sup> whose perimeters are in the ratio of 6 : 11. What is 20% of the difference (in m<sup>2</sup>) in areas of both triangles?

- (a) 164 (b) 149 (c) 153 (d) 157

**Q.22.** Two equal circles are drawn on a square in such a way that opposite side of the square forms the diameter of each circle. If the remaining area of the square is 42 cm<sup>2</sup>, what is the measurement (in cm) of the diameter of each circle?

- (a) 14 (b) 2.5 (c) 7 (d) 3.5

**Q.23.** The ratio of the diameters of two spheres is given as 1 : 4. The larger sphere is melted and 125 identical spheres are made out of the molten

material. The smaller sphere is melted and 27 identical spheres are made out of the molten material. If the ratio of the volume of each of the 125 identical spheres to the volume of each of the 27 identical spheres is given as 1 : m, what is the value of m?

- (a)  $\frac{1728}{125}$  (b)  $\frac{125}{1728}$  (c)  $\frac{27}{8000}$  (d)  $\frac{64}{3375}$

**Q.24.** Two rectangular sheets of paper, each 60 cm × 36 cm, are made into two right circular cylinders, one by rolling the paper along its length and the other along the breadth. The ratio of the volumes of the two cylinders, thus formed, is:

- (a) 5 : 6 (b) 8 : 3 (c) 7 : 4 (d) 5 : 3

**Q.25.** Three circles each of radius 5 cm touch one another. The area (in cm<sup>2</sup>) subtended between them is:

- (a)  $50(\sqrt{3} - \frac{\pi}{2})$  (b)  $25(\sqrt{3} + \frac{\pi}{2})$   
(c)  $25(2\sqrt{3} - \frac{\pi}{2})$  (d)  $25(\sqrt{3} - \frac{\pi}{2})$

**Q.26.** A gear 15 cm in diameter is turning a gear 21 cm in diameter. When the smaller gear has 105 revolutions, how many has the larger one made?

- (a) 75 (b) 100 (c) 85 (d) 90

**Q.27.** A solid metal cylinder of 12 cm height and 7 cm radius is melted and recast into two cones in the proportion of 1 : 2 (volume), keeping the height 12 cm. What would be the percentage change in the flat surface area before and after

[use  $\pi = \frac{22}{7}$  ]?

- (a) 35% (b) 20% (c) 45% (d) 50%

**Q.28.** Three solid iron cubes of edges 8 cm, 10 cm and 12 cm are melted together to make a new cube. It is observed that 496 cm<sup>3</sup> of the melted material is lost due to improper handling. The area (in cm<sup>2</sup>) of the whole surface of the newly formed cube is :

- (a) 1331 (b) 1176 (c) 2197 (d) 3375

**Q.29.** If the semi-perimeter and area of a rectangular field whose length and breadth are 'x' and 'y' is 12 cm and 28 cm<sup>2</sup>, respectively, then find the value of  $x^4 + x^2y^2 + y^4$ .

- (a) 6690 (b) 6960 (c) 6609 (d) 6906

**Q.30.** The base of a right prism is an equilateral triangle with each side measuring 4 cm. If the lateral surface area is 120 cm<sup>2</sup>, find the volume (in cm<sup>3</sup>) of the prism.

- (a)  $30\sqrt{3}$  (b)  $40\sqrt{3}$  (c)  $10\sqrt{3}$  (d)  $20\sqrt{3}$

**Q.31.** Some ice pieces, spherical in

shape, of diameter 6 cm are dropped in a cylindrical container containing some juice and are fully submerged. If the diameter of the container is 18 cm and level of juice rises by 40 cm, then how many ice pieces are dropped in the container?

- (a) 90 (b) 80 (c) 85 (d) 95

**Q.32.** Some medicine in liquid form is prepared in a hemispherical container of diameter 36 cm. When the container is full of medicine, the medicine is transferred to small cylindrical bottles of diameter 6 cm and height 6 cm. How many bottles are required to empty the container?

- (a) 70 (b) 75 (c) 72 (d) 76

**Q.33.** The length and the breadth of the floor of a rectangular hall are 126 feet and 90 feet, respectively. What will be the area (in square feet) of each of the largest identical square tiles that can be used to tile this floor in a way that no part of the floor remains uncovered?

- (a) 196 (b) 256 (c) 324 (d) 484

**Q.34.** The radii of the ends of a frustum of a solid right-circular cone 45 cm high are 28 cm and 7 cm. If this frustum is melted and reconstructed into a solid right circular cylinder whose radius of base and height are in the ratio 3 : 5, find the curved surface area (in cm<sup>2</sup>) of this cylinder. [Use  $\pi = \frac{22}{7}$ .]

- (a) 4610 (b) 4620 (c) 4580 (d) 4640

**Q.35.** A square and a rhombus have the same base and the rhombus is inclined at 45°, then what will be the ratio of the area of the square to the area of the rhombus?

- (a)  $\sqrt{3} : 1$  (b)  $1 : \sqrt{3}$  (c)  $\sqrt{2} : \sqrt{3}$  (d)  $\sqrt{2} : 1$

**Q.36.** A closed wooden box measures externally 10 cm long, 8 cm broad and 6 cm high. If the thickness of the wood is 0.5 cm, then the volume of wood required is

- (a) 165 cm<sup>3</sup> (b) 300 cm<sup>3</sup>  
(c) 230 cm<sup>3</sup> (d) 150 cm<sup>3</sup>

**Q.37.** The weight of a cube varies directly as the product of its volume and its density. The ratio of densities of the materials of the first cube and second cube is 27:16. If the weight of the first cube is 4 times the weight of the second cube, then what is the ratio of the edges of the first cube and second cube?

- (a) 2 : 3 (b) 3 : 2 (c) 3 : 4 (d) 4 : 3

**Q.38.** The weight of a circular disc varies directly as the product of the square of



the radius and its thickness. Two identical/similar discs have their thickness in the ratio of 16 : 9. What is the ratio of their radii if the weight of the first is four times that of the second ?

- (a) 3 : 2 (b) 2 : 3 (c) 1 : 2 (d) 2 : 1

**Q.39.** A hemispherical dome of a building needs to be painted. If the circumference of the base of the dome is 154 cm, then find the cost of painting it if the cost of painting is ₹4 per 100 cm<sup>2</sup> (use  $\pi = \frac{22}{7}$ ).

- (a) ₹150.92 (b) ₹ 150.66  
(c) ₹ 105.29 (d) ₹ 105.66

**Q.40.** The area of a sector of a circle is 616 cm<sup>2</sup> with a central angle of 10°. The radius of the circle is \_\_\_\_\_. (use  $\pi = \frac{22}{7}$ )

- (a) 84 cm (b) 21 cm (c) 48 cm (d) 28 cm

**Q.41.** Find the volume of the largest right circular cone that can be cut out from a cube whose edge is  $3\frac{1}{2}$  cm, correct to two places of decimals (use  $\pi = \frac{22}{7}$ ).

- (a) 13.21 cm<sup>3</sup> (b) 21.31 cm<sup>3</sup>  
(c) 11.23 cm<sup>3</sup> (d) 12.13 cm<sup>3</sup>

**Q.42.** The sum of the radius of the base and the height of a cylinder is 42 m. If the total surface area of the cylinder is 6336 m<sup>2</sup>, find the curved surface area of the cylinder correct to two places of decimals (use  $\pi = \frac{22}{7}$ )

- (a) 2157.43 m<sup>2</sup> (b) 2571.43 m<sup>2</sup>  
(c) 2715.43 m<sup>2</sup> (d) 2517.43 m<sup>2</sup>

**Q.43.** Three cubes with sides in the ratio of 3 : 4 : 5 are melted to form a single cube whose diagonal is  $18\sqrt{3}$  cm. The sides of the three cubes are:

- (a) 21 cm, 28 cm and 35 cm  
(b) 9 cm, 12 cm and 15 cm  
(c) 18 cm, 24 cm and 30 cm  
(d) 12 cm, 16 cm and 20 cm

**Q.44.** The perimeter of equilateral triangle is  $3 \times (\sqrt[4]{\frac{16}{3}})$  units. Determine the area of the triangle.

- (a) 4 unit<sup>2</sup> (b) 1 unit<sup>2</sup> (c) 3 unit<sup>2</sup> (d) 2 unit<sup>2</sup>

**Q.45.** A spherical ball of diameter 8 cm is cut into two equal parts. The curved area of one such part has to be painted with green colour, while the other part has to be painted with red colour. The bases of both the hemispheres are to be painted with blue colour. The cost of painting with blue is ₹2/cm<sup>2</sup>, while the cost of painting the curved area is

₹3/cm<sup>2</sup>. What will be the cost (in ₹) of painting the hemispheres? Take  $\pi = 3.14$

- (a) ₹451.92 (b) ₹492.92  
(c) ₹803.84 (d) ₹401.92

**Q.46.** A cube of side 5 cm, has been assembled using 125 cubes each of 1 cm edge. One cube is removed from the middle of each of the faces of the cube. What is the surface area (in cm<sup>2</sup>) of the remaining solid?

- (a) 155 (b) 144 (c) 150 (d) 174

**Q.47.** A prism and a pyramid have the same base and the same height. Find the ratio of the volumes of the prism and the pyramid.

- (a) 2 : 3 (b) 3 : 1 (c) 1 : 3 (d) 3 : 2

**Q.48.** How many metres of 2 m wide cloth will be required to make a conical tent with the diameter of the base as 14 m and slant height as 9 m to ignore wastage ?

- (a) 66 m (b) 88 m (c) 99 m (d) 77 m

**Q.49.** The minute hand of a clock is 20 cm long. Find the area on the face of the clock swept by the minute hand between 8 a.m. and 8:45 a.m.

- (a)  $\frac{6600}{7}$  cm<sup>2</sup> (b)  $\frac{6600}{9}$  cm<sup>2</sup>  
(c)  $\frac{6600}{18}$  cm<sup>2</sup> (d)  $\frac{6600}{14}$  cm<sup>2</sup>

**Q.50.** The external diameter of an iron pipe is 20 cm and its length is 12 cm. If the thickness of the pipe is 1 cm, find the surface area of the pipe (take  $\pi = \frac{22}{7}$ )

correct to two places of decimal.

- (a) 1,662.67 cm<sup>2</sup> (b) 1,552.57 cm<sup>2</sup>  
(c) 1,442.48 cm<sup>2</sup> (d) 1,772.76 cm<sup>2</sup>

**Q.51.** If the length of a certain rectangle is decreased by 4 cm and breadth is increased by 2 cm, it would result in a square of the same area. What is the perimeter of the original rectangle ?

- (a) 15 cm (b) 24 cm (c) 20 cm (d) 10 cm

**Q.52.** Two identical circles each of radius 30 cm intersect each other such that the circumference of each one passes through the center of the other. What is the area of the intersecting region?

- (a)  $400\pi - 250\sqrt{3}$  cm<sup>2</sup>  
(b)  $300\pi - 150\sqrt{3}$  cm<sup>2</sup>  
(c)  $500\pi - 350\sqrt{3}$  cm<sup>2</sup>  
(d)  $600\pi - 450\sqrt{3}$  cm<sup>2</sup>

**Q.53.** If the sum of the diagonals of a rhombus is L and the perimeter is 4P, find the area of the rhombus?

- (a)  $\frac{1}{4}(L^2 - P^2)$  (b)  $\frac{1}{4}(L^2 - 4P^2)$

- (c)  $\frac{1}{2}(L^2 - 4P^2)$  (d)  $\frac{1}{4}(L^2 + 3P^2)$

**Q.54.** Water is flowing through a cylindrical canal with an internal diameter of 7 m at the speed of 18 km/h. Find the volume of water flowing through the canal in 30 minutes. (take  $\pi = \frac{22}{7}$ )

- (a) 3,76,500 m<sup>3</sup> (b) 3,56,500 m<sup>3</sup>  
(c) 3,66,500 m<sup>3</sup> (d) 3,46,500 m<sup>3</sup>

**Q.55.** ABCDEF is a regular hexagon. Side of the hexagon is 36 cm. What is the area of the triangle ABC ?

- (a)  $324\sqrt{3}$  cm<sup>2</sup> (b)  $360\sqrt{3}$  cm<sup>2</sup>  
(c)  $240\sqrt{3}$  cm<sup>2</sup> (d)  $192\sqrt{3}$  cm<sup>2</sup>

**Q.56.** The frustum of a right circular cone has the radius of the base as 5 cm, radius of the top as 3 cm, and height as 6 cm. What is its volume ?

- (a)  $98\pi$  cm<sup>3</sup> (b)  $100\pi$  cm<sup>3</sup>  
(c)  $96\pi$  cm<sup>3</sup> (d)  $90\pi$  cm<sup>3</sup>

**Q.57.** Three sides of a triangle are  $\sqrt{a^2 + b^2}$ ,  $\sqrt{(2a)^2 + b^2}$  and  $\sqrt{a^2 + (2b)^2}$  units. What is the area (in unit squares) of a triangle?

- (a)  $\frac{5}{2} ab$  (b)  $3ab$  (c)  $4ab$  (d)  $\frac{3}{2} ab$

**Q.58.** A square park has been divided into two rectangles of equal area. If the perimeter of each of these rectangles is 39 m, then what will be the perimeter of the square park ?

- (a) 104 m (b) 39 m (c) 78 m (d) 52 m

**Q.59.** The circumference of a circle is 'at' units and the area of the circle is 'bπ' square units. If a:b is equal to 4:5, then the radius of the circle is:

- (a) 3 cm (b) 2.5 cm (c) 5 cm (d) 2 cm

**Q.60.** The area of a triangular field whose sides are 65m, 72m and 97m is equal to the area of a rectangular park whose sides are in the ratio of 5 : 13. What is the perimeter (in m) of the rectangular park?

- (a) 108 (b) 180 (c) 216 (d) 144

**Q.61.** A bicycle wheel has a radius of 42 cm. It makes 40 revolutions in 25 seconds. What is its speed (in kmph, up to one decimal place) ?

- (a) 3.5 (b) 11.6 (c) 15.2 (d) 9.5

**Q.62.** A wire is bent to form a square of area 169 cm<sup>2</sup>. If the same wire is bent to form a circle, then what is its area in cm<sup>2</sup> (to the nearest whole number) ?

- (a) 215 (b) 227 (c) 532 (d) 531

**Q.63.** A sector of radius 10.5 cm with the central angle 120° is folded to form a



cone by joining the two bounding radii of the sector. What is the volume (in  $cm^3$ ) of the cone so formed ?

- (a)  $\frac{343\sqrt{2}}{6} \pi$  (b)  $\frac{343\sqrt{3}}{6} \pi$   
(c)  $\frac{343\sqrt{3}}{12} \pi$  (d)  $\frac{343\sqrt{2}}{12} \pi$

**Q.64.** The base of a right prism is a triangle with sides 20 cm, 21 cm and 29 cm. If its volume is  $7560cm^3$ , then its lateral surface area (in  $cm^2$ ) is :

- (a) 2484 (b) 2556 (c) 2520 (d) 2448

**Q.65.** If the radius of a sphere is increased by 4 cm, its surface area is increased by  $464\pi cm^2$ . What is the volume (in  $cm^3$ ) of the original sphere ?

- (a)  $\frac{15625}{6} \pi$  (b)  $\frac{35937}{8} \pi$   
(c)  $\frac{11979}{2} \pi$  (d)  $\frac{15625}{8} \pi$

**Q.66.** The base of right prism is a trapezium whose parallel sides are 11 cm and 15cm and the distance between them is 9 cm. If the volume of the prism is  $1731.6 cm^3$ , then the height (in cm) of the prism will be :

- (a) 15.6 (b) 15.2 (c) 14.8 (d) 14.2

**Q.67.** A ribbon of equal width and length  $l$  is wrapped around the curved surface of a right circular cylinder to cover it completely. If the base circumference of this cylinder is  $c$ , then how many times the ribbon was wrapped around the cylinder ?

- (a)  $\frac{l}{4c}$  (b)  $\frac{l}{c}$  (c)  $\frac{l}{2c}$  (d)  $\frac{2l}{c}$

**Q.68.** ABCD is a rhombus with each side being equal to 8 cm. If  $BD = 10$  cm,  $AC = 2\sqrt{x}$  cm, what is the value of  $\sqrt{x + 10}$ ?

- (a)  $2\sqrt{3}$  (b)  $3\sqrt{2}$  (c) 7 (d) 5

**Q.69.** The two diagonals of a rhombus are respectively, 14 cm and 48 cm. The perimeter of the rhombus is equal to :

- (a) 120cm (b) 160cm (c) 80cm (d) 100cm

**Q.70.** The area of a sector of a circle with central angle  $60^\circ$  is A. The circumference of the circle is C. Then A is equal to :

- (a)  $\frac{c^2}{6\pi}$  (b)  $\frac{c^2}{18\pi}$  (c)  $\frac{c^2}{24\pi}$  (d)  $\frac{c^2}{4\pi}$

**Q.71.** Equilateral triangles are drawn on the hypotenuse and one of the perpendicular sides of a right-angled isosceles triangle. Their areas are H and

A respectively.  $\frac{A}{H}$  is equal to :

- (a)  $\frac{1}{4}$  (b)  $\frac{1}{2}$  (c)  $\frac{1}{\sqrt{2}}$  (d)  $\frac{1}{\sqrt[3]{2}}$

**Q.72.** Twelve sticks, each of length one unit, are used to form an equilateral triangle. The area of the triangle is :

- (a)  $3\sqrt{3}$  (b)  $2\sqrt{3}$  (c)  $4\sqrt{3}$  (d)  $8\sqrt{3}$

**Q.73.** If the length of a rectangle is decreased by 11% and the breadth is increased by 11%, its area will undergo :

- (a) 13.13% increase (b) 1.21% increase  
(c) 1.21% decrease (d) 13.13% decrease

**Q.74.** In triangle ABC, the length of BC is less than twice the length of AB by 2 cm. The length of AC exceeds the length of AB by 10 cm. The perimeter is 32cm. The length (in cm) of the smallest side of the triangle is :

- (a) 4 (b) 10 (c) 8 (d) 6

**Q.75.** The parallel sides of a trapezium are 20 cm and 10 cm and its non-parallel sides are equal to each other. If its area is  $180 cm^2$ , then what is the length (in cm) of each non parallel side?

- (a) 11 (b) 13 (c) 12 (d) 15

**Q.76.** The length of a rectangular park is 20m more than its breadth. If the cost of fencing the park at Rs. 53 per metre is Rs. 21,200, then what is the area (In square metres) of the park ?

- (a) 9504 (b) 8925 (c) 9240 (d) 9900

**Q.77.** A wire is in the shape of a rectangle whose sides are in the ratio 7 : 4. It was initially in the shape of a circle of radius, very nearly equal to 31.5 cm. The length of smaller side of the rectangle is : (Take  $\pi = \frac{22}{7}$ )

- (a) 44 cm (b) 36 cm (c) 40 cm (d) 32cm

**Q.78.** A circle circumscribes a rectangle whose sides are in the ratio 4 : 3. If the perimeter of the rectangle is 56 cm, then what is the area (In square cm) of the circle ?

- (a)  $70\pi$  (b)  $96\pi$  (c)  $90\pi$  (d)  $100\pi$

**Q.79.** The curved surface area and volume of a cylindrical pole are 132 square metres and 528 cubic metres, respectively. What is the height (In m) of

the pole ? (Take  $\pi = \frac{22}{7}$ )

- (a)  $2\frac{1}{2}$  (b)  $3\frac{5}{8}$  (c)  $3\frac{1}{2}$  (d)  $2\frac{5}{8}$

**Q.80.** The area of a field in the shape of a hexagon is  $2400\sqrt{3}$  square meters. What will be the cost(Rs.) of fencing it at Rs. 18.50 per meter?

- (a) 4440 (b) 5920 (c) 5550 (d) 5180

**Q.81.** Six cubes, each of edge 2 cm, are joined end to end. What is the total surface area of the resulting cuboid

- (a)  $96cm^2$  (b)  $144cm^2$   
(c)  $104cm^2$  (d)  $128cm^2$

**Q.82.** The radius of a sphere is reduced by 40%. By what percent, will its volume decrease?

- (a) 60% (b) 64% (c) 72.5% (d) 78.4%

**Q.83.** The area of a field in the shape of a triangle with each side x meter is equal to the area of another triangular field having sides 50m, 70m and 80m. The value of x is closest to :

- (a) 65.5 (b) 63.2 (c) 62.4 (d) 61.8

**Q.84.** A sector is cut out from a circle of diameter 42 cm. If the angle of the sector is  $150^\circ$ , then its area ( in square cm ) is :

(Take  $\pi = \frac{22}{7}$ )

- (a) 564 (b) 574 (c) 580.6 (d) 577.5

**Q.85.** The radii of three concentric circles are in the ratio of 4:5:7. What is the ratio of the area between the two inner circles to that between the two outer circles ?

- (a) 4 : 7 (b) 5 : 9 (c) 4 : 5 (d) 3 : 8

**Q.86.** The length of the metallic pipe is 7.56 m. Its external and internal radii are 2.5 cm and 1.5 cm respectively. If 1 cubic cm of the metal weigh 7.5 g, then the weight(kg) of the pipe is

- (a) 72.82 (b) 70.14 (c) 71.28 (d) 69.68

**Q.87.** The volume of a metallic cylindrical pipe is  $7480 cm^3$ . If its length is 1.4 m and its external radius is 9 cm, then its thickness (given  $\pi = \frac{22}{7}$ ) is:

- (a) 1 cm (b) 0.8 cm (c) 0.9 cm (d) 1.2cm

**Q.88.** The areas of the three adjacent faces of a cuboid are  $32cm^2$ ,  $24cm^2$  and  $48cm^2$ . What is the volume (in  $cm^3$ ) of the cuboid ?

- (a) 192 (b) 256 (c) 288 (d) 128

**Q.89.** The area of a triangle is 15 sq cm and the radius of its incircle is 3 cm. Its perimeter is equal to :

- (a) 12 cm (b) 20 cm (c) 5 cm (d) 10 cm

**Q.90.** How much iron sheet (in  $m^2$ ) will be needed to construct a rectangular tank measuring  $10 m \times 8 m \times 6 m$ , if a circular opening of radius one meter is to be left at the top of the tank? (correct to one decimal place)

- (a) 371.6 (b) 370.4 (c) 372.9 (d) 370.8

**Q.91.** A solid cube of volume  $13824 cm^3$  is cut into 8 cubes of equal volumes. The ratio of the surface area of the original cube to the sum of the surface areas of three of the smaller cubes is :



(a) 2 : 3 (b) 4 : 3 (c) 8 : 3 (d) 2 : 1

**Q.92.** The unequal side of an isosceles triangle is 2cm. The medians drawn to the equal sides are perpendicular. The area of the triangle is :

(a)  $2\text{cm}^2$  (b)  $3\text{cm}^2$  (c)  $5\text{cm}^2$  (d)  $1\text{cm}^2$

**Q.93.** A steel vessel has a base of length 60 cm and breadth 30 cm. Water is poured in the vessel. A cubical steel box having an edge of 30 cm is immersed completely in the vessel. By how much will the water rise?

(a) 12 cm (b) 15 cm (c) 10 cm (d) 9 cm

**Q.94.** Original breadth of a rectangular box is 20cm. The box was then remade in such a way that its length increased by 30% but the breadth decreased by 20% and the area increased by  $100\text{cm}^2$ . What is the new area(in  $\text{cm}^2$ ) of the box ?

(a) 2500 (b) 2200 (c) 2600 (d) 2400

**Q.95.** Find the cost of carpeting a room which is 11m long and 6m broad by a carpet which is 60cm broad at the rate of rs 112.50 per meter.

(a) 12,375 (b) 13,280

(c) 11,695 (d) 12,040

**Q.96.** A field is  $119\text{m} \times 18\text{m}$  in dimension. A tank  $17\text{m} \times 6\text{m} \times 3\text{m}$  is dug out in the middle and the soil removed is evenly spread over the remaining part of the field. The increase in the level on the remaining part of the field is: (in cm)

(a) 14 cm (b) 13 cm (c) 15 cm (d) 12 cm

**Q.97.** A swimming pool is 40 m in length, 30 m in breadth and 2.2 m in depth. The cost of cementing its floor and the four sides at Rs. 25/ $\text{m}^2$  is:

(a) Rs.43,980 (b) Rs.37,540

(c) Rs.34,260 (d) Rs.37,700

**Q.98.** A square cardboard with side 3 m is folded through one of its diagonal to make a triangle, the height(m) of the triangle is :

(a)  $\frac{3}{\sqrt{2}}$  (b)  $2\sqrt{3}$  (c)  $3\sqrt{2}$  (d)  $\frac{2}{\sqrt{3}}$

**Q.99.** If the area of a regular hexagon is  $108\sqrt{3}\text{cm}^2$ , its perimeter (in cm) is:

(a)  $36\sqrt{2}$  (b)  $42\sqrt{3}$  (c)  $28\sqrt{3}$  (d) 24

**Q.100.** A square piece of cardboard with side 12 cm has a small square of 2 cm cut out from each of the corners. The resulting flaps are turned up to make a box 2 cm deep. The volume (in  $\text{cm}^3$ ) of the box is:

(a) 128 (b) 94 (c) 102 (d) 11

**Q.101.** The area (in  $\text{cm}^2$ ) of a right angled triangle having base 24 cm and hypotenuse 25 cm is :

(a) 72 (b) 108 (c) 92 (d) 84

**Q.102.** The length of the longest pole that can be placed in a room 16 m long, 8 m wide and 11 m high is:

(a) 21 m (b) 20 m (c) 18 m (d) 19 m

**Q.103.** If the height of an equilateral triangle is  $10\sqrt{3}$  cm, the area (in  $\text{cm}^2$ ) is:

(a)  $124\sqrt{3}$  (b)  $75\sqrt{3}$  (c)  $80\sqrt{3}$  (d)  $100\sqrt{3}$

**Q.104.** The dimensions of a swimming pool are  $66\text{m} \times 35\text{m} \times 3\text{m}$ . How many hours will it take to fill the pool by a pipe of diameter 35 cm with water flowing at speed 8 m/s ?

(a) 2.75 h (b) 3.5 h (c) 2.5 h (d) 3.2 h

**Q.105.** Which of the following has the maximum number of vertices ?

(a) Cuboid

(b) Triangular Prism

(c) Hexagonal pyramid

(d) Tetrahedron

**Q.106.** A diagonal of a quadrilateral is 40 cm. The length of the perpendicular to the opposite ends is 7.5 cm and 8.6 cm. What is the area (in  $\text{cm}^2$ ) of quadrilateral?

(a) 434 (b) 322 (c) 368 (d) 28

**Q.107.** The area of each square of a chessboard having 64 squares is  $4\text{cm}^2$ . If there is a border on all the sides of the chessboard of 2 cm, then the perimeter of the chessboard is:

(a) 128cm (b) 80cm (c) 70cm (d) 256cm

**Q.108.** The sides of a triangle are 24cm, 45cm and 51cm. At each of its vertices, circles of radius 10.5cm are drawn. What is the area of the triangle, excluding the portion covered by sectors of the circles?

(a)  $244.75\text{cm}^2$  (b)  $366.75\text{cm}^2$

(c)  $464.75\text{cm}^2$  (d)  $327.75\text{cm}^2$

**Q.109.** The radius of a cylinder is increased by 150 % and its height is decreased by 20 %. What is the percentage increase in its volume?

(a) 400% (b) 600% (c) 500% (d) 80%

## Examination wise Questions

### SSC CGL 2025 Tier - 1

**Q.110.** A circular pizza having a radius of 21 cm. If 80% of it is eaten, what is the area of pizza remaining ?

SSC CGL 12/09/2025 (Shift 1)

(a)  $240.42\text{cm}^2$  (b)  $356.2\text{cm}^2$

(c)  $277.2\text{cm}^2$  (d)  $266.2\text{cm}^2$

**Q.111.** A ring-shaped disc has outer radius 10 cm and inner radius 8 cm. What is the ratio of the ring's area to the whole outer circle ?

SSC CGL 12/09/2025 (Shift 1)

(a) 9 : 25 (b) 6 : 25 (c) 9 : 20 (d) 3 : 15

**Q.112.** A truck wheel having a radius of 42 cm. What percentage of its circumference is covered in a quarter turn ? (Use  $\pi = \frac{22}{7}$ )

SSC CGL 12/09/2025 (Shift 1)

(a) 35% (b) 25% (c) 40% (d) 45%

**Q.113.** A circular park having a radius of 15 m. If a 1.5 m wide path is built around it, what is the approximate area of the path?

SSC CGL 12/09/2025 (Shift 2)

(a)  $148.5\text{m}^2$  (b)  $158.5\text{m}^2$

(c)  $138.5\text{m}^2$  (d)  $118.5\text{m}^2$

**Q.114.** A right circular cone has a diameter of 14 cm and a height of 24 cm. What is the slant height ?

SSC CGL 12/09/2025 (Shift 3)

(a) 25 cm (b) 30 cm (c) 35 cm (d) 40 cm

**Q.115.** A triangular prism has base area of  $25\text{cm}^2$ . If height is increased by 25 %, what is the new volume (original height was 16 cm) ?

SSC CGL 12/09/2025 (Shift 3)

(a)  $550\text{cm}^3$  (b)  $575\text{cm}^3$

(c)  $500\text{cm}^3$  (d)  $225\text{cm}^3$

**Q.116.** A circular disc's radius is decreased by 20% What is the percentage decrease in its area ?

SSC CGL 12/09/2025 (Shift 3)

(a) 20% (b) 38% (c) 36% (d) 30%

**Q.117.** A man has 160 m of fencing wire. He can use this wire to enclose a circular garden or a square garden. If he wants to maximize the enclosed area, what is the approximate ratio of the area of the largest possible circular garden to the area of the largest possible square garden, using the entire 160 m of wire ?

SSC CGL 12/09/2025 (Shift 3)

(a) 1.3 : 1 (b) 1.27 : 1 (c) 1.7 : 1 (d) 1.6 : 1

**Q.118.** A vertical cylindrical container is filled with oil. A solid hemispherical stone of radius 6 cm is immersed completely, and the oil level rises by 2.5 cm. What is the radius of the cylinder ?

SSC CGL 12/09/2025 (Shift 3)

(a) 6.29 cm. (b) 7.59 cm

(c) 8.79 cm (d) 11.41 cm



**Q.119.** A sector of a circle has a central angle of  $135^\circ$  and a radius of 9 cm. Another sector of the same circle has a central angle of  $\frac{3\pi}{4}$  radians. What is the ratio of the area of the first sector to the area of the second sector?

SSC CGL 12/09/2025 (Shift 3)

- (a) 4 : 3 (b) 5 : 4 (c) 1 : 1 (d) 6 : 7

**Q.120.** The area of a sector of a circle with radius 21 cm and central angle  $120^\circ$  is:

SSC CGL 13/09/2025 (Shift 1)

- (a)  $693 \text{ cm}^2$  (b)  $715 \text{ cm}^2$   
(c)  $645 \text{ cm}^2$  (d)  $750 \text{ cm}^2$

**Q.121.** What is the area of a sector with a central angle of  $60^\circ$  and radius 4 cm?

SSC CGL 13/09/2025 (Shift 1)

- (a)  $\frac{8\pi}{3} \text{ cm}^2$  (b)  $\frac{5\pi}{3} \text{ cm}^2$   
(c)  $\frac{\pi}{4} \text{ cm}^2$  (d)  $\frac{5\pi}{6} \text{ cm}^2$

**Q.122.** A ring-shaped metallic sheet has outer and inner radii 14 cm and 7 cm. What percentage of the full circle's area is the ring?

SSC CGL 13/09/2025 (Shift 2)

- (a) 36% (b) 48% (c) 64% (d) 75%

**Q.123.** A circular pond is surrounded by a 2 m wide path. If the total area (pond + path) is  $804.25 \text{ m}^2$  and the area of the pond alone is  $615.75 \text{ m}^2$ , what is the radius (approximately) of the pond?

SSC CGL 13/09/2025 (Shift 2)

- (a) 10 m (b) 14 m (c) 12 m (d) 11 m

**Q.124.** The ratio of the circumferences of two circular flower beds is 5 : 4. If the smaller flower bed has a radius of 10 m, what is the radius of the larger flower bed?

SSC CGL 13/09/2025 (Shift 2)

- (a) 12.5 m (b) 7 m  
(c) 9 m (d) 14.5 m

**Q.125.** A cylinder and a cone have the same base radius and volume. Find the ratio of the height of the cylinder to the height of the cone.

SSC CGL 13/09/2025 (Shift 3)

- (a) 1 : 3 (b) 3 : 1 (c) 2 : 1 (d) 1 : 2

**Q.126.** Two circular ponds have circumferences in the ratio 6 : 10. If the smaller pond has an area of  $160 \text{ m}^2$ , what is the approximate area of the larger pond?

SSC CGL 13/09/2025 (Shift 3)

- (a)  $444 \text{ m}^2$  (b)  $450 \text{ m}^2$   
(c)  $438 \text{ m}^2$  (d)  $449 \text{ m}^2$

**Q.127.** If the base of a prism is a regular hexagon of side 3 cm and the height is 8 cm, what is its volume?

SSC CGL 13/09/2025 (Shift 3)

- (a)  $106\sqrt{3} \text{ cm}^3$  (b)  $104\sqrt{3} \text{ cm}^3$   
(c)  $107\sqrt{3} \text{ cm}^3$  (d)  $108\sqrt{3} \text{ cm}^3$

**Q.128.** Three concentric circles are designed as part of a target practice board. Their diameters form an arithmetic progression. If the smallest circle has a circumference of  $8\pi \text{ cm}$  and the largest circle has a circumference of  $16\pi \text{ cm}$ , what is the circumference of the middle circle?

SSC CGL 13/09/2025 (Shift 3)

- (a)  $14\pi \text{ cm}$  (b)  $12\pi \text{ cm}$   
(c)  $10\pi \text{ cm}$  (d)  $11\pi \text{ cm}$

**Q.129.** A sector has a central angle of  $225^\circ$  and a radius of 6 cm. Another sector of the same circle has a central angle of  $\frac{5\pi}{4}$  radians. What is the ratio of the area of the first sector to the area of the second sector?

SSC CGL 13/09/2025 (Shift 3)

- (a) 5 : 7 (b) 1 : 1 (c) 4 : 3 (d) 3 : 7

**Q.130.** A circular garden having a diameter of 20 m is surrounded by a concrete path that is 2 m wide. Calculate the percentage increase in area resulting from the addition of the path.

SSC CGL 14/09/2025 (Shift 1)

- (a) 24% (b) 20% (c) 44% (d) 54%

**Q.131.** A sector of a circle having a radius 14 cm has area  $77\text{cm}^2$ . Find the angle of the sector.

SSC CGL 14/09/2025 (Shift 1)

- (a)  $90^\circ$  (b)  $45^\circ$  (c)  $60^\circ$  (d)  $180^\circ$

**Q.132.** A circular signboard has a radius of 4m. If painting costs ₹30 per  $\text{m}^2$  and 80 % of the board painted, what is the total painting cost? [use  $\pi = 3.14$ ]

SSC CGL 14/09/2025 (Shift 1)

- (a) ₹1658.2 (b) ₹1205.7  
(c) ₹1206.7 (d) ₹1681.2

**Q.133.** A cone is such that the area of its base equals its lateral surface area. If slant height is  $l$ , find the radius  $r$ .

SSC CGL 14/09/2025 (Shift 2)

- (a)  $l$  (b)  $2l$  (c)  $3l$  (d)  $4l$

**Q.134.** A square field of side 24 m has a circular pond in the center. If the area of the pond is  $484 \text{ m}^2$ , what is the area of the remaining field?

SSC CGL 14/09/2025 (Shift 2)

- (a)  $48 \text{ m}^2$  (b)  $92 \text{ m}^2$  (c)  $78 \text{ m}^2$  (d)  $82 \text{ m}^2$

**Q.135.** If the height of a right prism is increased by 60% and base area remains

the same, what is the percentage increase in volume?

SSC CGL 14/09/2025 (Shift 2)

- (a) 25% (b) 60% (c) 75% (d) 100%

**Q.136.** A circular decorative wall clock with a radius of 30 cm has a design where a chord connects two points on the circumference. This chord, along with two radii, forms an equilateral triangle at the center of the clock face. The smaller segment created by this chord is painted in a contrasting color. What percentage of the clock's total area does this smaller painted segment represent?

(Use  $\pi \approx 3.14$ ,  $\sqrt{3} \approx 1.732$ )

SSC CGL 14/09/2025 (Shift 2)

- (a) 2.88% (b) 5.6% (c) 9.08% (d) 12.5%

**Q.137.** A cylinder ( $r = 8 \text{ cm}$ ,  $h = 20 \text{ cm}$ ) is bored through by a hole ( $r = 4 \text{ cm}$ , full height). What percentage of original volume is removed?

SSC CGL 14/09/2025 (Shift 2)

- (a) 25% (b) 50% (c) 75% (d) 37.5%

**Q.138.** A sector of a circle with a radius of 12 cm has a central angle of  $60^\circ$ . Another sector of the same circle has a central angle of  $\pi/3$  radians. What is the ratio of the area of the first sector to the area of the second sector?

SSC CGL 14/09/2025 (Shift 2)

- (a) 1 : 2 (b) 2 : 3 (c) 1 : 1 (d) 3 : 4

**Q.139.** A circular path of 3 m width runs around a circular park with radius 7 m. What is the area of the path?

SSC CGL 14/09/2025 (Shift 3)

- (a)  $160.28 \text{ m}^2$  (b)  $161.29 \text{ m}^2$   
(c)  $158.02 \text{ m}^2$  (d)  $164.05 \text{ m}^2$

**Q.140.** A sector of a circle having a radius of 5 cm and has a central angle of  $\frac{\pi}{3}$  radians. What is the area of the sector?

SSC CGL 14/09/2025 (Shift 3)

- (a)  $13.09 \text{ cm}^2$  (b)  $8.33 \text{ cm}^2$   
(c)  $10.47 \text{ cm}^2$  (d)  $12.5 \text{ cm}^2$

**Q.141.** A metal sphere having a radius of 20 centimeters is melted down and molded into 64 identical smaller solid spheres. What is the ratio of the surface area of the original sphere to the total surface area of all 64 smaller spheres?

SSC CGL 15/09/2025 (Shift 1)

- (a) 1 : 2 (b) 2 : 1 (c) 1 : 4 (d) 1 : 1

**Q.142.** How many hemispheres with a radius of 6 cm can be produced by melting down a hemisphere with a radius of 24 cm?

SSC CGL 15/09/2025 (Shift 1)



(a) 32 (b) 64 (c) 12 (d) 54

**Q.143.** A cylinder and hemisphere have the same radius. Their combined height is 35 cm. If the cylinder and hemisphere have equal volumes, find the radius.

SSC CGL 15/09/2025 (Shift 1)

(a) 14 cm (b) 13 cm (c) 10 cm (d) 21 cm

**Q.144.** A pyramid has a base area of 120 cm<sup>2</sup> and height 18 cm. What is its volume ?

SSC CGL 15/09/2025 (Shift 1)

(a) 720 cm<sup>3</sup> (b) 780 cm<sup>3</sup>  
(c) 540 cm<sup>3</sup> (d) 460 cm<sup>3</sup>

**Q.145.** A right triangle has sides 6, 8, and 10. A smaller triangle is drawn inside it with its vertices on the sides of the larger triangle, such that it is similar to the larger triangle. If its perimeter is 12, what is its area ?

SSC CGL 15/09/2025 (Shift 1)

(a) 3.5 (b) 2.2 (c) 3 (d) 6

**Q.146.** What is the area of the segment formed by a chord in a circle of radius 14 cm, if the angle subtended at the center is 90° ?

SSC CGL 15/09/2025 (Shift 1)

(a)  $84\pi - 49$  (b)  $49\pi - 98$   
(c)  $60\pi - 72$  (d)  $36\pi - 72$ 

**Q.147.** A right circular cone has a radius of 8 cm and a height of 15 cm. A sphere is placed inside the cone touching the base and the slanted surface. Find the radius of this inscribed sphere.

SSC CGL 15/09/2025 (Shift 2)

(a) 5.2 cm (b) 4.8 cm  
(c) 3.8 cm (d) 6.2 cm

**Q.148.** If the perimeter of a regular octagon is 160 cm, what is the side length ?

SSC CGL 15/09/2025 (Shift 2)

(a) 20 cm (b) 16 cm (c) 24 cm (d) 12 cm

**Q.149.** A circular wheel is divided into 4 equal sectors. If the area of one sector is 90 cm<sup>2</sup>, what is the radius of the wheel ?

SSC CGL 15/09/2025 (Shift 2)

(a) 8.6 cm (b) 10.7 cm  
(c) 6.5 cm (d) 7.5 cm

**Q.150.** If the lateral surface area of a prism is 320 cm<sup>2</sup> and the height is 16 cm, what is the perimeter of the base ?

SSC CGL 15/09/2025 (Shift 2)

(a) 20 cm (b) 24 cm (c) 36 cm (d) 40 cm

**Q.151.** A sector of a circle with radius 8 cm has a central angle of 45°. What is the area of the corresponding segment ?

SSC CGL 15/09/2025 (Shift 2)

(a)  $(8\pi - 9\sqrt{2})$  sq. cm(b)  $(8\pi - 12\sqrt{2})$  sq. cm(c)  $(8\pi - 16\sqrt{2})$  sq. cm(d)  $(8\pi - 18\sqrt{2})$  sq. cm

**Q.152.** The cost of building a wall around a circular field at the rate of ₹120 per metre is ₹3,768. Find the diameter of the field.

SSC CGL 15/09/2025 (Shift 3)

(a) 5 m (b) 10 m (c) 15 m (d) 20 m

**Q.153.** A circular logo of radius 7 cm is fitted perfectly inside a square plaque. Find the area of unused square space around the circle.

SSC CGL 15/09/2025 (Shift 3)

(a) 42 cm<sup>2</sup> (b) 48 cm<sup>2</sup>  
(c) 50.54 cm<sup>2</sup> (d) 55 cm<sup>2</sup>

**Q.154.** A circular clock has a radius of 28 cm. Calculate the distance covered by the tip of the minute hand in 30 minutes, assuming it is positioned at the edge of the clock face ? (Use  $\pi = \frac{22}{7}$ )

SSC CGL 15/09/2025 (Shift 3)

(a) 88 cm (b) 84 cm (c) 82 cm (d) 44 cm

**Q.155.** If the radius of a hemisphere is increased to twice its original size, what is the ratio of the new surface area to the original surface area, as well as the proportion of the new volume to the original volume ?

SSC CGL 16/09/2025 (Shift 1)

(a) 4 : 1 and 8 : 1 (b) 6 : 1 and 8 : 1  
(c) 8 : 1 and 2 : 1 (d) 2 : 1 and 4 : 1

**Q.156.** A hemisphere and a cone share the same base and have equal volumes. Given that their common radius is R, determine the height of the cone.

SSC CGL 16/09/2025 (Shift 1)

(a) R (b) 2R (c) 5R (d) 3R

**Q.157.** A solid sphere of radius 8 cm is melted into four cones of equal height. If each cone's radius is 4 cm, what is the height of each cone ?

SSC CGL 16/09/2025 (Shift 1)

(a) 40 cm (b) 32 cm (c) 50 cm (d) 52 cm

**Q.158.** If base perimeter = 30 cm, slant height = 10 cm, what is lateral surface area of square pyramid ?

SSC CGL 16/09/2025 (Shift 1)

(a) 160 cm<sup>2</sup> (b) 200 cm<sup>2</sup>  
(c) 150 cm<sup>2</sup> (d) 152 cm<sup>2</sup>

**Q.159.** Two circular tracks have radii in the ratio 2 : 5. If the smaller track has an area of  $100\pi$  cm<sup>2</sup>, what is the area of the larger track ?

SSC CGL 16/09/2025 (Shift 1)

(a)  $215\pi$  cm<sup>2</sup> (b)  $235\pi$  cm<sup>2</sup>(c)  $625\pi$  cm<sup>2</sup> (d)  $240\pi$  cm<sup>2</sup>

**Q.160.** A sphere is melted and recast into 8 identical cones, each with diameter 6 cm and height 4 cm. What was the radius of the original sphere ?

SSC CGL 16/09/2025 (Shift 2)

(a) 4.16 cm (b) 3.26 cm  
(c) 4.26 cm (d) 5.56 cm

**Q.161.** The area of a regular octagon is made of how many isosceles triangles ?

SSC CGL 16/09/2025 (Shift 2)

(a) 6 (b) 7 (c) 8 (d) 9

**Q.162.** A circular fountain with a diameter of 14 m is encircled by a grass bed that is 3.5 m wide. Determine the ratio of the area of the grass bed to the area of the fountain.

SSC CGL 16/09/2025 (Shift 2)

(a) 5 : 3 (b) 3 : 4 (c) 5 : 4 (d) 4 : 1

**Q.163.** A tent designed as a triangular prism features a triangular base with an area of 30 m<sup>2</sup> and a height of 7 m. What is the volume of this tent ?

SSC CGL 16/09/2025 (Shift 2)

(a) 210 m<sup>3</sup> (b) 200 m<sup>3</sup>  
(c) 220 m<sup>3</sup> (d) 140 m<sup>3</sup>

**Q.164.** A chord in a circle with radius 20 cm subtends an angle of 90° at the center. What is the area of the minor segment ?

SSC CGL 16/09/2025 (Shift 2)

(a)  $(25\pi - 50)$  sq. cm  
(b)  $(100\pi - 150)$  sq. cm  
(c)  $(50\pi - 100)$  sq. cm  
(d)  $(100\pi - 200)$  sq. cm

**Q.165.** A sphere is completely contained within a cylinder, with the height and diameter of the cylinder matching the diameter of the sphere. Given that the volume of the sphere is  $288\pi$  cm<sup>3</sup>, what is the volume of the vacant space inside the cylinder?

SSC CGL 16/09/2025 (Shift 3)

(a)  $122\pi$  cm<sup>3</sup> (b)  $169\pi$  cm<sup>3</sup>  
(c)  $144\pi$  cm<sup>3</sup> (d)  $196\pi$  cm<sup>3</sup>

**Q.166.** A solid hemisphere is melted to form a cone whose radius is half that of the hemisphere. Find the height of the resulting cone.

SSC CGL 16/09/2025 (Shift 3)

(a) 4R (b) 6R (c) 8R (d) 10R

**Q.167.** Two solid hemispheres of radii 5 cm and 9 cm respectively are melted and recast into a single hemisphere. What is the approximate radius of the new hemisphere formed?

SSC CGL 16/09/2025 (Shift 3)

(a) 9.5 cm (b) 10.2 cm



- (c) 10.5 cm (d) 11 cm

**Q.168.** A regular right pyramid has a square base with a side length of 16 cm. Its slant height is 10 cm. What is its volume?

SSC CGL 16/09/2025 (Shift 3)

- (a) 512 cm<sup>3</sup> (b) 640 cm<sup>3</sup>  
(c) 768 cm<sup>3</sup> (d) 800 cm<sup>3</sup>

**Q.169.** The ratio of the radii of two circular tyres is  $1:\sqrt{3}$ . If the area of the smaller tyre is 120 cm<sup>2</sup>, what is the area of the larger tyre?

SSC CGL 16/09/2025 (Shift 3)

- (a) 240 cm<sup>2</sup> (b) 360 cm<sup>2</sup>  
(c) 320 cm<sup>2</sup> (d) 500 cm<sup>2</sup>

**Q.170.** A hollow metallic sphere has outer radius 14 cm and is melted to make 25 smaller solid spheres of radius 4 cm. What is the inner radius of the original sphere?

SSC CGL 17/09/2025 (Shift 1)

- (a) 10.45 cm (b) 17.48 cm  
(c) 15.46 cm (d) 16.44 cm

**Q.171.** The volume of a hemisphere is numerically equal to five times its curved surface area. Find its radius.

SSC CGL 17/09/2025 (Shift 1)

- (a) 13 cm (b) 15 cm (c) 20 cm (d) 11 cm

**Q.172.** What is the minimum number of cuts required to divide a cuboid into 16 equal cuboids?

SSC CGL 17/09/2025 (Shift 1)

- (a) 3 (b) 4 (c) 5 (d) 6

**Q.173.** If the height of a cone is the fourth time and the radius is double, how does the volume change?

SSC CGL 17/09/2025 (Shift 1)

- (a) Doubles  
(b) Becomes sixteen time of the original  
(c) Becomes one-third of the original  
(d) Remains unchanged

**Q.174.** A sector of a circle has a central angle of 90° and a radius of 10 cm. Another sector of the same circle has a central angle of  $\frac{\pi}{2}$  radians. What is the

ratio of the area of the first sector to the area of the second sector?

SSC CGL 17/09/2025 (Shift 1)

- (a) 2 : 3 (b) 1 : 2 (c) 1 : 1 (d) 1 : 3

**Q.175.** A spherical ball is submerged in water in a cylindrical container. The radius of the cylindrical container is 6 cm, and the height is 35 cm. What is the volume of water displaced by the ball if its radius is 6 cm?

SSC CGL 17/09/2025 (Shift 2)

(a)  $\frac{840}{3} \pi \text{ cm}^3$  (b)  $\frac{620}{3} \pi \text{ cm}^3$

(c)  $\frac{760}{3} \pi \text{ cm}^3$  (d)  $\frac{256 \pi}{3} \pi \text{ cm}^3$

**Q.176.** A hemispherical dome covers a circular area of diameter 12 m. What is the approximate curved surface area of this dome?

SSC CGL 17/09/2025 (Shift 2)

- (a) 226m<sup>2</sup> (b) 220m<sup>2</sup> (c) 408m<sup>2</sup> (d) 762m<sup>2</sup>

**Q.177.** A hollow hemisphere has uniform thickness. Its inner radius is r, and outer radius is R. If  $R = 4r$ , find the ratio of outer to inner curved surface areas.

SSC CGL 17/09/2025 (Shift 2)

- (a) 16 : 1 (b) 8 : 5 (c) 6 : 5 (d) 2 : 5

**Q.178.** A square pyramid has its lateral surface area equal to twice the area of its base. If the side of the base is 8 cm, find its slant height.

SSC CGL 17/09/2025 (Shift 2)

- (a) 8 cm (b) 9 cm (c) 5 cm (d) 2 cm

**Q.179.** A circular garden has a diameter of 42 m. If a gardener wants to fence 25% of the garden's circumference, what length of fencing is needed? (Use  $\pi = \frac{22}{7}$ )

SSC CGL 17/09/2025 (Shift 2)

- (a) 28 m (b) 33 m (c) 32 m (d) 31 m

**Q.180.** In an equilateral triangle with side length 'b', what is the ratio of the inradius to the circumradius?

SSC CGL 17/09/2025 (Shift 2)

- (a) 2 : 1 (b) 3 : 1 (c) 4 : 1 (d) 1 : 2

**Q.181.** A cone and a sphere have equal volumes. If the cone's height is thrice its radius, what is the ratio of the cone's radius to the sphere's radius?

SSC CGL 17/09/2025 (Shift 3)

- (a)  $\sqrt[3]{2} : \sqrt[3]{3}$  (b)  $\sqrt[3]{4} : \sqrt[3]{3}$   
(c)  $\sqrt[3]{4} : \sqrt[3]{2}$  (d)  $\sqrt[3]{3} : \sqrt[3]{4}$

**Q.182.** A sphere and cylinder of equal radii have equal total surface areas. Find the ratio of cylinder height to radius.

SSC CGL 17/09/2025 (Shift 3)

- (a) 1 : 1 (b) 7 : 4 (c) 6 : 5 (d) 2 : 5

**Q.183.** A large cubical cake needs to be cut into 27 smaller, identical cubical pieces. What is the minimum number of straight cuts required?

SSC CGL 17/09/2025 (Shift 3)

- (a) 8 (b) 6 (c) 4 (d) 5

**Q.184.** The radius of a cone is raised by 25%, while the height is reduced by 10%. What is the percentage change in the volume of the cone?

SSC CGL 17/09/2025 (Shift 3)

- (a) 27.4% decrease (b) 40.6% increase  
(c) 40.8% increase (d) 51.2% increase

**Q.185.** A sector has a central angle of 180° and a radius of 24 cm. Another

sector has a central angle of  $\frac{\pi}{2}$  radians and the same radius. What is the ratio of the area of the first sector to the area of the second sector?

SSC CGL 17/09/2025 (Shift 3)

- (a) 3 : 2 (b) 2 : 1 (c) 1 : 2 (d) 1 : 3

**Q.186.** The ratios of the perimeters of two similar triangles are expressed as 5 : 7. Together, their areas total 74 cm<sup>2</sup>. What is the area of the larger triangle?

SSC CGL 17/09/2025 (Shift 3)

- (a) 39 cm<sup>2</sup> (b) 45 cm<sup>2</sup>  
(c) 49 cm<sup>2</sup> (d) 35 cm<sup>2</sup>

**Q.187.** Two identical solid hemispheres, each with a radius of 27 cm, are melted and recast into a single sphere. What is the radius of the new sphere formed?

SSC CGL 18/09/2025 (Shift 1)

- (a) 21 cm (b) 27 cm (c) 24 cm (d) 29 cm

**Q.188.** A decorative column is made in the shape of a regular hexagonal prism. The side length of its base is 8 cm. The column is built from multiple sections, and their heights form an arithmetic progression: the first section is 8 cm tall, and each subsequent section is 3 cm taller than the previous one. If the column has 4 sections in total, what is the total volume of the column?

SSC CGL 18/09/2025 (Shift 1)

- (a)  $4200\sqrt{3} \text{ cm}^3$  (b)  $4800\sqrt{3} \text{ cm}^3$   
(c)  $5500\sqrt{3} \text{ cm}^3$  (d)  $4600\sqrt{3} \text{ cm}^3$

**Q.189.** A tile is spherical in shape with a radius of 6 cm. If the tile is placed inside a cube with side length 12 cm, what is the volume of the cube that is not occupied by the tile?

SSC CGL 18/09/2025 (Shift 2)

- (a) 459 cm<sup>3</sup> (b) 798.16 cm<sup>3</sup>  
(c) 350 cm<sup>3</sup> (d) 822.86 cm<sup>3</sup>

**Q.190.** Two hemispheres of radii 9 cm and 10 cm respectively are melted and recast into another hemisphere. What is the approximate total surface area of the newly formed hemisphere?

SSC CGL 18/09/2025 (Shift 2)

- (a) 1633 cm<sup>2</sup> (b) 1425 cm<sup>2</sup>  
(c) 1358 cm<sup>2</sup> (d) 1385 cm<sup>2</sup>

**Q.191.** A cone is sliced into two sections by a plane that is parallel to its base, with the upper section representing  $\frac{1}{27}$  of the overall volume. What is the ratio of the heights of the smaller cone to the



original cone ?

SSC CGL 18/09/2025 (Shift 2)

(a) 1 : 4 (b) 1 : 6 (c) 5 : 1 (d) 1 : 3

**Q.192.** A sector of a circle has a central angle of  $150^\circ$  and a radius of 6 cm. Another sector of the same circle has a central angle of  $180^\circ$ . What is the ratio of the area of the first sector to the area of the second sector ?

SSC CGL 18/09/2025 (Shift 2)

(a) 6 : 5 (b) 3 : 4 (c) 2 : 3 (d) 5 : 6

**Q.193.** The inner and outer diameter of a hollow hemispherical shell are 14 cm and 28 cm respectively. Find the approximate volume of the metal used.

SSC CGL 18/09/2025 (Shift 3)

(a) 4986.25 cm<sup>3</sup> (b) 5030.66 cm<sup>3</sup>  
(c) 4532.66 cm<sup>3</sup> (d) 5530.25 cm<sup>3</sup>

**Q.194.** A right circular cone is divided into two sections by a plane that is parallel to the base, with the volumes of the frustum and the smaller cone having a ratio of 19 : 8. Determine the ratio of their heights.

SSC CGL 18/09/2025 (Shift 3)

(a) 1 : 1 (b) 1 : 2 (c) 5 : 6 (d) 2 : 1

**Q.195.** If the side of a regular hexagon is tripled, what happens to its area ?

SSC CGL 18/09/2025 (Shift 3)

(a) Doubled (b) Tripled  
(c) Quadrupled (d) Increases 9 times

**Q.196.** A cylindrical storage container has a base diameter of 6 cm. It is designed with sections whose heights form an arithmetic progression, starting from 2 cm for the bottom section and reaching 10 cm for the topmost section, covering a total of 5 sections. What is the total volume of the container ?

(Use  $\pi \approx 3.14$ )

SSC CGL 18/09/2025 (Shift 3)

(a) 853.6 cm<sup>3</sup> (b) 847.8 cm<sup>3</sup>  
(c) 842.8 cm<sup>3</sup> (d) 613.4 cm<sup>3</sup>

**Q.197.** In triangle ABC, medians AD, BE, and CF intersect at the Centroid G. If the area of triangle ABC is 96 cm<sup>2</sup>, what is the area of triangle GBC ?

SSC CGL 18/09/2025 (Shift 3)

(a) 16 cm<sup>2</sup> (b) 24 cm<sup>2</sup>  
(c) 48 cm<sup>2</sup> (d) 32 cm<sup>2</sup>

**Q.198.** A hemisphere of radius 8 cm is carved from a sphere. What is the volume of the hemisphere ?

SSC CGL 19/09/2025 (Shift 1)

(a)  $\frac{1024\pi}{3}$  cm<sup>3</sup> (b)  $\frac{1205\pi}{6}$  cm<sup>3</sup>  
(c)  $\frac{1250\pi}{3}$  cm<sup>3</sup> (d)  $\frac{275\pi}{6}$  cm<sup>3</sup>

**Q.199.** A solid Ball is placed inside a cube such that it touches all six faces. What percentage of the cube's volume is occupied by the Ball ?

SSC CGL 19/09/2025 (Shift 1)

(a) 47.67% (b) 48.56%  
(c) 49.28% (d) 52.33%

**Q.200.** The sides of a rectangular prism are in the ratio 2 : 5 : 3. If the volume of the prism is 240 cm<sup>3</sup>, what is the length of its longest side ?

SSC CGL 19/09/2025 (Shift 1)

(a) 12 cm (b) 10 cm (c) 8 cm (d) 16 cm

**Q.201.** A conical bucket is cut at one - fourth its height from the top. What is the ratio of the volumes of the two parts (top part to bottom part) ?

SSC CGL 19/09/2025 (Shift 1)

(a) 1 : 64 (b) 1 : 63 (c) 1 : 27 (d) 1 : 7

**Q.202.** A sector of a circle has a central angle of  $45^\circ$  and a radius of 9 cm. Another sector of the same circle has a central angle of  $\frac{\pi}{4}$  radians. What is the ratio of the area of the first sector to the area of the second sector?

SSC CGL 19/09/2025 (Shift 1)

(a) 1 : 1 (b) 2 : 3 (c) 3 : 5 (d) 5 : 6

**Q.203.** If the curved surface area of a sphere is 154 cm<sup>2</sup>, find its diameter.

SSC CGL 19/09/2025 (Shift 2)

(a) 9 cm (b) 8 cm (c) 7 cm (d) 5 cm

**Q.204.** A cone has height  $h$  and radius  $r$ . The cone is melted and recast into a smaller cone whose height is  $\frac{h}{4}$  and radius is  $\frac{r}{2}$ . What fraction of the original

volume is unused ?

SSC CGL 19/09/2025 (Shift 2)

(a)  $\frac{17}{18}$  (b)  $\frac{15}{16}$  (c)  $\frac{12}{17}$  (d)  $\frac{16}{19}$

**Q.205.** If a regular hexagon and a square have the same side length, what will be the ratio of their areas ?

SSC CGL 19/09/2025 (Shift 2)

(a)  $3\sqrt{3} : 2$  (b)  $2 : 3\sqrt{3}$  (c)  $\sqrt{3} : 1$  (d)  $1 : 6$

**Q.206.** A prism has an equilateral triangular base with a side length of 6 cm. Its height starts at 1 cm and increases by 1 cm for each subsequent layer, forming 5 layers in total. What is the total volume of the prism ?

(Use  $\sqrt{3} = 1.732$ )

SSC CGL 19/09/2025 (Shift 2)

(a) 207.84 cm<sup>3</sup> (b) 233.82 cm<sup>3</sup>  
(c) 259.80 cm<sup>3</sup> (d) 303.10 cm<sup>3</sup>

**Q.207.** A solid cylinder has a radius of 8 cm and a height of 15 cm. A smaller cylindrical hole of radius 4 cm is drilled coaxially through its entire length. What is the approximate volume of the remaining solid ?

SSC CGL 19/09/2025 (Shift 3)

(a) 2825 cm<sup>3</sup> (b) 2785 cm<sup>3</sup>  
(c) 2502 cm<sup>3</sup> (d) 2263 cm<sup>3</sup>

**Q.208.** A tank is in the shape of a rectangular parallelepiped of size 30m × 20m × 5 m. Its capacity in kilolitres is

SSC CGL 19/09/2025 (Shift 3)

(a) 7,000 kilolitres (b) 3,000 kilolitres  
(c) 5,500 kilolitres (d) 3,500 kilolitres

**Q.209.** A solid sphere of radius  $P$  is melted and recast into  $n$  smaller spheres, each of radius  $q$ . Find the value of  $n$ .

SSC CGL 19/09/2025 (Shift 3)

(a)  $(\frac{P}{q})^3$  (b)  $(\frac{3P}{2q})^3$  (c)  $(\frac{q}{3P})^3$  (d)  $(\frac{q}{P})^3$

**Q.210.** A sector has a central angle of  $60^\circ$  and a radius of 18 cm. Another sector has a central angle of  $\frac{2\pi}{3}$  radians. What is the ratio of the area of the first sector to the area of the second sector ?

SSC CGL 19/09/2025 (Shift 3)

(a) 4 : 3 (b) 1 : 2 (c) 7 : 2 (d) 2 : 1

**Q.211.** Three metallic spheres with radii 4 cm, 2 cm, and 7 cm respectively are melted together and recast into a single solid sphere. What is the radius of the new sphere ?

SSC CGL 20/09/2025 (Shift 1)

(a) 12.5 cm (b) 9.5 cm  
(c) 6.5 cm (d) 7.5 cm

**Q.212.** In an urban project, triangular plots with side ratios 8 : 15 : 17 are allocated. If the perimeter is 120 m, find the area.

SSC CGL 20/09/2025 (Shift 1)

(a) 380 m<sup>2</sup> (b) 320 m<sup>2</sup>  
(c) 400 m<sup>2</sup> (d) 540 m<sup>2</sup>

**Q.213.** A prism with a square base of side length 6 cm is constructed in layers. Its height follows an arithmetic progression, increasing from 6 cm in the first layer to 24 cm in the fourth layer. What is the total volume of the prism ?

SSC CGL 20/09/2025 (Shift 1)

(a) 1800 cm<sup>3</sup> (b) 1950 cm<sup>3</sup>  
(c) 1210 cm<sup>3</sup> (d) 2160 cm<sup>3</sup>

**Q.214.** A circle of radius of 14 cm. Two radii are drawn such that the length of the chord connecting their endpoints is 14 cm. What is the area of the minor segment formed ?



SSC CGL 20/09/2025 (Shift 1)

- (a)  $(\frac{98\pi}{3} - 49\sqrt{3})$  sq. cm  
 (b)  $(\frac{98\pi}{3} - 59\sqrt{3})$  sq. cm  
 (c)  $(\frac{78\pi}{3} - 49\sqrt{3})$  sq. cm  
 (d)  $(\frac{68\pi}{3} - 49\sqrt{3})$  sq. cm

**Q.215.** A solid cone is to be painted. If paint covers 3 m<sup>2</sup> per litre and the total surface area is 72 m<sup>2</sup>, how much paint is required, including 10% extra for retouching?

SSC CGL 20/09/2025 (Shift 2)

- (a) 24.2 litres (b) 26.4 litres  
 (c) 21.5 litres (d) 22.0 litres

**Q.216.** An L-shaped floor is made of two rectangles (7 m × 4 m and 7 m × 6 m). Tiles cost ₹500/m<sup>2</sup>, but a 10% discount is given if more than 60 m<sup>2</sup> is tiled. What is the total cost after discount?

SSC CGL 20/09/2025 (Shift 2)

- (a) ₹31,500 (b) ₹39,160  
 (c) ₹30,240 (d) ₹31,540

**Q.217.** A triangular traffic sign has sides measuring 9 m, 10 m, and 17 m. If the cost to paint the sign is ₹7 per square meter, what will be the total expense to paint the entire sign?

SSC CGL 20/09/2025 (Shift 2)

- (a) ₹422 (b) ₹380 (c) ₹252 (d) ₹462

**Q.218.** The base area of a prism is increased by 35%, while the height remains the same. By what percentage does the volume increase?

SSC CGL 20/09/2025 (Shift 2)

- (a) 10% (b) 20% (c) 35% (d) 30%

**Q.219.** The number of triangles formed by diagonals from one vertex of an hexagon is:

SSC CGL 20/09/2025 (Shift 2)

- (a) 4 (b) 5 (c) 8 (d) 6

**Q.220.** A hemisphere is inscribed in a cube of side 12 cm. What is the volume of the hemisphere?

SSC CGL 20/09/2025 (Shift 3)

- (a) 356.55 cm<sup>3</sup> (b) 354.24 cm<sup>3</sup>  
 (c) 350.26 cm<sup>3</sup> (d) 452.16 cm<sup>3</sup>

**Q.221.** A triangular window has dimensions such that its area is 35% of a square window of area 300 m<sup>2</sup>. If its base is 20 m, find its height.

SSC CGL 20/09/2025 (Shift 3)

- (a) 12.5 m (b) 18.6 m  
 (c) 10.5 m (d) 24.5 m

**Q.222.** A display stand is shaped like a right triangular prism. The triangular

base is a right-angled triangle with perpendicular sides measuring 8 cm and 15 cm. The height of the prism is 5 cm. What is the ratio of the area of the largest rectangular face to the area of the smallest rectangular face of the prism?

SSC CGL 20/09/2025 (Shift 3)

- (a) 13 : 8 (b) 17 : 8 (c) 15 : 13 (d) 5 : 14

**Q.223.** A company makes a gift box in the shape of a right pyramid with a square base. The base side of the box is 8 cm, and its height is 15 cm. Due to safety limits, the box can be filled only up to 80% of its total volume. If 1 cm<sup>3</sup> of material costs ₹0.60, what will be the cost to fill one such box?

SSC CGL 21/09/2025 (Shift 1)

- (a) ₹153.6 (b) ₹163.6  
 (c) ₹182.4 (d) ₹192.3

**Q.224.** Consider a right trapezoidal field. Its two parallel sides are 40 meters and 60 meters long. One of its non-parallel sides, which acts as the height, measures 20 meters. What is the area of this field?

SSC CGL 21/09/2025 (Shift 1)

- (a) 900 square meters  
 (b) 1000 square meters  
 (c) 1220 square meters  
 (d) 1100 square meters

**Q.225.** A regular hexagon is drawn inside a circle of radius 14 cm. Find the approximate area of the hexagon.

SSC CGL 21/09/2025 (Shift 1)

- (a) 500 cm<sup>2</sup> (b) 509 cm<sup>2</sup>  
 (c) 520 cm<sup>2</sup> (d) 530 cm<sup>2</sup>

**Q.226.** A triangle has sides 4 cm and 8 cm, and the angle between them is 120°. What is its area?

SSC CGL 21/09/2025 (Shift 1)

- (a)  $10\sqrt{3}$  cm<sup>2</sup> (b)  $8\sqrt{3}$  cm<sup>2</sup>  
 (c)  $6\sqrt{3}$  cm<sup>2</sup> (d)  $12\sqrt{3}$  cm<sup>2</sup>

**Q.227.** A right prism has a base in the shape of a trapezium with parallel sides 8 cm and 10 cm, and height 6 cm. If the prism height is 18 cm, what is the volume?

SSC CGL 21/09/2025 (Shift 1)

- (a) 972 cm<sup>3</sup> (b) 776 cm<sup>3</sup>  
 (c) 820 cm<sup>3</sup> (d) 962 cm<sup>3</sup>

**Q.228.** If the surface area of a sphere is equal to twice its volume, find the radius of the sphere.

SSC CGL 21/09/2025 (Shift 2)

- (a) 1 unit (b) 3 units  
 (c) 2 units (d) 1.5 units

**Q.229.** A triangular flag has base and height in the ratio 3 : 4. If the area is 216 cm<sup>2</sup>, what are the base and height?

SSC CGL 21/09/2025 (Shift 2)

- (a) 16 cm and 12 cm  
 (b) 18 cm and 24 cm  
 (c) 20 cm and 25 cm  
 (d) 18 cm and 20 cm

**Q.230.** A water tank is shaped like a composite prism: the bottom part is a triangular prism with base area 80 cm<sup>2</sup> and height 1.5 m, while the top part is a rectangular prism with base 50 cm × 35 cm and height 0.9 m. What is the total volume of the tank in liters?

(1 m<sup>3</sup> = 1000 L)

SSC CGL 21/09/2025 (Shift 2)

- (a) 160.8 L (b) 178.5 L  
 (c) 169.5 L (d) 201.5 L

**Q.231.** A pyramid with an equilateral triangular base of side 15 cm and height 20 cm is placed on top of a cube with side length 15 cm. Find the total volume of the composite solid.

SSC CGL 21/09/2025 (Shift 3)

- (a) 2102.3 cm<sup>3</sup> (b) 1202.5 cm<sup>3</sup>  
 (c) 4024.5 cm<sup>3</sup> (d) 6202.3 cm<sup>3</sup>

**Q.232.** A triangular field with base 80 m and height 60 m shares a side with a rectangle of length 80 m and breadth 50 m. Find the combined area and percentage of the triangle's area in the total.

SSC CGL 21/09/2025 (Shift 3)

- (a) 1, 450 m<sup>2</sup> and 24.5%  
 (b) 2, 450 m<sup>2</sup> and 25.5%  
 (c) 6, 400 m<sup>2</sup> and 37.5%  
 (d) 2, 500 m<sup>2</sup> and 25%

**Q.233.** A regular hexagon is inscribed inside a circle of radius 18 cm. Find the area of the hexagon.

SSC CGL 21/09/2025 (Shift 3)

- (a) 438.5 cm<sup>2</sup> (b) 841.8 cm<sup>2</sup>  
 (c) 598.4 cm<sup>2</sup> (d) 638.7 cm<sup>2</sup>

**Q.234.** A steel plate is in the shape of a right triangle. The lengths of the legs are in the ratio 8 : 15. If the hypotenuse is 17 m, what is the area?

SSC CGL 21/09/2025 (Shift 3)

- (a) 30 m<sup>2</sup> (b) 60 m<sup>2</sup> (c) 78 m<sup>2</sup> (d) 50 m<sup>2</sup>

**Q.235.** In a right prism with a square base, volume = 1296 cm<sup>3</sup> and height = 9 cm. What is the side of the square base?

SSC CGL 21/09/2025 (Shift 3)

- (a) 5 cm (b) 8 cm (c) 10 cm (d) 12 cm

**Q.236.** A spherical balloon is inflated, causing its radius to grow by 5%. By what percentage does its surface area



increase?

SSC CGL 22/09/2025 (Shift 1)

- (a) 10.25% (b) 20% (c) 21.25% (d) 25%

**Q.237.** A rectangular sheet has an area equal to a square plot. If the sheet measures 36 m in length and 25 m in breadth, and the square plot has an area equal to the sheet, find the perimeter of the square plot.

SSC CGL 22/09/2025 (Shift 1)

- (a) 110 m (b) 120 m  
(c) 140 m (d) 220 m

**Q.238.** Two right prisms have the same height of 12 cm. One has a regular hexagonal base with side 5 cm, and the other has a square base with side 5 cm. What is the ratio of their volumes?

SSC CGL 22/09/2025 (Shift 1)

- (a)  $\sqrt{3} : 1$  (b)  $3\sqrt{3} : 2$   
(c)  $3\sqrt{3} : 5$  (d)  $3 : 2\sqrt{3}$

**Q.239.** A pyramid is inscribed inside a cube with edge 9 cm, sharing the base and apex at center of top. Find volume of pyramid.

SSC CGL 22/09/2025 (Shift 2)

- (a) 180 cm<sup>3</sup> (b) 220 cm<sup>3</sup>  
(c) 243 cm<sup>3</sup> (d) 270 cm<sup>3</sup>

**Q.240.** A rectangular park has an area of 720 square meters. If its length is increased by 8 meters and its width is decreased by 3 meters, the area remains unchanged. Find the original dimensions of the park.

SSC CGL 22/09/2025 (Shift 2)

- (a) Length = 45 m, Width = 16 m  
(b) Length = 40 m, Width = 18 m  
(c) Length = 48 m, Width = 15 m  
(d) Length = 36 m, Width = 20 m

**Q.241.** A garden is in the shape of a regular hexagon, each side 30 m. If 5% of it is occupied by a circular fountain, find the remaining area.

SSC CGL 22/09/2025 (Shift 2)

- (a) 2,131.5 m<sup>2</sup> (b) 2,351.9 m<sup>2</sup>  
(c) 2,221.3 m<sup>2</sup> (d) 2,421.7 m<sup>2</sup>

**Q.242.** A triangular field has sides in the ratio 3 : 5 : 7 and a perimeter of 90 m. What is the area of the field?

SSC CGL 22/09/2025 (Shift 2)

- (a) 196 m<sup>2</sup> (b) 234 m<sup>2</sup>  
(c) 186 m<sup>2</sup> (d) 182 m<sup>2</sup>

**Q.243.** A storage box is in the shape of a right rectangular prism with internal dimensions 60 cm × 50 cm × 40 cm. How many liters can it hold?

SSC CGL 22/09/2025 (Shift 2)

- (a) 150 (b) 120 (c) 140 (d) 170

**Q.244.** The volume of a sphere is  $288\pi$  cm<sup>3</sup>. If the radius of the sphere is tripled, by what percentage does the volume increase?

SSC CGL 22/09/2025 (Shift 3)

- (a) 800% (b) 2600% (c) 900% (d) 2700%

**Q.245.** A rectangular plot has a length of 15.75 m and a width of 8.2 m. What is its area?

SSC CGL 22/09/2025 (Shift 3)

- (a) 129.15 m<sup>2</sup> (b) 128.9 m<sup>2</sup>  
(c) 180.2 m<sup>2</sup> (d) 181.5 m<sup>2</sup>

**Q.246.** The base of a prism is a rectangle whose length is twice its width. The perimeter of the base is 54 cm. The prism's height decreases in an arithmetic progression from 17 cm in the first layer to 11 cm in the fourth layer. What is the total volume of the prism?

SSC CGL 22/09/2025 (Shift 3)

- (a) 9000 cm<sup>3</sup> (b) 9500 cm<sup>3</sup>  
(c) 9060 cm<sup>3</sup> (d) 9072 cm<sup>3</sup>

**Q.247.** A sphere is inscribed inside a cube. The volume of the cube is 3375 cm<sup>3</sup>. What is the total surface area of the sphere?

SSC CGL 23/09/2025 (Shift 1)

- (a)  $225\pi$  cm<sup>2</sup> (b)  $254\pi$  cm<sup>2</sup>  
(c)  $256\pi$  cm<sup>2</sup> (d)  $298\pi$  cm<sup>2</sup>

**Q.248.** A large wooden box, open at the top, needs to be lined with fabric on its inner floor and inner four side walls. The inner dimensions of the box are 5 m length, 3.6 m wide, and 2.4 m high. If the fabric comes in square pieces of side 100 cm, and 30% extra fabric is needed for cuts and overlaps, how many approximate fabric pieces should be purchased?

SSC CGL 23/09/2025 (Shift 1)

- (a) 77 (b) 88 (c) 62 (d) 98

**Q.249.** A square of side 12cm has a smaller square (of side 5 cm) removed from one side. What is the ratio of the remaining area to the original?

SSC CGL 23/09/2025 (Shift 1)

- (a) 129 : 136 (b) 119 : 144  
(c) 114 : 119 (d) 117 : 125

**Q.250.** A rectangular park is 80 m by 40 m. A semicircular pond of diameter 20 m is inside. What percent of the park's area does the pond occupy?

SSC CGL 23/09/2025 (Shift 1)

- (a) 6.91% (b) 4.91% (c) 5.81% (d) 3.00%

**Q.251.** A regular hexagon has a perimeter of 84 cm. What is its area?

SSC CGL 23/09/2025 (Shift 1)

- (a) 508.62 cm<sup>2</sup> (b) 450.00 cm<sup>2</sup>

- (c) 600.12 cm<sup>2</sup> (d) 480.36 cm<sup>2</sup>

**Q.252.** A triangular prism has a base with area 56 cm<sup>2</sup> and height 15 cm. If 20% of the prism is hollowed out for wiring, what is the volume of the solid part?

SSC CGL 23/09/2025 (Shift 1)

- (a) 672 cm<sup>3</sup> (b) 680 cm<sup>3</sup>  
(c) 690 cm<sup>3</sup> (d) 700 cm<sup>3</sup>

**Q.253.** A circular jogging track has a radius of 60 m. A runner covers a curved path that subtends an angle of 150° at the center. What is the length of the arc she runs?

SSC CGL 23/09/2025 (Shift 1)

- (a) 70π m (b) 120π m  
(c) 100π m (d) 50π m

**Q.254.** A rectangular fish tank with a base of 80 cm × 60 cm contains water up to a height of 25 cm. If 4 identical spherical stones, each with a radius of 5 cm, are dropped into the tank and fully submerged, what is the rise in the water level?

SSC CGL 23/09/2025 (Shift 2)

- (a)  $\frac{\pi}{8}$  cm (b)  $\frac{5\pi}{36}$  cm  
(c)  $\frac{\pi}{5}$  cm (d)  $\frac{2\pi}{3}$  cm

**Q.255.** A square field has a diagonal of  $\sqrt{48}$  m. What is its area?

SSC CGL 23/09/2025 (Shift 2)

- (a) 46 m<sup>2</sup> (b) 24 m<sup>2</sup> (c) 40 m<sup>2</sup> (d) 12 m<sup>2</sup>

**Q.256.** A square has a perimeter of 80 cm. What is the area of the square?

SSC CGL 23/09/2025 (Shift 2)

- (a) 352 cm<sup>2</sup> (b) 692 cm<sup>2</sup>  
(c) 400 cm<sup>2</sup> (d) 425 cm<sup>2</sup>

**Q.257.** A spherical tank is filled with water. The radius of the sphere is 12 meters. The tank is used to fill cylindrical containers, each with a radius of 2 meter and a height of 24 meters. How many containers can be filled?

SSC CGL 23/09/2025 (Shift 3)

- (a) 24 (b) 28 (c) 54 (d) 30

**Q.258.** A perfect cube with a side length of 4 cm has the largest possible sphere fitted inside it. What is the volume of the empty space within the cube?

SSC CGL 23/09/2025 (Shift 3)

- (a)  $64 - (\frac{32\pi}{3})$  cm<sup>3</sup> (b)  $12 - (\frac{64\pi}{3})$  cm<sup>3</sup>  
(c)  $64 - (\frac{56\pi}{3})$  cm<sup>3</sup> (d)  $\frac{256\pi}{3}$  cm<sup>3</sup>

**Q.259.** A banner is shaped like a trapezium with bases 10 m and 6 m, and



height 8 m. If printing costs ₹500/m<sup>2</sup>, what is the total cost?

SSC CGL 23/09/2025 (Shift 3)

- (a) ₹32,000 (b) ₹30,000  
(c) ₹28,000 (d) ₹25,000

**Q.260.** A farmer's land is two adjoining trapezoids: T1 with bases 35 m & 25 m, height 10 m; T2 with bases 25 m & 15 m, height 10 m. Land costs ₹30/m<sup>2</sup>. What is the total land cost?

SSC CGL 23/09/2025 (Shift 3)

- (a) ₹15,000 (b) ₹30,000  
(c) ₹24,000 (d) ₹25,000

**Q.261.** The ratio of areas of two regular polygons with the same number of sides is 16 : 25. What is the ratio of their side lengths?

SSC CGL 23/09/2025 (Shift 3)

- (a) 5 : 4 (b) 4 : 5 (c) 2 : 3 (d) 4 :  $\sqrt{5}$

**Q.262.** A prism with a parallelogram base has base area 160 cm<sup>2</sup> and height 24 cm. If its lateral surface area is 768 cm<sup>2</sup>, what is the perimeter of the base?

SSC CGL 23/09/2025 (Shift 3)

- (a) 26 cm (b) 22 cm (c) 36 cm (d) 32 cm

**Q.263.** The angle of a sector is  $\frac{\pi}{4}$  radians, and the radius of the circle is 16 cm. What is the area of the sector?

SSC CGL 23/09/2025 (Shift 3)

- (a)  $6\pi$  cm<sup>2</sup> (b)  $22\pi$  cm<sup>2</sup>  
(c)  $32\pi$  cm<sup>2</sup> (d)  $28\pi$  cm<sup>2</sup>

**Q.264.** A rectangular tank measures 15 cm long, 14 cm wide, and 9 cm high. If a spherical ball of the largest possible size is placed inside this tank, what is the volume of the space not occupied by the ball?

SSC CGL 24/09/2025 (Shift 1)

- (a)  $(1790 - 121.5\pi)$  cm<sup>3</sup>  
(b)  $(1880 - 121.5\pi)$  cm<sup>3</sup>  
(c)  $(1999 - 121.5\pi)$  cm<sup>3</sup>  
(d)  $(1890 - 121.5\pi)$  cm<sup>3</sup>

**Q.265.** Mohit walks a square plot of area 196 m<sup>2</sup>. If he walks once around its boundary, how far does he walk?

SSC CGL 24/09/2025 (Shift 1)

- (a) 59 m (b) 52 m (c) 47 m (d) 56 m

**Q.266.** A spherical shell has an outer diameter of 24 cm and is 3 cm thick. What is the volume of the material in the shell? (Use  $\pi = 3.14$ ).

SSC CGL 24/09/2025 (Shift 2)

- (a) 4,782.48 cm<sup>3</sup> (b) 4,182.48 cm<sup>3</sup>  
(c) 2,182.48 cm<sup>3</sup> (d) 3,182.58 cm<sup>3</sup>

**Q.267.** A rectangular tank measuring 80 cm long and 40 cm wide contains water with an initial level of 15 cm. If 5 identical

metal spheres each 10 cm in diameter are gently lowered into the tank, how much does the water level rise? (Assume all spheres are completely submerged.)

SSC CGL 24/09/2025 (Shift 2)

- (a)  $\frac{25\pi}{86}$  cm (b)  $\frac{24\pi}{93}$  cm  
(c)  $\frac{25\pi}{96}$  cm (d) 25  $\pi$  cm

**Q.268.** A warehouse floor is 16.5 meters long and 16 meters wide. If 2.5 meter  $\times$  2.2 meter rectangular tiles are used to completely cover the floor, how many tiles will be needed?

SSC CGL 24/09/2025 (Shift 2)

- (a) 48 (b) 68 (c) 72 (d) 80

**Q.269.** A large advertisement board is made on a 10 m  $\times$  6 m rectangle. From this, a 3 m  $\times$  2 m rectangular portion is cut out and removed for the advertisement display. If the cost of painting the remaining open area (the part left after cutting) is ₹250 per square meter, what will be the total cost of painting that open area?

SSC CGL 24/09/2025 (Shift 2)

- (a) ₹13,240 (b) ₹13,500  
(c) ₹13,960 (d) ₹14,200

**Q.270.** A regular hexagonal nut has an outer side of 8 cm and an inner hollow side of 4 cm. What is the approximate area of the material in the cross-section?

SSC CGL 24/09/2025 (Shift 2)

- (a) 124.7 cm<sup>2</sup> (b) 120.2 cm<sup>2</sup>  
(c) 121.1 cm<sup>2</sup> (d) 125.6 cm<sup>2</sup>

**Q.271.** A triangular storage prism has a base in the shape of a right triangle and its side lengths are in the ratio 3 : 4 : 5. If the total perimeter of the base is 54 cm and the height of the prism is 8 cm, find the lateral surface area of the prism.

SSC CGL 24/09/2025 (Shift 2)

- (a) 360 cm<sup>2</sup> (b) 300 cm<sup>2</sup>  
(c) 480 cm<sup>2</sup> (d) 432 cm<sup>2</sup>

**Q.272.** A sector of a circle having a radius of 20 cm and has a central angle of  $\frac{6\pi}{7}$  radians. What is the area of the sector?

SSC CGL 24/09/2025 (Shift 2)

- (a)  $141.4\pi$  cm<sup>2</sup> (b)  $171.4\pi$  cm<sup>2</sup>  
(c)  $145.5\pi$  cm<sup>2</sup> (d)  $175.8\pi$  cm<sup>2</sup>

**Q.273.** A bathroom floor measures 4.5 m in length and 3 m in width. One of its walls, which is to be tiled fully, measures 2 m in length and 2.5 m in height. If square tiles of side 25 cm are used, and an extra 12% of tiles are purchased to cover cuts and wastage, how many tiles are needed in total?

SSC CGL 24/09/2025 (Shift 3)

- (a) 90 (b) 70 (c) 95 (d) 80

**Q.274.** A square tile of area 289 cm<sup>2</sup> is cut into two equal rectangular tiles. What will be the perimeter of each new tile?

SSC CGL 24/09/2025 (Shift 3)

- (a) 48 cm (b) 52 cm (c) 51 cm (d) 50 cm

**Q.275.** Two similar hexagons have their perimeters in the ratio 3:4. If the smaller hexagon has an area of 90 cm<sup>2</sup>, what is the area of the larger hexagon?

SSC CGL 24/09/2025 (Shift 3)

- (a) 140 cm<sup>2</sup> (b) 155 cm<sup>2</sup>  
(c) 160 cm<sup>2</sup> (d) 175 cm<sup>2</sup>

**Q.276.** What is the length of a diagonal of a rectangle with length 30 cm and width 16 cm?

SSC CGL 24/09/2025 (Shift 3)

- (a) 32 cm (b) 38 cm (c) 36 cm (d) 34 cm

**Q.277.** The volume of a sphere is  $36\pi$  cm<sup>3</sup>. If the radius is tripled, what is the new volume?

SSC CGL 25/09/2025 (Shift 1)

- (a)  $972\pi$  cm<sup>3</sup> (b)  $729\pi$  cm<sup>3</sup>  
(c)  $1296\pi$  cm<sup>3</sup> (d)  $1881\pi$  cm<sup>3</sup>

**Q.278.** A rectangular room measures 10 m long and 8 m wide. Its floor is to be covered with square tiles of side 50 cm. If 15% extra tiles are purchased for cutting and breakage, how many tiles are needed in total?

SSC CGL 25/09/2025 (Shift 1)

- (a) 400 (b) 350 (c) 468 (d) 368

**Q.279.** A rectangular lawn measures 60 m by 40 m. A 5 m wide path is constructed all around the inside of the lawn. What percentage of the lawn's area is occupied by the path?

SSC CGL 25/09/2025 (Shift 1)

- (a) 35% (b) 37.5% (c) 16.67% (d) 47.5%

**Q.280.** An L-shaped floor is formed by adjoining a 8m  $\times$  5m and a 5m  $\times$  3m rectangle. Tiles cost ₹200/m<sup>2</sup>. What is the total cost to tile the floor?

SSC CGL 25/09/2025 (Shift 1)

- (a) ₹19,600 (b) ₹21,700  
(c) ₹22,000 (d) ₹11000

**Q.281.** An equilateral triangle and a square have equal perimeters. What is the ratio of their sides?

SSC CGL 25/09/2025 (Shift 1)

- (a) 3 : 4 (b) 1 : 2 (c) 4 : 3 (d) 2 : 1

**Q.282.** A prism has a base perimeter of 20 cm and height 8 cm. If its total surface area is 200 cm<sup>2</sup>, what is the area of both bases combined?

SSC CGL 25/09/2025 (Shift 1)



- (a) 52 cm<sup>2</sup> (b) 46 cm<sup>2</sup>  
(c) 40 cm<sup>2</sup> (d) 62 cm<sup>2</sup>

**Q.283.** The surface area of a cuboid is 736 cm<sup>2</sup>, and its dimensions are in the ratio 1 : 3 : 5. Find its volume.

SSC CGL 25/09/2025 (Shift 2)

- (a) 960 cm<sup>3</sup> (b) 948 cm<sup>3</sup>  
(c) 860 cm<sup>3</sup> (d) 528 cm<sup>3</sup>

**Q.284.** A building has 4 square pyramidal turrets, each of base 5 m and height 7 m. If each needs 1.5 times volume for structural fill, find total fill needed.

SSC CGL 25/09/2025 (Shift 2)

- (a) 350 m<sup>3</sup> (b) 260 m<sup>3</sup>  
(c) 280 m<sup>3</sup> (d) 180 m<sup>3</sup>

**Q.285.** If the height of a cylinder is equal to its diameter and radius is 6 cm, find its volume.

SSC CGL 25/09/2025 (Shift 2)

- (a)  $432\pi$  cm<sup>3</sup> (b)  $238\pi$  cm<sup>3</sup>  
(c)  $500\pi$  cm<sup>3</sup> (d)  $316\pi$  cm<sup>3</sup>

**Q.286.** A right triangular prism has a base area of 35 cm<sup>2</sup> and height 8 cm. What is its volume ?

SSC CGL 25/09/2025 (Shift 2)

- (a) 120 cm<sup>3</sup> (b) 240 cm<sup>3</sup>  
(c) 280 cm<sup>3</sup> (d) 360 cm<sup>3</sup>

**Q.287.** What is the area of the segment formed by a chord in a circle of radius 4 cm, if the angle subtended at the center is 90° ?

SSC CGL 25/09/2025 (Shift 2)

- (a)  $4\pi - 8$  (b)  $8\pi - 4$  (c)  $16\pi - 8$  (d)  $4\pi - 4$

**Q.288.** If the surface area of a sphere increases by 26.56%, by what percentage does its volume increase ?

SSC CGL 25/09/2025 (Shift 3)

- (a) 21.23% (b) 11.89%  
(c) 33.1% (d) 42.4%

**Q.289.** A kitchen wall, measuring 5.6 m in length and 2.4 m in height, needs to be tiled. The tiles are rectangular, 70 cm × 60 cm. If an additional 12% of tiles are required to account for errors and cuts, how many tiles should be purchased ?

SSC CGL 25/09/2025 (Shift 3)

- (a) 36 (b) 38 (c) 40 (d) 50

**Q.290.** A square and a rectangle have equal areas. The square's side is 22 cm, and the rectangle's length is 44 cm. Find its breadth.

SSC CGL 25/09/2025 (Shift 3)

- (a) 15 cm (b) 16 cm (c) 14 cm (d) 11 cm

**Q.291.** A composite wall consists of a rectangle 20m by 8 m with a triangular window of base 15 m and height 8 m cut out. If the wall needs to be painted at ₹111 per m<sup>2</sup>, find the painting cost after

deducting the window area.

SSC CGL 25/09/2025 (Shift 3)

- (a) 12,528 (b) 111,00  
(c) 120,00 (d) 11,520

**Q.292.** A regular hexagonal mirror has a side of 20 cm. If 10% of the mirror area is covered by the frame, what is the visible glass area ?

SSC CGL 25/09/2025 (Shift 3)

- (a) 690.2 cm<sup>2</sup> (b) 710.5 cm<sup>2</sup>  
(c) 941.15 cm<sup>2</sup> (d) 935.28 cm<sup>2</sup>

**Q.293.** A large storage box in the shape of a right prism with square base is 2 m tall. If the box holds 720 L of water, what is the side of the base (in cm)?

SSC CGL 25/09/2025 (Shift 3)

- (a) 40 cm (b) 45 cm (c) 60 cm (d) 50 cm

**Q.294.** If the radius of a cylinder is doubled, what happens to its volume (assuming height remains constant)?

SSC CGL 26/09/2025 (Shift 1)

- (a) Volume is halved  
(b) Volume is doubled  
(c) Volume becomes four times  
(d) Volume remains the same

**Q.295.** A hollow rectangular parallel piped has external dimensions 8 cm × 6 cm × 5 cm and a thickness of 0.5 cm. Find its volume.

SSC CGL 26/09/2025 (Shift 1)

- (a) 100 cm<sup>3</sup> (b) 120 cm<sup>3</sup>  
(c) 160 cm<sup>3</sup> (d) 240 cm<sup>3</sup>

**Q.296.** A triangular prism has base side lengths 3 cm, 4 cm, and 5 cm and height 8 cm. What is its lateral surface area ?

SSC CGL 26/09/2025 (Shift 1)

- (a) 84 cm<sup>2</sup> (b) 90 cm<sup>2</sup>  
(c) 96 cm<sup>2</sup> (d) 100 cm<sup>2</sup>

**Q.297.** A cone of radius 6 cm and height 9 cm is cut into two parts by a plane parallel to its base, at a height of 6 cm from the vertex. Find the volume of the frustum formed.

SSC CGL 26/09/2025 (Shift 1)

- (a)  $43\pi$  cm<sup>3</sup> (b)  $76\pi$  cm<sup>3</sup>  
(c)  $68\pi$  cm<sup>3</sup> (d)  $72\pi$  cm<sup>3</sup>

**Q.298.** A cuboid has a volume of 1400 cm<sup>3</sup> and its length is twice its height. If the breadth is 7 cm, find its dimensions.

SSC CGL 26/09/2025 (Shift 2)

- (a) 18 cm × 7 cm × 9 cm  
(b) 12 cm × 7 cm × 6 cm  
(c) 20 cm × 7 cm × 10 cm  
(d) 10 cm × 7 cm × 5 cm.

**Q.299.** A plot has an area of 576 m<sup>2</sup>. If a footpath 1 m wide is constructed around it, what is the total area including the footpath ?

SSC CGL 26/09/2025 (Shift 2)

- (a) 729 m<sup>2</sup> (b) 676 m<sup>2</sup>  
(c) 784 m<sup>2</sup> (d) 756 m<sup>2</sup>

**Q.300.** A rectangle has a length that is four times its width. If the perimeter is 200 cm, what is the area of the rectangle?

SSC CGL 26/09/2025 (Shift 2)

- (a) 1400 cm<sup>2</sup> (b) 1500 cm<sup>2</sup>  
(c) 1540 cm<sup>2</sup> (d) 1600 cm<sup>2</sup>

**Q.301.** A rectangular parallelepiped has dimensions in arithmetic progression. If the sum of its dimensions is 42 cm and the volume is 1344 cm<sup>3</sup>, find its dimensions.

SSC CGL 26/09/2025 (Shift 3)

- (a) 8 cm, 14 cm, 20 cm  
(b) 6 cm, 14 cm, 22 cm  
(c) 10 cm, 12 cm, 20 cm  
(d) 4 cm, 14 cm, 24 cm

**Q.302.** The volumes of two right circular cylinders have a ratio of 7 : 9. If the radii of these cylinders are in the ratio of 1 : 3, what is the ratio of their heights ?

SSC CGL 26/09/2025 (Shift 3)

- (a) 7 : 1 (b) 3 : 4 (c) 7 : 3 (d) 8 : 3

**Q.303.** If the base area of a prism is doubled while its height remains the same, what happens to the volume?

SSC CGL 26/09/2025 (Shift 3)

- (a) Tripled (b) Doubled  
(c) Quadrupled (d) Remains same

**Q.304.** A cone is melted and recast into another cone with half the radius. Find the new height if the original height was 24 cm.

SSC CGL 26/09/2025 (Shift 3)

- (a) 68 cm (b) 96 cm (c) 64 cm (d) 98 cm

**Q.305.** What is the area of the segment formed by a chord in a circle of radius 5 cm, if the angle subtended at the center is 90° ?

SSC CGL 26/09/2025 (Shift 3)

- (a)  $\frac{25\pi}{4} - 12.5$  (b)  $\frac{50\pi}{4} - 12.5$   
(c)  $100\pi - 12.5$  (d)  $25\pi - 12.5$

**Q.306.** A triangular field with base 70 m and height 96 m shares a side with a rectangle of length 70 m and breadth 35 m. Find the combined area and percentage of the triangle's area in the total.

SSC CGL 14/10/2025 (Shift 1)

- (a) 4, 450 m<sup>2</sup> and 34.5%  
(b) 3, 450 m<sup>2</sup> and 25 %  
(c) 5810 m<sup>2</sup> and 57.83%  
(d) 5, 500 m<sup>2</sup> and 59 %



**Q.307.** The areas of three adjacent faces of a cuboid are  $48 \text{ cm}^2$ ,  $54 \text{ cm}^2$ , and  $72 \text{ cm}^2$ . What is the volume of the cuboid?

SSC CGL 14/10/2025 (Shift 2)

- (a)  $360 \text{ cm}^3$  (b)  $432 \text{ cm}^3$   
(c)  $504 \text{ cm}^3$  (d)  $576 \text{ cm}^3$

**Q.308.** A solid hemisphere has a radius of 9 cm. It is melted into a cylinder of height 6 cm. Find the radius of the cylinder.

SSC CGL 14/10/2025 (Shift 2)

- (a) 9 cm (b) 6 cm (c) 7 cm (d) 5 cm

**Q.309.** A hemisphere, cone, and cylinder have the same base radius and equal volumes. If the height of the cylinder is 3h, find the height of the cone.

SSC CGL 14/10/2025 (Shift 2)

- (a) 9h (b) 10h (c) 5h (d) 6h

**Q.310.** The longest diagonal of a cuboid is equal to the diameter of a sphere in which it is inscribed. If the sphere's radius is 13 cm, and two dimensions of the cuboid are 9 cm and 15 cm, what is the third dimension?

SSC CGL 14/10/2025 (Shift 2)

- (a) 10.1 cm (b) 12.3 cm  
(c) 16.5 cm (d) 19.2 cm

**Q.311.** A regular right pyramid has a square base with side length 10 cm and height 12 cm. Find the slant height of one triangular face.

SSC CGL 14/10/2025 (Shift 2)

- (a) 12 cm (b) 13 cm (c) 14 cm (d) 16 cm

**Q.312.** A rhombus has diagonals in the ratio 5 : 6. If its area is  $135 \text{ cm}^2$ , find the diagonals.

SSC CGL 14/10/2025 (Shift 2)

- (a) 18 cm and 24 cm  
(b) 12 cm and 16 cm  
(c) 15 cm and 18 cm  
(d) 27 cm and 36 cm

**Q.313.** A circular plate having a radius of 12 cm and is surrounded by a decorative rim that is 3 cm wide. What is the area of the rim?

SSC CGL 14/10/2025 (Shift 2)

- (a)  $238.23 \text{ cm}^2$  (b)  $254.34 \text{ cm}^2$   
(c)  $258.51 \text{ cm}^2$  (d)  $230.69 \text{ cm}^2$

**Q.314.** A square-shaped piece of metal has a circular hole cut out exactly from its center, touching all four sides. What percentage of the metal sheet's area was removed by the hole? (Use  $\pi \approx 3.14$ )

SSC CGL 14/10/2025 (Shift 2)

- (a) 21.5% (b) 25% (c) 75% (d) 78.5%

**Q.315.** A spherical ball is submerged in water in a cylindrical container. The

radius of the cylindrical container is 6 cm, and the height is 35 cm. What is the volume of water displaced by the ball if its radius is 7 cm?

SSC CGL 14/10/2025 (Shift 3)

- (a)  $\frac{1842}{3} \pi \text{ cm}^3$  (b)  $\frac{1622}{3} \pi \text{ cm}^3$   
(c)  $\frac{1762}{3} \pi \text{ cm}^3$  (d)  $\frac{1372}{3} \pi \text{ cm}^3$

**Q.316.** A right circular cone is cut parallel to its base, forming a frustum. The height of the original cone is 30 cm, and the cut is made 6 cm from the top. What is the ratio of the volume of the frustum to the volume of the entire cone?

SSC CGL 14/10/2025 (Shift 3)

- (a) 124 : 125 (b) 125 : 124  
(c) 125 : 126 (d) 126 : 125

### SSC Selection Post (XIII)

**Q.317.** A solid cone is melted and recast into smaller cones of the same height but one-third of the radius. How many smaller cones can be made?

Matriculation Level 24/07/2025 (Shift - 1)

- (a) 3 (b) 6 (c) 9 (d) 27

**Q.318.** A sphere and a cylinder have the same radius and height. The ratio of their volumes is:

Matriculation Level 24/07/2025 (Shift - 1)

- (a) 1 : 2 (b) 2 : 3 (c) 4 : 3 (d) 2 : 1

**Q.319.** If the volume of a rectangular parallelepiped is  $280 \text{ cm}^3$  and its base area is  $20 \text{ cm}^2$ . What is its height?

Matriculation Level 24/07/2025 (Shift - 1)

- (a) 6 cm (b) 14 cm (c) 10 cm (d) 12 cm

**Q.320.** The radius of the base of a cylinder is 7 cm and its height is 10 cm.

What is its volume? (Take  $\pi = \frac{22}{7}$ ).

Matriculation Level 25/07/2025 (Shift - 1)

- (a)  $1450 \text{ cm}^3$  (b)  $1540 \text{ cm}^3$   
(c)  $1600 \text{ cm}^3$  (d)  $1400 \text{ cm}^3$

**Q.321.** If a cylinder and a cone have the same base radius and same height, then the volume of the cylinder is \_\_\_ of the volume of the cone.

Matriculation Level 25/07/2025 (Shift - 1)

- (a) Double (b) Three times  
(c) Equal (d) Half

**Q.322.** A box has dimensions  $12 \text{ cm} \times 6 \text{ cm} \times 4 \text{ cm}$ . What is its volume?

Matriculation Level 25/07/2025 (Shift - 1)

- (a)  $144 \text{ cm}^3$  (b)  $250 \text{ cm}^3$   
(c)  $288 \text{ cm}^3$  (d)  $240 \text{ cm}^3$

**Q.323.** What is the area of a triangle with base 14 cm and height 12 cm?

Matriculation Level 26/07/2025 (Shift - 1)

- (a)  $84 \text{ cm}^2$  (b)  $42 \text{ cm}^2$   
(c)  $80 \text{ cm}^2$  (d)  $96 \text{ cm}^2$

**Q.324.** A conical party hat has a slant height of 8 cm and a radius of 5 cm. What is its curved surface area? (Use  $\pi = 3.14$ )

Matriculation Level 26/07/2025 (Shift - 1)

- (a)  $115.4 \text{ cm}^2$  (b)  $125.6 \text{ cm}^2$   
(c)  $136.2 \text{ cm}^2$  (d)  $145.8 \text{ cm}^2$

**Q.325.** A solid hemisphere with radius 6 cm is painted. Find its total surface area.

Matriculation Level 26/07/2025 (Shift - 1)

- (a)  $339.12 \text{ cm}^2$  (b)  $226.08 \text{ cm}^2$   
(c)  $452.16 \text{ cm}^2$  (d)  $113.04 \text{ cm}^2$

**Q.326.** The perimeter of a triangle is 30 cm. If two sides are 9 cm and 11 cm, what is the third side?

Matriculation Level 28/07/2025 (Shift - 1)

- (a) 8 cm (b) 10 cm (c) 12 cm (d) 14 cm

**Q.327.** An ice cream cone is shaped like a cone having radius of 4 cm and a height of 10 cm. How much ice cream (in  $\text{cm}^3$ ) can it hold?

Matriculation Level 28/07/2025 (Shift - 1)

- (a)  $167.61 \text{ cm}^3$  (b)  $160 \text{ cm}^3$   
(c)  $165.5 \text{ cm}^3$  (d)  $158 \text{ cm}^3$

**Q.328.** A tank in the shape of a hemisphere is filled with milk. The milk is then poured into a cylindrical container of the same radius but height equal to 4r. What fraction of the cylinder will be filled?

Matriculation Level 28/07/2025 (Shift - 1)

- (a)  $\frac{1}{3}$  (b)  $\frac{2}{3}$  (c)  $\frac{1}{6}$  (d)  $\frac{1}{4}$

**Q.329.** A rectangular parallelepiped has dimensions  $6 \text{ cm} \times 5 \text{ cm} \times 4 \text{ cm}$ . Find its total surface area.

Matriculation Level 28/07/2025 (Shift - 1)

- (a)  $148 \text{ cm}^2$  (b)  $150 \text{ cm}^2$   
(c)  $140 \text{ cm}^2$  (d)  $135 \text{ cm}^2$

**Q.330.** A conical tent has a radius of 6 m and a slant height of 14 m. Find its lateral surface area.

Matriculation Level 29/07/2025 (Shift - 1)

- (a)  $72\pi \text{ m}^2$  (b)  $84\pi \text{ m}^2$   
(c)  $90\pi \text{ m}^2$  (d)  $56\pi \text{ m}^2$

**Q.331.** A hemisphere is placed on top of a cylinder with the same base radius. If their combined height is 12 cm and radius is 4 cm, find the total surface area.

Matriculation Level 29/07/2025 (Shift - 1)

- (a)  $785.97 \text{ cm}^2$  (b)  $351.68 \text{ cm}^2$   
(c)  $720 \text{ cm}^2$  (d)  $785.97 \text{ cm}^2$

**Q.332.** Find the surface area of a brick with dimensions  $10 \text{ cm} \times 4 \text{ cm} \times 3 \text{ cm}$ .



Matriculation Level 29/07/2025 (Shift - 1)

- (a) 160 cm<sup>2</sup> (b) 155 cm<sup>2</sup>  
(c) 164 cm<sup>2</sup> (d) 144 cm<sup>2</sup>

**Q.333.** A triangle has an area of 175 cm<sup>2</sup> and a base of 25 cm. What is its height?  
Matriculation Level 30/07/2025 (Shift - 1)  
(a) 15 cm (b) 16 cm (c) 17 cm (d) 14 cm

**Q.334.** An overhead water tank is made by placing a hemisphere over a cylinder. The cylinder has a height of 6 m, and the radius of both shapes is 3 m. What is the total volume of the tank in cubic meters? (Use  $\pi = 3.14$ )  
Matriculation Level 30/07/2025 (Shift - 1)

- (a) 287.43 m<sup>3</sup> (b) 124.56 m<sup>3</sup>  
(c) 226.08 m<sup>3</sup> (d) 126.08 m<sup>3</sup>

**Q.335.** A wooden block has dimensions 16 cm × 10 cm × 6 cm. What is its volume?  
Matriculation Level 30/07/2025 (Shift - 1)

- (a) 865 cm<sup>3</sup> (b) 920 cm<sup>3</sup>  
(c) 960 cm<sup>3</sup> (d) 678 cm<sup>3</sup>

**Q.336.** Find the volume of a right circular cone whose base radius is 21 cm and height is 36 cm.

$$[\pi = \frac{22}{7}]$$

Matriculation Level 30/07/2025 (Shift - 1)  
(a) 16632 cm<sup>3</sup> (b) 17845 cm<sup>3</sup>  
(c) 26252 cm<sup>3</sup> (d) 16327 cm<sup>3</sup>

**Q.337.** From a cuboid of dimension 12 cm × 6 cm × 3 cm, small cubes of edge 3 cm are cut. How many small cubes are formed from the cuboid?  
Matriculation Level 31/07/2025 (Shift - 1)

- (a) 8 (b) 12 (c) 24 (d) 16

**Q.338.** The total surface area of a hemisphere is 1,539 m<sup>2</sup>. If the radius of the hemisphere is halved, what will be the new curved surface area of the hemisphere? (Use  $\pi = \frac{22}{7}$ )  
Matriculation Level 31/07/2025 (Shift - 1)

- (a) 192.5 m<sup>2</sup> (b) 255.82 m<sup>2</sup>  
(c) 481.5 m<sup>2</sup> (d) 314 m<sup>2</sup>

**Q.339.** If the volume of a cone is 84  $\pi$  cm<sup>3</sup> and the height is 7 cm. Find the radius of its base in nearest one decimal place.  
Matriculation Level 31/07/2025 (Shift - 1)

- (a) 6 cm (b) 5 cm (c) 4 cm (d) 7 cm

**Q.340.** Find the total surface area of a cube, the length of whose diagonal is  $\sqrt{75}$  cm.  
Matriculation Level 01/08/2025 (Shift - 1)

- (a) 150 cm<sup>2</sup> (b) 180 cm<sup>2</sup>  
(c) 210 cm<sup>2</sup> (d) 250 cm<sup>2</sup>

**Q.341.** A bottle is made by attaching a hemisphere to a cylinder of the same radius. Radius = 3.5 cm, height of cylinder = 12 cm. Find the total surface area of the bottle.  
Matriculation Level 01/08/2025 (Shift - 1)

- (a) 315.5 cm<sup>2</sup> (b) 341 cm<sup>2</sup>  
(c) 397 cm<sup>2</sup> (d) 309.5 cm<sup>2</sup>

**Q.342.** A cone has a slant height of 25 cm and a vertical height of 20 cm. What is the radius of the base?  
Matriculation Level 01/08/2025 (Shift - 1)

- (a) 15 cm (b) 12 cm (c) 18 cm (d) 9 cm

**Q.343.** A hollow hemisphere has an inner radius of 6 cm and an outer radius of 8 cm. Find its total surface area.  
Matriculation Level 01/08/2025 (Shift - 2)

- (a) 678.36 cm<sup>2</sup> (b) 754.29 cm<sup>2</sup>  
(c) 804.02 cm<sup>2</sup> (d) 716.56 cm<sup>2</sup>

**Q.344.** A cone has a base radius of 9 cm and height of 12 cm. What is its slant height?  
Matriculation Level 01/08/2025 (Shift - 2)

- (a) 15 cm (b) 14 cm (c) 13 cm (d) 16 cm

**Q.345.** Find the length of the longest pole that can be placed in a cubical room of edge 23 cm.  
Matriculation Level 01/08/2025 (Shift - 2)

- (a)  $16\sqrt{2}$  cm (b)  $23\sqrt{2}$  cm  
(c)  $23\sqrt{3}$  cm (d)  $16\sqrt{3}$  cm

**Q.346.** A cuboid with dimensions (12 × 6 × 3) meters is melted and recast into a cube. Find the total surface area of the cube.  
Matriculation Level 01/08/2025 (Shift - 3)

- (a) 216 m<sup>2</sup> (b) 324 m<sup>2</sup>  
(c) 486 m<sup>2</sup> (d) 600 m<sup>2</sup>

**Q.347.** The ratio of the radius of Hemisphere A to Hemisphere B is 3 : 1. If the curved surface area of Hemisphere A is equal to the volume of Hemisphere B, find the radius of Hemisphere B.  
Matriculation Level 01/08/2025 (Shift - 3)

- (a) 31 cm (b) 23 cm (c) 27 cm (d) 25 cm

**Q.348.** A wheel of a bicycle has a radius of 56 cm. How many revolutions will it make to cover a distance of 1.32 km?  
Matriculation Level 01/08/2025 (Shift - 3)

- (a) 276 revolutions (b) 352 revolutions  
(c) 375 revolutions (d) 415 revolutions

**Q.349.** The height and slant height of the cone are 20 cm and 29 cm respectively. Find the volume of the cone.  
Matriculation Level 01/08/2025 (Shift - 3)

- (a) 8160 cm<sup>3</sup> (b) 9240 cm<sup>3</sup>  
(c) 8820 cm<sup>3</sup> (d) 9360 cm<sup>3</sup>

**Q.350.** Find the volume of a cylinder whose radius is 7 cm and height is 14 cm. ( $\pi = \frac{22}{7}$ )  
Matriculation Level 02/08/2025 (Shift - 1)

- (a) 2156 cm<sup>3</sup> (b) 2464 cm<sup>3</sup>  
(c) 2695 cm<sup>3</sup> (d) 3080 cm<sup>3</sup>

**Q.351.** Which of the following statements is true about Rectangle?  
Matriculation Level 02/08/2025 (Shift - 1)

- (a) all sides are equal  
(b) All angles are right angles  
(c) Diagonals are not equal  
(d) opposite sides are unequal

**Q.352.** What is the area of a rhombus which has diagonals 12 cm and 6 cm?  
Matriculation Level 02/08/2025 (Shift - 1)

- (a) 35 cm<sup>2</sup> (b) 40 cm<sup>2</sup>  
(c) 36 cm<sup>2</sup> (d) 50 cm<sup>2</sup>

**Q.353.** A triangle has sides of length 5 cm, 9 cm, and 13 cm. Find its perimeter.  
Matriculation Level 02/08/2025 (Shift - 1)

- (a) 27 cm (b) 28 cm  
(c) 32 cm (d) 25 cm

**Q.354.** If the height of a cylinder is tripled and the radius remains the same, what will be the change in volume?  
Matriculation Level 02/08/2025 (Shift - 1)

- (a) Doubles (b) Triples  
(c) Halves (d) Stays the same

**Q.355.** A cylindrical concrete pillar has radius 0.7 m and height 5 m. If the curved surface is to be plastered at the rate of ₹200 per square metre, what will be the total cost of plastering for 10 such pillars? (Take  $\pi = 22/7$ )  
Matriculation Level 02/08/2025 (Shift - 1)

- (a) ₹38,500 (b) ₹42,000  
(c) ₹44,000 (d) ₹46,200

**Q.356.** How will the volume of a cube change if its side is doubled?  
Matriculation Level 02/08/2025 (Shift - 1)

- (a) Twice (b) Four times  
(c) Six times (d) Eight times

**Q.357.** A sphere and a hemisphere have the same radius. The ratio of their total surface areas is:  
Matriculation Level 02/08/2025 (Shift - 1)

- (a) 5 : 2 (b) 9 : 1 (c) 4 : 3 (d) 7 : 8

**Q.358.** A square and a rectangle have the same perimeter. The length of the rectangle is three times its width. If the side of the square is 16 meters, then what is the difference between the area of the square and the area of the rectangle?  
Matriculation Level 02/08/2025 (Shift - 1)



(a) 32 m<sup>2</sup> (b) 48 m<sup>2</sup> (c) 64 m<sup>2</sup> (d) 72 m<sup>2</sup>

**Q.359.** A regular hexagon has a side length of 8 cm. What is its perimeter?

Matriculation Level 29/08/2025 (Shift - 3)  
(a) 56 cm (b) 64 cm (c) 48 cm (d) 54 cm

**Q.360.** Which of the following statements is always true for any pyramid?

Matriculation Level 29/08/2025 (Shift - 3)  
(a) All faces are similar  
(b) The base is a polygon  
(c) All lateral faces are circular  
(d) All angles are right angles

**Q.361.** What is the slant height of a cone, base has a radius of 9 cm and a vertical height of 12 cm?

Matriculation Level 29/08/2025 (Shift - 3)  
(a) 14 cm (b) 15 cm (c) 16 cm (d) 18 cm

**Q.362.** If the height of a cone is Triples while keeping the radius the same, how does the volume change?

Matriculation Level 29/08/2025 (Shift - 3)  
(a) Doubles (b) Triples  
(c) Stays the same (d) Becomes half

**Q.363.** The radius of a spherical ball increases from 3 cm to 9 cm. How many times does its surface area increase?

Matriculation Level 29/08/2025 (Shift - 3)  
(a) 2 (b) 3 (c) 6 (d) 9

**Q.364.** A company manufactures hollow spherical balls with internal radius of 10 cm. If the thickness of the rubber is 0.5 cm, what is the volume of rubber used in making one ball? (Use  $\pi = \frac{22}{7}$ )

Matriculation Level 29/08/2025 (Shift - 3)  
(a) 560.50 cm<sup>3</sup> (b) 660.52 cm<sup>3</sup>  
(c) 720.00 cm<sup>3</sup> (d) 840.25 cm<sup>3</sup>

**Q.365.** A bowl is in the shape of a hemisphere with radius 14 cm. What is its total surface area? ( $\pi = 3.14$ )

Matriculation Level 29/08/2025 (Shift - 3)  
(a) 1772 cm<sup>2</sup> (b) 1848 cm<sup>2</sup>  
(c) 1648 cm<sup>2</sup> (d) 1856 cm<sup>2</sup>

**Q.366.** A circle and a square have the same perimeter. Which of these will have larger area?

Matriculation Level 29/08/2025 (Shift - 3)  
(a) circle  
(b) Square  
(c) Both are equal  
(d) Depends on dimensions

**Q.367.** The area of a trapezium with bases 12 cm, 8 cm, and height 5 cm is:

Matriculation Level 29/08/2025 (Shift - 3)  
(a) 45 cm<sup>2</sup> (b) 48 cm<sup>2</sup>  
(c) 50 cm<sup>2</sup> (d) 52 cm<sup>2</sup>

**Q.368.** Two rectangles have the same area. The first has sides 18cm and 22 cm. The second has one side 12 cm. What is the other side of the second rectangle?

Matriculation Level 29/08/2025 (Shift - 3)  
(a) 33 cm (b) 37 cm (c) 41 cm (d) 43 cm

**Q.369.** A cylinder is cut horizontally into two equal parts. How will its volume change?

Matriculation Level 29/08/2025 (Shift - 3)  
(a) doubles (b) becomes half  
(c) remains the same (d) increase of 25%

**Q.370.** The parallel sides of a trapezium differ by 4 cm. Its height is 6 cm and the area is 60 cm<sup>2</sup>. What are the lengths of the parallel sides?

Higher Secondary 24/07/2025 (Shift - 2)  
(a) 6 cm and 10 cm (b) 7 cm and 11 cm  
(c) 8 cm and 12 cm (d) 5 cm and 9 cm

**Q.371.** Calculate the volume of a prism standing on a trapezoidal base. The parallel sides of the base are 5 cm and 7 cm, and they are 4 cm apart. The height of the prism is 10 cm.

(a) 480 cm<sup>3</sup> (b) 120 cm<sup>3</sup>  
(c) 240 cm<sup>3</sup> (d) 360 cm<sup>3</sup>

**Q.372.** If the lateral surface area of a right circular cylinder is 264 cm<sup>2</sup> and radius is 3.5 cm, find the height.

Higher Secondary 24/07/2025 (Shift - 2)  
(a) 12 cm (b) 11 cm (c) 10 cm (d) 13 cm

**Q.373.** A cylinder with a base radius of 8 cm is melted and converted into 16 smaller cylinders of the same height. What is the radius of the base of each small cylinder?

(a) 4 cm (b) 1 cm (c) 2 cm (d) 0.5 cm

**Q.374.** If the side of a square is reduced to half of its original length, the area of the square becomes:

Higher Secondary 25/07/2025 (Shift - 2)  
(a) One-half (b) One-fourth  
(c) Two times (d) Four times

**Q.375.** A room has a floor in the shape of a rectangle (25 m × 18 m), with a triangular extension on one end. The triangle has a base of 15 m and height of 10 m. What is the total area of the floor?

Higher Secondary 25/07/2025 (Shift - 2)  
(a) 252 m<sup>2</sup> (b) 525 m<sup>2</sup>  
(c) 550 m<sup>2</sup> (d) 335 m<sup>2</sup>

**Q.376.** A solid sphere of radius 10 cm is melted and recast into smaller spheres of radius 2 cm. How many such small spheres will be formed?

Higher Secondary 25/07/2025 (Shift - 2)  
(a) 100 (b) 125 (c) 500 (d) 250

**Q.377.** Determine the volumetric measure of a cuboidal solid with a length of 24 meters, a breadth of 14 meters, and a height of 10 meters.

Higher Secondary 26/07/2025 (Shift - 2)  
(a) 3360m<sup>3</sup> (b) 3350m<sup>3</sup>  
(c) 3260m<sup>3</sup> (d) 3920m<sup>3</sup>

**Q.378.** A rectangular parallelepiped tank has dimensions 8 m × 6 m × 5 m. If it is filled with water and the cost of water is ₹50 per 1000 liters, what is the total cost to fill the tank? (1 m<sup>3</sup> = 1000 liters)

Higher Secondary 26/07/2025 (Shift - 2)  
(a) ₹16750 (b) ₹17000  
(c) ₹12000 (d) ₹13000

**Q.379.** A cuboidal block of metal is melted and recast into 12 identical cubes. If the original block had dimensions 24 cm × 18 cm × 6 cm, what is the edge of each cube?

Higher Secondary 28/07/2025 (Shift - 2)  
(a) 8 cm (b) 5 cm (c) 6 cm (d) 4 cm

**Q.380.** Five cubes, each with an edge of 4 cm are joined end to end. Find the total surface area of the resulting cuboid.

Higher Secondary 28/07/2025 (Shift - 2)  
(a) 320 cm<sup>2</sup> (b) 384 cm<sup>2</sup>  
(c) 352 cm<sup>2</sup> (d) 480 cm<sup>2</sup>

**Q.381.** One diagonal of a rhombus is double the length of the other. If the area of the rhombus is 100 cm<sup>2</sup>, what are the lengths of the diagonals?

(a) 10 cm and 20 cm (b) 5 cm and 10 cm  
(c) 8 cm and 16 cm (d) 20 cm and 40 cm

**Q.382.** Two right prisms have the same height, but their bases are triangles with areas in the ratio 2 : 5. What is the ratio of their volumes?

Higher Secondary 29/07/2025 (Shift - 2)  
(a) 4 : 25 (b) 2 : 5 (c) 8 : 125 (d) 5 : 2

**Q.383.** A right circular cylinder has height 10 cm and curved surface area 220 cm<sup>2</sup>.

What is its volume? (use  $\pi = \frac{22}{7}$ )

Higher Secondary 29/07/2025 (Shift - 2)  
(a) 385 cm<sup>3</sup> (b) 350 cm<sup>3</sup>  
(c) 420 cm<sup>3</sup> (d) 770 cm<sup>3</sup>

**Q.384.** A cuboidal tank has dimensions: Length = 30 cm, Breadth = 25 cm, and Height = 24 cm. If the length is increased by 25%, the breadth is decreased by 12.5%, and the height is increased by 50%, then by what percentage does the volume of the tank change?



Higher Secondary 30/07/2025 (Shift - 2)  
 (a) 61.25% increase (b) 60% decrease  
 (c) 64.06% increase (d) 78.25% decrease

**Q.385.** How much iron sheet (in m<sup>2</sup>) will be needed to construct a rectangular tank measuring 8m × 10m × 11m, if a circular opening of radius 2.1 meters is to be left at the top of the tank?

Higher Secondary 30/07/2025 (Shift - 2)  
 (a) 330.82 m<sup>2</sup> (b) 561.04 m<sup>2</sup>  
 (c) 542.14 m<sup>2</sup> (d) 646.98 m<sup>2</sup>

**Q.386.** Which of the following is NOT a polynomial?

Higher Secondary 30/07/2025 (Shift - 2)  
 (a)  $ab^3 + a^3b$  (b)  $y^2 + 3y + 2$   
 (c)  $\sqrt{y}, y^{\frac{1}{3}}, \text{ or } y^{-3}$  (d)  $bc + cd + ef + 9a^2$

**Q.387.** The total length around a regular 9-sided polygon is 72 cm. If the length of each side is increased by 1.5 cm, what will be the new perimeter?

Higher Secondary 31/07/2025 (Shift - 2)  
 (a) 82.5 cm (b) 85.5 cm  
 (c) 88.5 cm (d) 90.0 cm

**Q.388.** A metallic sphere of radius 14 cm is melted to form a cone of height 56 cm. Find the radius of the cone.

Higher Secondary 31/07/2025 (Shift - 2)  
 (a) 14 cm (b) 21 cm (c) 17 cm (d) 11 cm

**Q.389.** What is the ratio of surface areas of two spheres whose radii are in the ratio 4 : 5?

Higher Secondary 02/08/2025 (Shift - 2)  
 (a) 25 : 16 (b) 9 : 4 (c) 9 : 16 (d) 16 : 25

**Q.390.** A cylindrical container with internal radius 9 cm and height 36 cm is completely filled with water. The water is then poured into identical conical vessels, each having a radius of 4.5 cm and height of 8 cm. How many such conical vessels can be completely filled with the water from the cylinder?

Higher Secondary 02/08/2025 (Shift - 2)  
 (a) 55 (b) 54 (c) 65 (d) 72

**Q.391.** The length of the base and height of a triangle is 30 cm and 15 cm, respectively. If the area of the triangle is equal to the area of a rectangle whose length is 25 cm, then find the breadth of the same rectangle.

Higher Secondary 02/08/2025 (Shift - 2)  
 (a) 8 cm (b) 11 cm (c) 9 cm (d) 15 cm

**Q.392.** A decorative candle consists of two parts: a cylindrical base and a conical top. The radius of both parts is 7 cm. The cylindrical part is 15 cm high, while the conical part rises to a height of

24 cm above it. Determine the estimated total surface area of the candle. (take  $\pi = \frac{22}{7}$ )

Higher Secondary 29/08/2025 (Shift - 2)  
 (a) 1364cm<sup>2</sup> (b) 1540cm<sup>2</sup>  
 (c) 1200 cm<sup>2</sup> (d) 1384 cm<sup>2</sup>

**Q.393.** If the perimeter of an equilateral triangle is 24 units, what is its area (in square units)?

Higher Secondary 29/08/2025 (Shift - 2)  
 (a)  $144\sqrt{2}$  (b)  $36\sqrt{2}$  (c)  $144\sqrt{3}$  (d)  $16\sqrt{3}$

**Q.394.** Each side of a square park is 89 meters. a 4-meter-wide uniform path inside the park along its boundary is formed. Find the difference between the perimeter of the park and the path now.

Higher Secondary 29/08/2025 (Shift - 2)  
 (a) 35 meters (b) 32 meters  
 (c) 30 meters (d) 34 meters

**Q.395.** The sides of a triangle are in the ratio 15 : 17 : 27. If the perimeter of the triangle is 2655 cm, what is the length of the smallest side in metres?

Higher Secondary 29/08/2025 (Shift - 2)  
 (a) 675 m (b) 680 m (c) 695 m (d) 685 m

**Q.396.** A solid metallic sphere of radius  $R$  is melted and recast into a solid cone of height  $R$  and base radius  $r$ . What is the ratio  $r : R$ ?

Graduate Level 24/07/2025 (Shift - 3)  
 (a) 1 : 2 (b) 2 : 1 (c) 1 : 4 (d) 4 : 1

**Q.397.** A right prism has a rectangular base with dimensions 8 cm and 12 cm. The height of the prism is 20 cm. Find the total surface area of the prism.

Graduate Level 24/07/2025 (Shift - 3)  
 (a) 800 cm<sup>2</sup> (b) 896 cm<sup>2</sup>  
 (c) 992 cm<sup>2</sup> (d) 1024 cm<sup>2</sup>

**Q.398.** A rectangular garden 30 m × 20 m has four full circular flowerbeds of radius 3.5 m placed within it. What is the area left for grass?

Graduate Level 24/07/2025 (Shift - 3)  
 (a) 420 m<sup>2</sup> (b) 446 m<sup>2</sup>  
 (c) 456 m<sup>2</sup> (d) 512 m<sup>2</sup>

**Q.399.** The base of a right prism is a regular hexagon with a side length of 12 cm. The height of the prism is 16 cm. Find the volume of the prism.

Graduate Level 25/07/2025 (Shift - 3)  
 (a) 5536 cm<sup>3</sup> (b) 5986 cm<sup>3</sup>  
 (c) 5048 cm<sup>3</sup> (d) 5304 cm<sup>3</sup>

**Q.400.** A composite shape has a rectangle (15 cm × 6 cm) with a semicircle on one short side. What is the total area of the shape?

Graduate Level 25/07/2025 (Shift - 3)  
 (a) 104.13 cm<sup>2</sup> (b) 158.12 cm<sup>2</sup>  
 (c) 111.28 cm<sup>2</sup> (d) 106.12 cm<sup>2</sup>

**Q.401.** The base of a right prism is a regular hexagon with a side length of 6 cm. The height of the prism is 13 cm. Find the volume of the prism.

Graduate Level 26/07/2025 (Shift - 3)  
 (a) 1536.9 cm<sup>3</sup> (b) 1215.9cm<sup>3</sup>  
 (c) 2048.9 cm<sup>3</sup> (d) 2304.9cm<sup>3</sup>

**Q.402.** A composite shape has a rectangle (20 cm × 8cm) with a semicircle on one short side. What is the total area of the shape?

Graduate Level 26/07/2025 (Shift - 3)  
 (a) 154.22 cm<sup>2</sup> (b) 158.34 cm<sup>2</sup>  
 (c) 185.14 cm<sup>2</sup> (d) 156.42 cm<sup>2</sup>

**Q.403.** The base of a right prism is a square of side 15 cm. If the height of the prism is 8 cm, find its volume in cubic centimetres.

Graduate Level 28/07/2025 (Shift - 3)  
 (a) 1710 (b) 1800 (c) 1720 (d) 1690

**Q.404.** A sphere and a right circular cone have the same volume and radius. Find the ratio of their curved surface areas (sphere: Cone).

Graduate Level 28/07/2025 (Shift - 3)  
 (a)  $4 : \sqrt{17}$  (b)  $3 : \sqrt{17}$   
 (c)  $4 : \sqrt{15}$  (d)  $7 : \sqrt{5}$

**Q.405.** A solid metallic sphere of radius 10 cm is melted and recast into 125 identical spheres. What is the ratio of the surface area of the original sphere to the total surface area of 6 smaller spheres so formed?

Graduate Level 28/07/2025 (Shift - 3)  
 (a) 49 : 108 cm (b) 25 : 6 cm  
 (c) 109 : 84 cm (d) 25 : 96 cm

**Q.406.** A solid hemisphere is melted into a smaller hemisphere and a cone of equal radii. The radius of both the smaller hemisphere and the cone is  $r$ , and the height of the cone is also  $r$ . Find the radius  $R$  of the original hemisphere.

Graduate Level 29/07/2025 (Shift - 3)  
 (a)  $\sqrt[3]{1.2} \times r$  (b)  $\sqrt[3]{1.5} \times r$   
 (c)  $\sqrt[3]{2} \times r$  (d)  $\sqrt[3]{3} \times r$

**Q.407.** The base of a right prism is a rectangle with sides 6 cm and 8 cm. The height of the prism is 14 cm. What is the volume?

Graduate Level 29/07/2025 (Shift - 3)  
 (a) 672 cm<sup>3</sup> (b) 456 cm<sup>3</sup>  
 (c) 628 cm<sup>3</sup> (d) 700 cm<sup>3</sup>

**Q.408.** Two semicircles of radius 14 cm are attached to opposite ends of a



rectangle of 21 cm × 14 cm. What is the total area of the figure?

Graduate Level 29/07/2025 (Shift - 3)

- (a) 951.94 cm<sup>2</sup> (b) 769.5 cm<sup>2</sup>  
(c) 885.52 cm<sup>2</sup> (d) 909.44 cm<sup>2</sup>

**Q.409.** The base of a right prism is a right-angled triangle with legs 40 cm and 41 cm. The height of the prism is 16 cm. Find the lateral surface area of the prism.  
Graduate Level 31/07/2025 (Shift - 3)

- (a) 1440 cm<sup>2</sup> (b) 1660 cm<sup>2</sup>  
(c) 1672 cm<sup>2</sup> (d) 1684 cm<sup>2</sup>

**Q.410.** The total surface area of a solid in the shape of a hemisphere mounted on a cylinder both with equal radius and height of cylinder being twice its radius is equal to.  
Graduate Level 31/07/2025 (Shift - 3)

- (a)  $8\pi r^2$  (b)  $4\pi r^2$  (c)  $7\pi r^2$  (d)  $(2\pi rh + 2\pi r^2)$

**Q.411.** A right prism has a regular pentagonal base of side length 8 cm and height 13 cm. Find the lateral surface area.  
Graduate Level 02/08/2025 (Shift - 3)

- (a) 250 cm<sup>2</sup> (b) 580 cm<sup>2</sup>  
(c) 300 cm<sup>2</sup> (d) 520 cm<sup>2</sup>

**Q.412.** A hemisphere has radius R. A sphere of maximum possible radius is inscribed in the hemisphere such that it touches the curved surface internally and rests on the hemisphere's base. Find the ratio of their volume.  
Graduate Level 02/08/2025 (Shift - 3)

- (a) 4 : 1 (b) 5 : 4 (c) 6 : 5 (d) 7 : 5

**Q.413.** An equilateral triangle and a regular hexagon share the same perimeter. What is the ratio of their areas?  
Graduate Level 02/08/2025 (Shift - 3)

- (a) 1 : 5 (b) 2 : 3 (c) 1 : 2 (d) 2 : 5

**Q.414.** Find the radius of a sphere whose volume is numerically equal to its curved surface area.  
Graduate Level 29/08/2025 (Shift - 1)

- (a) 3 units (b) 4 units  
(c) 2 units (d) 1 units

**Q.415.** Two solid hemispheres of radii 2R and 3R are melted and recast into a sphere and a cone. The cone has the same radius as the new sphere, and its height is three times the sphere's radius. Find the radius of the newly formed sphere.  
Graduate Level 29/08/2025 (Shift - 1)

- (a)  $\sqrt[3]{8} \times R$  (b)  $\sqrt[3]{10} \times R$   
(c)  $\sqrt[3]{12} \times R$  (d)  $\sqrt[3]{15} \times R$

**Q.416.** A company produces plastic signs shaped like regular hexagons of side 10 cm. If plastic costs ₹4/cm<sup>2</sup> and design costs ₹250 per piece, what is the total cost per sign?  
Graduate Level 29/08/2025 (Shift - 1)

- (a) ₹1288 (b) ₹2564 (c) ₹2500 (d) ₹4000

**Q.417.** A running track is comprised of a rectangular shape with semicircles at each of its shorter ends. The combined area of the two semicircles is 80% of the area of the central rectangular section. What is the approximate ratio of the length of the rectangular section to its width? (Use  $\pi = 3.14$ )  
Graduate Level 29/08/2025 (Shift - 1)

- (a) 1 : 3.14 (b) 15.7 : 16  
(c) 1 : 2 (d) 3.14 : 1

### eduquity-based pattern (ebp) Questions

**Q.418.** The surface area of a cube is 1728 cm<sup>2</sup>. Find its volume.

- (a)  $6453\sqrt{2}$  cm<sup>3</sup> (b)  $5436\sqrt{2}$  cm<sup>3</sup>  
(c)  $3456\sqrt{2}$  cm<sup>3</sup> (d)  $4356\sqrt{2}$  cm<sup>3</sup>

**Q.419.** A cylinder has a radius  $\frac{8}{\pi}$  cm and its volume is 448 cm<sup>3</sup>. What is the height of the cylinder (taken  $\pi = \frac{22}{7}$ )?

- (a) 24 cm (b) 21 cm (c) 23 cm (d) 22 cm

**Q.420.** The radii of the two spheres are in the ratio 4 : 5. Their volume will be in the ratio:

- (a) 64 : 125 (b) 125 : 64  
(c) 16 : 25 (d) 4 : 5

**Q.421.** For making a toy, a hemisphere is attached at one end of a cylinder and a cone is attached at the other end of the cylinder. The cylinder, the cone and the hemisphere have a common radius of 4.2 cm. The height of the cylinder and that of the cone is 7 cm. Find the volume (in cm<sup>3</sup>) of the toy. (Use  $\pi = \frac{22}{7}$ )

- (a) 358.8 (b) 762.255  
(c) 672.672 (d) 863.25

**Q.422.** The area of a regular hexagon is  $2048\sqrt{3}$  cm<sup>2</sup>. What is the length (in cm) of each side of the hexagon?

- (a)  $32\sqrt{3}$  (b)  $\frac{64\sqrt{3}}{3}$  (c)  $\frac{32\sqrt{3}}{3}$  (d)  $64\sqrt{3}$

**Q.423.** A right circular cone of height h and radius R is cut by a plane parallel to the base at a distance  $\frac{h}{4}$  from the base.

The ratio of the curved surface areas of the resulting cone to the frustum is:

- (a) 7 : 16 (b) 16 : 9 (c) 16 : 7 (d) 9 : 7

**Q.424.** What is the number of gold coins, each having a radius of 0.875 cm and a thickness of 2 mm, that need to be melted to create a cuboid with dimensions 5.5 cm × 5 cm × 3.5 cm?

(Use  $\pi = \frac{22}{7}$ )

- (a) 250 (b) 175 (c) 200 (d) 225

**Q.425.** A cylindrical tank of height 10 m and radius 5 m has a small circular cut of 0.5 m radius on its top surface. What would be the total surface area of the tank?

- (a)  $\frac{599\pi}{4}$  m<sup>2</sup> (b)  $156\pi$  m<sup>2</sup>  
(c)  $\frac{299\pi}{2}$  m<sup>2</sup> (d)  $150\pi$  m<sup>2</sup>

**Q.426.** A solid cylinder with a height of 8 units and a radius of 3.5 units is to be covered with a paper sheet. If the cost of the paper sheet is ₹40 per 5 square units, the cost (in ₹) of covering the cylinder is \_\_\_\_\_ (Take  $\pi = \frac{22}{7}$ )

- (a) 2268 (b) 1664 (c) 2024 (d) 1012

**Q.427.** A hemispherical bowl is 264 cm round at the brim. Suppose that the bowl is half-full of beverage. If the beverage is served in hemispherical glasses 6 cm in the diameter at the top, how many people would be served?

- (a) 1170 (b) 1372 (c) 1270 (d) 1470

**Q.428.** What is the area of the remaining portion of a circle of radius 14 cm when a square with maximum possible area inscribed in it is removed from it? (Use  $\pi = \frac{22}{7}$ )

- (a) 518 cm<sup>2</sup> (b) 520 cm<sup>2</sup>  
(c) 224 cm<sup>2</sup> (d) 616 cm<sup>2</sup>

**Q.429.** The volume of the hemisphere is  $486\pi$  cm<sup>3</sup>. If the radius is tripled, then what will be its curved surface area (in cm<sup>2</sup>)?

- (a) 1458 (b)  $1025\pi$  (c)  $1458\pi$  (d) 1546

**Q.430.** A hemispherical bowl of internal radius 15 cm contains a liquid. The liquid is to be filled into cylindrical shaped bottles of diameter 5 cm and height 6 cm. The number of bottles required to empty the bowl is:

- (a) 40 (b) 30 (c) 60 (d) 50

**Q.431.** A solid cone of base radius 7 cm and height 18 cm is melted and recast into a hollow metal hemisphere of internal and external radii 20 cm and R



cm, respectively. What is the value of R (in cm)?

- (a)  $\sqrt[3]{8144}$  (b)  $\sqrt[3]{8441}$   
(c)  $\sqrt[3]{8448}$  (d)  $\sqrt[3]{8568}$

**Q.432.** A solid spherical item is melted into the shape of a cylinder mounted by a cone having same radius. The diameters of the cylinder and sphere are 21 cm each. What is the height (in cm) of the cone if the height of the cylinder is 7 cm?

(Take  $\pi = \frac{22}{7}$ )

- (a) 21 (b) 10.5 (c) 14 (d) 7

**Q.433.** ABCDEF is a regular hexagon, and  $m(\overline{AD}) = 12$  cm. What is the area (in  $\text{cm}^2$ ) of the given hexagon?

- (a)  $216\sqrt{3}$  (b)  $54\sqrt{3}$  (c)  $108\sqrt{3}$  (d)  $162\sqrt{3}$

**Q.434.** A right circular cone of height  $h$  and radius  $R$  is cut by a parallel to the base at a distance  $\frac{h}{5}$  from the base. The ratio of the curved surface area of the frustum and total surface area of the resulting cone is:

- (a)  $16(R+L) : 9L$  (b)  $9(R+L) : 16L$   
(c)  $16L : 9(R+L)$  (d)  $9L : 16(R+L)$

**Q.435.** The radius of the base and height of a right circular cone are in the ratio 8 : 15. Find the slant height of the cone if its volume is  $1005\frac{5}{7} \text{ cm}^3$

- (a) 12 cm (b) 17 cm (c) 21 cm (d) 15 cm

**Q.436.** A rectangular sheet of length 28 cm and breadth 16 cm is folded to meet the edges so that it is made into a cylindrical pipe in two possible ways – one along the length and the other along the breadth. The larger volume is how much percentage of smaller volume?

- (a) 165% (b) 180% (c) 185% (d) 175%
- Q.437.** A conical tent is to accommodate 12 persons. Each person must have  $5 \text{ m}^2$  of the space on the ground and  $20 \text{ m}^3$  of air to breathe. The height of the tent should be:

- (a) 12 m (b) 13 m (c) 12.5 m (d) 22 m

**Q.438.** A medicine capsule is shaped like a cylinder with two hemispheres attached to each of its ends. The length of the entire capsule is 15 mm, and its diameter is 6 mm. Calculate the surface area (in  $\text{mm}^2$ , rounded to the nearest integer) of the capsule. (Use  $\pi = \frac{22}{7}$ )

- (a) 286 (b) 283 (c) 289 (d) 280

**Q.439.** A hemispherical container of total

surface area  $7392 \text{ cm}^2$  is filled with oil.

However,  $\frac{1}{4}$  of the container is empty.

This oil is to be transferred in small hemispherical bottles having radius of 6.3 cm. How many bottles will be required (rounded off to the nearest whole number)? (Take  $\pi = \frac{22}{7}$ .)

- (a) 351 (b) 66 (c) 75 (d) 156

**Q.440.** A cone of radius 16 cm and height 12 cm is cut, parallel to the base, at some height closer to the top. If the ratio of the heights of the solids is 1 : 3, then the ratio of the total surface areas of the two solids formed is approximately given by:

- (a) 9 : 143 (b) 1 : 8 (c) 2 : 15 (d) 7 : 180

**Q.441.** A conical cavity is drilled in a solid circular cylinder of 7 cm height and 48 cm base diameter. The height and base diameter of the cone are the same as those of the cylinder. What is the total surface area of the remaining solid?

- (a)  $1512\pi \text{ cm}^2$  (b)  $1076\pi \text{ cm}^2$   
(c)  $1420\pi \text{ cm}^2$  (d)  $1654\pi \text{ cm}^2$

**Q.442.** The side of a square exceeds the side of another square by 3 cm and the sum of the areas of the two squares is  $225 \text{ cm}^2$ . Find the sum of the perimeter (in cm) of both squares.

- (a) 84 (b) 74 (c) 64 (d) 94

**Q.443.** The external curved surface area of a metallic hemispherical bowl is  $308 \text{ cm}^2$ . If the density of metal is  $8.7 \text{ gm per cm}^3$ , then the mass of the bowl is equal to: (Take  $\pi = \frac{22}{7}$ )

- (a) 6254.2 gm (b) 6252.4 gm  
(c) 5262.4 gm (d) 5264.2 gm

**Q.444.** From a square with area  $a^2$  another square made on mid points of the sides of the square is removed. What is the area left in the square?

- (a)  $\frac{a^2}{2}$  (b)  $\frac{2a^2}{3}$  (c)  $\frac{a^2}{4}$  (d)  $\frac{3a^2}{4}$

**Q.445.** In a lead pipe 4.2 m long, the external diameter is 5.4 cm and the thickness of the pipe is 3 mm. Find the weight of the pipe, if  $1 \text{ cm}^3$  of lead mass is 25 grams. (Use  $\pi = \frac{22}{7}$ )

- (a) 54.90 kg (b) 45.09 kg  
(c) 59.40 kg (d) 50.49 kg

**Q.446.** A solid sphere of iron having radius 15 cm is to be melted and recast in the shape of a cylindrical pipe having internal and external radii 4 cm and 6 cm,

respectively. Find the length (in m) of the pipe thus formed.

- (a) 2.15 (b) 2.25 (c) 2.45 (d) 2.35

**Q.447.** The sum of the radius and slant height of a cone is square of the height of the cone, while the difference between the radius and slant height is 1 cm only. What is the total surface area (in  $\text{cm}^2$ ) of the cone having height of 9 cm? (Take  $\pi = \frac{22}{7}$ ) (Rounded off to 2 decimal places)

- (a) 10,437.43 (b) 11,282.85  
(c) 10,182.86 (d) 10,128.85

**Q.448.** A cylinder of radius 10 cm and height 30 cm is cut, parallel to the base, into two pieces of different sizes. The total surface area of the larger piece formed is 70% of that of the given cylinder. The smaller piece is melted and made into a sphere. The radius (in cm) of the sphere is:

- (a)  $\sqrt[3]{821}$  (b)  $\sqrt[3]{900}$  (c)  $\sqrt[3]{986}$  (d)  $\sqrt[3]{844}$

**Q.449.** Find the volume of the largest right circular cone which is inscribed in a hollow cube of length 6 units and is standing on its base.

- (a)  $18\pi \text{ units}^3$  (b)  $9\pi \text{ units}^3$   
(c)  $21\pi \text{ units}^3$  (d)  $10\pi \text{ units}^3$

**Q.450.** A cylindrical vessel of height 14 cm has outer and inner radii 14 cm and 10 cm, respectively. It is to be filled by transferring water in it by using a small cylinder of height 7 cm and outer and inner radii 7 cm and 5 cm, respectively. What will be the unfilled height of the big vessel after transferring water 6 times by using the smaller cylinder? (Use  $\pi = \frac{22}{7}$ )

- (a) 2.2 cm (b) 3.5 cm  
(c) 2.9 cm (d) 3.8 cm

**Q.451.** A cylindrical tube, which is open at both ends, is crafted from a metal. The internal diameter of the tube is 12 cm, and its length is 35 cm. The metal throughout has a thickness of 2 cm. Determine the total surface area (in  $\text{cm}^2$ ) of the metal tube. (Take  $\pi = \frac{22}{7}$ )

- (a) 3696 (b) 3256 (c) 3854 (d) 3145

**Q.452.** The perimeter of a rhombus is 160 cm and one of its diagonals has a length 50 cm. Find the area (in  $\text{cm}^2$ ) of the rhombus.

- (a)  $280\sqrt{39}$  (b)  $300\sqrt{39}$   
(c)  $250\sqrt{39}$  (d)  $175\sqrt{39}$

**Q.453.** The diagonals of two squares are in the ratio of 5 : 6. What is the ratio of their areas?



(a) 18 : 20 (b) 25 : 36 (c) 23 : 18 (d) 2 : 3

**Q.454.** An ice-cream cone with a lid has a total surface area equal to the area of another circle of certain radius. If the radius of the circle is equal to average of the slant height and the radius of the cone, then the ratio of the slant height of the cone to the radius of the circle is:

(a) 3 : 1 (b) 2 : 3 (c) 3 : 2 (d) 1 : 3

**Q.455.** The diameter of a hollow hemisphere is 30 cm. Water is filled in it up to its brim. When spherical lead shots each of radius 0.75 cm are dropped into the hemisphere, two-fifth of the water flows out. How many lead shots were dropped into the hemisphere ?

(a) 1575 (b) 2250 (c) 1600 (d) 1380

**Q.456.** In a conical tent of diameter 13.2 m and vertical height 14 m, how many full bags of wheat can be emptied, if the space required for the wheat in each bag

is  $29.04 \text{ m}^3$  ? [Use  $\pi = \frac{22}{7}$ ]

(a) 25 (b) 28 (c) 22 (d) 21

**Q.457.** What is the radius of the largest right circular cone that can be cut out from a cube of edge 8.4 cm ?

(a) 4.1 cm (b) 4.2 cm  
(c) 2.2 cm (d) 2.1 cm

**Q.458.** A right circular solid metallic cone is 7.2 cm high and radius of its base 3.5 cm. It is melted and recast into two right circular solid metallic cones equal in all respects, each with base radius 2.1 cm and height 'h'. What is the value of 'h' (in cm) ?

(a) 14 (b) 10 (c) 21 (d) 12

**Q.459.** The radius of a circle is equal to the side of a square. The perimeters of the square and circle are in the ratio 7 : 11, then the ratio of the areas will be :

(Use  $\pi = \frac{22}{7}$ )

(a) 14 : 22 (b) 22 : 14 (c) 22 : 7 (d) 7 : 22

**Q.460.** The internal and external diameters of a bowl in the shape of a hemisphere are 26.6 cm and 28 cm, respectively. What is the cost (in ₹, to the nearest rupee) of painting it, all over, at the rate of ₹3.75/cm<sup>2</sup> ? (Take  $\pi = \frac{22}{7}$ ).

(a) 9,380 (b) 9,015 (c) 8,950 (d) 9,000

**Q.461.** If the radii of two right circular cylinders are in the ratio of 2 : 3 and their heights are in the ratio of 5 : 4, then find the ratio of their curved surface areas ?

(a) 6 : 5 (b) 5 : 6 (c) 6 : 10 (d) 10 : 6

**Q.462.** A hollow hemispherical steel bowl can contain  $\frac{686}{3}\pi$  cubic unit of

liquid. The outer radius is 9 units. How much steel is used in making the bowl ?

(a)  $\frac{772}{3}\pi$  cubic units

(b)  $\frac{433}{3}\pi$  cubic units

(c)  $\frac{727}{3}\pi$  cubic units

(d)  $\frac{343}{3}\pi$  cubic units

**Q.463.** Air is leaking from a spherical-shaped advertising balloon at the rate of 24 cubic feet per minute. If the radius of the balloon is 8 feet, how long would it take for the balloon to empty? Round your answer to the nearest minute. Use the approximate value of  $\pi$ , that is 3.14.

(a) 89 minutes (b) 80 minutes

(c) 65 minutes (d) 94 minutes

**Q.464.** A solid hemisphere of radius 12 cm is melted into N identical cones, each of radius 6 cm and height 8 cm. What is the value of N ?

(a) 12 (b) 8 (c) 10 (d) 15

**Q.465.** The radii of the two cones are in the ratio of 2 : 5 and their volumes are in the ratio of 3 : 5. What is the ratio of their heights?

(a) 15 : 4 (b) 11 : 15 (c) 13 : 11 (d) 4 : 11

**Q.466.** What is the volume (in cm<sup>3</sup>) of a cylinder if the radius of the cylinder is 8 cm and the height is 14 cm ?

(Take  $\pi = \frac{22}{7}$ )

(a) 2686 (b) 2816 (c) 2784 (d) 2456

**Q.467.** From a circle with the radius of 15.75 cm, a sector with the arc length of 11 cm is cut off. Find the area (in cm<sup>2</sup>) of this sector.

(a) 86.525 (b) 86.625

(c) 86.875 (d) 86.125

**Q.468.** A right-angled isosceles triangle has an area of 50 square units. Its hypotenuse is (in units):

(a)  $10\sqrt{3}$  (b)  $5\sqrt{5}$  (c)  $10\sqrt{2}$  (d)  $5\sqrt{2}$

**Q.469.** What is the perimeter (in cm, rounded off to one decimal place) of the sector of a circle with a radius of 10 cm and of angle 30°? (Use  $\pi = 3.14$ )

(a) 20.4 (b) 35.7 (c) 82.8 (d) 25.2

**Q.470.** If a right circular cone of height 24 cm has a volume of 1232 cm<sup>3</sup>, then the total surface area of the cone is (use

$\pi = \frac{22}{7}$ ) :

(a) 806 cm<sup>2</sup> (b) 704 cm<sup>2</sup>

(c) 904 cm<sup>2</sup> (d) 608 cm<sup>2</sup>

**Q.471.** A rectangular piece of paper is 50 cm long and 22 cm wide. If a cylinder is formed by rolling the paper along its breadth, then the volume of the cylinder is:

(a) 2,125 cm<sup>3</sup> (b) 1,925 cm<sup>3</sup>

(c) 2,025 cm<sup>3</sup> (d) 1,825 cm<sup>3</sup>

**Q.472.** A rectangular courtyard is 4m 95cm long and 16m 65cm broad. It is to be paved with the square tiles of the same size. Find the least number of such square tiles required to pave the rectangular courtyard.

(a) 877 (b) 944 (c) 388 (d) 407

**Q.473.** The inner and outer radii of a circular track are 28 m and 25 m, respectively. The cost (in ₹) of levelling the track at ₹28 per m<sup>2</sup> is: (Use  $\pi = \frac{22}{7}$ )

(a) 15,421

(b) 13,992

(c) 11,229

(d) 17,355

**Q.474.** If the perimeter of an equilateral triangle is 36 units, then its area is (in square units):

(a)  $144\sqrt{2}$  (b)  $36\sqrt{2}$  (c)  $144\sqrt{3}$  (d)  $36\sqrt{3}$

**Q.475.** If the ratio of the areas of two similar triangles is 25 : 144, the ratio of their corresponding sides is:

(a) 5 : 12 (b) 5 : 144 (c) 25 : 12 (d) 12 : 5

**Q.476.** A cuboid of dimensions 50 cm, 150 cm, 175 cm can be divided into how many identical largest cubes ?

(a) 75 (b) 84 (c) 85 (d) 90

**Q.477.** The height of a cylinder is 20 cm. The lateral surface area is 1760 cm<sup>2</sup>. Its volume is:

(a) 12032 cm<sup>3</sup> (b) 12302 cm<sup>3</sup>

(c) 12203 cm<sup>3</sup> (d) 12320 cm<sup>3</sup>

**Q.478.** Two cylinders have the radii of their bases in the ratio 4 : 5 and their heights are in the ratio 5 : 7. Find the ratio of their curved surface areas.

(a) 5 : 4 (b) 4 : 7 (c) 1 : 2 (d) 4 : 5

**Q.479.** The circumference of the base of a conical tent of height 8 m is 32  $\pi$  m. Find its curved surface area.

(a)  $164\sqrt{5} \pi \text{ m}^2$  (b)  $162\sqrt{3} \pi \text{ m}^2$

(c)  $132\sqrt{3} \pi \text{ m}^2$  (d)  $128\sqrt{5} \pi \text{ m}^2$

**Q.480.** A circular arc whose radius is 12 cm makes an angle of 30° at the centre. Find the perimeter (in cm) of the sector formed. (Use  $\pi = 3.14$ )

(a) 32.38 (b) 30.28 (c) 28.64 (d) 26.24



**Q.481.** One-quarter of a circular pizza of radius 21 cm was removed from the whole pizza. What is the perimeter (in cm)

of the remaining pizza? (Use  $\pi = \frac{22}{7}$ )

- (a) 99 (b) 141 (c) 128 (d) 131

**Q.482.** What is the height (in m) of a conical water tank having radius of 70 m and volume of  $46,200 \text{ m}^3$ ? (Take  $\pi = \frac{22}{7}$ )

- (a) 27 (b) 9 (c) 18 (d) 54

**Q.483.** A and B are centres of two circles with radii 2 cm and 1 cm respectively, where  $AB = 5$  cm. C is the centre of another circle of radius  $r$  cm, which touches each of the above two circles externally. If  $\angle ACB = 90^\circ$ , then the value of  $r$  is:

- (a) 5 cm (b) 2 cm (c) 3 cm (d) 4 cm

**Q.484.** From a circle of radius 7 units, an arc length is cut by a chord of length 7 units. What is the arc length of the smaller portion (in units)?

- (a)  $\frac{8}{3}\pi$  (b)  $\frac{7}{6}\pi$  (c)  $\frac{7}{3}\pi$  (d)  $\frac{7}{4}\pi$

**Q.485.** A chord of the larger among two concentric circles is of length 20 cm and it is tangent to the smaller circle. What is the area (in  $\text{cm}^2$ ) of the annular portion between the two circles?

- (a)  $100\pi$  (b)  $164\pi$  (c)  $122\pi$  (d)  $175\pi$

**Q.486.** What is the volume in cubic units of a cylinder with a height equal to the diameter and a radius is 4 units?

- (a)  $128\pi$  (b)  $\frac{64\pi}{3}$  (c)  $32\pi$  (d)  $\frac{32\pi}{3}$

**Q.487.** The angle turned by minutes hand in 25 minutes in a clock is:

- (a)  $\pi$  (b)  $\frac{\pi}{2}$  (c)  $\frac{5\pi}{3}$  (d)  $\frac{5\pi}{6}$

**Q.488.** Find the measure of the central angle of a sector if its area is  $16\pi$  and the radius is 8.

- (a)  $75^\circ$  (b)  $60^\circ$  (c)  $108^\circ$  (d)  $90^\circ$

**Q.489.** The circumference of the base of a cylindrical vessel is 44 cm and its height is 25 cm. How many liters of water can it hold? (Take  $\pi = \frac{22}{7}$ )

- (a) 38.5 (b) 3.58 (c) 3.85 (d) 83.5

**Q.490.** M and N are, respectively, the centres of two circles of radii 12 cm and 8 cm. QR is the common tangent to the circles. If  $MN = 16$  cm, then what will be the length of QR?

- (a)  $5\sqrt{17}$  cm (b)  $4\sqrt{15}$  cm  
(c)  $3\sqrt{19}$  cm (d)  $6\sqrt{8}$  cm

**Q.491.** AB and CD are two parallel chords drawn in a circle with centre O. The distance between the two chords is 21 cm. If the lengths of AB and CD are 24 cm and 18 cm, respectively, then the radius of the circle is equal to:

- (a) 15 cm (b) 20 cm (c) 18 cm (d) 24 cm

**Q.492.** Find the length of the longest stick that can be fitted in a cubical vessel of edge 70 cm.

- (a)  $35\sqrt{3}$  cm (b)  $70\sqrt{3}$  cm  
(c)  $10\sqrt{3}$  cm (d)  $70\sqrt{2}$  cm

**Q.493.** In a rhombus PQRS, O is any interior point such that  $OP = OR$ . What is the degree measure of  $\angle SOQ$ ?

- (a)  $120^\circ$  (b)  $180^\circ$  (c)  $240^\circ$  (d)  $200^\circ$

**Q.494.** If the height and diameter of a cylinder are 12 cm and 28 cm, respectively, then find the ratio of the total surface area to the curved surface area.

- (a) 13 : 5 (b) 10 : 3 (c) 13 : 6 (d) 11 : 6

**Q.495.** A circular arc makes an angle of  $15^\circ$  at the centre. If the radius of the circular arc is 14 cm, what is the perimeter (correct up to two decimal places) of the sector so formed? ( $\pi = 3.14$ )?

- (a) 17.66 cm (b) 21.32 cm  
(c) 35.32 cm (d) 31.66 cm

**Q.496.** The perimeter of a rectangular garden is 48 m. If the length is 6 m more than the breadth, the area (in  $\text{m}^2$ ) of the garden is:

- (a) 135 (b) 96 (c) 84 (d) 112

**Q.497.** If three cubes of volume  $512 \text{ cm}^3$  each are joined end to end, find the surface area of the resulting cuboid.

- (a)  $896 \text{ cm}^2$  (b)  $986 \text{ cm}^2$   
(c)  $895 \text{ cm}^2$  (d)  $869 \text{ cm}^2$

**Q.498.** The radius of the base of a right circular cone is 9 cm and its slant height is 49 cm. Find the curved surface area of the cone.

- (a)  $1386 \text{ cm}^2$  (b)  $1265 \text{ cm}^2$   
(c)  $1262 \text{ cm}^2$  (d)  $1490 \text{ cm}^2$

**Q.499.** The area of a parallelogram with base 26 cm and height 18 cm is:

- (a)  $234 \text{ cm}^2$  (b)  $936 \text{ cm}^2$   
(c)  $932 \text{ cm}^2$  (d)  $468 \text{ cm}^2$

**Q.500.** The base radius of a circular cylinder is 10.5 cm. If the area of its curved surface is  $792 \text{ cm}^2$ , then what is the volume of the cylinder?

- (a)  $4185 \text{ cm}^3$  (b)  $4518 \text{ cm}^3$   
(c)  $4152 \text{ cm}^3$  (d)  $4158 \text{ cm}^3$

**Q.501.** In a rectangular field, the sum of the length and the breadth is 9 cm and area is  $18 \text{ cm}^2$ . If the length is 'a' and breadth is 'b', then find the value of  $(a^2 - ab + b^2)$ .

- (a) 36 (b) 24 (c) 18 (d) 27

**Q.502.** If the area of a sector of a circle is  $462 \text{ cm}^2$  and the central angle is  $120^\circ$ , then its radius is \_\_\_\_\_.

- (a) 18 cm (b) 17 cm (c) 21 cm (d) 14 cm

**Q.503.** The total surface area of a closed cube is given as  $1152 \text{ cm}^2$ . What is the length (in cm) of each side of the cube?

- (a)  $9\sqrt{2}$  (b)  $4\sqrt{13}$  (c)  $8\sqrt{3}$  (d)  $8\sqrt{2}$

**Q.504.** The length, breadth and height of a room is 15 m, 9 m, and 5 m, respectively. From each can of paint 40 square metre of area is painted. How many cans of paint will be needed to paint only the walls of the room?

- (a) 12 (b) 6 (c) 4 (d) 8

**Q.505.** The volume of a right circular cylindrical box of radius 30 cm is  $600 \text{ cm}^3$ . Find the height of the box.

- (a) 0.61 cm (b) 0.31 cm  
(c) 0.51 cm (d) 0.21 cm

**Q.506.** The radii of two circles are 8 cm and 15 cm. The perimeter of the circle having area equal to the sum of the areas of the two circles (in cm) is:

- (a)  $32\pi$  (b)  $30\pi$  (c)  $28\pi$  (d)  $34\pi$

**Q.507.** The ratio of the length, the breadth, and the height of a solid cuboid is given as 3 : 2 : 1, and the breadth of this cuboid is given as 12 cm. Find the volume (in  $\text{cm}^3$ ) of the cuboid.

- (a) 1250 (b) 1296 (c) 1200 (d) 1500

**Q.508.** The circumference of the base of the cylindrical vessel is 154 cm and its height is 49 mm. How many litres of water can it hold? (correct to three places of decimals, use  $\pi = \frac{22}{7}$ )

- (a) 9.244 (b) 2.439 (c) 3.924 (d) 4.329

**Q.509.** A cylindrical tank of radius 14 cm is full of water. If 616 litres of water are drawn out, then the water level in the tank is dropped by \_\_\_ m. (Take  $\pi = \frac{22}{7}$ )

- (a) 1000 (b) 100 (c) 1 (d) 10

**Q.510.** Find the diameter (in cm) of a circle whose circumference is equal to the sum of the circumferences of two circles of radii 12 cm and 16 cm.

- (a) 32 (b) 56 (c) 64 (d) 28

**Q.511.** A well with 14 cm radius is dug 23 cm deep. Find the volume of the earth taken out of it.

- (a) 14,178 cm<sup>3</sup> (b) 14,108 cm<sup>3</sup>  
(c) 14,168 cm<sup>3</sup> (d) 15,168 cm<sup>3</sup>

**Q.512.** What is the total surface area of a cone whose curved surface area is 550 cm<sup>2</sup> and radius is 7 cm? (Take  $\pi = \frac{22}{7}$ )

- (a) 764.71 cm<sup>2</sup> (b) 704.00 cm<sup>2</sup>  
(c) 682.60 cm<sup>2</sup> (d) 714.05 cm<sup>2</sup>

**Q.513.** From a point A, which is at a distance of 17 cm from the centre C of a circle with radius 8 cm, the pair of tangents AB and AD to the circle are drawn. The area of the quadrilateral ABCD is \_\_\_\_\_ cm<sup>2</sup>.

- (a) 192 (b) 120 (c) 60 (d) 360

**Q.514.** The central angle of a sector is 80° and whose length is 96π. What is the radius of the circle?

- (a) 196 units (b) 204 units  
(c) 116 units (d) 216 units

**Q.515.** A largest possible sphere is carved from a cube of side 14 cm. What is its volume in cm<sup>3</sup>?

- (a)  $205\frac{1}{3}$  (b)  $1707\frac{1}{3}$   
(c)  $1437\frac{1}{3}$  (d)  $1600\frac{1}{3}$

**Q.516.** The respective ratio between numerical values of the curved surface area and the volume of a right circular cylinder is 2 : 3. If the respective ratio between the radius and the height of the cylinder is 3 : 7, what is the total surface area of the cylinder?

- (a) 62 π cm<sup>2</sup> (b) 45 π cm<sup>2</sup>  
(c) 65 π cm<sup>2</sup> (d) 60 π cm<sup>2</sup>

**Q.517.** The surface areas of two spheres are in the ratio 25 : 36. What is the ratio of their volumes?

- (a)  $\frac{36}{115}$  (b)  $\frac{125}{216}$  (c)  $\frac{36}{125}$  (d)  $\frac{25}{36}$

**Q.518.** The sides of a cuboid are in the ratio 3 : 4 : 5 and its surface area is equal to the surface area of the cube with side 6 units. What is the approximate volume of the Cuboid?

- (a) 240 (b) 210 (c) 150 (d) 180

**Q.519.** The radius of a roller is 49 cm and its length is 200 cm. It takes 700 complete revolutions to move once over to level a playground. Find the area of the playground (Use  $\pi = \frac{22}{7}$ )

- (a)  $4312 \times 10^4$  cm<sup>2</sup> (b)  $4312 \times 10^2$  cm<sup>2</sup>  
(c)  $4322 \times 10^4$  cm<sup>2</sup> (d)  $4312 \times 10^3$  cm<sup>2</sup>

**Q.520.** The curved surface area of a cone is 2200 cm<sup>2</sup> and its radius is 28 cm, what is the slant height (in cm) of the cone? (Use  $\pi = \frac{22}{7}$ )

- (a) 22 (b) 23 (c) 24 (d) 25

**Q.521.** If the side of a cube is 3, which of the following options is equal to the ratio between the surface area and the volume of the cube?

- (a) 2 : 3 (b) 2 : 1 (c) 2 : 5 (d) 2 : 7

**Q.522.** The surface area of a sphere is 2464 cm<sup>2</sup>. Find its volume.

- (a)  $13121\frac{1}{3}$  cm<sup>3</sup> (b)  $11498\frac{2}{3}$  cm<sup>3</sup>  
(c)  $11892\frac{2}{3}$  cm<sup>3</sup> (d)  $11621\frac{1}{3}$  cm<sup>3</sup>

**Q.523.** Which of the following are the sides of a right - angled triangle?

- (a) 16 cm, 31 cm, 34 cm  
(b) 12 cm, 16 cm, 20 cm  
(c) 6 cm, 7 cm, 10 cm  
(d) 5 cm, 11 cm, 13 cm

**Q.524.** Find the diameter of a spherical ball whose volume is 38.808 cm<sup>3</sup>.

(Use  $\pi = \frac{22}{7}$ )

- (a) 4.8 cm (b) 4.4 cm  
(c) 4.6 cm (d) 4.2 cm

**Q.525.** The perimeter of a minor sector of a circle of radius 4 units subtending an angle of 45° is:

- (a) 4 - π units (b) 8 + π units  
(c) 8 - π units (d) 4 + π units

**Q.526.** If the diameter of a hemisphere is 21 cm, then the curved surface area (in cm<sup>2</sup>) of the hemisphere is: Take  $\pi = \frac{22}{7}$

- (a) 639 (b) 939 (c) 693 (d) 963

**Q.527.** The capacity of a cylinder tank is 8316 cm<sup>3</sup>. If the radius of its base is 21 cm, then find the depth of the tank.

- (a) 6 cm (b) 8 cm (c) 4 cm (d) 10 cm

**Q.528.** The volume of a cube is reduced to 72.9%. By how much is the side of the cube reduced?

- (a) 9.5% (b) 10% (c) 9% (d) 8.5%

**Q.529.** A hemisphere has a radius of 6.2 cm. Find its total surface area (use  $\pi = 3.14$ ) (round off to two decimal places).

- (a) 276.35 cm<sup>2</sup> (b) 282.25 cm<sup>2</sup>  
(c) 358.52 cm<sup>2</sup> (d) 362.10 cm<sup>2</sup>

**Q.530.** If the side of a square is increased by 5%, then the percentage increase in its area will be:

- (a) 11% (b) 10.25% (c) 10% (d) 10.5%

**Q.531.** If the brick size is 25 cm × 12 cm × 9 cm, then how many bricks are required to construct a wall of length 10 m, breadth 22.5 cm and height 6 m?

- (a) 4500 (b) 6000 (c) 8000 (d) 5000

**Q.532.** An iron box with external dimensions 60 cm, 40 cm, and 20 cm is made of 1 cm thick sheet. If 1 cm<sup>3</sup> of iron weighs 50 gm, the weight of the empty box is:

- (a) 416.40 kg (b) 240 kg  
(c) 214.05 kg (d) 400 kg

**Q.533.** If the diameter of a hemisphere is 42 cm, then the volume of hemisphere (in cm<sup>3</sup>) is: Take  $\pi = \frac{22}{7}$

- (a) 19154 (b) 19254 (c) 19444 (d) 19404

**Q.534.** Find the difference between the curved surface area and total surface area of a right circular cylinder having a length of 68 cm and base diameter of 28 cm.

- (a) 1432 cm<sup>2</sup> (b) 1332 cm<sup>2</sup>  
(c) 1132 cm<sup>2</sup> (d) 1232 cm<sup>2</sup>

**Q.535.** The perimeter of a triangle is 36 units. Its area cannot be more than \_\_\_\_\_ square units.

- (a)  $12\sqrt{7}$  (b)  $36\sqrt{2}$  (c)  $36\sqrt{3}$  (d)  $18\sqrt{5}$

**Q.536.** The outer surface of a sphere having diameter 10 m is painted at the rate of ₹  $\frac{80}{\pi}$  per m<sup>2</sup>. What is the cost of painting?

- (a) ₹8,000 (b) ₹9,000  
(c) ₹4,000 (d) ₹6,000

**Q.537.** The triangle has sides 3 cm, 4 cm and 5 cm. What is the length of the perpendicular from the opposite vertex to the side whose length is 5 cm?

- (a) 3.5 cm (b) 2.4 cm  
(c) 2.2 cm (d) 1.4 cm

**Q.538.** The radius and height of a cylinder are in the ratio 6 : 7 and its volume is 792 cm<sup>3</sup>. Calculate its curved surface area in cm<sup>2</sup>

- (a) 262 (b) 490 (c) 264 (d) 226

**Q.539.** The internal length, breadth and height of a cardboard box are 6 cm, 10 cm and 12 cm, respectively. How many boxes are needed in which cubes whose volume is 2160 cm<sup>3</sup> can be packed?

- (a) 6 (b) 5 (c) 3 (d) 4

**Q.540.** The heights of two cones are in the ratio of 5 : 3 and their radii are in the ratio 6 : 5. Find the ratio of their volumes.

- (a) 10 : 5 (b) 13 : 5 (c) 12 : 5 (d) 11 : 5

**Q.541.** If the radius of a sphere is doubled, then the volume of the sphere becomes \_\_\_ times the previous volume.  
(a) two (b) one (c) eight (d) four

**Q.542.** The curved surface area of a right circular cylinder of height 14 cm, is 88 cm<sup>2</sup>. Find the total surface area of the cylinder.

- (a)  $\frac{360}{7}$  cm<sup>2</sup> (b)  $\frac{660}{7}$  cm<sup>2</sup>  
(c)  $\frac{560}{7}$  cm<sup>2</sup> (d)  $\frac{460}{7}$  cm<sup>2</sup>

**Q.543.** The area of a rectangular field is 3600 m<sup>2</sup>. If the field is 45 m long, what is its perimeter ?

- (a) 255 m (b) 260 m (c) 250 m (d) 265 m

**Q.544.** If the surface area of cuboid is 108 m<sup>2</sup>, then find the height of the cuboid, whose length is 6 m and breadth is 3 m.

- (a) 5.5 m (b) 5 m (c) 4.5 m (d) 4 m

**Q.545.** For a given circle of radius 4 cm, the angle of its sector is 45°. Find the area (in cm<sup>2</sup>) of the sector. (Use  $\pi = 3.14$ .)

- (a) 7.28 (b) 6.28 (c) 6.18 (d) 7.18

**Q.546.** If the height of a right circular cone is 24 cm and radius of its base is 10.5 cm, then the volume of right circular cone (in cm<sup>3</sup>) is : Take  $\pi = \frac{22}{7}$

- (a) 2727 (b) 2772 (c) 2277 (d) 2237

**Q.547.** If the radius of a sphere is increased by 12 percent, then the volume will be increased by approximately:

- (a) 41.49 percent (b) 45.49 percent  
(c) 40.49 percent (d) 42.5 percent

**Q.548.** A cylindrical iron rod whose height is twelve times its radius is melted and cast into spherical balls, the radius of each of which is one third of the radius of the cylinder. Find the number of spherical balls.

- (a) 243 (b) 198 (c) 224 (d) 212

**Q.549.** If the area of a triangle whose base measures 6 cm is 18 cm<sup>2</sup>, then its height is:

- (a) 6 cm (b) 3.1 cm (c) 9 cm (d) 2.2 cm

**Q.550.** The edges of a rectangular box are in the ratio of 3 : 5 : 8, and its surface area is 632 cm<sup>2</sup>.

What will be the volume of the box ?

- (a) 801 cm<sup>3</sup> (b) 960 cm<sup>3</sup>  
(c) 990 cm<sup>3</sup> (d) 510 cm<sup>3</sup>

**Q.551.** Three solid cubes of metal have sides 6 cm, 8 cm and 10 cm, respectively. They are melted to form a

new cube. What is the surface area of this new cube?

- (a) 812 cm<sup>2</sup> (b) 950 cm<sup>2</sup>  
(c) 915 cm<sup>2</sup> (d) 864 cm<sup>2</sup>

**Q.552.** If the diameter of the base and curved surfaced area of a cylinder are 14cm and 352 cm<sup>2</sup>, respectively, then the total surface area of the cylinder is equal to:

- (a) 660 cm<sup>2</sup> (b) 654 cm<sup>2</sup>  
(c) 760 cm<sup>2</sup> (d) 754 cm<sup>2</sup>

**Q.553.** The radius of a sphere is increased by 25%. What is the increase percent in surface area?

- (a) 15.55% (b) 36.45%  
(c) 56.25% (d) 25.25%

**Q.554.** What is the curved surface area of cylinder whose radius is 3cm and

height is 14cm ? (Take  $\pi = \frac{22}{7}$ )

- (a) 262 cm<sup>2</sup> (b) 264 cm<sup>2</sup>  
(c) 263 cm<sup>2</sup> (d) 261 cm<sup>2</sup>

**Q.555.** The total surface area of a solid hemisphere of diameter 28 cm is:

- (a) 1848 cm<sup>2</sup> (b) 1784 cm<sup>2</sup>  
(c) 1488 cm<sup>2</sup> (d) 1648 cm<sup>2</sup>

**Q.556.** Find the capacity of an overhead cylindrical water tank, in litres, whose radius is 2.1 m and height is 6.3 m.

- (a) 61802 (b) 611726  
(c) 87318 (d) 65622

**Q.557.** The diameter of a cylinder is 14 cm and its height is 9 cm. Find the total surface area of the cylinder.

- (a) 704 cm<sup>2</sup> (b) 714 cm<sup>2</sup>  
(c) 708 cm<sup>2</sup> (d) 700 cm<sup>2</sup>

**Q.558.** From a circle, a sector with a central angle of 30° is cut off. If the length of the arc of this sector is 5.5 cm, then find the radius (in cm) of the circle.

[Use  $\pi = \frac{22}{7}$ ]

- (a) 12 (b) 10 (c) 11 (d) 10.5

**Q.559.** If the height of a solid right circular cylinder is 27 cm and diameter of its base is 30 cm, the total surface area (in cm<sup>2</sup>) of the right circular cylinder is:

(Take  $\pi = \frac{22}{7}$ )

- (a) 3978 (b) 3960 (c) 3976 (d) 3964

**Q.560.** 3 solid spheres of diameters 6 cm, 8 cm and 10 cm are melted together to form a single solid sphere. Find its radius.

- (a) 3 cm (b) 12 cm (c) 9 cm (d) 6 cm

**Q.561.** The ratio of the lateral surface area to the total surface area of a

cylinder with base diameter 60 cm and height 10 cm is:

- (a) 1 : 9 (b) 1 : 3 (c) 1 : 7 (d) 1 : 4

**Q.562.** The length, breadth and height of a cuboid are in ratio of 3 : 4 : 5. If the volume of the cuboid is 1620 m<sup>3</sup>, find the total surface area of the cuboid.

- (a) 646 m<sup>2</sup> (b) 846 m<sup>2</sup>  
(c) 423 m<sup>2</sup> (d) 712 m<sup>2</sup>

**Q.563.** A cylindrical oil tank has a radius of 5 m and a total surface area of 628 m<sup>2</sup>, what would be its curved surface area? (round off to the second decimal place)

- (a) 401.43 m<sup>2</sup> (b) 371.83 m<sup>2</sup>  
(c) 470.8 m<sup>2</sup> (d) 481.43 m<sup>2</sup>

**Q.564.** Rajesh is playing with a football having diameter of 28 cm. He wants to get a photo of his favourite player painted on the entire ball. The painter charges ₹2/cm<sup>2</sup>. What will be the cost (in ₹) of painting the ball (take  $\pi = \frac{22}{7}$ ) ?

- (a) 4,928 (b) 2,414 (c) 4,828 (d) 2,464

**Q.565.** A solid copper sphere of radius 9 cm is drawn into a wire of radius 4 mm. Find the length of the wire

- (a) 70.25 m (b) 60.75 m  
(c) 20.75 m (d) 65.25 m

**Q.566.** Find the height of the triangle whose base is two-thirds of its height and whose area is 363 cm<sup>2</sup>

- (a) 33 cm (b) 27 cm (c) 29 cm (d) 31 cm

**Q.567.** Find the percentage increase in surface area of a cube by triple each side.

- (a) 700% (b) 800% (c) 650% (d) 500%

**Q.568.** If the length, breadth and total surface area of a cuboid are 30 cm, 20 cm and 2600 cm<sup>2</sup>, respectively, then find its height.

- (a) 50 cm (b) 30 cm (c) 14 cm (d) 15 cm

**Q.569.** Two similar triangles have area 490 sq . cm. and 640 sq . cm. If the length of the one side of the first triangle is 21 cm, then what is the length of the corresponding side of the second triangle?

- (a) 16 cm (b) 14 cm (c) 24 cm (d) 21 cm

**Q.570.** The diameter of a roller is 35 cm and its length is 100 cm. It takes 200 complete revolutions to level a playground. Find the area of the playground in m<sup>2</sup>

- (a) 220 (b) 2,20,00,000  
(c) 1,10,000 (d) 110



**Q.571.** The perimeter of a rectangle is 70 cm and its diagonal is 25 cm. The area of the rectangle is :

- (a)  $425 \text{ cm}^2$  (b)  $420 \text{ cm}^2$   
(c)  $300 \text{ cm}^2$  (d)  $625 \text{ cm}^2$

**Q.572.** A solid sphere of radius 8 cm is melted and then recast into small spherical balls each of diameter 4 cm. Find the number of balls thus obtained (used  $\pi = \frac{22}{7}$ ).

- (a) 64 (b) 48 (c) 32 (d) 24

**Q.573.** If the radius of a sphere is increased by 3.5 cm, its surface area increases by  $550 \text{ cm}^2$ . What was the original radius of the sphere ?

(use  $\pi = \frac{22}{7}$ )

- (a) 4 cm (b) 4.5 cm (c) 6 cm (d) 3.5 cm

**Q.574.** The perimeter of a square whose area is the same as the area of a semi-circle of radius  $2\sqrt{2}$  cm is:

- (a)  $8\sqrt{\pi}$  cm (b)  $8\sqrt{2\pi}$  cm  
(c)  $16\sqrt{\pi}$  cm (d)  $4\sqrt{\pi}$  cm

**Q.575.** A metallic solid cuboid of dimensions  $36 \text{ cm} \times 22 \text{ cm} \times 12 \text{ cm}$  is melted and recast in the form of cubes of side 6 cm. Find the number of the cubes so formed.

- (a) 45 (b) 46 (c) 43 (d) 44

**Q.576.** The area of an isosceles right angled triangle is  $16 \text{ m}^2$ . Its hypotenuse is \_\_\_\_\_.

- (a) 0.8 m (b) 8 m (c) 4 m (d) 8 cm

**Q.577.** Find the cost of painting a ball that is in the shape of a sphere with a radius of 14 cm. The painting cost of the ball is ₹5 per square centimetre (take  $\pi = \frac{22}{7}$ ).

- (a) ₹12,230 (b) ₹12,320  
(c) ₹13,022 (d) ₹13,220

**Q.578.** Two cubes, each of edge 5 cm, are joined face to face. Find the surface area of the cuboid thus formed.

- (a)  $250 \text{ cm}^2$  (b)  $300 \text{ cm}^2$   
(c)  $125 \text{ cm}^2$  (d)  $200 \text{ cm}^2$

**Q.579.** The length of a rectangular shaped park exceeds its breadth by 15 m. If the perimeter of the park is 110 m, find the length of the park (in m).

- (a) 35 (b) 40 (c) 25 (d) 20

**Q.580.** If the lateral surface area of a cube is  $144 \text{ cm}^2$ , then find its volume.

- (a)  $864 \text{ cm}^3$  (b)  $1728 \text{ cm}^3$   
(c)  $117.58 \text{ cm}^3$  (d)  $216 \text{ cm}^3$

**Q.581.** If the areas of three adjacent faces of a cuboid are  $5 \text{ cm}^2$ ,  $15 \text{ cm}^2$ ,  $27 \text{ cm}^2$ , respectively, the surface area of the cuboid is:

- (a)  $75 \text{ cm}^2$  (b)  $47 \text{ cm}^2$   
(c)  $45 \text{ cm}^2$  (d)  $94 \text{ cm}^2$

**Q.582.** A closed cylindrical tank of radius 1.5 m and height 3 m is made from a sheet of metal, how much sheet of metal is required to do it ?

- (a) 42.24 square metre  
(b) 24.24 square metre  
(c) 42.42 square metre  
(d) 24.42 square metre

**Q.583.** Two cubes have their volumes in the ratio 1 : 125. Find the ratio of their surface areas.

- (a) 1 : 125 (b) 1 : 25 (c) 1 : 5 (d) 25 : 1

**Q.584.** If the radius of the base and slant height of the cone is 6 cm and 10 cm, respectively, then the volume of the cone is equal to:

- (a)  $108 \pi \text{ cm}^3$  (b)  $98 \pi \text{ cm}^3$   
(c)  $106 \pi \text{ cm}^3$  (d)  $96 \pi \text{ cm}^3$

**Q.585.** Find the number of paper balls, each 2 cm in diameter, that can be made from a sphere of diameter 18 cm.

- (a) 1020 (b) 484 (c) 625 (d) 729

**Q.586.** A cuboidal water purifier can carry 10 litres of water. Its breadth is  $\frac{1}{5}$

of its height and length is  $\frac{1}{3}$  of the sum of the breadth and height. The height of the tank is:

- (a) 20 cm (b) 10 cm (c) 80 cm (d) 50 cm

**Q.587.** A cube of edge 4 cm is cut into cubes each of edge of 2 cm. The ratio of the total surface area of one of the small cubes to that of the large cube is equal to:

- (a) 1 : 4 (b) 1 : 64 (c) 1 : 8 (d) 1 : 16

**Q.588.** The curved surface area of a right circular cone is  $20\pi \text{ cm}^2$ . If its radius is 4 cm, then find the volume of cone.

- (a)  $21 \pi \text{ cm}^3$  (b)  $20 \pi \text{ cm}^3$   
(c)  $16 \pi \text{ cm}^3$  (d)  $18 \pi \text{ cm}^3$

**Q.589.** The ratio of the outer and the inner circumference of a circular path is 7 : 6. If the path is 45 m wide, then what is the radius (in m) of the inner circle?

- (a) 240 (b) 270 (c) 150 (d) 120

**Q.590.** A solid metallic cube having surface area of  $54 \text{ cm}^2$  is melted to form smaller cubes of surface area  $\frac{8}{12} \text{ cm}^2$  each. Find the number of smaller cubes.

- (a) 512 (b) 343 (c) 216 (d) 729

**Q.591.** Find the surface area (in  $\text{cm}^2$ ) of a cardboard box in the shape of a cuboid whose length is 60 cm, breadth is half its length and height is one-third of its breadth.

- (a) 2700 (b) 10800 (c) 8100 (d) 5400

**Q.592.** From a circle of radius  $r$  units, the largest equilateral triangle is cut out. What is the length (in units) of the side of the triangle?

- (a)  $\frac{r}{\sqrt{2}}$  (b)  $\sqrt{2}r$  (c)  $\sqrt{3}r$  (d)  $\frac{r}{\sqrt{3}}$

**Q.593.** If the area of a rhombus is  $70 \text{ m}^2$  and its height is 5 m, its perimeter is \_\_\_\_.

- (a) 64 m (b) 35 m (c) 56 m (d) 25 m

**Q.594.** The measure of each side of an equilateral triangle is 16 cm. What is the ratio of its perimeter to the sum of its medians?

- (a)  $\sqrt{3} : 2$  (b)  $2\sqrt{3} : 1$  (c)  $2 : \sqrt{3}$  (d)  $1 : 2\sqrt{3}$

**Q.595.** A hemispherical bowl is made of silver and has an inner diameter of 4 cm if the thickness of silver is 0.5 cm. Find the volume of silver used in making the bowl correct to two places of decimals (use  $\pi = 3.14$  )

- (a)  $15.69 \text{ cm}^3$  (b)  $51.96 \text{ cm}^3$   
(c)  $15.96 \text{ cm}^3$  (d)  $51.69 \text{ cm}^3$

**Q.596.** The volume of the rectangular block is  $12288 \text{ m}^3$ . Its dimensions are in the ratio of 4 : 3 : 2. If the entire surface is polished at the rate 2 paise per  $\text{m}^2$ , then find the total cost of polishing :

- (a) ₹44.42 (b) ₹11.14  
(c) ₹66.56 (d) ₹33.28

**Q.597.** A hemispherical bowl is 88 cm round the brim. Assuming it to be full, how many persons may be served from it in hemispherical glasses, 7 cm in diameter at the top ? (Where  $\pi = \frac{22}{7}$ )

- (a) 74 (b) 68 (c) 64 (d) 70

**Q.598.** The chord length of a chord made on an arc of a circle is equal to the radius of the circle. The length of the arc is \_\_\_\_\_ (in units), if the radius of the circle

is 21 units. (Take  $\pi = \frac{22}{7}$  )

- (a) 20 (b) 22 (c) 24 (d) 21

**Q.599.** The base of a parallelogram is twice its height. If the area of the parallelogram is  $338 \text{ cm}^2$ , then find its height (in cm).

- (a) 13 (b) 11 (c) 14 (d) 12

**Q.600.** What is the total surface area of a cone whose radius is  $\frac{r}{4}$  and slant height is  $4l$  ?

- (a)  $8\pi r(l + r)$  (b)  $\pi r(l + \frac{r}{16})$   
(c)  $\pi r(l + \frac{r}{4})$  (d)  $4\pi r(l + r)$

**Q.601.** How many metres of cloth will be required to make a conical tent, the radius of whose base is 21 metres and height is 28 metres. The width of the cloth is 5 metres. (Where  $\pi = \frac{22}{7}$ )

(a) 470 (b) 462 (c) 456 (d) 478

**Q.602.** The volume of a cube is four times the volume of a cuboid. If the sides of the cuboid are 32 cm, 8 cm and 4 cm, then find the ratio of the total surface area of the cube to that of the cuboid.

(a) 23 : 13 (b) 24 : 13  
(c) 21 : 13 (d) 22 : 13

**Q.603.** A copper sphere of diameter 12 cm is drawn into a wire of diameter 4 mm. What is the length (in cm) of the wire? (Where  $\pi = \frac{22}{7}$ )

(a) 7200 (b) 7823 (c) 8342 (d) 9000

**Q.604.** The length of the largest possible rod that can be placed in a cubical room is  $42\sqrt{3}$  m. The surface area (in  $m^2$ ) of the largest possible sphere that fit within the cubical room is: [Use  $\pi = \frac{22}{7}$ ]

(a) 3590 (b) 4589 (c) 2564 (d) 5544

**Q.605.** Find the area of a minor sector of a circle whose circumference is 88 cm and the length of its minor arc is 22 cm (use  $\pi = \frac{22}{7}$ ).

(a)  $154 \text{ cm}^2$  (b)  $451 \text{ cm}^2$   
(c)  $415 \text{ cm}^2$  (d)  $145 \text{ cm}^2$

**Q.606.** A rectangular sheet of metal is 60 cm by 25 cm. Equal squares of side 5 cm are cut off at the corners and the remainder is folded up to form an open rectangular box. Find the volume of the box.

(a)  $2750 \text{ cm}^3$  (b)  $5750 \text{ cm}^3$   
(c)  $3750 \text{ cm}^3$  (d)  $4750 \text{ cm}^3$

**Q.607.** A drainage tile is a cylindrical shell 42 cm long. The inside and outside diameters are 8 cm and 14 cm, respectively. What is the volume (in  $\text{cm}^3$ ) of clay required for the tile? (Use  $\pi = \frac{22}{7}$ )

(a) 5241 (b) 4356 (c) 4881 (d) 4125

**Q.608.** One laddoo with a radius of 5 cm belongs to a shopkeeper. How many laddoos of radius 2.5 cm may be created from the same one laddoo ?

(a) 2 (b) 6 (c) 4 (d) 8

**Q.609.** The cost of painting a spherical vessel of diameter 14 cm is ₹21,560. What is the cost of painting (in ₹) per square centimetre ? (Use  $\pi = \frac{22}{7}$ )

(a) 30 (b) 32 (c) 28 (d) 35

**Q.610.** The radius of the base and height of a cone are 5 cm and 6 cm, respectively, whereas the radius of the base and height of a cylinder are 2.5 cm and 3 cm, respectively. The ratio of the volume of the cone to that of the cylinder is: (Where  $\pi = \frac{22}{7}$ )

(a) 8 : 5 (b) 9 : 4 (c) 8 : 3 (d) 3 : 5

**Q.611.** A cylinder of radius 7 cm has a curved surface area of  $2200 \text{ cm}^2$ . Find its total surface area. (Use  $\pi = \frac{22}{7}$ )

(a)  $2508 \text{ cm}^2$  (b)  $2260 \text{ cm}^2$   
(c)  $2550 \text{ cm}^2$  (d)  $1580 \text{ cm}^2$

**Q.612.** A hollow sphere of external and internal diameters of 10 cm and 6 cm, respectively, is melted and made into another solid in the shape of a right circular cone of base diameter 10 cm. Find the height of the cone.

(a) 13.68 cm (b) 14.68 cm  
(c) 16.68 cm (d) 15.68 cm

**Q.613.** A spherical box has a surface area  $528 \text{ cm}^2$ . Find the radius of the spherical box.

(a)  $\sqrt{37}$  cm (b)  $\sqrt{42}$  cm  
(c)  $\sqrt{29}$  cm (d)  $\sqrt{35}$  cm

**Q.614.** The distance between the parallel sides of a trapezium is 18 cm. If the area of the trapezium is  $1188 \text{ cm}^2$ , then what is the sum of the lengths of the parallel sides?

(a) 150 cm (b) 115 cm  
(c) 126 cm (d) 132 cm

**Q.615.** The area of the canvas cloth needed to erect a right conical tent of height 20ft. and circular base of circumference  $30\pi$  ft. is:

(a)  $455\pi \text{ sq. ft.}$  (b)  $375\pi \text{ sq. ft.}$   
(c)  $589\pi \text{ sq. ft.}$  (d)  $470\pi \text{ sq. ft.}$

**Q.616.** The volume of a right circular cone is  $150\pi \text{ cm}^3$  and its height is 18 cm. Find its slant height (in cm) (correct to 2 decimal places).

(a) 18.68 (b) 25.35 (c) 28.68 (d) 15.25

**Q.617.** A school has two rectangular playgrounds, which are similar in shape. If the length and the breadth of the first playground are given as 224 m and 160 m, respectively, then which of the following options CANNOT be the dimensions of the second playground ?

(a) Length = 273 m; Breadth = 195 m  
(b) Length = 168 m; Breadth = 120 m  
(c) Length = 210 m; Breadth = 150 m  
(d) Length = 343 m; Breadth = 255 m

**Q.618.** The height and radius of a cone-shaped storage tank are 9 ft. and 7 ft., respectively. Find the volume of water the tank can hold ? (Use  $\pi = \frac{22}{7}$ )

(a) 462 cubic ft. (b) 527 cubic ft.  
(c) 1386 cubic ft. (d) 1423 cubic ft.

**Q.619.** The breadth of a room is twice its height and half its length. The volume of the room is  $1728 \text{ m}^3$ . The length of the room is:

(a) 20 m (b) 24 m (c) 34 m (d) 28 m

**Q.620.** A woman working in a company was supplied a rectangular solid (cuboid shape) of wax with dimensions 70 cm, 44 cm and 20 cm to prepare cylindrical candles, each 14 cm in diameter and 20 cm of height. Find the number of candles.

(a) 15 (b) 20 (c) 10 (d) 30

**Q.621.** A cone of slant height 4.2 units has a lateral surface area  $13.2 \text{ units}^2$ . The radius of the base is:

(a) 1 unit (b) 1.5 units  
(c) 2.5 units (d) 2 units

**Q.622.** The in-radius of a triangle is 8 cm, and the sum of the lengths of its sides is 125 cm. The area of the triangle (in  $\text{cm}^2$ ) is:

(a) 480 (b) 1000 (c) 250 (d) 500

**Q.623.** Melting a metal ball of radius 3 units, 27 identical dice and a ball of radius 1 units are made. What is the side of dice?

(a) 1.7 units (b) 1.5 units  
(c) 1.6 units (d) 1.4 units

**Q.624.** The ratio of the slant height and the height of a cone is 4 : 3. If the curved surface area of the cone is  $4\sqrt{7}\pi$  square units, then the radius of the cone is \_\_\_\_\_ units.

(a)  $7\sqrt{7}$  (b) 7 (c)  $\sqrt{7}$  (d)  $\frac{\sqrt{7}}{7}$

**Q.625.** The length of a side of an equilateral triangle is 18 cm. The area (in  $\text{cm}^2$ ) of the region lying between the

circumcircle and the incircle of the triangle is [ use  $\pi = \frac{22}{7}$  ]

- (a)  $254\frac{3}{7}$  (b)  $254\frac{1}{7}$   
(c)  $254\frac{2}{7}$  (d)  $254\frac{4}{7}$

**Q.626.** The area of a circle is  $256\pi$  square cm. The length of its longest chord (in cm) is:

- (a) 34 (b) 36 (c) 6 (d) 32

**Q.627.** The perimeter of one face of a cube is 40 cm. Its volume will be:

- (a)  $10\text{ cm}^3$  (b)  $100\text{ cm}^3$   
(c)  $1000\text{ cm}^3$  (d)  $10000\text{ cm}^3$

**Q.628.** If the length of each of the two equal sides of an isosceles triangle is 15 cm and the adjacent angle is  $30^\circ$ , then the area of the triangle is:

- (a)  $66.25\text{ cm}^2$  (b)  $36.25\text{ cm}^2$   
(c)  $56.25\text{ cm}^2$  (d)  $26.25\text{ cm}^2$

**Q.629.** The lateral surface area of a cone is  $550\text{ cm}^2$ . If the diameter of the cone is 14 cm, then the height of the cone is:

- (a) 24 cm (b) 12.5 cm  
(c) 12 cm (d) 25 cm

**Q.630.** The radius of a hemisphere is 9 cm. Its volume will be:

- (a)  $712\pi\text{ cm}^3$  (b)  $243\pi\text{ cm}^3$   
(c)  $486\pi\text{ cm}^3$  (d)  $972\pi\text{ cm}^3$

**Q.631.** If the cylinder height is 9 cm and the base area is  $25\pi\text{ cm}^2$ , then find the total surface area.

- (a)  $240\text{ cm}^2$  (b)  $140\text{ cm}^2$   
(c)  $440\text{ cm}^2$  (d)  $340\text{ cm}^2$

**Q.632.** The height of one cone is 3 times the height of another cone, while its radius is half of the radius of the other cone. If their total volume is  $100\text{ unit}^3$ , then the difference in the volumes of the cones is \_\_\_\_\_  $\text{unit}^3$ .

- (a) 12.5 (b) 14.3 (c) 13.4 (d) 15.5

**Q.633.** A cube is having surface area of 5400 square units. Its volume would increase approximately \_\_\_\_\_ if the side is increased by 5 units.

- (a) 59% (b) 45% (c) 55% (d) 50%

**Q.634.** If the radius of a sphere is increased by 2 cm, its surface area increases by  $704\text{ cm}^2$ . What was the radius of the sphere before the increase?

(Use  $\pi = \frac{22}{7}$ )

- (a) 14 cm (b) 11 cm (c) 12 cm (d) 13 cm

**Q.635.** What must be the total percentage change in the volume of a cuboid if its length and breadth are

decreased by 10% and 20%, respectively, while its height is increased by 40%?

- (a) 0.8% increase (b) 0.8% decrease  
(c) 1.8% increase (d) 1.8% decrease

**Q.636.** There is a large cone with a diameter of 14 m and height of 12 m. It is being filled with water at a rate of  $2\text{ m}^3$  in every 20 seconds. How long will it take to fill the cone?

- (a) 313 sec (b) 6160 sec  
(c) 616 sec (d) 12320 sec

**Q.637.** The volume of a cuboid is twice the volume of a cube. If the dimensions of the cuboid are 12 cm, 16 cm and 18 cm, then the total surface area of the cube is:

- (a)  $625\text{ cm}^2$  (b)  $864\text{ cm}^2$   
(c)  $484\text{ cm}^2$  (d)  $764\text{ cm}^2$

**Q.638.** The height of a solid cylinder is 4 times its radius. Its curved surface area is \_\_\_\_\_ times its base area.

- (a) 2 (b) 6 (c) 8 (d) 4

**Q.639.** The lateral surface area of a cylinder is  $3862.2\text{ cm}^2$  and its height is 15 cm. Find its volume (take  $\pi = 3.14$ ) correct to one decimal place.

- (a)  $79175.1\text{ cm}^3$  (b)  $88275.2\text{ cm}^3$   
(c)  $78275.2\text{ cm}^3$  (d)  $89175.1\text{ cm}^3$

**Q.640.** The length of cold storage of cuboidal shape is double its breadth and its height is half of its breadth. If its volume is  $216\text{ m}^3$ , then what are the dimensions of the cold storage?

- (a)  $12\text{ m} \times 6\text{ m} \times 3\text{ m}$  (b)  $12\text{ m} \times 9\text{ m} \times 5\text{ m}$   
(c)  $14\text{ m} \times 7\text{ m} \times 5\text{ m}$  (d)  $8\text{ m} \times 4\text{ m} \times 2\text{ m}$

**Q.641.** The volume of a right circular cylinder of base radius  $r$  is obtained by multiplying its curved surface area by:

- (a)  $\frac{2}{r}$  (b)  $\frac{1}{2r}$  (c)  $2r$  (d)  $\frac{r}{2}$

**Q.642.** How many cubes of length 50 cm can be accommodated in a container of 5 m in length, 2 m in width and 5 m in height?

- (a) 400 (b) 500 (c) 450 (d) 250

**Q.643.** Find the area of a right angle whose base is 15 cm and hypotenuse is 17 cm.

- (a)  $40\text{ cm}^2$  (b)  $55\text{ cm}^2$   
(c)  $60\text{ cm}^2$  (d)  $75\text{ cm}^2$

**Q.644.** The dimensions of a rectangular field when extended by 6 metres are in the ratio of 4 : 3 and when decreased by 6 metres are in the ratio of 2 : 1. Find the dimensions of the field.

- (a) 16 metres, 12 metres  
(b) 10 metres, 6 metres  
(c) 18 metres, 12 metres

(d) 14 metres, 10 metres

**Q.645.** The radius of a circle is increased in such a way that its circumference increases by 7%. As a result, the area of the circle also increases by:

- (a) 18.69% (b) 10.28%  
(c) 14.49% (d) 12.20%

**Q.646.** The radii of the two concentric circles are 27 cm and 18 cm. The area (in  $\text{cm}^2$ ) of the closed figure bounded by the boundaries of the circle is:

- (a)  $505\pi$  (b)  $405\pi$  (c)  $288\pi$  (d)  $504\pi$

**Q.647.** A hemispherical bowl has a 21 cm radius. It is to be painted inside as well as outside. The cost of painting it at the rate of ₹0.05 per  $\text{cm}^2$  and assuming that the thickness of the bowl is negligible, is: (Use  $\pi = \frac{22}{7}$ )

- (a) ₹410.10 (b) ₹188.30  
(c) ₹277.20 (d) ₹388.20

**Q.648.** If the area of a regular pentagon is  $3920\sqrt{3}\text{ cm}^2$ , then how long is its each side?

- (a) 38 cm (b) 56 cm (c) 58 cm (d) 46 cm

**Q.649.** The base of a parallelogram is thrice its height. If the area of the parallelogram is  $1020\text{ cm}^2$ , then what is its height?

- (a)  $2\sqrt{85}\text{ cm}$  (b)  $\sqrt{89}\text{ cm}$   
(c)  $\sqrt{82}\text{ cm}$  (d)  $2\sqrt{84}\text{ cm}$

**Q.650.** A sphere and a cube have equal surface areas. The ratio of the volume of the sphere to that of the cube is:

- (a)  $\sqrt{6} : \sqrt{\pi}$  (b)  $\sqrt{\pi} : \sqrt{8}$   
(c)  $\sqrt{3} : \sqrt{\pi}$  (d)  $\sqrt{2} : \sqrt{\pi}$

**Q.651.** The surface area of a sphere is  $2464\text{ cm}^2$ . Calculate its volume.

(Use  $\pi = \frac{22}{7}$ )

- (a)  $11598.67\text{ cm}^3$  (b)  $11478.67\text{ cm}^3$   
(c)  $11488.67\text{ cm}^3$  (d)  $11498.67\text{ cm}^3$

**Q.652.** The area of the rhombus (in  $\text{cm}^2$ ) having each side equal to 13 cm and one of its diagonals equal to 24 cm is:

- (a) 110 (b) 120 (c) 60 (d) 130

**Q.653.** Find the area (in  $\text{cm}^2$ ) of a circle with a maximum radius that can be inscribed in a rectangle of length 18 cm and breadth 12 cm.

- (a)  $72\pi$  (b)  $28\pi$  (c)  $136\pi$  (d)  $36\pi$

**Q.654.** The difference between the total surface area and lateral surface area of a cube, if the side of the cube is 11 cm (in square cm), is \_\_\_\_\_.

- (a) 244 (b) 243 (c) 242 (d) 241



**Q.655.** If the radius of the right circular cone is increased by 20% and its height is decreased by 25%, then the volume of the right circular cone will be increased by:

- (a) 15% (b) 12% (c) 10% (d) 8%

**Q.656.** Half the perimeter of a rectangular garden, whose length is 8 cm more than its width, is 42 cm. Find the area of the rectangular garden.

- (a) 524 cm<sup>2</sup> (b) 425 cm<sup>2</sup>  
(c) 254 cm<sup>2</sup> (d) 542 cm<sup>2</sup>

**Q.657.** The volume of a cube is 64 cm<sup>3</sup>. The total surface area of the cube is:

- (a) 16 cm<sup>2</sup> (b) 64 cm<sup>2</sup>  
(c) 96 cm<sup>2</sup> (d) 128 cm<sup>2</sup>

**Q.658.** The diameter of a sphere is increased by 30%. What is the percentage increase in its volume?

- (a) 69% (b) 119.7% (c) 41.7% (d) 90.5%

**Q.659.** The diameter of a hemispherical bowl is 21 cm. The volume of the bowl is equal to: (use  $\pi = \frac{22}{7}$ )

- (a) 2,624.5 cm<sup>3</sup> (b) 2,425.5 cm<sup>3</sup>  
(c) 2,725.5 cm<sup>3</sup> (d) 2,524.5 cm<sup>3</sup>

**Q.660.** The radius and height of a right circular cone is increased by 25%. Find the increase in percentage of the volume of the cone (rounded off to 2 decimal places).

- (a) 94.31% (b) 95.31%  
(c) 94.51% (d) 93.51%

**Q.661.** A hollow cube is made of paper to have a volume of 512 cubic units. How much paper in square units will be required to make the cube?

- (a) 328 (b) 348 (c) 288 (d) 384

**Q.662.** A right rectangular box's measurements are 1.6 m long, 90 cm wide, 60 cm in height. Soap cakes of measurement 6 cm × 5 cm × 40 mm are to be packed in the box, so that no space is left. How many cakes can be packed in the box?

- (a) 6000 (b) 5600 (c) 6500 (d) 7200

**Q.663.** The ratio of the length, width and height of a closed cuboid is given as 6 : 3 : 2. The total surface area of this cuboid is given as 1800 cm<sup>2</sup>. Find the volume (in cm<sup>3</sup>) of this cuboid.

- (a) 4650 (b) 4800 (c) 4500 (d) 4200

**Q.664.** A solid cube, whose each edge is of length 48 cm, is melted. Identical solid cubes, each of volume 64 cm<sup>3</sup>, are made out of this molten cube, without any

wastage. How many such small cubes are obtained?

- (a) 1728 (b) 1718 (c) 1748 (d) 1738

**Q.665.** The cost of painting the total surface area of a 30 m high solid right circular cylinder at the rate of ₹25 per m<sup>2</sup> is ₹18,425. What is the volume (in m<sup>3</sup>) of this cylinder [use  $\pi = \frac{22}{7}$ ]?

- (a) 1210 (b) 1155 (c) 1145 (d) 1122

**Q.666.** What will be the difference between the total surface area and the curved surface area of a hemisphere having 4 cm diameter?

- (a) 4.4π cm<sup>2</sup> (b) 5π cm<sup>2</sup>  
(c) 4π cm<sup>2</sup> (d) 8π cm<sup>2</sup>

**Q.667.** A conical shape vessel has a radius of 21 cm and has a slant height of 25 cm. If the curved part of the vessel is to be painted white, find the cost (in ₹) of painting at the rate of ₹1.5 per cm<sup>2</sup>.

- (a) 2475 (b) 825 (c) 1650 (d) 1250

**Q.668.** A solid sphere made of wax of radius 12 cm is melted and cast into solid hemispheres of radius 4 cm each. Find the number of such solid hemispheres.

- (a) 14 (b) 54 (c) 28 (d) 27

**Q.669.** Find the area of a rhombus if the perimeter of the rhombus is 52 cm, and one of its diagonals is 10 cm long.

- (a) 164 cm<sup>2</sup> (b) 144 cm<sup>2</sup>  
(c) 160 cm<sup>2</sup> (d) 120 cm<sup>2</sup>

**Q.670.** The total surface area of a square-based right pyramid is 1536 m<sup>2</sup>, of which 37.5% is the area of the base of the pyramid. What is the volume (in m<sup>3</sup>) of this pyramid?

- (a) 3108 (b) 3048 (c) 3072 (d) 3144

**Q.671.** If the diameter of a sphere is reduced to its half, then the volume would be:

- (a) reduced by  $\frac{1}{8}$  of the former volume  
(b) increased by  $\frac{1}{4}$  of the former volume  
(c) reduced by  $\frac{1}{4}$  of the former volume  
(d) increased by  $\frac{1}{8}$  of the former volume

**Q.672.** The heights of two cones are in the ratio 7 : 5 and their diameters are in the ratio 10 : 21. What is the ratio of their volumes? (Where  $\pi = \frac{22}{7}$ )

- (a) 17 : 21 (b) 14 : 19  
(c) 20 : 63 (d) 26 : 47

**Q.673.** The diameter of a hemisphere is equal to the diagonal of a rectangle of length 4 cm and breadth 3 cm. Find the

total surface area (in cm<sup>2</sup>) of the hemisphere.

- (a)  $25\pi$  (b)  $\frac{50\pi}{4}$  (c)  $\frac{25\pi}{4}$  (d)  $\frac{75\pi}{4}$

**Q.674.** The height and the radius of the base of a right circular cone are in the ratio of 12 : 5. If its volume is 314 cm<sup>3</sup>, then what is the slant height of the cone? (Use  $\pi = 3.14$ )

- (a) 11 cm (b) 14 cm (c) 12 cm (d) 13 cm

**Q.675.** A right square pyramid having lateral surface area is 624 cm<sup>2</sup>. If the length of the diagonal of the square is  $24\sqrt{2}$ , then the volume of the pyramid is:

- (a) 780 cm<sup>3</sup> (b) 1150 cm<sup>3</sup>  
(c) 1083 cm<sup>3</sup> (d) 960 cm<sup>3</sup>

**Q.676.** A cylinder has some water in it at a height of 16 cm. If a sphere of radius 9 cm is put into it, then find the rise in the height of the water if the radius of the cylinder is 12 cm.

- (a) 1.6 cm (b) 8.75 cm  
(c) 8 cm (d) 6.75 cm

**Q.677.** What will be the area of a plot of quadrilateral shape, one of whose diagonals is 20 m and lengths of the perpendiculars from the opposite vertices on it are 12 m and 18 m, respectively?

- (a) 250 m<sup>2</sup> (b) 400 m<sup>2</sup>  
(c) 200 m<sup>2</sup> (d) 300 m<sup>2</sup>

**Q.678.** Two metallic balls P and Q are such that the diameter of P is four times the diameter of Q. What is the ratio between the volumes of P and Q?

- (a) 8 : 1 (b) 16 : 1 (c) 64 : 1 (d) 32 : 1

**Q.679.** If the slant height of a cone is 29 cm and its height is 20 cm, find the ratio between the magnitudes of total surface area and the volume.

- (a) 3 : 7 (b) 3 : 14 (c) 5 : 14 (d) 7 : 15

**Q.680.** The radius of a metallic spherical ball is 3 cm. If the metallic ball is melted and recast into x number of hemispheres of radius equal to half the radius of the metallic spherical ball, then find the value of x.

- (a) 14 (b) 16 (c) 13 (d) 15

**Q.681.** A hollow cylindrical tube made of plastic is 3 cm thick. If the external diameter is 20 cm and length of the tube is 49 cm, find the volume of the plastic.

- (a) 7644 cm<sup>3</sup> (b) 7238 cm<sup>3</sup>  
(c) 7854 cm<sup>3</sup> (d) 7546 cm<sup>3</sup>

**Q.682.** The diameter of a metallic sphere is 6 cm. It is melted and drawn into a wire having a diameter of the cross



section as 0.4 cm. Find the length of the wire (in meters).

- (a) 18 (b) 9 (c) 12 (d) 10

**Q.683.** From a cuboid of dimension of  $8 \times 6 \times 6$  units, small cubes of dimension  $0.5 \times 0.5 \times 0.5$  unit are cut. How many small cubes can be cut from the big cuboid?

- (a) 576 (b) 1152 (c) 2304 (d) 1552

**Q.684.** The radius and the height of a right circular cone are in the ratio 3 : 4. If its volume is  $301.44\text{m}^3$ , find the slant height (use  $\pi = 3.14$ ).

- (a) 5 cm (b) 20 cm (c) 10 cm (d) 15 cm

**Q.685.** Water is pouring into a cuboidal reservoir at the rate of 50 litres per minute. The volume of the reservoir is  $120\text{ m}^3$ . How many hours will it take to fill the reservoir?

- (a) 35 (b) 45 (c) 40 (d) 30

**Q.686.** Using the middle points of sides of a square with an area of 36 square units, another square is made. What is the area of a new square made in square units?

- (a) 24 (b) 27 (c) 18 (d) 21

**Q.687.** The length of the median of an equilateral triangle is  $6\sqrt{3}$  cm. The area of the triangle is :

- (a)  $36\sqrt{3}\text{ cm}^2$  (b)  $36\text{ cm}^2$   
(c)  $72\text{ cm}^2$  (d)  $72\sqrt{3}\text{ cm}^2$

**Q.688.** The areas of three adjacent faces of a cuboid are  $p$ ,  $q$  and  $r\text{ unit}^2$ , respectively. What is its volume?

- (a)  $p^3 q^3 r^3\text{ unit}^3$  (b)  $\sqrt{pqr}\text{ unit}^3$   
(c)  $pqr\text{ unit}^3$  (d)  $\sqrt[3]{pqr}\text{ unit}^3$

**Q.689.** A cylinder has a total surface area of  $64\pi$  square units, where the height of the cylinder is equal to its radius. If the height is doubled and the radius is halved, then the total surface area of the new cylinder is \_\_\_\_\_ square units.

- (a)  $54\pi$  (b)  $40\pi$  (c)  $64\pi$  (d)  $48\pi$

**Q.690.** A hollow metal sphere of internal and external radii 5 cm and 6 cm, respectively, is melted into a solid cone of base radius 5.2 cm. What is the height (in cm, rounded off to 1 decimal place) of the cone?

- (a) 11.8 (b) 12.8 (c) 13.5 (d) 14.5

**Q.691.** Find the area (in  $\text{cm}^2$ ) of the rhombus, lengths of whose diagonals are, respectively, 7.5 cm and 5.6 cm.

- (a) 21 (b) 28 (c) 27 (d) 30

**Q.692.** If a hollow sphere of an external and internal diameters of 6 cm and 4 cm, respectively, is melted into a cone of base diameter 8 cm, then the height of the cone is:

- (a)  $4\frac{3}{4}\text{ cm}$  (b)  $3\frac{1}{4}\text{ cm}$   
(c)  $5\frac{1}{4}\text{ cm}$  (d)  $2\frac{3}{4}\text{ cm}$

**Q.693.** A conical tent is set to accommodate 15 persons. Each person must have 5 square metres of space on the ground and 30 cubic metres of air to breathe. What is the height of the cone?

- (a) 15 m (b) 22 m (c) 13 m (d) 18 m

**Q.694.** A solid metallic cuboid of length 11 cm, breadth 8 cm and height 6 cm is melted and reshaped to form a cone of height 14 cm. What is the radius of the cone thus formed?

- (a) 6 cm (b) 9 cm (c) 7 cm (d) 8 cm

**Q.695.** The length, breadth and height of a cuboid are in the ratio of 6 : 4 : 3. If the total surface area of the cuboid is  $972\text{ cm}^2$ , what is the length (in cm) of the diagonal of the cuboid?

- (a)  $5\sqrt{65}$  (b)  $3\sqrt{61}$  (c)  $3\sqrt{65}$  (d)  $5\sqrt{61}$

**Q.696.** A triangle has its sides 5 cm, 12 cm and 13 cm. What will be the area of the triangle formed by joining the mid point of the three sides?

- (a)  $7\frac{1}{2}\text{ cm}^2$  (b)  $19\frac{1}{2}\text{ cm}^2$   
(c)  $13\frac{1}{2}\text{ cm}^2$  (d)  $15\text{ cm}^2$

**Q.697.** The height of a right circular cone is 20 cm, and its base diameter is 30 cm. What is the curved surface area and the total surface area of the right circular cone, respectively?

- (a)  $335\pi\text{ cm}^2$  and  $525\pi\text{ cm}^2$   
(b)  $390\text{ cm}^2$  and  $475\pi\text{ cm}^2$   
(c)  $450\text{ cm}^2$  and  $625\pi\text{ cm}^2$   
(d)  $375\pi\text{ cm}^2$  and  $600\pi\text{ cm}^2$

**Q.698.** Aman wants to get his spherical overhead water tank having a diameter of 7 m, painted. If the cost of painting is ₹13 per square metre, find the cost of painting Aman's spherical overhead water tank. (Use  $\pi = \frac{22}{7}$ )

- (a) ₹1,872 (b) ₹1,984  
(c) ₹2,002 (d) ₹1,926

**Q.699.** If the height and slant height of a cone are 21 cm and 29 cm, respectively.

Find the volume. (use  $\pi = \frac{22}{7}$ )

- (a)  $8800\text{ cm}^3$  (b)  $8780\text{ cm}^3$   
(c)  $8440\text{ cm}^3$  (d)  $8080\text{ cm}^3$

**Q.700.** The perimeter of a parallelogram is 48 cm. If the height of the parallelogram is 6 cm and the length of the adjacent side is 8 cm. find its area.

- (a)  $90\text{ cm}^2$  (b)  $80\text{ cm}^2$   
(c)  $84\text{ cm}^2$  (d)  $96\text{ cm}^2$

**Q.701.** In an isosceles triangle, if the unequal side is 8 cm and equal side is 5 cm, then the area of the triangle is:

- (a)  $12\text{ cm}^2$  (b)  $25\text{ cm}^2$   
(c)  $6\text{ cm}^2$  (d)  $11\text{ cm}^2$

**Q.702.** Find the volume of the largest possible sphere circumscribed by a cube of edge 7 cm.

- (a)  $179.67\text{ cm}^3$  (b)  $169.67\text{ cm}^3$   
(c)  $189.67\text{ cm}^3$  (d)  $199.67\text{ cm}^3$

**Q.703.** If a map of a city was drawn on a scale of 0.2 cm to a km, what area would represent a country if it has an area of  $22500\text{ km}^2$ ?

- (a)  $2500\text{ cm}^2$  (b)  $2250\text{ cm}^2$   
(c)  $5625\text{ cm}^2$  (d)  $900\text{ cm}^2$

**Q.704.** The sides of the three cubes of metal are 15 cm, 18 cm and 25 cm, respectively. Find the side (in cm) of the new cube formed by melting these cubes together

- (a)  $9\sqrt[3]{388}$  (b)  $6\sqrt[3]{388}$  (c)  $7\sqrt[3]{388}$  (d)  $4\sqrt[3]{388}$

**Q.705.** A shopkeeper usually makes one big besan laddoo (spherical in shape) with a radius of 9 cm. With the same laddoo, How many besan laddoos of radius 3 cm can be made? (Use  $\pi = \frac{22}{7}$ )

- (a) 9 (b) 18 (c) 36 (d) 27

**Q.706.** The area of a triangle is half the area of a square. The perimeter of the square is 172 cm. What is the area of the triangle (in  $\text{cm}^2$ )?

- (a) 924.5 (b) 1984 (c) 1849 (d) 992

**Q.707.** A cuboidal water tank is 12 m long, 9 m wide and 7.5 m deep. How many litres of water can it hold?

- (a) 840000 (b) 805000  
(c) 780000 (d) 810000

**Q.708.** What is the area of a triangular flower bed of a garden whose perimeter is 20 metres and two sides are 5 metres and 6 metres.

- (a)  $10\sqrt{2}\text{ m}^2$  (b)  $2\sqrt{10}\text{ m}^2$   
(c)  $5\sqrt{2}\text{ m}^2$  (d)  $4\sqrt{5}\text{ m}^2$

**Q.709.** The perimeter of a sector of a circle of radius 7.4 cm is 20.6 cm. The area of this sector is:



- (a)  $42.92 \text{ cm}^2$  (b)  $5.8 \text{ cm}^2$   
 (c)  $10.73 \text{ cm}^2$  (d)  $21.46 \text{ cm}^2$

**Q.710.** The perimeter of a minor sector of a circle of radius 5 cm is 15.5 cm. Find the area of the minor sector (use  $\pi = \frac{22}{7}$ ).

- (a)  $17.53 \text{ cm}^2$  (b)  $13.75 \text{ cm}^2$   
 (c)  $13.57 \text{ cm}^2$  (d)  $17.35 \text{ cm}^2$

**Q.711.** A copper wire is bent in the form of a square and it encloses an area of  $30.25 \text{ cm}^2$ . If the same wire is bent to form a circle, then find the area of the circle. (use  $\pi = \frac{22}{7}$ ).

- (a)  $38.50 \text{ cm}^2$  (b)  $42.25 \text{ cm}^2$   
 (c)  $35 \text{ cm}^2$  (d)  $30.25 \text{ cm}^2$

**Q.712.** A hemispherical bowl is made of 1 cm thick steel. The inside radius of the bowl is 4 cm. The volume of the steel (in  $\text{cm}^3$ ) used in making the bowl is:

- (a)  $40\frac{2}{3}\pi$  (b)  $27\frac{1}{3}\pi$   
 (c)  $41\frac{1}{3}\pi$  (d)  $35\frac{1}{3}\pi$

**Q.713.** A rod of 5 cm radius and 52 cm length is converted into a wire of 100 cm length of uniform thickness. The radius of the wire (in cm) is:

- (a)  $\sqrt{13}$  (b)  $\sqrt{15}$  (c)  $\sqrt{23}$  (d)  $\sqrt{17}$

**Q.714.** Ranu carries water to school in a cylindrical flask with diameter 12 cm and height 21 cm. Determine the amount of water that she can carry in the flask.

(Use  $\pi = \frac{22}{7}$ )

- (a)  $2372 \text{ cm}^3$  (b)  $2370 \text{ cm}^3$   
 (c)  $2376 \text{ cm}^3$  (d)  $2374 \text{ cm}^3$

**Q.715.** If the area of a circle is  $616 \text{ cm}^2$  and a chord  $XY = 10 \text{ cm}$ , then find the perpendicular distance from the center of the circle to the chord  $XY$ .

- (a)  $\sqrt{171} \text{ cm}$  (b)  $\sqrt{161} \text{ cm}$   
 (c)  $\sqrt{117} \text{ cm}$  (d)  $\sqrt{181} \text{ cm}$

**Answer Key :-**

|        |        |        |        |
|--------|--------|--------|--------|
| 1.(b)  | 2.(d)  | 3.(a)  | 4.(a)  |
| 5.(a)  | 6.(d)  | 7.(a)  | 8.(a)  |
| 9.(b)  | 10.(a) | 11.(b) | 12.(c) |
| 13.(d) | 14.(b) | 15.(b) | 16.(a) |
| 17.(d) | 18.(d) | 19.(b) | 20.(c) |
| 21.(c) | 22.(a) | 23.(b) | 24.(d) |
| 25.(d) | 26.(a) | 27.(d) | 28.(b) |
| 29.(b) | 30.(b) | 31.(a) | 32.(c) |
| 33.(c) | 34.(b) | 35.(d) | 36.(a) |

|         |         |         |         |
|---------|---------|---------|---------|
| 37.(d)  | 38.(a)  | 39.(a)  | 40.(a)  |
| 41.(c)  | 42.(c)  | 43.(b)  | 44.(b)  |
| 45.(c)  | 46.(d)  | 47.(b)  | 48.(c)  |
| 49.(a)  | 50.(b)  | 51.(c)  | 52.(d)  |
| 53.(b)  | 54.(d)  | 55.(a)  | 56.(a)  |
| 57.(d)  | 58.(d)  | 59.(b)  | 60.(c)  |
| 61.(c)  | 62.(a)  | 63.(d)  | 64.(c)  |
| 65.(a)  | 66.(c)  | 67.(b)  | 68.(c)  |
| 69.(d)  | 70.(c)  | 71.(b)  | 72.(c)  |
| 73.(c)  | 74.(d)  | 75.(b)  | 76.(d)  |
| 77.(b)  | 78.(d)  | 79.(d)  | 80.(a)  |
| 81.(c)  | 82.(d)  | 83.(b)  | 84.(d)  |
| 85.(d)  | 86.(c)  | 87.(a)  | 88.(a)  |
| 89.(d)  | 90.(c)  | 91.(b)  | 92.(b)  |
| 93.(b)  | 94.(c)  | 95.(a)  | 96.(c)  |
| 97.(d)  | 98.(a)  | 99.(a)  | 100.(a) |
| 101.(d) | 102.(a) | 103.(d) | 104.(c) |
| 105.(a) | 106.(b) | 107.(b) | 108.(b) |
| 109.(a) | 110.(c) | 111.(a) | 112.(b) |
| 113.(a) | 114.(a) | 115.(c) | 116.(c) |
| 117.(b) | 118.(b) | 119.(c) | 120.(a) |
| 121.(a) | 122.(d) | 123.(b) | 124.(a) |
| 125.(a) | 126.(a) | 127.(d) | 128.(b) |
| 129.(b) | 130.(c) | 131.(b) | 132.(b) |
| 133.(a) | 134.(b) | 135.(b) | 136.(a) |
| 137.(a) | 138.(c) | 139.(a) | 140.(a) |
| 141.(c) | 142.(b) | 143.(d) | 144.(a) |
| 145.(d) | 146.(b) | 147.(b) | 148.(a) |
| 149.(b) | 150.(a) | 151.(c) | 152.(b) |
| 153.(a) | 154.(a) | 155.(a) | 156.(b) |
| 157.(b) | 158.(c) | 159.(c) | 160.(a) |
| 161.(c) | 162.(c) | 163.(a) | 164.(d) |
| 165.(c) | 166.(c) | 167.(a) | 168.(a) |
| 169.(b) | 170.(a) | 171.(b) | 172.(c) |
| 173.(b) | 174.(c) | 175.(d) | 176.(a) |
| 177.(a) | 178.(a) | 179.(b) | 180.(d) |
| 181.(b) | 182.(a) | 183.(b) | 184.(b) |
| 185.(b) | 186.(c) | 187.(b) | 188.(b) |
| 189.(d) | 190.(c) | 191.(d) | 192.(d) |
| 193.(b) | 194.(b) | 195.(d) | 196.(b) |
| 197.(d) | 198.(a) | 199.(d) | 200.(b) |
| 201.(b) | 202.(a) | 203.(c) | 204.(b) |
| 205.(a) | 206.(b) | 207.(d) | 208.(b) |
| 209.(a) | 210.(b) | 211.(d) | 212.(d) |
| 213.(d) | 214.(a) | 215.(b) | 216.(a) |
| 217.(c) | 218.(c) | 219.(a) | 220.(d) |

|         |         |         |         |
|---------|---------|---------|---------|
| 221.(c) | 222.(b) | 223.(a) | 224.(b) |
| 225.(b) | 226.(b) | 227.(a) | 228.(d) |
| 229.(b) | 230.(c) | 231.(c) | 232.(c) |
| 233.(b) | 234.(b) | 235.(d) | 236.(a) |
| 237.(b) | 238.(b) | 239.(c) | 240.(b) |
| 241.(c) | 242.(b) | 243.(b) | 244.(b) |
| 245.(a) | 246.(d) | 247.(a) | 248.(a) |
| 249.(b) | 250.(b) | 251.(a) | 252.(a) |
| 253.(d) | 254.(b) | 255.(b) | 256.(c) |
| 257.(a) | 258.(a) | 259.(a) | 260.(a) |
| 261.(b) | 262.(d) | 263.(c) | 264.(d) |
| 265.(d) | 266.(b) | 267.(c) | 268.(a) |
| 269.(b) | 270.(a) | 271.(d) | 272.(b) |
| 273.(a) | 274.(c) | 275.(c) | 276.(d) |
| 277.(a) | 278.(d) | 279.(b) | 280.(d) |
| 281.(c) | 282.(c) | 283.(a) | 284.(a) |
| 285.(a) | 286.(c) | 287.(a) | 288.(d) |
| 289.(a) | 290.(d) | 291.(b) | 292.(d) |
| 293.(c) | 294.(c) | 295.(a) | 296.(c) |
| 297.(b) | 298.(c) | 299.(b) | 300.(d) |
| 301.(d) | 302.(a) | 303.(b) | 304.(b) |
| 305.(a) | 306.(c) | 307.(b) | 308.(a) |
| 309.(a) | 310.(d) | 311.(b) | 312.(c) |
| 313.(b) | 314.(d) | 315.(d) | 316.(a) |
| 317.(c) | 318.(c) | 319.(b) | 320.(b) |
| 321.(b) | 322.(c) | 323.(a) | 324.(b) |
| 325.(a) | 326.(b) | 327.(a) | 328.(c) |
| 329.(a) | 330.(b) | 331.(b) | 332.(c) |
| 333.(d) | 334.(c) | 335.(c) | 336.(a) |
| 337.(a) | 338.(b) | 339.(a) | 340.(a) |
| 341.(b) | 342.(a) | 343.(d) | 344.(a) |
| 345.(c) | 346.(a) | 347.(c) | 348.(c) |
| 349.(b) | 350.(a) | 351.(b) | 352.(c) |
| 353.(a) | 354.(b) | 355.(c) | 356.(d) |
| 357.(c) | 358.(c) | 359.(c) | 360.(b) |
| 361.(b) | 362.(b) | 363.(d) | 364.(b) |
| 365.(b) | 366.(a) | 367.(c) | 368.(a) |
| 369.(b) | 370.(c) | 371.(c) | 372.(a) |
| 373.(c) | 374.(b) | 375.(b) | 376.(b) |
| 377.(a) | 378.(c) | 379.(c) | 380.(c) |
| 381.(a) | 382.(b) | 383.(a) | 384.(c) |
| 385.(c) | 386.(c) | 387.(b) | 388.(a) |
| 389.(d) | 390.(b) | 391.(c) | 392.(a) |
| 393.(d) | 394.(b) | 395.(a) | 396.(b) |
| 397.(c) | 398.(b) | 399.(b) | 400.(a) |
| 401.(b) | 402.(c) | 403.(b) | 404.(a) |

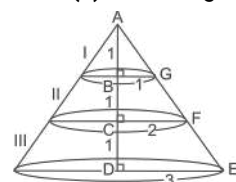


|         |         |         |         |
|---------|---------|---------|---------|
| 405.(b) | 406.(b) | 407.(a) | 408.(d) |
| 409.(a) | 410.(c) | 411.(d) | 412.(a) |
| 413.(b) | 414.(a) | 415.(b) | 416.(a) |
| 417.(b) | 418.(c) | 419.(d) | 420.(a) |
| 421.(c) | 422.(b) | 423.(d) | 424.(c) |
| 425.(a) | 426.(c) | 427.(b) | 428.(c) |
| 429.(c) | 430.(c) | 431.(b) | 432.(a) |
| 433.(b) | 434.(d) | 435.(b) | 436.(d) |
| 437.(a) | 438.(b) | 439.(b) | 440.(a) |
| 441.(a) | 442.(a) | 443.(b) | 444.(a) |
| 445.(d) | 446.(b) | 447.(c) | 448.(b) |
| 449.(a) | 450.(b) | 451.(b) | 452.(c) |
| 453.(b) | 454.(c) | 455.(c) | 456.(c) |
| 457.(b) | 458.(b) | 459.(d) | 460.(b) |
| 461.(b) | 462.(a) | 463.(a) | 464.(a) |
| 465.(a) | 466.(b) | 467.(b) | 468.(c) |
| 469.(d) | 470.(b) | 471.(b) | 472.(d) |
| 473.(b) | 474.(d) | 475.(a) | 476.(b) |
| 477.(d) | 478.(b) | 479.(d) | 480.(b) |
| 481.(b) | 482.(b) | 483.(b) | 484.(c) |
| 485.(a) | 486.(a) | 487.(d) | 488.(d) |
| 489.(c) | 490.(b) | 491.(a) | 492.(b) |
| 493.(b) | 494.(c) | 495.(d) | 496.(a) |
| 497.(a) | 498.(a) | 499.(d) | 500.(d) |
| 501.(d) | 502.(c) | 503.(c) | 504.(b) |
| 505.(d) | 506.(d) | 507.(b) | 508.(a) |
| 509.(d) | 510.(b) | 511.(c) | 512.(b) |
| 513.(b) | 514.(d) | 515.(c) | 516.(d) |
| 517.(b) | 518.(b) | 519.(a) | 520.(d) |
| 521.(b) | 522.(b) | 523.(b) | 524.(d) |
| 525.(b) | 526.(c) | 527.(a) | 528.(b) |
| 529.(d) | 530.(b) | 531.(d) | 532.(a) |
| 533.(d) | 534.(d) | 535.(c) | 536.(a) |
| 537.(b) | 538.(c) | 539.(c) | 540.(c) |
| 541.(c) | 542.(b) | 543.(c) | 544.(d) |
| 545.(b) | 546.(b) | 547.(c) | 548.(a) |
| 549.(a) | 550.(b) | 551.(d) | 552.(a) |
| 553.(c) | 554.(b) | 555.(a) | 556.(c) |
| 557.(a) | 558.(d) | 559.(b) | 560.(d) |
| 561.(d) | 562.(b) | 563.(c) | 564.(a) |
| 565.(b) | 566.(a) | 567.(b) | 568.(c) |
| 569.(c) | 570.(a) | 571.(c) | 572.(a) |
| 573.(b) | 574.(a) | 575.(d) | 576.(b) |
| 577.(b) | 578.(a) | 579.(a) | 580.(d) |
| 581.(d) | 582.(c) | 583.(b) | 584.(d) |
| 585.(d) | 586.(d) | 587.(a) | 588.(c) |

|         |         |         |         |
|---------|---------|---------|---------|
| 589.(b) | 590.(d) | 591.(d) | 592.(c) |
| 593.(c) | 594.(c) | 595.(c) | 596.(c) |
| 597.(c) | 598.(b) | 599.(a) | 600.(b) |
| 601.(b) | 602.(b) | 603.(a) | 604.(d) |
| 605.(a) | 606.(c) | 607.(b) | 608.(d) |
| 609.(d) | 610.(c) | 611.(a) | 612.(d) |
| 613.(b) | 614.(d) | 615.(b) | 616.(a) |
| 617.(d) | 618.(a) | 619.(b) | 620.(b) |
| 621.(a) | 622.(d) | 623.(c) | 624.(c) |
| 625.(d) | 626.(d) | 627.(c) | 628.(c) |
| 629.(a) | 630.(c) | 631.(c) | 632.(b) |
| 633.(a) | 634.(d) | 635.(a) | 636.(b) |
| 637.(b) | 638.(c) | 639.(a) | 640.(a) |
| 641.(d) | 642.(a) | 643.(c) | 644.(c) |
| 645.(c) | 646.(b) | 647.(c) | 648.(b) |
| 649.(a) | 650.(a) | 651.(d) | 652.(b) |
| 653.(d) | 654.(c) | 655.(d) | 656.(b) |
| 657.(c) | 658.(b) | 659.(b) | 660.(b) |
| 661.(d) | 662.(d) | 663.(c) | 664.(a) |
| 665.(b) | 666.(c) | 667.(a) | 668.(b) |
| 669.(d) | 670.(c) | 671.(a) | 672.(c) |
| 673.(d) | 674.(d) | 675.(d) | 676.(d) |
| 677.(d) | 678.(c) | 679.(c) | 680.(b) |
| 681.(c) | 682.(b) | 683.(c) | 684.(c) |
| 685.(c) | 686.(c) | 687.(a) | 688.(b) |
| 689.(b) | 690.(c) | 691.(a) | 692.(a) |
| 693.(d) | 694.(a) | 695.(b) | 696.(a) |
| 697.(d) | 698.(c) | 699.(a) | 700.(d) |
| 701.(a) | 702.(a) | 703.(d) | 704.(d) |
| 705.(d) | 706.(a) | 707.(d) | 708.(a) |
| 709.(d) | 710.(b) | 711.(a) | 712.(a) |
| 713.(a) | 714.(c) | 715.(a) |         |

**Solutions :-****Sol.1.(b)** According to question,

$$\text{Ratio} = 12 \times \left(6 \times \frac{\sqrt{3}}{4} a^2\right) : 2 \times \left(6 \times \frac{\sqrt{3}}{4} a^2\right) = 6 : 1$$

**Sol.2.(d)** According to question,

$$\begin{aligned} \text{Ratio} &\rightarrow \text{I} : \text{II} : \text{III} \\ \text{Volume} &\rightarrow \frac{1}{3}\pi(1)^3 \times 5 : \frac{1}{3}\pi(2^3 - 1^3) \times 5 : \frac{1}{3}\pi(3^3 - 2^3) \times 5 \\ &= 1 : 7 : 19 \end{aligned}$$

**Sol.3.(a)** According to the question,

$$\text{Ratio of heights} = \frac{h_{\text{small}}}{h_{\text{full}}} = \frac{5}{20} = \frac{1}{4}$$

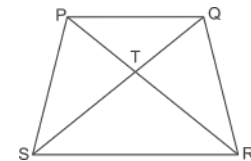
The truncated cone and the original cone are similar, their volumes are

The ratio will be equal to the cube of the ratio of their heights:

$$V_{\text{small}} : V_{\text{full}} = \left(\frac{1}{4}\right)^3 = \frac{1}{64}$$

$$\text{volume of frustum} = 1 - \frac{1}{64} = \frac{63}{64}$$

$$V_{\text{frustum}} : V_{\text{full}} = 63 : 64$$

**Sol.4.(a)**

When the diagonals of a trapezium intersect, the ratio of the areas of the triangles formed by those diagonals with the parallel sides is equal to the square of the ratio of the parallel sides.

$$\frac{\text{ar. } \Delta PQT}{\text{ar. } \Delta SRT} = \left(\frac{PQ}{SR}\right)^2$$

**Sol.5.(a)**

$$\text{Area of sector} = \frac{90^\circ}{360^\circ} \times \pi \times (10)^2 = 25\pi$$

$$\text{Area of triangle} = \frac{1}{2} \times (10)^2 \times \sin 90^\circ = 50$$

$$\text{Area of minor segment} = 25\pi - 50 \text{ cm}^2$$

**Sol.6.(d)** According to question,

$$\text{Volume of cylinder} = \pi r^2 h$$

$$= \pi \left(\frac{R}{4}\right)^2 \times \frac{R}{2} = \pi \times \frac{R^3}{32}$$

Volume of remaining hemisphere

$$\begin{aligned} &= \frac{2}{3}\pi \left(\frac{R}{2}\right)^3 - \pi \times \frac{R^3}{32} = \frac{8\pi R^3 - 3\pi R^3}{96} \\ &= \frac{5\pi R^3}{96} \end{aligned}$$

$$\text{So, Required ratio} = \pi \times \frac{R^3}{32} : \frac{5\pi R^3}{96}$$

$$= 3 : 5$$

**Sol.7.(a)** Side of triangle = 7 m, 8 m, 9 m

Semi-perimeter of triangle(s)

$$= \frac{7+8+9}{2} = 12 \text{ m}$$

Area of triangle

$$= \sqrt{s(s-a)(s-b)(s-c)}$$

$$= \sqrt{12 \times 5 \times 4 \times 3} = 12\sqrt{5} \text{ m}^2$$

$$\text{Sol.8.(a)} \text{ Radius (r)} = \frac{2.8}{2}$$

Maximum area rolled

$$= \text{CSA} \times \text{Speed} \times \text{time} = 2\pi r h \times 12 \times 60$$

$$= 2 \times \frac{22}{7} \times \frac{2.8}{2} \times 1.5 \times 12 \times 60 = 9,504 \text{ m}^2$$

**Sol.9.(b) Radius** =  $\frac{2.8}{2} = 1.4$  cm

Semicircle circumference =  $\pi r$

=  $\frac{22}{7} \times 1.4 = 4.4$

Total length = Semicircle circumference + diameter  
=  $4.4 + 2.8 = 7.2$  m

**Sol.10.(a)**

**All edges (sides) are of equal length**

**Explanation:** A rectangular parallelepiped has length, breadth, and height. If all three dimensions are equal, it forms a cube, making it a special case of a rectangular parallelepiped.

**Sol.11.(b)** Let the arc length be  $l$

According to the question,

$$l = \frac{2\pi r}{2} \Rightarrow l = \pi r$$

We know that,

$$\Rightarrow \text{Area of a circle} = \frac{1}{2} \times r \times l$$

$$\Rightarrow \pi r^2 \times \frac{\theta}{360^\circ} = \frac{1}{2} \times r \times \pi r$$

$$\Rightarrow \theta = \frac{360^\circ}{2} = 180^\circ$$

**Sol.12.(c)** Area of circle =  $\frac{\theta}{360} \times \pi r^2$

$$= 209 \Rightarrow \frac{\theta}{360} = \frac{209}{\pi r^2}$$

And perimeter of sector

$$= \frac{\theta}{360} \times 2\pi r + 2r = 63 \Rightarrow \frac{\theta}{360}$$

$$= \frac{63 - 2r}{2\pi r}$$

On equating both:

$$\frac{209}{\pi r^2} = \frac{63 - 2r}{2\pi r} \Rightarrow \frac{209}{r} = \frac{63 - 2r}{2}$$

$$418 = 63r - 2r^2 \Rightarrow 2r^2 - 63r + 418 = 0$$

$$2r^2 - 44r - 19r + 418 = 0$$

$$2r(r - 22r) - 19(r - 22) = 0$$

$$(r - 22)(2r - 19) \Rightarrow r = 22 \text{ or } \frac{19}{2}$$

Now

$$\text{perimeter of sector} = \frac{\theta}{360} \times 2\pi r + 2r$$

$$63 = \frac{\theta}{360} \times 2 \times \frac{22}{7} \times 22 + 44$$

$$\Rightarrow \theta = 49.5^\circ$$

**Sol.13.(d)** Area of square =  $(\text{side})^2$

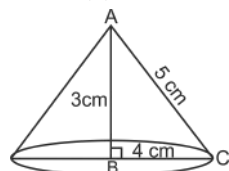
$$= 16x^2 + 40x + 25 = (4x + 5)^2$$

$$\Rightarrow \text{Side of square} = (4x + 5)$$

$$\Rightarrow \text{Perimeter of square} = 4 \times (4x + 5)$$

$$= 2(8x + 10)$$

**Sol.14.(b)**



Volume of the cone =  $\frac{1}{3}\pi r^2 h$

$$= \frac{1}{3} \times \pi \times 4 \times 4 \times 3 = 16\pi \text{ cm}^3$$

**Sol.15.(b)**

$$\text{Perimeter of sector} = 2r + \frac{\theta}{360} \times 2\pi r$$

$$\Rightarrow 2r + \frac{\theta}{360} \times 2\pi r = \frac{64}{3}$$

$$\Rightarrow 2r + \frac{1}{6} \times 2 \times \frac{22}{7} \times r = \frac{64}{3}$$

$$\Rightarrow 42r + 22r = 64 \times 7$$

$$\Rightarrow 64r = 64 \times 7 \Rightarrow r = 7$$

Now,

$$2r + l = \frac{64}{3} \text{ (given)}$$

$$\Rightarrow 2 \times 7 + l = \frac{64}{3} \Rightarrow l = \frac{64}{3} - 14 = \frac{22}{3}$$

$$\text{Hence, area of sector} = \frac{1}{2}lr$$

$$= \frac{1}{2} \times \frac{22}{3} \times 7 = \frac{77}{3} \text{ cm}^2$$

**Sol.16.(a)**

$$\text{New length of cuboid} = \frac{20}{2} = 10 \text{ cm}$$

$$\text{New breadth of cuboid} = \frac{12}{2} = 6 \text{ cm}$$

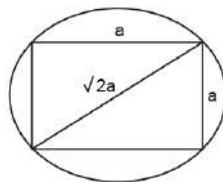
$$\text{New height of cuboid} = \frac{10}{2} = 5 \text{ cm}$$

TSA of all the pieces

$$= 8 \times 2(10 \times 6 + 6 \times 5 + 5 \times 10)$$

$$= 8 \times 2(60 + 30 + 50) = 2240 \text{ cm}^2$$

**Sol.17.(d)** As we know, Rectangle with the largest area cut out of the circle is square.



$$\text{Area of circle} = \pi \times 5^2 = 25\pi \text{ square units}$$

$$\text{Diameter of circle} = 5 \times 2 = 10 \text{ units}$$

$$\text{Diagonal of square} = \sqrt{2}a = 10$$

$$\Rightarrow a = \frac{10}{\sqrt{2}} = 5\sqrt{2} \text{ unit}$$

$$\text{Area of square} = (5\sqrt{2})^2 = 50 \text{ square units}$$

$$\text{Area of remaining portion} = \text{Area of circle} - \text{Area of square}$$

$$25\pi - 50 = 25(\pi - 2) \text{ square units}$$

**Sol.18.(d)** According to the question,

$$\text{CSA of the cylinder} = 2\pi r h$$

$$2\pi r h \times \frac{350}{100} = 2\pi R H$$

$$h : H = 6 : 7 \text{ (where radii ratio are, } 3 : 1)$$

$$\text{Volume of the cylinder} = \pi r^2 h$$

$$\text{Small cylinder volume} = \pi(1)^2 \times 6$$

$$\text{Big cylinder volume} = \pi(3)^2 \times 7$$

$$\text{required\%} = \frac{\pi(1)^2 \times 6}{\pi(3)^2 \times 7} \times 100 = 9.52\%$$

**Sol.19.(b)**

$$\text{Outer area of hemispherical bowl} = 2\pi$$

$$(R)^2 = 2\pi(8)^2 = 128\pi \text{ cm}^2$$

$$\text{Inner area of hemispherical bowl} = 2\pi(r)^2$$

$$= 2\pi(6)^2 = 72\pi \text{ cm}^2$$

$$\text{Ring area of the top} = \pi(R^2 - r^2)$$

$$= \pi(64 - 36) = 28\pi \text{ cm}^2$$

$$\text{Total area to polished}$$

$$= 128\pi + 72\pi + 28\pi = 228\pi \text{ cm}^2$$

$$\text{Hence, cost to polish the bowl} = 228\pi$$

$$= 228 \times 50 = \text{Rs. } 11400$$

**Sol.20.(c)** Option 'c' is false

$$\text{Length of an arc} = \frac{\theta}{360^\circ} \times 2\pi r$$

**Sol.21.(c)** Sum of the area of similar triangles =  $706.5 \times 2 = 1413 \text{ m}^2$

$$\frac{\text{1st triangle's perimeter}}{\text{2nd triangle's perimeter}}$$

$$= \frac{\text{side of 1st triangle}}{\text{side of 2nd triangle}} = \frac{6}{11}$$

$$\frac{\text{area of 1st triangle}}{\text{area of 2nd triangle}}$$

$$= \left(\frac{\text{side of 1st triangle}}{\text{side of 2nd triangle}}\right)^2 = \left(\frac{6}{11}\right)^2 = \frac{36}{121}$$

$$\text{Sum of both triangles area} = 36 + 121$$

$$= 157 \text{ units} \Rightarrow 157 \text{ units} = 1413 \text{ m}^2$$

$$\text{Area of 1st triangle (36 units)}$$

$$= \frac{1413}{157} \times 36 = 324 \text{ m}^2$$

$$\text{Area of 2nd triangle (121 units)}$$

$$= \frac{1413}{157} \times 121 = 1089 \text{ m}^2$$

$$\text{Required value} = (1089 - 324) \times \frac{1}{5}$$

$$= 153 \text{ m}^2$$

**Sol.22.(a)** Area of remaining area = area of square -  $2 \times$  area of semicircle

According to the question,

$$42 = (2r)^2 - 2 \times \frac{1}{2} \times \pi \times (r)^2$$

$$42 = 4r^2 - \frac{22}{7} \times (r)^2$$

$$42 = r^2 \left(4 - \frac{22}{7}\right) \Rightarrow 42 = r^2 \left(\frac{6}{7}\right)$$

$$r^2 = 42 \times \frac{7}{6} = 49 \Rightarrow r = 7$$

$$\text{So, diameter of each circle (2r)} = 14 \text{ cm}$$

**Sol.23.(b)** Let the radius of larger and smaller spheres  $2x$  and  $\frac{x}{2}$  respectively.

**1st condition :-** Volume of larger spheres =  $125 \times$  volume of identical spheres

$$\frac{4}{3} \pi R_1^3 = 125 \times \frac{4}{3} \pi r_1^3$$

$$(2x)^3 = 125 \times r_1^3 \Rightarrow r_1^3 = \frac{(2x)^3}{125}$$

**2nd condition :-**

Volume of smaller spheres =  $27 \times$  volume of identical spheres

$$\frac{4}{3} \pi (R_2^3) = 27 \times \frac{4}{3} \pi r_2^3$$

$$\frac{4}{3}\pi\left(\frac{x}{2}\right)^3 = 27 \times \frac{4}{3}\pi r_2^3 \Rightarrow r_2^3 = \frac{\left(\frac{x}{2}\right)^3}{27}$$

$$\text{Ratio of the volume} = \frac{(2x)^3}{125} : \frac{\left(\frac{x}{2}\right)^3}{27}$$

$$= \frac{(2x)^3}{125} : \frac{(x)^3}{27 \times 8} = 1728 : 125$$

According to question

$$\frac{1}{m} = \frac{1728}{125}$$

$$\text{Hence the value of } m = \frac{125}{1728}$$

**Alternate method :**

$$\frac{\text{volume of larger sphere}}{\text{volume of smaller sphere}} = \frac{1}{m}$$

$$\frac{\frac{4}{3}\pi \times (2)^3}{\frac{4}{3}\pi \times \left(\frac{1}{2}\right)^3} = \frac{1}{m}$$

$$\Rightarrow \frac{(2)^3 \times 27}{\left(\frac{1}{2}\right)^3 \times 125} = \frac{1}{m}$$

$$\Rightarrow \frac{8 \times 27 \times 8}{125} = \frac{1}{m} \Rightarrow m = \frac{125}{1728}$$

#### Sol.24.(d)

**Case 1:** When the sheet of paper is rolled along its length to form a cylinder, then the circumference of the base of the cylinder = length of the rectangular sheet  
height of the cylinder = breadth of the rectangular sheet.

Let radius =  $r$  and height =  $h = 36$  cm

Circumference of base = 60 cm

$$2\pi r = 60 \text{ cm} \Rightarrow r = \frac{30}{\pi} \text{ cm}$$

Volume of the cylinder formed ( $V_1$ )

$$= \pi r^2 h = \pi \times \frac{30}{\pi} \times \frac{30}{\pi} \times 36 \dots\dots (i)$$

**Case 2:** When the sheet of paper is rolled along its breadth to form a cylinder, then the circumference of the base of the cylinder = the breadth of the rectangular sheet  
height of the cylinder = length of the rectangular sheet.

Let radius =  $R$  and height =  $H = 60$  cm

Circumference of base = 36 cm

$$2\pi R = 36 \text{ cm} \Rightarrow R = \frac{18}{\pi} \text{ cm}$$

Volume of the cylinder formed ( $V_2$ )

$$= \pi R^2 H = \pi \times \frac{18}{\pi} \times \frac{18}{\pi} \times 60 \dots\dots (ii)$$

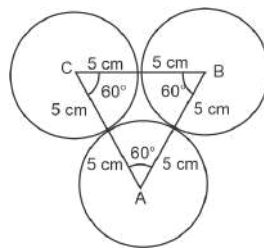
Now,

$$V_1 : V_2 = \pi \times \frac{30}{\pi} \times \frac{30}{\pi} \times 36 : \pi \times \frac{18}{\pi} \times \frac{18}{\pi} \times 60$$

$$\frac{18}{\pi} \times 60 \Rightarrow V_1 : V_2 = 5 : 3$$

Thus, the ratio of the volumes of the two cylinders so formed is 5 : 3.

#### Sol.25.(d)



Center distance between circle = side of triangle =  $2 \times 5 = 10$  cm

Since all sides are of 10 cm so,  $\Delta ABC$  is an equilateral triangle

Area subtended between three circle = area of triangle -  $3 \times$  area of sector

$$= \frac{\sqrt{3}}{4} \times 10 \times 10 - 3 \times \frac{\theta}{360^\circ} \times \pi r^2$$

$$= 25\sqrt{3} - 3 \times \frac{60^\circ}{360^\circ} \times \pi \times 5 \times 5$$

$$= 25\sqrt{3} - \frac{\pi}{2} \times 25 = 25\left(\sqrt{3} - \frac{\pi}{2}\right)$$

#### Sol.26.(a)

Circumference of Larger gear ( $C_1$ ) =  $2\pi R$

$$= 2 \times \frac{22}{7} \times \frac{21}{2}$$

Circumference of smaller gear ( $C_2$ ) =  $2\pi r$

$$= 2 \times \frac{22}{7} \times \frac{15}{2}$$

$$C_1 : C_2 = 7 : 5$$

Ratio  $\rightarrow$  Larger : Smaller

Circumference  $\rightarrow 7 : 5$

Revolution  $\rightarrow 5 : 7$

7 unit = 105 revolution

$$5 \text{ unit} = \frac{105}{7} \times 5 = 75$$

Hence, larger gear has 75 revolutions.

**Sol.27.(d)** Volume of cylinder =  $\pi r^2 h$

$$= \frac{22}{7} \times 7 \times 7 \times 12 = 1848 \text{ cm}^3$$

Let the two new recast cones volume be  $x$  and  $2x$

So,

$$\Rightarrow 3x = 1848$$

$$\Rightarrow x = 616 \text{ cm}^3 \text{ and } 2x = 1232 \text{ cm}^3$$

According to question,

$$\Rightarrow \frac{616}{1232} = \frac{\frac{1}{3}\pi r_1^2 h}{\frac{1}{3}\pi r_2^2 h}$$

$$\Rightarrow \frac{616}{1232} = \frac{\frac{1}{3}\pi r_1^2 \times 12}{\frac{1}{3}\pi r_2^2 \times 12}$$

$$\Rightarrow \frac{r_1^2}{r_2^2} = \frac{1}{2} \Rightarrow r_1 : r_2 = 1 : \sqrt{2}$$

$$\text{Now, } \Rightarrow \pi r_1^2 : \pi r_2^2 = 1 : 2$$

$$\text{Required percentage} = \frac{2-1}{2} \times 100$$

$$= 50\%$$

**Sol.28.(b)** Volume of cubes =  $8^3 + 10^3 +$

$$12^3 = 512 + 1000 + 1728 = 3240 \text{ cm}^3$$

Remaining volume of cube

$$= 3240 - 496 = 2744 \text{ cm}^3$$

$$a^3 = 2744 \Rightarrow a = 14 \text{ cm}$$

Area of whole surface of the newly formed cube ( $6a^2$ )

$$= 6 \times 14 \times 14 = 1176 \text{ cm}^2$$

#### Sol.29.(b)

$$\text{Semiperimeter} = \frac{2(x+y)}{2} = x + y = 12$$

Area of rectangle = 28

$$xy = 28$$

$$\text{Now, } (x+y)^2 = x^2 + y^2 + (xy + xy)$$

$$(x+y)^2 - xy = x^2 + y^2 + xy$$

$$12^2 - 28 = x^2 + y^2 + xy$$

$$116 = x^2 + y^2 + xy$$

$$\text{Again, } (x+y)^2 = x^2 + y^2 + (3xy - xy)$$

$$(x+y)^2 - 3xy = x^2 + y^2 - xy$$

$$12^2 - 28 \times 3 = x^2 + y^2 - xy$$

$$60 = x^2 + y^2 - xy$$

$$\text{then, } x^4 + x^2 y^2 + y^4 = (x^2 + y^2 + xy)(x^2 + y^2 - xy) = 116 \times 60 = 6960$$

**Sol.30.(b)** Lateral surface area =  $120 \text{ cm}^2$

Lateral surface area of prism = perimeter of base  $\times$  height

$$120 = 3 \times 4 \times \text{height} \Rightarrow \text{Height} = 10 \text{ cm}$$

Now,

volume of prism = area of base  $\times$  height

$$= \frac{\sqrt{3}}{4} \times 4 \times 4 \times 10 = 40\sqrt{3} \text{ cm}^3$$

**Sol.31.(a)** According to the question,

Let the number of ice pieces be  $n$

Volume of spherical ice pieces  $\times n$

= volume of cylindrical container

$$\frac{4}{3} \times \frac{22}{7} \times 3^3 \times n = \frac{22}{7} \times 9^2 \times 40$$

$$n = 90$$

**Sol.32.(c)** According to the question,

Volume of hemispherical container

= volume of cylindrical bottles

Let the number of bottles =  $x$

$$\frac{2}{3}\pi r^3 = \pi r^2 h \times x$$

$$= \frac{2}{3} \times 18 \times 18 \times 18 = 3 \times 3 \times 6 \times x$$

$$(\text{Number of bottles}) x = 72$$

**Sol.33.(c)** H.C.F of 90 and 126 = 18

Largest possible side of the tiles

$$= 18 \text{ feet}$$

$$\text{Area of tiles} = 18 \times 18 = 324 \text{ feet}^2$$

#### Sol.34.(b)

$$\text{Volume of frustum} = \frac{\pi H}{3}(R^2 + r^2 + Rr)$$

$$= \frac{22 \times 45}{7 \times 3}(28^2 + 7^2 + 28 \times 7)$$

$$= \frac{22 \times 45}{7 \times 3} \times 1029 = 48510$$

Now,

Volume of cylinder = Volume of frustum



$$\frac{22}{7} \times 3x \times 3x \times 5x = 48510$$

$$x^3 = 343 \Rightarrow x = 7$$

So, curved surface area of cylinder  
 $= 2\pi rh$

$$= 2 \times \frac{22}{7} \times 21 \times 35 = 4620 \text{ cm}^2$$

**Sol.35.(d)** Area of square = Base<sup>2</sup>

Area of rhombus = Base<sup>2</sup>  $\times \sin \theta$  ( $\theta$  = angle between the sides)

The ratio of area of the square to the

$$\text{area of rhombus} = \frac{1}{\sin 45} = \sqrt{2} : 1$$

**Sol.36.(a)** Volume of the wood Required

= (Total volume - inner volume)

$$(10 \times 8 \times 6) - [(10 - 1) \times (8 - 1) \times (6 - 1)]$$

$$= 480 - 315 = 165 \text{ cm}^3$$

**Sol.37.(d)** According to the question,  
 Weight  $\propto$  (density  $\times$  volume)

$$\text{Volume} = k \times \frac{\text{weight}}{\text{density}}$$

Ratio - 1st cube : 2nd cube

Density - 27 : 16

Weight - 4 : 1

$$\text{Volume} - \frac{4}{27} : \frac{1}{16} \rightarrow 64 : 27$$

$$\text{Side} = \sqrt[3]{64} : \sqrt[3]{27} \rightarrow 4 : 3$$

**Sol.38.(a)** According to the question,

weight = radii<sup>2</sup>  $\times$  thickness

Ratio of weight = 4 : 1

Ratio of thickness = 16 : 9

$$\text{Ratio of their radii} = \sqrt{\frac{4}{16}} : \sqrt{\frac{1}{9}}$$

$$= \frac{2}{4} : \frac{1}{3} = 3 : 2$$

**Sol.39.(a)** Circumference of dome ( $2\pi r$ )

$$= 154 \text{ cm} \Rightarrow r = 154 \times \frac{7}{44} = 24.5 \text{ cm}$$

Curved surface area of hemispherical

$$\text{dome} = 2 \times \frac{22}{7} \times 24.5 \times 24.5$$

$$\text{Required cost} = \frac{2 \times 22 \times 24.5 \times 24.5}{7 \times 100} \times 4$$

$$= \frac{2 \times 22 \times 3.5 \times 24.5}{25} = 150.92 \text{ ₹}$$

**Sol.40.(a)** Area of sector =  $\frac{\theta}{360^\circ} \times \pi r^2$

$$616 \text{ cm}^2 = \frac{10^\circ}{360^\circ} \times \frac{22}{7} \times r^2$$

$$r^2 = \frac{616 \times 36 \times 7}{22} = 28 \times 36 \times 7$$

$$r = 84 \text{ cm}$$

**Sol.41.(c)** Volume of cone

$$= \frac{1}{3} \times \frac{22}{7} \times \frac{7}{4} \times \frac{7}{4} \times \frac{7}{2}$$

$$= \frac{11 \times 7 \times 7}{3 \times 16} = \frac{539}{48} = 11.23 \text{ cm}^3$$

**Sol.42.(c)**

Total surface area of cylinder =  $2\pi r(h + r)$

$$6336 = 2 \times \frac{22}{7} \times r \times 42$$

$$r = \frac{6336}{22 \times 12} = 24 \text{ m}$$

Curved surface area of cylinder =  $2\pi rh$

$$= 2 \times \frac{22}{7} \times 24 \times 18 = 2715.43 \text{ m}^2$$

**Sol.43.(b)**

Let the side of cubes is  $3x, 4x, 5x$

$$\text{Side of newly formed cube} = \frac{\text{diagonal}}{\sqrt{3}}$$

$$\text{Side of newly formed cube} = \frac{18\sqrt{3}}{\sqrt{3}} = 18$$

Equating the volume of cubes :-

$$(3x)^3 + (4x)^3 + (5x)^3 = 18 \times 18 \times 18$$

$$x^3 = \frac{18 \times 18 \times 18}{216} = 27 \Rightarrow x = 3$$

Sides are :- 9, 12 and 15

**Sol.44.(b)**

Perimeter of equilateral angle triangle =  $3a$

$$\text{Area of equilateral angle triangle} = \frac{\sqrt{3}}{4} a^2$$

$$\text{According to question, } a = \sqrt[4]{\frac{16}{3}}$$

$$\text{Therefore, required area} = \frac{\sqrt{3}}{4} \times \left(\sqrt[4]{\frac{16}{3}}\right)^2$$

$$= \frac{\sqrt{3}}{4} \times \frac{4}{\sqrt{3}} = 1 \text{ unit}^2$$

**Sol.45.(c)**

Curved surface area of hemisphere =  $2\pi r^2$

Cost for green colour =

$$3 \times (2 \times 3.14 \times 4 \times 4) = 301.44 \text{ Rs.}$$

Cost for red colour

$$= 3 \times (2 \times 3.14 \times 4 \times 4) = 301.44 \text{ Rs.}$$

$$\text{Area of two bases} = \pi r^2 + \pi r^2 = 2\pi r^2$$

$$= (2 \times 3.14 \times 4 \times 4) = 100.48$$

The cost of painting with blue

$$= 2 \times 100.48 = 200.96 \text{ Rs.}$$

$$\text{Total cost} = 301.44 + 301.44 + 200.96$$

$$= 803.84 \text{ Rs.}$$

**Sol.46.(d)**

$$\text{Total surface area of cube} = (6a^2)$$

$$= 6 \times 5^2 = 150 \text{ cm}^2$$

$$\text{Increased area} = 6 \times 5 \text{ cm}^2 - 6 \times 1 \text{ cm}^2$$

$$= 24 \text{ cm}^2$$

$$\text{Total increased surface area} = 150 + 24$$

$$= 174 \text{ cm}^2$$

**Sol.47.(b)**

Volume of Prism = Area of Base  $\times$  Height

Volume of Pyramid

$$= \frac{1}{3} \times \text{Area of Base} \times \text{Height}$$

Required ratio

$$\Rightarrow \text{Volume of Prism} : \text{Volume of Pyramid}$$

$$= 3 : 1$$

**Sol.48.(c)** ATQ, CSA of cone = total area of rectangular cloth

$$\frac{22}{7} \times \frac{14}{2} \times 9 = 2 \times (\text{length of cloth})$$

$$\Rightarrow 198 = 2 \times l \Rightarrow l = \frac{198}{2} = 99 \text{ m}$$

**Sol.49.(a)** Angle covered in 45 minutes

$$\text{by minute hand} = \frac{360}{60} \times 45 = 270^\circ$$

$$\text{Required area} = \frac{270^\circ}{360^\circ} \times \frac{22}{7} \times 20 \times 20$$

$$= \frac{6600}{7} \text{ cm}^2$$

**Sol.50.(b)** External radius (R) = 10 cm

Internal radius (r) = 9 cm and

length (h) = 12 cm

Total surface area of hollow cylinder pipe

$$= 2\pi(R + r)\{h + R - r\}$$

$$= 2 \times \frac{22}{7} \times 19 \times 13$$

$$= \frac{44 \times 247}{7} = 1552.57 \text{ cm}^2$$

**Sol.51.(c)** Let the length of rectangle is 'l' and breadth of rectangle is 'b'

Area of rectangle = length  $\times$  breadth =  $l \times b$

$$(\text{Area of rectangle}) = (lb) \dots\dots \text{e.q. (1)}$$

According to question,

(Area of resulting square)

$$= (l - 4) \times (b + 2) \dots\dots \text{e.q. (2)}$$

Side of square are equal, so

$$l - 4 = b + 2 \Rightarrow l - b = 6 \dots\dots \text{e.q. (3)}$$

Equating e.q. (1) and e.q. (2), we get

$$(lb) = lb + 2l - 4b - 8$$

$$l - 2b = 4 \dots\dots \text{e.q. (4)}$$

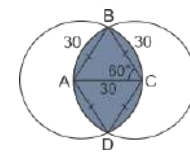
From e.q. (3) and e.q. (4), we get

$$b = 2, l = 8$$

So, Perimeter of the original rectangle

$$= 2(8 + 2) = 20 \text{ cm}$$

**Sol.52.(d)**



$$\text{Area of sector ACB} = \frac{60^\circ}{360^\circ} \times \pi r^2 = \frac{900\pi}{6}$$

$$\text{Area of sector BAC} = \frac{60^\circ}{360^\circ} \times \pi r^2 = \frac{900\pi}{6}$$

Area of equilateral triangle ABC

$$= \frac{\sqrt{3}}{4} r^2 = \frac{900\sqrt{3}}{4}$$

Upper shaded region = area of sector ACB + area of sector BAC - area of equilateral  $\Delta ABC$

Area of total shaded region

= 2  $\times$  area of upper half shaded region

Required shaded area

$$= 2 \times \left\{ \frac{900\pi}{6} + \frac{900\pi}{6} - \frac{900\sqrt{3}}{4} \right\}$$

$$= 2 \times \left\{ \frac{1800\pi}{6} - \frac{900\sqrt{3}}{4} \right\}$$

$$= (600\pi - 450\sqrt{3}) \text{ cm}^2$$

**Short Trick:-**

$$\text{Area of shaded region} = r^2 \left( \frac{2}{3} \pi - \frac{\sqrt{3}}{2} \right)$$

$$\text{Area} = 900\left(\frac{2}{3}\pi - \frac{\sqrt{3}}{2}\right)$$

$$= (600\pi - 450\sqrt{3}) \text{ cm}^2$$

**Sol.53.(b)** Let side of the rhombus be 'a'  
Perimeter of rhombus =  $4a = 4P \Rightarrow a = P$   
As we know that,

$$D_1^2 + D_2^2 = 4a^2, \quad D_1 + D_2 = L$$

$$\text{Then, } D_1 D_2 = \frac{L^2 - 4a^2}{2}$$

$$\text{Area of Rhombus} = \frac{1}{2} \times D_1 \times D_2$$

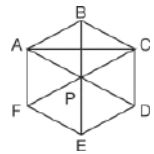
$$= \frac{1}{2} \times \frac{L^2 - 4a^2}{2} = \frac{1}{4} \times (L^2 - 4P^2)$$

**Sol.54.(d)** Volume of water flowing through the canal in 30 min

$$= \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \times (18 \times \frac{5}{18}) \times (30 \times 60)$$

$$= 3,46,500 \text{ m}^3$$

**Sol.55.(a)** According to the question,



As ABCDEF is a regular hexagon,

$$\text{Area of } \square ABCP = 2 \times \frac{\sqrt{3}}{4} \times (\text{side})^2$$

$$= 2 \times \frac{\sqrt{3}}{4} \times 36^2 = 648\sqrt{3} \text{ cm}^2$$

$$\text{So, area of } \triangle ABC = \frac{1}{2} \times \text{Area of } \square ABCP$$

$$= \frac{1}{2} \times 648\sqrt{3} = 324\sqrt{3} \text{ cm}^2$$

**Sol.56.(a)**

$$\text{Volume of frustum} = \frac{1}{3}\pi h (R^2 + r^2 + Rr)$$

$$= \frac{1}{3}\pi \times 6 (5^2 + 3^2 + 15)$$

$$= 2\pi (25 + 9 + 15) = 98\pi \text{ cm}^3$$

**Sol.57.(d)** Let  $a = b = 2$  units

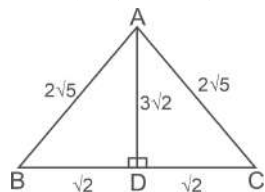
$$BC = \sqrt{a^2 + b^2} = 2\sqrt{2} \text{ units}$$

$$AB = \sqrt{(2a)^2 + b^2} = 2\sqrt{5} \text{ units}$$

$$AC = \sqrt{a^2 + (2b)^2} = 2\sqrt{5} \text{ units}$$

It becomes an isosceles triangle as below.

$$AD = \sqrt{(2\sqrt{5})^2 - (\sqrt{2})^2} = 3\sqrt{2} \text{ units}$$

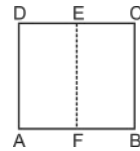


Area of triangle

$$= \frac{1}{2} \times 2\sqrt{2} \times 3\sqrt{2} = 6 \text{ sq units}$$

$$\text{In option (d), } \frac{3}{2} ab = \frac{3}{2} \times 2 \times 2 = 6$$

**Sol.58.(d)**



Let the side of square =  $a$

$$AB = BC = CD = DA = EF = a$$

Perimeter of rectangle ADEF + Perimeter of rectangle BCFE =  $39 + 39 = 78$

$$AB + BC + CD + DA + EF + EF = 78$$

$$6a = 78 \text{ cm} \Rightarrow a = 13 \text{ cm}$$

$$\text{Perimeter of Square} = 4a = 52 \text{ cm}$$

**Sol.59.(b)** Circumference of a circle =  $2\pi r$   
 $2\pi r = a\pi \Rightarrow 2r = a$

$$\text{Area} = b\pi \Rightarrow \pi r^2 = b\pi \Rightarrow b = r^2$$

$$a : b = 2r : r^2 \text{ or } 2 : r$$

$$4 : 5 = 2 : r \Rightarrow r = 2.5 \text{ cm}$$

**Sol.60.(c)** 65 m, 72 m and 97 m are triplets of a right-angled triangle.

So area of triangular park

$$= \frac{1}{2} \times 65 \times 72 = 2340$$

Let sides of the rectangle be  $5x$  and  $13x$ .

$$\text{Area of rectangle} = 65x^2$$

$$\text{ATQ, } 65x^2 = 2340 \Rightarrow x = 6$$

$$\text{Perimeter of rectangle} = 2(5x + 13x)$$

$$= 36x = 216 \text{ m}$$

**Sol.61.(c)** Radius of bicycle = 0.42 m

Distance travelled in 25 seconds

$$= 2\pi r \times 40 \Rightarrow 2 \times \frac{22}{7} \times 0.42 \times 40$$

$$= 105.6 \text{ m}$$

$$\text{Speed of bicycle (in m/s)} = \frac{105.6}{25}$$

$$= 4.224$$

$$\text{Speed of bicycle (in kmph)} = 4.224 \times \frac{18}{5}$$

$$= 15.2$$

**Sol.62.(a)** Area of the square =  $a^2 = 169$ .

$$\text{So, } a = 13, \text{ Perimeter} = 4a = 4 \times 13 = 52$$

$$\text{Circumference of the circle} = 2\pi r = 52$$

$$r = \frac{7}{44} \times 52 = \frac{91}{11}$$

$$\text{Area} = \pi r^2 = \frac{22}{7} \times \frac{91}{11} \times \frac{91}{11}$$

$$= \frac{2366}{11} = 215.09 \approx 215 \text{ cm}^2$$

**Sol.63.(d)** Slant height of the cone

$$= \text{radius of the sector And } \frac{\theta}{360} \times$$

perimeter of the circle

= perimeter of the base of the cone

Let the radius of cone =  $r$

$$\frac{120}{360} \times 2 \times \pi \times 10.5 = 2 \times \pi \times r$$

$$\Rightarrow r = 3.5$$

$$\text{Height of the cone} = \sqrt{l^2 - r^2}$$

$$= \sqrt{(10.5)^2 - (3.5)^2} = 7\sqrt{2}$$

Volume of the cone

$$= \frac{1}{3} \times \pi \times 3.5 \times 3.5 \times 7\sqrt{2}$$

$$= \frac{343\sqrt{2}}{12} \pi \text{ cm}^3$$

**Sol.64.(c)** 20, 21 and 29 are triplet pairs, clearly a given triangle is a right angle triangle.

$$\text{Area of the triangle} = \frac{1}{2} \times 20 \times 21$$

$$= 210 \text{ cm}^2$$

Volume of the prism = Base Area  $\times$  Height

$$\Rightarrow 7560 = 210 \times \text{Height} \Rightarrow \text{Height} = 36$$

Lateral surface area of the prism

$$= \text{Base perimeter} \times \text{Height}$$

$$= (20 + 21 + 29) \times 36 = 2520 \text{ cm}^2$$

**Sol.65.(a)** Let the original radius be  $r$ .

According to the question

$$4\pi(r + 4)^2 = 4\pi r^2 + 464\pi$$

$$\Rightarrow r^2 + 16 + 8r = r^2 + 116 \Rightarrow r = 12.5$$

$$\text{Volume of the sphere} = \frac{4}{3}\pi r^3$$

$$= \frac{4}{3} \times 12.5 \times 12.5 \times 12.5 \times \pi$$

$$= \frac{15625}{6} \pi \text{ cm}^3$$

**Sol.66.(c)**

Volume of prism = base area  $\times$  height

$$1731.6 = \frac{1}{2} \times (15 + 11) \times 9 \times \text{height}$$

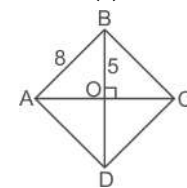
$$\Rightarrow \text{height} = 14.8$$

**Sol.67.(b)** Let  $n$  be the number of turns.

Total length of the ribbon = number of turns  $\times$  circumference of the base of

$$\text{cylinder } l = n \times c \Rightarrow n = \frac{l}{c}$$

**Sol.68.(c)**



$$BO = \frac{BD}{2} = \frac{10}{2} = 5$$

$$AO = \frac{AC}{2} = \frac{2\sqrt{x}}{2} = \sqrt{x}$$

$$\text{Now, } AB^2 = AO^2 + BO^2$$

$$8^2 = (\sqrt{x})^2 + 5^2 \Rightarrow x = 39$$

$$\sqrt{x + 10} \Rightarrow \sqrt{39 + 10} = 7 \text{ cm}$$

**Sol.69.(d)** Perimeter of rhombus

$$= 2\sqrt{d_1^2 + d_2^2} = 2\sqrt{14^2 + 48^2} = 100 \text{ cm}$$

**Sol.70.(c)** Circumference of the circle(C)

$$= 2\pi r \Rightarrow r = \frac{C}{2\pi}$$

$$\text{Area of the sector (A)} = \frac{60}{360} \times \pi \times r^2$$



$$= \frac{1}{6} \times \pi \times \left(\frac{C}{2\pi}\right)^2 \Rightarrow \frac{C^2}{24\pi}$$

**Sol.71.(b)** Let the length of the perpendicular and base of the right angle triangle is a cm.

$\Rightarrow$  Hypotenuse of the triangle =  $a\sqrt{2}$

Area of  $\Delta$  formed on hypotenuse(H)

$$= \frac{\sqrt{3}}{4} \times (a\sqrt{2})^2 = \frac{\sqrt{3}a^2}{2}$$

Area of  $\Delta$  formed on perpendicular(A)

$$= \frac{\sqrt{3}a^2}{4} \Rightarrow \frac{A}{H} = \frac{\frac{\sqrt{3}a^2}{4}}{\frac{\sqrt{3}a^2}{2}} = \frac{1}{2}$$

**Sol.72.(c)** Since it is an equilateral triangle. Each side will be formed of 4 sticks and its length will be 4 units.

Area of the equilateral triangle =  $\frac{\sqrt{3}}{4}a^2$

$$= \frac{\sqrt{3}}{4} \times 4 \times 4 = 4\sqrt{3}$$

**Sol.73.(c)** % age change in the area

$$= (-11) + 11 + \frac{(-11) \times (11)}{100} = -1.21\%$$

Here, - ve sign indicates that the area is decreasing.

**Sol.74.(d)** Let AB = a cm

$$AC = a + 10 \Rightarrow BC = 2a - 2$$

According to the question

$$a + a + 10 + 2a - 2 = 32 \Rightarrow 4a = 24$$

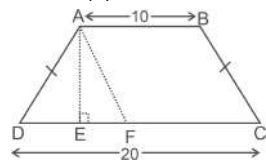
$$a = AB = 6 \text{ cm}$$

$$AC = 6 + 10 = 16 \text{ cm}$$

$$BC = 2(6) - 2 = 10 \text{ cm}$$

Clearly AB is the smallest side.

**Sol.75.(b)**



Let ABCD be the given trapezium. And AE be its height.

$$\text{Area of trapezium} = \frac{1}{2} \times (CD + AB) \times AE$$

$$180 = \frac{1}{2} \times (20 + 10) \times AE$$

$$AE = 12 \text{ cm} \quad \text{Draw } AF \parallel BC$$

$$\Rightarrow AB = FC = 10 \text{ cm and } AF = BC$$

$$\Rightarrow DF = DC - FC = 20 - 10 = 10 \text{ cm}$$

$$\text{In } \triangle ADF \Rightarrow AF = AD \quad \text{....(AD = BC, given)}$$

$$DE = EF = \frac{DF}{2} = 5 \text{ cm}$$

....(Perpendicular drawn on the base of an isosceles triangle)

$$AF^2 = AE^2 + EF^2, AF = \sqrt{AE^2 + EF^2}$$

$$= \sqrt{12^2 + 5^2} = 13$$

$$\text{Hence } AD = BC = 13 \text{ cm}$$

**Sol.76.(d)**

Let the breadth of the rectangle = b

Length of the rectangle (l) = b + 20

Perimeter of the rectangle =  $2(b + b + 20)$

$$= 4(b + 10)$$

According to the question

$$4(b + 10) \times 53 = 21200$$

$$b + 10 = 100 \Rightarrow b = 90 \text{ m}$$

$$l = 90 + 20 = 110 \text{ m}$$

$$\text{Area of the rectangle} = 90 \times 110 = 9900 \text{ m}^2$$

**Sol.77.(b)** Let the sides of the rectangle are 7x and 4x.

$$\text{Perimeter of rectangle} = 2(7x + 4x) = 22x$$

$$\text{Perimeter of circle} = 2\pi r \Rightarrow 2 \times \frac{22}{7} \times 31.5$$

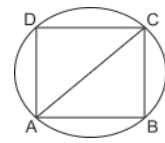
According to the question

$$22x = 2 \times \frac{22}{7} \times 31.5 \Rightarrow x = 9$$

$$\text{Smaller side of rectangle} = 4x$$

$$= 4 \times 9 = 36 \text{ cm}$$

**Sol.78.(d)**



Let the sides be 4 units and 3 units.

According to the question

$$2(4 \text{ unit} + 3 \text{ unit}) = 56$$

$$1 \text{ unit} = 4 \text{ cm, } 4 \text{ unit} = 16 \text{ cm, } 3 \text{ unit} = 12 \text{ cm}$$

Diagonal of Rectangle = Diameter of circle

$$= \sqrt{AB^2 + BC^2} = \sqrt{16^2 + 12^2} = 20 \text{ cm}$$

$$\text{Radius of the circle} = \frac{20}{2} = 10 \text{ cm}$$

$$\text{Area of circle} = \pi r^2 = \pi 10^2 = 100\pi$$

**Sol.79.(d)**

Curved Surface area of cylinder =  $2\pi rh$

$$\text{Volume of cylinder} = \pi r^2 h$$

According to the question

$$2\pi rh = 132 \text{.....(1), } \pi r^2 h = 528 \text{.....(2)}$$

$$\text{Dividing eq 2 by eq 1, } r = 8 \text{ cm}$$

$$\text{Put } r = 8 \text{ in eq (1)}$$

$$2\pi rh = 132$$

$$2 \times \frac{22}{7} \times 8 \times h = 132 \Rightarrow h = 2\frac{5}{8} \text{ m}$$

$$\text{Sol.80.(a) Area of Hexagon} = 6 \times \frac{\sqrt{3}}{4} a^2$$

$$\text{Here, } 6 \times \frac{\sqrt{3}}{4} a^2 = 2400\sqrt{3},$$

$$a^2 = 1600, a = 40$$

$$\text{Perimeter of the hexagonal field} = 6a$$

$$6 \times 40 = 240 \text{ metre}$$

$$\text{Required cost of fencing} = 240 \times 18.5$$

$$= \text{Rs. 4440}$$

**Sol.81.(c)** One of the sides of newly formed cuboid =  $6 \times 2 = 12 \text{ cm}$

Other two sides will be 2 cm.

$$\text{Surface area of cuboid} = 2(lb + bh + hl)$$

$$= 2(12 \times 2 + 2 \times 2 + 2 \times 12) = 104 \text{ cm}^2$$

$$\text{Sol.82.(d) } 40\% = \frac{2}{5}$$

Old : New

$$\text{Radius} \rightarrow 5 : 3$$

$$\text{Volume} (\propto r^3) \rightarrow 125 : 27$$

Let the volumes are 125 unit and 27 unit

$$\text{Required \%} = \frac{125 - 27}{125} \times 100 = 78.4\%$$

**Sol.83.(b)** Area of triangular field with

$$\text{side } x \text{ m} = \frac{\sqrt{3}}{4} x^2$$

Area of other triangular field

$$= \sqrt{s(s-a)(s-b)(s-c)}$$

$$\text{Where } s = \frac{a+b+c}{2}$$

$$\Rightarrow \frac{50 + 70 + 80}{2} = 100 \text{ m}$$

Area of field =

$$\sqrt{100(100-50)(100-70)(100-80)}$$

$$= \sqrt{100 \times 50 \times 30 \times 20} = 1000\sqrt{3} \text{ m}^2$$

$$\text{According to question, } \frac{\sqrt{3}}{4} x^2 = 1000\sqrt{3}$$

$$x^2 = 4000 \Rightarrow x = 63.25 \text{ m}^2$$

$$\text{Sol.84.(d) Area of Sector} = \frac{\theta}{360^\circ} \times \pi r^2$$

$$= \frac{150}{360} \times \frac{22}{7} \times 21^2 = 577.5 \text{ cm}^2$$

**Sol.85.(d)** Area of circle is  $\propto r^2$ .

$$\Rightarrow \text{Area of three circles will be } 4^2, 5^2, 7^2$$

$$\text{Desired Ratio} = (5^2 - 4^2) : (7^2 - 5^2)$$

$$= 9 : 24 = 3 : 8$$

**Sol.86.(c)**

$$\text{Volume of a cylinder} = \pi(R^2 - r^2)h$$

Where R = external radius

r = internal radius

h = height or length of the cylinder

$$\text{Volume} = \frac{22}{7} \times (2.5^2 - 1.5^2) \times 756$$

$$= 22 \times (2.5 + 1.5)(2.5 - 1.5) \times 108$$

$$= 9504 \text{ cm}^3$$

$$\text{Total Weight of pipe} = \text{Volume of pipe} \times \text{density of metal}$$

$$= 9504 \times 7.5 = 71280 \text{ g or } 71.28 \text{ kg}$$

**Sol.87.(a)**

$$\text{Volume of a cylinder} = \pi(R^2 - r^2)h$$

Where R = external radius

r = internal radius

h = height or length of the cylinder

$$7480 = \frac{22}{7} \times (9^2 - r^2) \times 140$$

$$17 = 81 - r^2 \Rightarrow r^2 = 64 \Rightarrow r = 8 \text{ cm}$$

$$\text{Thickness} = 9 - 8 = 1 \text{ cm}$$

$$\text{Sol.88.(a) Volume of cuboid} = \sqrt{a \times b \times c}$$

Here a, b and c are the area of three faces

$$= \sqrt{32 \times 24 \times 48} = 192 \text{ cm}^3$$

**Sol.89.(d)** We know that

$$\text{Inradius} = \frac{\text{area}}{\text{semiperimeter}}$$

$$3 = \frac{15}{\text{semiperimeter}} \Rightarrow \text{Semiperimeter} = 5$$

$$\text{Perimeter} = 2 \times 5 = 10 \text{ cm}$$

**Sol.90.(c)** Required area of sheet = Total surface area of rectangular tank - Area of circular opening

$$= 2 \times (80 + 48 + 60) - \pi \times 1^2$$

$$= 376 - 3.1 = 372.9 \text{ m}^2$$

**Sol.91.(b)** Let side of bigger cube = R

$$R^3 = 13824 \Rightarrow R = \sqrt[3]{13824} = 24$$

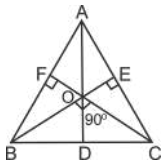
Let side of smaller cubes = r

$$13824 = 8 \times r^3 \Rightarrow 1728 = r^3 \Rightarrow 12 = r$$

$$\text{Required ratio} = (6 \times 24^2) : (3 \times 6 \times 12^2)$$

$$= 4 : 3$$

**Sol.92.(b)**



$$\triangle OBC, OB = OC = x \Rightarrow BC = 2$$

$$x^2 + x^2 = 2^2 \Rightarrow x^2 = 2 \Rightarrow x = \sqrt{2}$$

$$\text{In } \triangle OBD, BD = 1 \Rightarrow OB = \sqrt{2}$$

$$OD = \sqrt{(\sqrt{2})^2 - 1^2} = 1$$

$$\text{Now, } AO = 2 \times OD \Rightarrow 2 \times 1 = 2$$

$$AD = AO + OD = 3$$

$$\text{Area of } \triangle ABC = \frac{1}{2} \times 3 \times 2 = 3 \text{ cm}^2$$

**Sol.93.(b)** Volume of cubical box

$$= 30 \times 30 \times 30 = 27000 \text{ cm}^3$$

Required rise in water level

$$= \frac{27000}{60 \times 30} = 15 \text{ cm}$$

**Sol.94.(c)** Old : New

$$\text{Length} \rightarrow 10 : 13$$

$$\text{Breadth} \rightarrow 5 : 4$$

$$\text{Area} \rightarrow 50 : 52$$

$$52 - 50 = 2 \text{ unit} = 100 \Rightarrow 1 \text{ unit} = 50$$

$$\text{New area} = 52 \times 50 = 2600 \text{ cm}^2$$

**Sol.95.(a)** Required Cost

$$= \frac{11 \times 6 \times 100}{60} \times 112.50 = 12,375$$

**Sol.96.(c)** Remaining area

$$= 119 \times 18 - 17 \times 6 = 2040$$

$$\text{Volume of sand} = 17 \times 6 \times 3 = 306$$

Required height

$$= \frac{306 \times 100 \times 100 \times 100}{2040 \times 100 \times 100} = 15 \text{ cm}$$

**Sol.97.(d)** Total area to be painted

$$= 40 \times 30 + 2(40 \times 2.2 + 30 \times 2.2) = 1508$$

$$\text{Total cost} = 1508 \times 25 = 37700$$

**Sol.98.(a)** Since, the height of the triangle formed will be half of the diagonal of the square.

Therefore,

$$\text{Height} = \frac{3\sqrt{2}}{2} = \frac{3}{\sqrt{2}} \text{ m}$$

**Sol.99.(a)** ATQ,

$$\frac{3\sqrt{3}}{2} \times a^2 = 108\sqrt{3} \Rightarrow a = 6\sqrt{2}$$

$$\therefore \text{Perimeter} = 6 \times 6\sqrt{2} = 36\sqrt{2} \text{ cm}$$

**Sol.100.(a)**

$$\text{Volume} = (12 - 4) \times (12 - 4) \times 2$$

$$= 8 \times 8 \times 2 = 128 \text{ cu.cm}$$

**Sol.101.(d)** Given, in a right angle triangle

$$\text{Hypotenuse} = 25, \text{ Base} = 24$$

$$\text{Then, Height} = 7$$

$$\therefore \text{Area} = \frac{1}{2} \times 24 \times 7 = 84 \text{ cm}^2$$

**Sol.102.(a)**

$$\text{Longest pole} = \sqrt{16^2 + 8^2 + 11^2} = 21 \text{ m}$$

$$\text{Sol.103.(d)} \quad \frac{\sqrt{3}}{2} a = 10\sqrt{3} \Rightarrow a = 20$$

$$\text{Area} = \frac{\sqrt{3}}{4} a^2$$

$$= \frac{\sqrt{3}}{4} \times 20 \times 20 = 100\sqrt{3} \text{ cm}^2$$

**Sol.104.(c)**

$$\text{Required time} = \frac{66 \times 35 \times 3}{\frac{22}{7} \times \frac{35}{200} \times \frac{35}{200} \times 8}$$

$$= 9000 \text{ sec} = 2.5 \text{ h}$$

**Sol.105.(a)**

In the given solid figures. Cuboid has the maximum no. of vertex i.e. 8.

**Sol.106.(b)**

$$\text{Area} = 20 \times (7.5 + 8.6) = 322 \text{ cm}^2$$

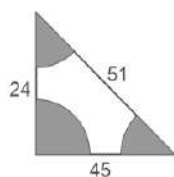
**Sol.107.(b)** Since the chessboard is composed of 64 squares of side 2cm and the border is of thickness 2 cm.

Therefore, the length of each side of chessboard =  $2 \times 8 + 4 = 20$

So, the perimeter =  $4 \times 20 = 80 \text{ cm}$

**Sol.108.(b)** Given

triangle is a right angled triangle:



$$\text{Area of triangle} = \frac{1}{2} \times 24 \times 45 = 540$$

$$\text{Area of all sectors of the circle covering}$$

$$\text{the triangle} = \frac{180}{360} \times \frac{22}{7} \times 10.5 \times 10.5$$

$$= 173.25$$

$$\text{Therefore, Area of triangle excluding the}$$

$$\text{portion covered} = 540 - 173.25$$

$$= 366.75 \text{ sq. cm}$$

**Sol.109.(a)** Old : New

$$\text{Radius} \rightarrow 2^2 : 5^2$$

$$\text{Height} \rightarrow 5 : 4$$

$$\text{Volume} \rightarrow 20 : 100$$

$$\text{Required \%} = \frac{100 - 20}{20} \times 100 = 400\%$$

**Sol.110.(c)**

$$\text{Required area of pizza} = \pi r^2 \times \frac{20}{100}$$

$$= \frac{22}{7} \times 21 \times 21 \times \frac{1}{5} = 277.2 \text{ cm}^2$$

**Sol.111.(a)** Area of the whole outer circle

$$= \pi r^2$$

$$= \pi \times 10 \times 10 = 100\pi \text{ cm}^2$$

$$\text{Area of ring} = \pi(r_2^2 - r_1^2) = \pi(100 - 64)$$

$$= 36\pi \text{ cm}^2$$

So,

$$\text{ratio} = 36\pi : 100\pi \text{ or } 9 : 25$$

**Sol.112.(b)** Let the circumference of the circle (wheel) = 100 units

Circumference in a quarter revolution = 25 units

So, Required percentage

$$= \frac{25}{100} \times 100 = 25 \%$$

**Sol.113.(a)** r = 15 m

$$\text{New radius} = 15 + 1.5 = 16.5 \text{ m}$$

$$\text{Area of path} = \pi(R^2 - r^2)$$

$$= \frac{22}{7} \times (272.25 - 225) = \frac{22}{7} \times 47.25$$

$$= 148.5 \text{ m}^2$$

$$\text{Sol.114.(a)} \quad \text{radius} = \frac{14}{2} = 7 \text{ cm}$$

$$\text{Slant height} = \sqrt{7^2 + 24^2}$$

$$= \sqrt{49 + 576} = \sqrt{625} = 25 \text{ cm}$$

**Sol.115.(c)**

$$\text{Volume} = \text{area of base} \times \text{height}$$

$$\text{Initial height} = 16 \text{ cm,}$$

$$\text{new height} = 16 \times \frac{5}{4} = 20 \text{ cm}$$

$$\text{Volume} = 25 \times 20 = 500 \text{ cm}^3$$

**Sol.116.(c)**

Circular area =  $\pi R^2$ , here  $\pi$  is constant.

Percentage decrease

$$= -20 - 20 + \frac{20 \times 20}{100} = -40\% + 4$$

$$= -36\% \text{ (-ve sign means decrease)}$$

**Sol.117.(b)** A/Q, Circumference of circle

$$= \text{perimeter of square} = 160 \text{ m}$$

$$\text{Now, } 2\pi r = 160 \Rightarrow r = \frac{80}{\pi}$$

$$\text{Area of circle } (\pi r^2) = \pi \times \frac{80}{\pi} \times \frac{80}{\pi}$$

$$= \frac{6400}{\pi}$$

$$\text{Now, } 4a = 160 \Rightarrow a = 40 \text{ m}$$

$$\text{Area of square} = 40 \times 40 = 1600 \text{ m}^2$$



$$\text{Required ratio} = \frac{\frac{6400}{\pi}}{1600} = \frac{4 \times 7}{22}$$

$$= \frac{1.27}{1} = 1.27 : 1$$

**Sol.118.(b)** A/Q, Volume of hemispherical stone = Volume of cylinder

$$\Rightarrow \frac{2}{3} \pi 6^3 = \pi R^2 \times 2.5$$

$$\Rightarrow \frac{2 \times 6 \times 6 \times 6}{3 \times 2.5} = R^2 \Rightarrow R^2 = \frac{144}{2.5} = 57.6$$

$$\Rightarrow R = \sqrt{57.6} = 7.59 \text{ cm (approx)}$$

**Sol.119.(c)**

$$\frac{3\pi}{4} \text{ radians} = \frac{180}{\pi} \times \frac{3\pi}{4} = 135^\circ$$

Angle of first sector is equal to angle of second sector and having same radius  
 $\therefore$  Ratio of the area of the first sector to the area of the second sector = 1 : 1

**Sol.120.(a)** Area of sector =  $\frac{\theta}{360} \times \pi r^2$

$$\frac{180^\circ}{360} \times \frac{22}{7} \times 21 \times 21 = 693 \text{ cm}^2$$

**Sol.121.(a)** Area of a sector =  $\frac{\theta}{360} \times \pi r^2$

$$= \frac{60^\circ}{360} \times \pi (4)^2 = \frac{8\pi}{3} \text{ cm}^2$$

**Sol.122.(d)**

Formula for Area of ring =  $\pi (R^2 - r^2)$

where, R = Outer radius = 14 cm

r = Inner radius = 7 cm

Area of the full (Outer) Circle =  $\pi R^2$

$$= \pi \times 14^2 = 196\pi$$

Area of ring =  $\pi (14^2 - 7^2) = \pi (196 - 49)$

$$= 147\pi$$

Therefore, Percentage of ring area to the

$$\text{full circle} = \frac{147\pi}{196\pi} \times 100 = 75\%$$

**Sol.123.(b)**

Let the radius of the pond = r meters.

Width of the path = 2 m  $\rightarrow$  Outer radius

= r + 2 meters

then,

Total area (pond + path) = 804.25 m<sup>2</sup>

Area of pond = 615.75 m<sup>2</sup>

$$\text{So, } \pi (r + 2)^2 - \pi r^2 = 804.25 - 615.75$$

$$\pi [(r + 2)^2 - r^2] = 188.5$$

$$\pi (4r + 4) = 188.5$$

$$4\pi (r + 1) = 188.5$$

$$r + 1 = \frac{188.5}{4\pi} = \frac{47.125}{\pi} = 14.9 \approx 15$$

$$r = 15 - 1 = 14$$

Radius of the pond = 14 m

**Sol.124.(a)** According to the question,

$$\Rightarrow \frac{C_1}{C_2} = \frac{2\pi r_1}{2\pi r_2}$$

$$\Rightarrow \frac{5}{4} = \frac{r_1}{10} = r_1 = 12.5$$

Therefore, the radius of the larger flower bed = 12.5 m

**Sol.125.(a)** According to question,

$\Rightarrow$  Volume of cylinder = Volume of cone

$$\Rightarrow \pi r^2 H = \frac{1}{3} \pi r^2 h \Rightarrow \frac{H}{h} = \frac{1}{3} = 1 : 3$$

**Sol.126.(a)**

$$\frac{\text{Perimeter of smaller circular pond}}{\text{Perimeter of bigger circular pond}} = \frac{6}{10}$$

$$\Rightarrow \frac{2\pi r_1}{2\pi r_2} = \frac{6}{10} \Rightarrow \frac{r_1}{r_2} = \frac{6}{10}$$

Now,

$$\Rightarrow \frac{\text{Area of smaller circular pond}}{\text{Area of bigger circular pond}} = \frac{160}{x}$$

$$\Rightarrow \frac{\pi \times 6 \times 6}{\pi \times 10 \times 10} = \frac{160}{x}$$

$$\Rightarrow x = 444.44 \approx 444 \text{ m}^2$$

**Sol.127.(d)**

Volume of prism = base area  $\times$  height

$$= 3 \times \frac{\sqrt{3}}{2} \times 3 \times 3 \times 8 = 108\sqrt{3} \text{ cm}^3$$

**Sol.128.(b)** We know that,

Smallest circumference =  $8\pi$

$$\Rightarrow \text{Smallest diameter} = \frac{8\pi}{\pi} = 8 \text{ cm}$$

Largest circumference =  $16\pi$

$$\Rightarrow \text{Largest diameter} = \frac{16\pi}{\pi} = 16 \text{ cm}$$

Middle diameter =  $(8 + 16) \div 2 = 12 \text{ cm}$

Circumference of middle circle =  $12\pi \text{ cm}$

**Sol.129.(b)**

$$\frac{5\pi}{4} \text{ radians} = \frac{180}{\pi} \times \frac{5\pi}{4} = 225^\circ$$

Angle of first sector is equal to angle of second sector and having same radius

$\therefore$  Ratio of the area of the first sector to the area of the second sector = 1 : 1

**Sol.130.(c)** Area of circular garden

$$= \pi \times 10 \times 10 = 100\pi$$

Area of circular garden with path

$$= \pi \times 12 \times 12 = 144\pi$$

Percentage increase

$$= \frac{144\pi - 100\pi}{100\pi} \times 100 = 44\%$$

**Sol.131.(b)** Area of sector =  $\frac{\theta}{360^\circ} \times \pi r^2$

$$\Rightarrow 77 = \frac{\theta}{360^\circ} \times \frac{22}{7} \times 14 \times 14$$

$$\Rightarrow \theta = \frac{77 \times 360^\circ \times 7}{22 \times 14 \times 14} = \frac{360^\circ}{8} = 45^\circ$$

**Sol.132.(b)** Area of signboard =  $\pi r^2$

$$= 3.14 \times 4 \times 4 = 50.24 \text{ m}^2$$

$$80\% \text{ of board painted} = 50.24 \times \frac{80}{100}$$

$$= 40.19 \text{ m}^2$$

Total cost of painting =  $40.19 \times 30$

$$= ₹1205.7$$

**Sol.133.(a)** According to question,

$\Rightarrow$  Area of base = Lateral surface area of cone

$$\Rightarrow \pi r^2 = \pi r l \Rightarrow r = l$$

So, the radius of the cone is l.

**Sol.134.(b)** Area of remaining field =

Area of square - Area of pond

$$= 24 \times 24 - 484 = 576 - 484 = 92 \text{ m}^2$$

**Sol.135.(b)** Volume of prism ( $V_1$ ) = base

Area  $\times$  height .....(i)

When height is increased 60 % then,

Volume of prism ( $V_2$ ) =

$$\text{base area} \times \text{height} \times \frac{160}{100} \text{ .....(ii)}$$

From eq. (i) and (ii),

$$\Rightarrow \frac{V_1}{V_2} = \frac{100}{160} = \frac{5}{8}$$

$$\text{So, Required \%} = \frac{3}{5} \times 100 = 60\%$$

**Sol.136.(a)** According to question,

Area of smaller segment = Area of sector - area of triangle

$$= \pi r^2 \times \frac{\theta}{360^\circ} - \frac{\sqrt{3}}{4} a^2$$

$$= 3.14 \times 30 \times 30 \times \frac{60^\circ}{360^\circ} - \frac{1.732}{4} \times 30$$

$$\times 30 = 471 - 389.7 = 81.3$$

Area of clock =  $\pi r^2$

$$= 3.14 \times 30 \times 30 = 2826$$

$$\text{So, Required \%} = \frac{81.3}{2826} \times 100 \approx 2.88\%$$

**Sol.137.(a)** Volume of cylinder =  $\pi r^2 h$

$$= \pi \times 8 \times 8 \times 20 = 1280\pi \text{ cm}^2$$

Volume of hole =  $\pi r_1^2 h = \pi \times 4 \times 4 \times 20$

$$= 320\pi$$

$$\text{So, Required \%} = \frac{320\pi}{1280\pi} \times 100 = 25\%$$

**Sol.138.(c)** Area of first sector : Area of

$$\text{second sector} = \pi r^2 \frac{\theta_1}{360^\circ} : \pi r^2 \frac{\theta_2}{360^\circ}$$

$$= 60^\circ : \frac{\pi}{3} = 60^\circ : \frac{180^\circ}{3} = 1 : 1$$

**Sol.139.(a)** Given,  $R_1 = 10$ ,  $R_2 = 7$

Area of the path =  $\pi (R_1^2 - R_2^2)$

$$\Rightarrow \frac{22}{7} (10^2 - 7^2) = \frac{22 \times 3 \times 17}{7}$$

$$= 160.28 \text{ m}^2$$

**Sol.140.(a)** We know that, area of sector

$$= \frac{\pi r^2 \theta}{360^\circ} \text{ and } 1 \text{ radian} = \frac{180^\circ}{\pi}$$

$$\frac{\pi}{3} \text{ radians} = \frac{\pi}{3} \times \frac{180^\circ}{\pi} = 60^\circ$$

$$= \frac{\frac{22}{7} \times 5^2 \times 60^\circ}{360^\circ}$$



$$= \frac{\frac{22}{7} \times 5^2}{6} = \frac{275}{21} = 13.09 \text{ cm}^2$$

**Sol.141.(c)** A/Q,

$$\frac{4}{3} \times \frac{22}{7} \times 8000 = 64 \times \frac{4}{3} \times \frac{22}{7} \times r^3$$

$$r^3 = \frac{8000}{64} = 125 \Rightarrow r = 5 \text{ cm}$$

Surface area of original sphere

$$= 4 \times \frac{22}{7} \times 400$$

Surface area of new sphere

$$= 64 \times 4 \times \frac{22}{7} \times 25$$

$$\text{Reqd. ratio} = 400 : 25 \times 64 = 1 : 4$$

**Sol.142.(b)** Let hemisphere produced = N  
A/Q

$$\frac{2}{3}\pi \times 24 \times 24 \times 24 = N \times \frac{2}{3}\pi \times 6 \times 6 \times 6$$

$$N = \frac{24 \times 24 \times 24}{6 \times 6 \times 6}$$

$$N = 4 \times 4 \times 4 = 64$$

**Sol.143.(d)** Let the common radius = r cm  
and cylinder height = (35 - r) cm  
A/Q,

$$\Rightarrow \pi r^2 h = \frac{2}{3}\pi r^3$$

$$\Rightarrow h = \frac{2}{3}r \Rightarrow (35 - r) = \frac{2}{3}r$$

$$\Rightarrow 35 = \frac{5}{3}r \Rightarrow r = 21 \text{ cm}$$

**Sol.144.(a)** Volume of pyramid

$$= \frac{1}{3} \times \text{Base Area} \times \text{Height}$$

$$\text{Volume} = \frac{1}{3} \times 120 \times 18 = 720 \text{ cm}^3$$

**Sol.145.(d)**

Perimeter of bigger triangle = 24 cm

$$\text{Area of bigger triangle} = \frac{1}{2} \times 6 \times 8$$

$$= 24 \text{ cm}^2$$

Perimeter of smaller triangle = 12 cm

When the perimeter is half then the side is also halved.

When side becomes half the area becomes  $\frac{1}{4}$

$$\text{Area of new triangle} = \frac{24}{4} = 6 \text{ cm}^2$$

**Sol.146.(b)** Area of segment = Area of sector - Area of triangle

$$\text{Area of sector} = \pi r^2 \times \frac{90^\circ}{360^\circ}$$

$$\text{Area of sector} = \pi \times 196 \times \frac{1}{4} = 49\pi \text{ cm}^2$$

$$\text{Area of triangle} = \frac{1}{2} r^2 \sin 90^\circ$$

$$\text{Area of triangle} = \frac{1}{2} \times 14 \times 14 \times 1$$

$$= 98 \text{ cm}^2$$

$$\text{Area of segment} = 49\pi - 98$$

**Sol.147.(b)** r = 8 cm, h = 15 cm

$$\text{Slant height} = \sqrt{8^2 + 15^2} = 17 \text{ cm}$$

$$\text{Used formula} = \frac{R \times H}{R + L}$$

$$\text{Radius of Sphere} = \frac{8 \times 15}{17 + 8} = \frac{120}{25}$$

$$= 4.8 \text{ cm}$$

**Sol.148.(a)** Perimeter = side × 8

$$\text{Side} \times 8 = 160$$

$$\text{Side} = 20 \text{ cm}$$

$$\text{Sol.149.(b)} \text{ Sector Area} = \pi r^2 \times \frac{\theta}{360}$$

$$90 = 3.14 \times r^2 \times \frac{90^\circ}{360^\circ}$$

$$r = \sqrt{\frac{360}{3.14}} = \sqrt{114.64} \Rightarrow r = 10.7 \text{ cm}$$

**Sol.150.(a)** Lateral surface area of prism

$$= \text{Perimeter of base} \times \text{Height}$$

$$320 = \text{Perimeter of base} \times 16$$

$$\text{Perimeter of base} = 20 \text{ cm}$$

$$\text{Sol.151.(c)} \text{ Sector Area} = \pi r^2 \times \frac{\theta}{360}$$

$$= \pi \times 8^2 \times \frac{45^\circ}{360^\circ} = 8\pi \text{ cm}^2$$

$$\text{Area of triangle} = \frac{1}{2} \times r^2 \times \sin 45^\circ$$

$$= \frac{1}{2} \times 8 \times 8 \times \frac{1}{\sqrt{2}} = 16\sqrt{2} \text{ cm}^2$$

$$\text{Segment area} = (8\pi - 16\sqrt{2}) \text{ cm}^2$$

**Sol.152.(b)**

$$\text{perimeter of circle} (2\pi r) = \frac{3768}{120}$$

$$\frac{2 \times 22 \times r}{7} = 31.4$$

$$\text{Radius} = \frac{31.4}{6.28} = 5$$

$$\text{Diameter} = 2r = 10 \text{ m}$$

**Sol.153.(a)** Length of the square's side = 14

$$\text{Area of square} = 196$$

$$\text{Area of the circle} = \pi r^2 = \frac{22}{7} \times 7 \times 7$$

$$= 154$$

$$\text{Unused square space area}$$

$$= 196 - 154 = 42 \text{ cm}^2$$

**Sol.154.(a)** The minute hand completes one full revolution ( $2\pi r$  distance) in one hour.

Therefore, in half an hour, it will cover ( $\pi r$ )

$$\text{distance} = \frac{22}{7} \times 28 = 88 \text{ cm}$$

**Sol.155.(a)**

$$\text{Surface area of hemisphere} = 3\pi r^2$$

Here  $3\pi$  is constant

Let Initial radius = 1 or new radius = 2

Required ratio = 4 : 1

Volume of hemisphere

$$= \frac{2}{3} \pi r^3, \text{ here } \frac{2}{3} \pi \text{ is constant}$$

$$\text{Required ratio} = 8 : 1$$

**Sol.156.(b)** A/Q,

volume of hemisphere = volume of cone

$$\Rightarrow \frac{2}{3} \pi R^3 = \frac{1}{3} \pi R^2 H \Rightarrow R = \frac{H}{2} \Rightarrow H = 2R$$

**Sol.157.(b)** A/Q, volume of sphere

$$= 4 \times \text{volume of cone}$$

$$\Rightarrow \frac{4}{3} \pi r^3 = 4 \times \frac{1}{3} \pi r^2 h$$

$$\Rightarrow \frac{4}{3} \pi \times 8 \times 8 \times 8 = \frac{4}{3} \pi \times 4 \times 4 \times h$$

$$\Rightarrow h = 32 \text{ cm}$$

**Sol.158.(c)**

Lateral surface area of square pyramid

$$= \frac{1}{2} \times \text{base perimeter} \times \text{Slant height}$$

$$= \frac{1}{2} \times 30 \times 10 = 150 \text{ cm}^2$$

**Sol.159.(c)** Area  $\propto (\text{radius})^2$

$$\text{Ratio of areas} = 2^2 : 5^2 = 4 : 25$$

$$\text{A/Q}, \frac{4}{25} = \frac{100\pi}{x} \Rightarrow x = 625\pi$$

$$\text{Hence area of the larger track} = 625\pi \text{ cm}^2$$

**Sol.160.(a)** According to question,

Volume of sphere = 8 × Volume of cone

$$\Rightarrow \frac{4}{3} \pi r^3 = 8 \times \frac{1}{3} \pi \times \frac{6}{2} \times \frac{6}{2} \times 4$$

$$\Rightarrow r^3 = 72 \Rightarrow r = 4.16 \text{ cm}$$

**Sol.161.(c)**

A regular octagon has 8 equal sides and 8 equal angles. If we draw straight lines from the center of the octagon to all its 8 vertices, the figure divides into 8 identical isosceles triangles.

$$\text{Sol.162.(c)} \text{ Fountain radius} = \frac{14}{2}$$

$$= 7 \text{ m, Outer radius} = 10.5 \text{ m}$$

Ratio = Area of grass bed : Area of the fountain

$$\pi(10.5^2 - 7^2) : \pi(7)^2$$

$$= 61.25\pi : \pi(7)^2$$

$$= 61.25 : 49 = 5 : 4$$

**Sol.163.(a)**

Volume = Area of base × Height

$$= 30 \times 7 = 210 \text{ m}^3$$

$$\therefore \text{Volume of tent} = 210 \text{ m}^3$$

**Sol.164.(d)**

$$\text{Area of sector} = \frac{90^\circ}{360^\circ} \times \pi \times (20)^2 = 100\pi$$

$$\text{Area of triangle} = \frac{1}{2} \times (20)^2 \times \sin 90^\circ$$

$$= 200$$

$$\text{Area of minor segment} = 100\pi - 200 \text{ cm}^2$$

**Sol.165.(c)** According to the question,



Volume of Sphere =  $288\pi$

$$\Rightarrow \frac{4}{3}\pi r^3 = 288\pi \Rightarrow r = 6$$

Volume of Cylinder =  $\pi r^2 h = \pi(6)^2 \times (2 \times 6)$   
 $= 432\pi \text{ cm}^3$

Therefore, volume of the vacant space inside the cylinder = Volume of cylinder - Volume of sphere  
 $= 432\pi - 288\pi = 144\pi \text{ cm}^3$

**Sol.166.(c)** According to the question ,

$$\Rightarrow \frac{2}{3}\pi R^3 = \frac{1}{3}\pi\left(\frac{R}{2}\right)^2 h$$

$$\Rightarrow 8R = h$$

Hence , the height of the resulting cone =  $8R$

**Sol.167.(a)** According to the question,

$$\Rightarrow R^3 = 5^3 + 9^3 = 125 + 729 = 854$$

$$\Rightarrow R = \sqrt[3]{854} \approx 9.5 \text{ cm (approx.)}$$

**Sol.168.(a) Given**

Base side length ( b ) = 16 cm

Slant height ( s ) = 10 cm

$$\text{Height (h)} = \sqrt{s^2 - \left(\frac{b}{2}\right)^2} = \sqrt{10^2 - 8^2} = 6 \text{ cm}$$

$$\text{Volume (V)} = \frac{1}{3} \times b^2 \times h = \frac{1}{3} \times 16^2 \times 6 = \frac{1}{3} \times 256 \times 6 = 512 \text{ cm}^3$$

**Sol.169.(b)** The area of a circle is proportional to the square of its radius.

According to the question,

$$\left(\frac{r_1}{r_2}\right)^2 = \left(\frac{1}{\sqrt{3}}\right)^2 = \frac{1}{3}$$

So, area of larger tyre

$$= 120 \times 3 = 360 \text{ cm}^2$$

**Sol.170.(a)** Let Internal radius =  $x$  cm

According to question,

$$\Rightarrow \frac{4}{3}\pi(R^3 - x^3) = 25 \times \frac{4}{3}\pi r^3$$

$$\Rightarrow (14^3 - x^3) = 25 \times 4^3$$

$$\Rightarrow 2744 - x^3 = 25 \times 64$$

$$\Rightarrow x = \sqrt[3]{2744 - 1600} = 10.45 \text{ cm}$$

**Sol.171.(b)** According to question

$$\Rightarrow \frac{2}{3}\pi r^3 = 5 \times 2\pi r^2 \Rightarrow r = 15 \text{ cm}$$

**Sol.172.(c)** Formula used:

Minimum number of cuts

$$= (n_x - 1) + (n_y - 1) + (n_z - 1)$$

Number of identical pieces  $\rightarrow 16$

$$= 2 \times 2 \times 4$$

Here,  $n_x = 2$ ,  $n_y = 2$  and  $n_z = 4$

Put all values in the given formula,

Minimum number of cuts

$$= (2 - 1) + (2 - 1) + (4 - 1) = 1 + 1 + 3 = 5$$

Hence, the minimum no. of cuts = 5

**Sol.173.(b)** Volume of cone(V) =  $\frac{1}{3}\pi r^2 h$

$$H' = 4h, r' = 2r$$

$$\text{Volume of cone} = \frac{1}{3}\pi(2r)^2 \times 4h$$

$$= 16 \times \left(\frac{1}{3}\pi r^2 h\right) = 16 \times (V)$$

Therefore, the new volume will be sixteen times the original volume.

**Sol.174.(c)**  $\frac{\text{area of sector}}{\text{area of sector in radian}}$

$$= \frac{\frac{1}{4}\pi r^2}{\frac{1}{2}r^2 \times \frac{\pi}{2}} = \frac{1}{1}$$

$$\text{Ratio} \rightarrow 1 : 1$$

**Sol.175.(d)** Volume of water displaced

= Volume of sphere

$$\text{Volume of sphere} = \frac{4}{3}\pi r^3$$

$$\text{Volume of sphere} = \frac{4}{3}\pi(4)^3$$

$$= \frac{256\pi}{3} \text{ cm}^3$$

**Sol.176.(a)** Diameter = 12cm

Radius = 6cm

CSA of hemisphere =  $2\pi r^2$

$$= 2 \times \frac{22}{7} \times 36 \approx 226 \text{ cm}^2$$

**Sol.177.(a)** CSA of hemisphere =  $2\pi r^2$

Outer curved surface area of hemisphere

$$= 2\pi R^2 = 2\pi(4r)^2$$

Outer curved surface area of hemisphere

$$= 32\pi r^2$$

Inner curved surface area =  $2\pi r^2$

Ratio of outer to inner surface area =

$$32\pi r^2 : 2\pi r^2 = 16 : 1$$

**Sol.178.(a)** LSA =  $2 \times$  Base area

$$\text{Base area} = (8)^2 = 64 \text{ cm}^2$$

$$\text{LSA} = 2 \times 64 = 128 \text{ cm}^2$$

LSA of square pyramid =  $2al$

$$2al = 128 \Rightarrow l = \frac{128}{2 \times 8} = 8 \text{ cm}$$

**Sol.179.(b)**

Diameter = 42 m. Radius = 21 m

$$\text{Circumference} = 2\pi r = 2 \times \frac{22}{7} \times 21$$

$$= 132 \text{ m}$$

Gardener wants of fence 25% of

$$\text{circumference} = 132 \times \frac{1}{4} = 33 \text{ m}$$

$$\text{Reqd. length} = 33 \text{ m}$$

**Sol.180.(d)** Inradius(r) =  $\frac{b}{2\sqrt{3}} \text{ cm}$

$$\text{Circumradius(R)} = \frac{b}{\sqrt{3}} \text{ cm}$$

$$\frac{\text{Inradius}}{\text{Circumradius}} = \frac{\frac{b}{2\sqrt{3}}}{\frac{b}{\sqrt{3}}} = \frac{1}{2}$$

$$\frac{\text{Inradius}}{\text{Circumradius}} = \frac{b}{2\sqrt{3}} \times \frac{\sqrt{3}}{b} = 1 : 2$$

**Sol.181.(b)**

Let the cone's radius =  $R$ , height =  $h = 3R$ , and the sphere's radius =  $r$

Volume of cone = volume of sphere

$$\frac{1}{3}\pi R^2 \times 3R = \frac{4}{3}\pi r^3$$

$$\frac{R}{r} = \frac{\sqrt[3]{4}}{\sqrt[3]{3}}$$

**Sol.182.(a)** According to the question,

Total area of sphere = Total area of Cylinder

$$\Rightarrow 4\pi r^2 = 2\pi r^2 + 2\pi r h$$

$$\Rightarrow 2\pi r^2 = 2\pi r h \Rightarrow h : r = 1 : 1.$$

Therefore, the ratio of the height to the radius of the cylinder is 1 : 1.

**Sol.183.(b) Formula used :**

Minimum number of cuts

$$= (n_x - 1) + (n_y - 1) + (n_z - 1)$$

Number of identical pieces

$$= 27 \rightarrow 3 \times 3 \times 3$$

$$\text{Here, } n_x = 3, n_y = 3, n_z = 3$$

Put all values in the given formula ,

Minimum number of cuts

$$= (3 - 1) + (3 - 1) + (3 - 1) = 6$$

**Sol.184.(b)**

Old : New

$$\text{Radius} \rightarrow 4^2 : 5^2$$

$$\text{Height} \rightarrow 10 : 9$$

$$\text{Volume} \rightarrow 160 : 225$$

$$\text{Required \%} = \frac{225 - 160}{160} \times 100$$

$$= 40.6\% \text{ increase}$$

**Sol.185.(b)** According to the question ,

$$\frac{\text{Area of the first sector}}{\text{Area of the second sector}}$$

$$= \frac{\frac{1}{2} \times 24^2 \times \pi}{\frac{1}{2} \times 24^2 \times \frac{\pi}{2}} = \frac{2}{1}$$

**Sol.186.(c)** According to the question,

$$\frac{5}{7} \Rightarrow \frac{5^2}{7^2} = \frac{25}{49}$$

$$74 \text{ unit} = 74 \text{ cm}^2$$

$$49 \text{ unit} = 49 \text{ cm}^2$$

**Sol.187.(b)** Volume of sphere =  $\frac{4}{3}\pi r^3$

According to question

$$2 \times \frac{2}{3}\pi r^3 = \frac{4}{3}\pi r^3$$

The volume of both is equal.

$$\therefore r = 27 \text{ cm}$$

**Sol.188.(b)**

$$\text{Total height} = 8 + 11 + 14 + 17 = 50 \text{ cm}$$

$$\text{Area of hexagon} = \frac{3\sqrt{3}}{2} \times a^2$$



$$= \frac{3\sqrt{3}}{2} \times 64 = 96\sqrt{3} \text{ cm}^2$$

$$\text{Volume of prism} = 96\sqrt{3} \times 50$$

$$= 4800\sqrt{3} \text{ cm}^3$$

**Sol.189.(d) Given:** radius = 6 cm

$$\text{Volume of spherical tile} = \frac{4}{3} \pi r^3$$

$$= \frac{4}{3} \times \frac{22}{7} \times 216 = \frac{4 \times 22 \times 72}{7}$$

$$= 905.14 \text{ cm}^3$$

$$\text{Volume of cube} = a^3$$

$$= 12^3 = 1728 \text{ cm}^3$$

$$\text{volume of the cube that is not occupied}$$

$$= 1728 - 905.14 = 822.86 \text{ cm}^3$$

**Sol.190.(c)**

$$\text{Volume of hemisphere} = \frac{2}{3} \pi r^3$$

$$\text{Volume of new hemisphere}$$

$$= \frac{2}{3} \pi (9^3 + 10^3)$$

$$\frac{2}{3} \pi R^3 = \frac{2}{3} \pi (729 + 1000)$$

$$R^3 = 1729$$

$$R = 12 (\text{whole number})$$

$$\text{Total surface area of the newly formed}$$

$$\text{hemisphere} = 3\pi r^2$$

$$= 3 \times \frac{22}{7} \times 12^2 = 1357.71 \approx 1358 \text{ cm}^2$$

$$\text{Sol.191.(d)} \quad \frac{V_{\text{small}}}{V_{\text{original}}} = \frac{1}{27}$$

For similar solids, the ratio of their volumes is equal to the cube of the ratio of their corresponding heights, radii, or slant heights.

$$\frac{V_{\text{small}}}{V_{\text{original}}} = \left(\frac{h}{H}\right)^3 \Rightarrow \frac{1}{27} = \left(\frac{h}{H}\right)^3$$

$$\frac{h}{H} = \frac{1}{3} \Rightarrow h : H = 1 : 3$$

**Sol.192.(d)**

$$\text{Area of the first sector} = \frac{\pi r^2 \theta}{360^\circ}$$

$$= \frac{\pi \times 36 \times 150^\circ}{360} = 15\pi$$

$$\text{Area of the second sector}$$

$$= \frac{\pi \times 36 \times 180^\circ}{360} = 18\pi$$

$$\text{Ratio} \rightarrow 15\pi : 18\pi = 5 : 6$$

**Sol.193.(b)**

$$\text{Volume of a hemisphere} = \frac{2}{3} \pi r^3$$

$$\text{Outer radius (R)} = \frac{28}{2} = 14 \text{ cm}$$

$$\text{and inner radius (r)} = \frac{14}{2} = 7 \text{ cm}$$

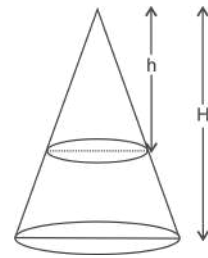
So metal volume:-

$$V = \frac{2}{3} \pi (R^3 - r^3) = \frac{2}{3} \pi (14^3 - 7^3)$$

$$= \frac{2}{3} \times \frac{22}{7} \times (2744 - 343)$$

$$= \frac{2}{3} \times \frac{22}{7} \times 2401 = 5030.66 \text{ cm}^3$$

**Sol.194.(b)**



Let the height of the original cone =  $H$   
and smaller (top) cone =  $h$

$$\text{Volume of a cone} \propto h^3$$

Then,

$$\text{Volume of frustum} = H^3 - h^3$$

According to question,

$$\frac{H^3 - h^3}{h^3} = \frac{19}{8} \Rightarrow 8H^3 = 27h^3 \Rightarrow \frac{H}{h} = \frac{3}{2}$$

Now,

$$\text{Height of frustum} = 3 - 2 = 1$$

So,

$$\text{Height ratio (frustum : small cone)} = 1 : 2$$

**Sol.195.(d)**

$$\text{Area of a regular hexagon} \propto (\text{side})^2$$

$$\text{New area} = (3)^2 = 9 \text{ times the original area.}$$

**Sol.196.(b)**

$$\text{Diameter} = 6 \text{ cm} \Rightarrow \text{Radius (r)} = 3 \text{ cm}$$

$$\text{Heights of sections are in A.P.:}$$

$$2, 4, 6, 8, 10$$

$$\text{Total height (h)} = 2 + 4 + 6 + 8 + 10 = 30 \text{ cm}$$

$$\text{Volume of cylinder (V)} = \pi r^2 h$$

$$= 3.14 \times (3)^2 \times 30 = 847.8 \text{ cm}^3$$

**Sol.197.(d) Area ( $\Delta GBC$ )**

$$= \frac{1}{3} \times \text{Area} (\Delta ABC) = \frac{1}{3} \times 96 = 32 \text{ cm}^2$$

**Sol.198.(a) Volume of the hemisphere**

$$= \frac{2}{3} \pi r^3 = \frac{2}{3} \times \pi \times 8^3 = \frac{1024\pi}{3} \text{ cm}^3$$

**Sol.199.(d) Side of cube = s**

$$\text{radius of ball} = \frac{s}{2}$$

$$\text{Volume of ball} = \frac{4}{3} \pi \left(\frac{s}{2}\right)^3$$

$$= \frac{4}{3} \times \pi \times \frac{s^3}{8} = \frac{\pi s^3}{6}$$

$$\text{Volume of cube} = s^3$$

$$\text{Percentage} = \frac{\frac{\pi s^3}{6}}{s^3} \times 100$$

$$= \frac{3.14 \times 100}{6} = 52.33\%$$

**Sol.200.(b) Let the sides be 2x, 5x, 3x**

$$\text{Volume of the prism} = (l \times b) \times h$$

$$(2x \times 5x) \times 3x = 240$$

$$x^3 = \frac{240}{30} = 8 \Rightarrow x = 2$$

$$\text{Longest side} = 5 \times 2 = 10 \text{ cm}$$

**Sol.201.(b)** The small cone and the large cone are similar.

Ratio of height  $\rightarrow$

$$\text{small cone} : \text{large cone} = 1 : 4$$

$$\text{volume} \propto (\text{height})^3$$

Ratio of volume  $\rightarrow$

$$\text{small cone} : \text{large cone} = 1 : 64$$

$$\text{Bottom volume} = 64 - 1 = 63$$

$$\frac{\text{upper part}}{\text{lower part}} = \frac{1}{63} = 1 : 63$$

**Sol.202.(a)** First Sector :-  $r = 9 \text{ cm}$

Central Angle

$$45^\circ = \frac{\pi}{4}$$

Second Sector :-  $r = 9 \text{ cm}$

$$\text{Central Angle} = \frac{\pi}{4}$$

Since both sectors have the same radius and same angle

Therefore the area will also be equal.

$$\text{Required ratio} = 1 : 1$$

**Sol.203.(c)**

$$4\pi r^2 = 154 \text{ cm}^2 \Rightarrow r = \frac{7}{2} \Rightarrow d(2r) = 7$$

$$\text{Sol.204.(b) volume of cone (V)} = \frac{1}{3} \pi r^2 h$$

According to question,

Volume of smaller cone

$$= \frac{1}{3} \pi \times \frac{r^2}{4} \times \frac{h}{4} = \frac{1}{48} \pi r^2 h \Rightarrow \frac{V}{16}$$

fraction of the original volume is unused

$$= V - \frac{V}{16} = \frac{15}{16} V$$

**Sol.205.(a)**

$$\text{Area of a regular hexagon} = \frac{3\sqrt{3}}{2} a^2$$

$$\text{Square of area} = a^2$$

$$\frac{\text{Area of hexagon}}{\text{Area of square}} = \frac{\frac{3\sqrt{3}}{2} a^2}{a^2} \Rightarrow 3\sqrt{3} : 2$$

**Sol.206.(b) Area of equilateral triangle**

$$= \frac{\sqrt{3}}{4} a^2 \Rightarrow \frac{\sqrt{3}}{4} \times 6^2 = 9\sqrt{3}$$

Its height starts from 1 cm and increases by 1 cm with each layer, forming a total of 5 layers.

Therefore, the total height (h) is

$$= 1 + 2 + 3 + 4 + 5 = 15 \text{ cm}$$

Now volume of prism =

$$\text{area of base} \times \text{height}$$

$$\Rightarrow 9\sqrt{3} \times 15 = 135 \times \sqrt{3} = 233.82$$

**Sol.207.(d)**

$$\text{Volume of big cylinder} = \pi r^2 h$$

$$= \pi \times 8^2 \times 15$$

$$= 960 \pi$$

$$\text{Volume of small cylinder} = \pi r^2 h$$



$$= \pi \times 4^2 \times 15$$

$$= 240\pi$$

volume of the remaining solid

$$= 960\pi - 240\pi \Rightarrow 720\pi$$

$$= 720 \times \frac{22}{7} \Rightarrow = 2262.85 \approx 2263 \text{ cm}^3$$

**Sol.208.(b)**

Volume of tank =  $l \times b \times h$

$$= 30 \times 20 \times 5$$

$$= 3000 \text{ m}^3$$

$$1 \text{ m}^3 = 1 \text{ kilolitres}$$

Capacity of tank = 3000 kilolitres

**Sol.209.(a)** Volume of big sphere

$$= \frac{4}{3}\pi P^3$$

$$\text{Volume of small sphere} = \frac{4}{3}\pi q^3$$

$$\frac{4}{3}\pi P^3 = n \times \frac{4}{3}\pi q^3 \Rightarrow n = \left(\frac{P}{q}\right)^3$$

$$\text{Sol.210.(b)} \quad \frac{2\pi}{3} = 120^\circ$$

$$\text{Area of sector} = \frac{\pi r^2 \theta}{360^\circ}$$

$$\text{Area of first sector} = \frac{\pi 18^2 60^\circ}{360^\circ} = 54\pi$$

$$\text{Area of second sector} = \frac{\pi 18^2 120^\circ}{360^\circ}$$

$$= 108\pi$$

$$\text{Ratio} \rightarrow 54\pi : 108\pi = 1 : 2$$

**Sol.211.(d)** Volume of the new sphere = sum of the volumes of the three smaller spheres

$$\frac{4}{3}\pi R^3 = \frac{4}{3}\pi r_1^3 + \frac{4}{3}\pi r_2^3 + \frac{4}{3}\pi r_3^3$$

$$R^3 = 4^3 + 2^3 + 7^3$$

$$R = \sqrt[3]{64 + 8 + 343} = \sqrt[3]{415} = 7.5 \text{ cm}$$

**Sol.212.(d)** Let the side of the triangle be =  $8x, 15x, 17x$

Parameter = 120 units

$$8x + 15x + 17x = 120 \text{ units}$$

$$x \text{ units} = \frac{120}{40} = 3 \text{ units}$$

24, 45, 51 is a Pythagorean triplet.

$$\text{Area of triangle} = \frac{1}{2} \times \text{base} \times \text{height}$$

$$= \frac{1}{2} \times 24 \times 45 = 12 \times 45 = 540 \text{ m}^2$$

**Sol.213.(d)**

Prism volume = base area  $\times$  total height

$$\text{Base area} = 6^2 = 36 \text{ cm}$$

First layer = 6 cm, fourth layer = 24 cm

Hence the heights of the layers: 6, 12, 18, 24

$$\text{Total height} = 6 + 12 + 18 + 24 = 60 \text{ cm}$$

$$\text{Volume} = 36 \times 60 = 2160 \text{ cm}^3$$

**Sol.214.(a)**

equilateral triangle then angle is  $60^\circ$

$$\text{Area of sector} = \frac{\theta}{360^\circ} \times \pi \times R^2$$

$$= \frac{1}{6} \times 14 \times 14 \times \pi = \frac{98\pi}{3} \text{ cm}$$

$$\text{Area of triangle} = \frac{1}{2} R^2 \sin \theta$$

$$= \frac{1}{2} \times 14 \times 14 \times \frac{\sqrt{3}}{2} = 49\sqrt{3}$$

Area of smaller segment

$$= \left(\frac{98\pi}{3} - 49\sqrt{3}\right) \text{ cm}^2$$

**Sol.215.(b)** Total surface area =  $72\text{m}^2$ 

Paint cover  $3\text{m}^2/\text{litre}$

$$\text{Paint needed} = \frac{72}{3} = 24 \text{ litre}$$

$$\text{So, Total paint} = 24 \times \frac{11}{10} = 26.4 \text{ litre}$$

**Sol.216.(a)** 1st Area =  $7 \times 4 = 28\text{m}$ 

$$\text{2nd Area} = 7 \times 6 = 42\text{m}$$

$$\text{Total area} = 42\text{m} + 28\text{m} = 70\text{m}^2$$

Area is greater than  $60\text{m}^2$ ,

so, 10% discount is applicable

$$\text{Total cost} = 70 \times 500 = \text{Rs.}35000$$

$$\text{After 10\% discount} = 35000 \times \frac{90}{100}$$

$$= \text{Rs.} 31500$$

$$\text{Final price} = \text{Rs.}31500$$

**Sol.217.(c)**

$$S = \frac{9 + 10 + 17}{2} = \frac{36}{2} = 18\text{m}$$

Area

$$= \sqrt{18 \times (18 - 9)(18 - 10)(18 - 17)}$$

$$\text{Area} = \sqrt{18 \times 9 \times 8} = \sqrt{162 \times 8} = 36\text{m}^2$$

$$\text{Total cost} = 36 \times 7 = \text{Rs.}252$$

**Sol.218.(c)**

Volume of Prism = Base area  $\times$  Height

Base area increases by 35% and height remains the same

so volume will also increase by 35%.

**Sol.219.(a)** For one vertex of  $n$  sided polygon the number of triangles formed =  $n - 2$

In hexagon number of side = 6

$$\text{Number of triangles formed} = 6 - 2 = 4$$

**Sol.220.(d)** Diameter of the hemisphere

= side of the cube = 12 cm

$$\text{Radius} = \frac{12}{2} = 6 \text{ cm}$$

$$\text{Volume of the hemisphere} = \frac{2}{3}\pi r^3$$

$$= \frac{2}{3} \times 3.14 \times 216 = 452.16 \text{ cm}^3$$

**Sol.221.(c)** Area of the triangular window

$$= 300 \times \frac{35}{100} = 105 \text{ m}^2$$

$$\text{Area of the triangle} = \frac{1}{2} \times \text{Base} \times \text{Height}$$

$$105 = \frac{1}{2} \times 20 \times \text{Height}$$

$$\text{Height} = 10.5 \text{ m}$$

**Sol.222.(b)** Sides of a right triangle

= 8, 15, 17 (Pythagorean triplet)

Area of rectangular face

= (side of triangle)  $\times$  (height of prism)

$$\text{Area}_1 = 8 \times 5 = 40$$

$$\text{Area}_2 = 15 \times 5 = 75$$

$$\text{Area}_3 = 17 \times 5 = 85$$

$$\text{Ratio} = 85 : 40 = 17 : 8$$

**Sol.223.(a)** area of the square base

$$= \text{side}^2 = 8^2 = 64 \text{ cm}^2$$

volume of a pyramid =  $\frac{1}{3}$  area of base  $\times$  height

$$= \frac{1}{3} 64 \times 15 = 320 \text{ cm}^3$$

Effective volume =  $80\% \times 320$

$$= 0.8 \times 320 = 256 \text{ cm}^3$$

total cost of material =  $256 \times 0.60$

$$= \text{₹}153.6$$

**Sol.224.(b)** The formula for the area of a

trapezoid is  $(A) = \frac{1}{2} (b_1 + b_2) \times h$

$$= \frac{1}{2} (40 + 60) \times 20 = 50 \times 20 = 1000 \text{ m}^2$$

**Sol.225.(b)**

The side length of the hexagon is equal to the radius of the circle.

$$\text{Area of a regular hexagon} = \frac{3\sqrt{3}}{2} r^2$$

$$= \frac{3\sqrt{3}}{2} \times 14^2 = \frac{3\sqrt{3}}{2} \times 196$$

$$= 294 \times \sqrt{3} = 509.2 \text{ cm}^2 \approx 509 \text{ cm}^2$$

**Sol.226.(b)**

$$\text{The area of a triangle} = \frac{1}{2} AB \sin \theta$$

$$= \frac{1}{2} \times 4 \times 8 \sin (120^\circ)$$

$$= 16 \times \frac{\sqrt{3}}{2} = 8\sqrt{3} \text{ cm}^2$$

**Sol.227.(a)**

$$\text{Area of the trapezium} = \frac{1}{2} (8 + 10) \times 6$$

$$= \frac{1}{2} \times 18 \times 6 = 54 \text{ cm}^2$$

volume of the prism =  $54 \text{ cm}^2 \times 18 \text{ cm}$

$$= 972 \text{ cm}^3$$

**Sol.228.(d)** According to question,

Surface area of sphere

=  $2 \times$  Volume of sphere

$$\Rightarrow 4\pi r^2 = 2 \times \frac{4}{3}\pi r^3 \Rightarrow 4 = \frac{8}{3}r$$

$$\Rightarrow r = 1.5 \text{ units}$$

**Sol.229.(b)**

Let base =  $3x$  and height =  $4x$

According to the formula,



$$\text{Area of a triangle} = \frac{1}{2} b \times h$$

$$216 = \frac{1}{2} \times 3x \times 4x \Rightarrow 216 = 6x^2$$

$$x^2 = \frac{216}{6} = 36 \Rightarrow x = 6$$

$$\text{Base (b)} = 3x = 3 \times 6 = 18 \text{ cm}$$

$$\text{Height (h)} = 4x = 4 \times 6 = 24 \text{ cm}$$

**Sol.230.(c) Given:**

$$\text{Base area} = \frac{80}{10000} \text{ m}^2 = 0.008 \text{ m}^2$$

$$\text{Volume of the triangular prism}$$

$$= 0.008 \text{ m}^2 \times 1.5 \text{ m} = 0.012 \text{ m}^3$$

$$\text{Volume of the rectangular prism}$$

$$= 0.5 \text{ m} \times 0.35 \text{ m} \times 0.9 \text{ m} = 0.1575 \text{ m}^3$$

$$\text{Total volume} = 0.012 \text{ m}^3 + 0.1575 \text{ m}^3$$

$$= 0.1695 \text{ m}^3$$

Convert total volume to liters

$$= 0.1695 \text{ m}^3 \times 1000 \frac{\text{L}}{\text{m}^3} = 169.5 \text{ L}$$

**Sol.231.(c) Volume of composite solid**

= Volume of cube + Volume of pyramid

$$= a^3 + \frac{1}{3} \times \text{base area} \times \text{height}$$

$$= 15 \times 15 \times 15 + \frac{1}{3} \times \frac{\sqrt{3}}{4} \times 15 \times 15 \times 20$$

$$= 3375 + 375 \times 1.732$$

$$= 3375 + 649.5 = 4024.5 \text{ cm}^3$$

**Sol.232.(c) Area of a triangle**

$$= \frac{1}{2} \times \text{Base} \times \text{height}$$

$$= \frac{1}{2} \times 80 \times 60 = 2400 \text{ m}^2$$

$$\text{Area of rectangle} = 80 \text{ m} \times 50 \text{ m} = 4000 \text{ m}^2$$

$$\text{Total area} = 2400 + 4000 = 6400 \text{ m}^2$$

$$\text{So, Required \%} = \frac{2400}{6400} \times 100 = 37.5 \%$$

**Sol.233.(b)**

$$\text{Area of regular hexagon} = 6 \times \frac{\sqrt{3}}{4} \times a^2$$

$$= 6 \times \frac{\sqrt{3}}{4} \times 18 \times 18$$

$$= 486 \times 1.732 \approx 841.8 \text{ cm}^2$$

**Sol.234.(b)**

(8, 15, 17) is the pythagorean triplet,  
It means the given triangle is a right angle triangle,

$$\text{Area of triangle} = \frac{1}{2} \times \text{base} \times \text{height}$$

$$= \frac{1}{2} \times 8 \times 15 = 60 \text{ m}^2$$

**Sol.235.(d)**

Volume of prism = Base area  $\times$  height

$$\Rightarrow 1296 = a^2 \times 9 \Rightarrow 144 = a^2 \Rightarrow a = 12 \text{ cm}$$

**Sol.236.(a)**

The surface area of a sphere is  $= 4\pi r^2$

Where  $4\pi$  is constant

Initial : Final

20 : 21

20 : 21

.....

400 : 441

Diff = 41

$$\frac{41}{400} \times 100 = 10.25 \%$$

**Sol.237.(b)** The area of a rectangle

= Length  $\times$  Breadth

$$= 36 \text{ m} \times 25 \text{ m} = 900 \text{ m}^2$$

Area of square plot =  $900 \text{ m}^2$

The area of a square is the square of its side

$$\text{side}^2 = 900 \Rightarrow \text{Side} = 30$$

The perimeter of a square is four times the length of its side

$$4 \times 30 \text{ m} = 120 \text{ m}$$

**Sol.238.(b)**

Volume of prisms = Base area  $\times$  height

According to the question ,

$$\Rightarrow \frac{3\sqrt{3}}{2} 5^2 \times 12 : 5^2 \times 12$$

$$\Rightarrow \frac{3\sqrt{3}}{2} : 1 \Rightarrow 3\sqrt{3} : 2$$

**Sol.239.(c)**

$$\text{volume of a pyramid} = \frac{1}{3} \times \text{base area} \times h$$

$$= \frac{1}{3} \times 9^2 \times 9 \Rightarrow 81 \times 3 = 243 \text{ cm}^3$$

**Sol.240.(b)**

Let , New length =  $l + 8$  , New width =  $b - 3$

By the option (b) :-

$$(l + 8) \times (b - 3) = 720$$

$$(40 + 8) \times (18 - 3) = 720$$

$$(48) \times (15) = 720$$

$$720 = 720$$

LHS = RHS (Proved)

**Sol.241.(c)** Area of garden (hexagon)

$$= \frac{3\sqrt{3}}{2} (l)^2 = \frac{3\sqrt{3}}{2} (30)^2 = 1350\sqrt{3} \text{ m}^2$$

Remaining area of garden

$$= 1350\sqrt{3} \times \frac{95}{100} = 1282.5\sqrt{3} \approx 2,221.3 \text{ m}^2$$

**Sol.242.(b)** According to question

Perimeter = 90 m

$$\Rightarrow 3x + 5x + 7x = 90$$

$$\Rightarrow 15x = 90 \Rightarrow x = 6$$

$$\text{then, } a = 3x = 18, b = 5x = 30, c = 7x = 42$$

$$\text{The semi-perimeter (s)} = \frac{18 + 30 + 42}{2}$$

$$= \frac{90}{2} = 45 \text{ m}$$

using Heron's formula

$$A = \sqrt{s(s-a)(s-b)(s-c)}$$

$$A = \sqrt{45(45-18)(45-30)(45-42)}$$

$$A = \sqrt{45 \times 27 \times 15 \times 3}$$

$$= \sqrt{5 \times 9 \times 9 \times 3 \times 5 \times 3 \times 3}$$

$$= 9 \times 5 \times 3 \sqrt{3} = 135 \times 1.732 \approx 234 \text{ m}^2$$

**Sol.243.(b)** volume of a rectangular prism = length  $\times$  width  $\times$  height.

$$= 60 \text{ cm} \times 50 \text{ cm} \times 40 \text{ cm}$$

$$= 120000 \text{ cm}^3$$

Convert the volume to liters

$$= \frac{120000 \text{ cm}^3}{1000 \text{ cm}^3} = 120 \text{ liter}$$

**Sol.244.(b)** Let the radius of sphere = 1

$$\text{Volume of Sphere (V)} = \frac{4}{3} \pi r^3$$

According to question,

$$\Rightarrow V_1 : V_2 = \frac{4}{3} \pi r_1^3 : \frac{4}{3} \pi r_2^3 = 1 : 27$$

$$\text{So, Required \%} = \frac{26}{1} \times 100 = 2600 \%$$

**Sol.245.(a)**

Area of rectangle = length  $\times$  breadth

$$= 15.75 \times 8.2 = 129.15 \text{ m}^2$$

**Sol.246.(d)** Given :  $l = 2b$

According to question,

$$\Rightarrow 2(l + b) = 54 \Rightarrow 2b + b = 27$$

$$\Rightarrow b = 9 \text{ cm and } l = 18 \text{ cm}$$

$$\text{Area of rectangle (base area)} = 9 \times 18 = 162 \text{ cm}^2$$

Height are in AP

$$a_1 = 17, a_4 = 11, n = 4$$

$$\text{Sum of height (s}_4\text{)} = s_n = \frac{n}{2} (a_1 + a_4)$$

$$s_4 = \frac{4}{2} (17 + 11) = 2 \times 28 = 56 \text{ cm}$$

So,

Volume of prism = base area  $\times$  Height

$$= 162 \times 56 = 9072 \text{ cm}^3$$

**Sol.247.(a)** we know that

Volume of cube =  $a^3$

$$3375 = a^3$$

$$a = \sqrt[3]{3375} \Rightarrow a = 15 \text{ cm}$$

For sphere diameter = side of cube

$$r = \frac{a}{2} = \frac{15}{2} = 7.5 \text{ cm}$$

Surface area of sphere =  $4\pi r^2$

$$= 4\pi(7.5)^2 = 225\pi \text{ cm}^2$$

**Sol.248.(a)** Total surface area of wooden

box =  $2h(l + b) + l \times b$

$$= 2 \times 2.4(5 + 3.6) + 5 \times 3.6 = 59.28 \text{ m}^2$$

30% extra for fabric,

$$= 59.28 \times \frac{130}{100} = 77.064 \text{ m}^2$$

Area of Square =  $100 \times 100$

$$= 10000 \text{ cm}^2 = 1 \text{ m}^2$$

So, the required fabric pieces

$$= \frac{77.064}{1} \approx 77$$

**Sol.249.(b)**



Area of large square =  $144 \text{ cm}^2$

Area of small square =  $25 \text{ cm}^2$

Area of remaining part of square

$$= 144 - 25 = 119 \text{ cm}^2$$

Ratio of area of remaining square and original square =  $119 : 144$

**Sol.250.(b)**

Area of park =  $80 \times 40 = 3200 \text{ m}^2$

diameter = 20  $\rightarrow$  radius = 10

Area of semicircle =  $\frac{1}{2}\pi r^2$

$$= \frac{1}{2} \times \frac{22}{7} \times 10 \times 10$$

$$= \frac{1}{2} \times \frac{22}{7} \times 100 = \frac{1100}{7} = 157.14 \text{ m}^2$$

$$\text{Reqd. \%} = \frac{157.14}{3200} \times 100 = 4.91\%$$

**Sol.251.(a) Given**

Perimeter of regular hexagon = 84 cm

Side of a hexagon =  $\frac{\text{Perimeter}}{6}$

$$= \frac{84}{6} = 14 \text{ cm}$$

Area of a regular hexagon =  $\frac{3\sqrt{3}}{2} a^2$

$$= \frac{3\sqrt{3}}{2} \times 196 = 3\sqrt{3} \times 98 = 294\sqrt{3}.$$

Using  $\sqrt{3} = 1.73$

$$= 294 \times 1.73 = 508.62 \text{ cm}^2$$

**Sol.252.(a)** Area of base =  $56 \text{ cm}^2$ 

height (h) = 15 cm

Volume of the full prism (V)

= base area  $\times$  height

$$V = 56 \times 15 = 840 \text{ cm}^3$$

Solid part = 80% of total volume

$$= 0.8 \times 840 = 672 \text{ cm}^3$$

**Sol.253.(d) Given**

Radii  $r = 60 \text{ cm}$

Angle subtended at center  $\theta = 150^\circ$

for arc length

$$L = \frac{\theta}{360^\circ} \times 2\pi r \Rightarrow L = \frac{150^\circ}{360^\circ} \times 2\pi \times 60$$

$$L = 150^\circ \times \frac{120\pi}{360^\circ} \Rightarrow L = 50\pi \text{ m}$$

**Sol.254.(b)**

Base area =  $80 \times 60 = 4800 \text{ cm}^2$

radius = 5 cm

No. of sphere = 4

Rise in water level

$$= \frac{\text{Total volume of spheres}}{\text{Base area of tank}}$$

$$= \frac{4 \times \frac{4}{3}\pi r^3}{4800} = \frac{4 \times \frac{4}{3}\pi \times 125}{4800}$$

$$= \frac{\frac{2000}{3}\pi}{4800} = \frac{5\pi}{36} \text{ cm}$$

**Sol.255.(b)**

The area of square using diagonal

$$A = \frac{d^2}{2}$$

Where  $d = \sqrt{48} \text{ m}$

$$\text{area} = \frac{(\sqrt{48})^2}{2} = \frac{48}{2} = 24 \text{ m}^2$$

**Sol.256.(c) Given,**

Perimeter of square = 80 cm

$$4a = 80 \Rightarrow a = 20 \text{ cm}$$

Area =  $a^2$

$$\text{Area} = (20)^2 = 400 \text{ cm}^2$$

**Sol.257.(a)** Volume of spherical tank =

$$\frac{4}{3}\pi R^3 = \frac{4}{3}\pi (12)^3 = 2304\pi \text{ m}^3$$

Volume of one cylindrical container

$$= \pi \times 2^2 \times 24 = 96\pi \text{ m}^3$$

Number of containers that can be filled

$$= \frac{2304\pi}{96\pi} = 24$$

**Sol.258.(a) Given:**

Cube side (a) = 4 = Diameter of sphere

$\Rightarrow$  Radius of sphere (r) = 2

Volume of empty space =  $a^3 - \frac{4}{3}\pi(r)^3$

$$= 4^3 - \frac{4}{3}\pi \times 2^3 \Rightarrow 64 - \frac{32\pi}{3} \text{ cm}^3$$

**Sol.259.(a)** Area of trapezium

$$\frac{1}{2} \times (\text{sum of parallel sides}) \times \text{height}$$

$$= \frac{1}{2} \times (10 + 6) \times 8 = 64 \text{ m}^2$$

$$\text{Total cost} = 64 \times 500 = ₹32,000.$$

**Sol.260.(a)**

$$\text{Area of } T_1 = \frac{1}{2} \times (35 + 25) \times 10 = 300 \text{ m}^2$$

$$\text{Area of } T_2 = \frac{1}{2} \times (25 + 15) \times 10 = 200 \text{ m}^2$$

$$\text{Total area} = 300 + 200 = 500 \text{ m}^2$$

$$\text{Total cost} = 500 \times 30 = ₹15,000.$$

**Sol.261.(b)** Area  $\propto$  (side)<sup>2</sup>

$$\frac{s_1^2}{s_2^2} = \frac{16}{25} \Rightarrow \frac{s_1}{s_2} = \frac{4}{5}$$

**Sol.262.(d)** Lateral Surface Area

= Perimeter of base  $\times$  Height of prism

$$\text{Perimeter of base} = \frac{768 \text{ cm}^2}{24} = \mathbf{32 \text{ cm.}}$$

**Sol.263.(c)** Area of sector (in radians)

$$= \frac{1}{2} r^2 \theta \Rightarrow \frac{1}{2} \times 16^2 \times \frac{\pi}{4} = 32\pi \text{ cm}^2.$$

**Sol.264.(d)** Volume of rectangular tank

$$= 15 \times 14 \times 9 = 1890 \text{ cm}^3$$

Volume of maximum shape spherical

$$\text{ball} = \frac{4}{3}\pi \times 4.5^3$$

$$= 4\pi \times 30.375 \Rightarrow 121.5\pi$$

space not occupied by the ball

$$= 1890 - 121.5\pi \text{ cm}^3$$

**Sol.265.(d)** Area of square = (side)<sup>2</sup>

$$196 = (\text{side})^2$$

$$\text{side} = 14 \text{ m}$$

Total walk distance(perimeter) =  $4 \times \text{side}$

$$= 4 \times 14 = 56 \text{ m}$$

**Sol.266.(b)** Given Outer diameter = 24 cm

Outer radius (R) = 12 cm

Thickness = 3 cm

Inner radius  $r = 12 - 3 = 9 \text{ cm}$

volume of the material in the shell  $V = \frac{4}{3}$

$$\pi(R^3 - r^3)$$

$$V = \frac{4}{3} \times 3.14 \times (12^3 - 9^3)$$

$$= \frac{4}{3} \times 3.14 \times (1728 - 729)$$

$$= \frac{4}{3} \times 3.14 \times 999 = 4,182.48 \text{ cm}^3$$

**Sol.267.(c)** Total volume of 5 spheres

$$= 5 \left[ \frac{4}{3}\pi r^3 \right] = \frac{20}{3}\pi \times 5^3 = \frac{2500\pi}{3} \text{ cm}^3$$

Rise in water level

$$= \frac{\text{Total volume of 5 spheres}}{\text{Area of the base of the tank}}$$

$$= \frac{\frac{2500\pi}{3}}{80 \times 40} = \frac{25\pi}{96} \text{ cm}$$

**Sol.268.(a)**

Area of the floor =  $16.5 \times 16 = 264 \text{ m}^2$

area of one tile =  $2.5 \times 2.2 = 5.5 \text{ m}^2$

$$\text{Number of tiles} = \frac{264}{5.5} = 48 \text{ tiles}$$

**Sol.269.(b)** Area of the large signboard

$$= 10 \text{ m} \times 6 \text{ m} = 60 \text{ m}$$

Area of the cutout =  $3 \text{ m} \times 2 \text{ m} = 6 \text{ m}$

Area of the open part =  $60 - 6 = 54 \text{ m}^2$

Total painting cost = Area  $\times$  Rate

$$= 54 \times 250 = ₹13,500$$

**Sol.270.(a)** Area of outer hexagon

$$= \frac{3\sqrt{3}}{2} \times (8)^2 = 96\sqrt{3} \text{ cm}^2$$

$$\text{Area of inner hexagon} = \frac{3\sqrt{3}}{2} \times (4)^2$$

$$= 24\sqrt{3} \text{ cm}^2$$

Therefore, area of the substance

$$= 96\sqrt{3} - 24\sqrt{3} = 72\sqrt{3} = 72 \times 1.732$$

$$= 124.7 \text{ cm}^2$$

**Sol.271.(d)** LSA of prism

= perimeter of base  $\times$  height

$$= 54 \times 8 = 432 \text{ cm}^2$$

**Sol.272.(b)** According to the question ,

$$\text{Area of the sector in radian} = \frac{1}{2} r^2 \theta$$



$$= \frac{1}{2} (20)^2 \times \frac{6\pi}{7} = \frac{200 \times 6\pi}{7}$$

$$= 171.4 \pi \text{ cm}^2$$

**Sol.273.(a)** Area of wall =  $2 \times 2.5 = 5 \text{ m}^2$   
 Side of each tile =  $25 \text{ cm} = 0.25 \text{ m}$   
 Area of one tile =  $0.25 \times 0.25 = 0.0625 \text{ m}^2$   
 Number of tiles =  $\frac{5}{0.0625} = 80$  tiles  
 Extra 12% for wastage =  $80 \times \frac{12}{100} = 9.6$   
 $\approx 10$  tiles  
 Total tiles =  $80 + 10 = 90$  tiles

**Sol.274.(c)** Area of square =  $289 \text{ cm}^2$   
 $\therefore$  Side =  $\sqrt{289} = 17 \text{ cm}$   
 When the square is cut into two equal rectangles,  
 Length = 17 and Breadth =  $\frac{17}{2} = 8.5 \text{ cm}$   
 Perimeter of each rectangle  
 $= 2 \times (17 + 8.5) = 2 \times 25.5 = 51 \text{ cm}$

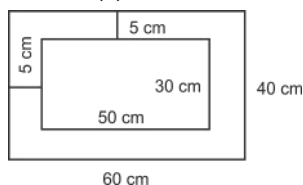
**Sol.275.(c)**  
 Perimeter ratio = linear (scale) ratio =  $3 : 4$   
 Area ratio = square of linear ratio  
 $= \left(\frac{3}{4}\right)^2 = \frac{9}{16}$   
 9 units  $\rightarrow 90$   
 16 units  $\rightarrow \frac{90}{9} \times 16 = 160$   
 Hence, area of the larger hexagon  
 $= 160 \text{ cm}^2$

**Sol.276.(d)**  
 Diagonal =  $\sqrt{(\text{Length})^2 + (\text{width})^2}$   
 $= \sqrt{(30)^2 + (16)^2}$   
 $= \sqrt{900 + 256} = \sqrt{1156} = 34 \text{ cm}$

**Sol.277.(a)**  
 volume of sphere =  $\frac{4}{3} \pi r^3 = 36\pi$   
 Volume of new sphere =  $3^3 \times 36\pi$   
 $= 27 \times 36\pi = 972\pi \text{ cm}^3$

**Sol.278.(d)** Length =  $8 \text{ m} = 800 \text{ cm}$   
 Breadth =  $10 \text{ m} = 1000 \text{ cm}$   
 Area of rectangular room =  $800 \times 1000$   
 $= 800000 \text{ cm}^2$   
 Area of square tile =  $(50)^2 = 2500 \text{ cm}^2$   
 number of tile =  $\frac{800000}{2500} = 320$   
 Extra tile =  $320 \times \frac{15}{100} = 48$   
 Total tile =  $320 + 48 = 368$

**Sol.279.(b)**



Area of large rectangular lawn =  $60 \times 40$   
 $= 2400 \text{ m}^2$

Area of Path =  $2W(l + b - 2W)$   
 $= 2 \times 5(60 + 40 - 10) = 900 \text{ m}^2$   
 percentage of the area of path

$$= \frac{900}{2400} \times 100 = 37.5\%$$

**Sol.280.(d)**  
 Area of first rectangle =  $8 \times 5 = 40 \text{ m}^2$   
 Area of second rectangle =  $5 \times 3 = 15 \text{ m}^2$   
 Total area of both rectangle =  $40 + 15 = 55 \text{ m}^2$   
 Total cost to tile the floor =  $55 \times 200$   
 $= ₹11000$

**Sol.281.(c)**  
 Let side of equilateral triangle =  $a$   
 perimeters of equilateral triangle =  $3a$   
 Let side of square =  $b$   
 Perimeters of square =  $4b$   
 ATQ,  
 perimeters of equilateral triangle  
 $=$  Perimeters of square  
 $3a = 4b$   
 $a : b = 4 : 3$

**Sol.282.(c)** Lateral Surface Area  
 $=$  Perimeter of base  $\times$  Height  
 $= 20 \times 8 = 160 \text{ cm}^2$   
 Total Surface Area of the Prism = Lateral  
 Surface Area +  $2 \times$  Base Area  
 $200 = 160 + 2 \times$  Base Area  
 $2 \times$  Base Area =  $40 \text{ cm}^2$

**Sol.283.(a)** Let sides be  $x$ ,  $3x$ , and  $5x$   
 Surface area =  $2(lb + bh + hl)$   
 $= 2(x \times 3x + 3x \times 5x + 5x \times x)$   
 $= 2(3x^2 + 15x^2 + 5x^2) = 2(23x^2) = 736$   
 $\Rightarrow 46x^2 = 736 \Rightarrow x^2 = 16 \therefore x = 4$   
 Volume =  $l \times b \times h = x \times 3x \times 5x$   
 $= 15x^3 = 15 \times 4^3 = 15 \times 64 = 960 \text{ cm}^3$

**Sol.284.(a)** A/Q, Volume of four pyramid  
 $= 4 \times \left(\frac{1}{3} \times \text{base area} \times \text{height}\right) \times 1.5$   
 $= 4 \times \frac{1}{3} \times 5 \times 5 \times 7 \times 1.5 = 350 \text{ m}^3$

**Sol.285.(a)** Radius ( $r$ ) =  $6 \text{ cm}$   
 Height ( $h$ ) = diameter =  $2r$   
 $= 2 \times 6 = 12 \text{ cm}$   
 Volume of cylinder =  $\pi r^2 h$   
 $= \pi \times 6^2 \times 12 = 432\pi \text{ cm}^3$

**Sol.286.(c)** Volume = Base Area  $\times$  Height  
 $= 35 \times 8 = 280 \text{ cm}^3$

**Sol.287.(a)** Given :- Radius ( $r$ ) =  $4 \text{ cm}$  and  
 central angle ( $\theta$ ) =  $90^\circ$   
 Area of Segment = Area of Sector - Area  
 of Triangle

$$\text{Area of Sector} = \frac{90^\circ}{360^\circ} \times \pi \times (4)^2$$

$$= \frac{1}{4} \times 16\pi = 4\pi \text{ cm}^2$$

$$\text{Area of Triangle} = \frac{1}{2} \times r^2 \sin 90^\circ$$

$$= \frac{1}{2} \times (4)^2 \times 1 = \frac{1}{2} \times 16 = 8 \text{ cm}^2$$

Now, Area of Segment =  $(4\pi - 8) \text{ cm}^2$

**Sol.288.(d)** The surface area of a sphere  
 $(4\pi r^2)$  increases by 26.56%  
 This is equivalent to two successive  
 increases of 12.5%, because  
 $\Rightarrow 12.5 + 12.5 + \frac{12.5 \times 12.5}{100} = 26.56\%$   
 That means the radius ( $r$ ) of the sphere  
 has increased by 12.5%.  
 Since the volume ( $V$ ) of a sphere  $\propto r^3$ , we  
 compare -

$$12.5\% = \frac{1}{8}$$

Old radius : new radius

$$\begin{array}{ccc} 8 & : & 9 \\ 8 & : & 9 \\ 8 & : & 9 \end{array}$$

$$\text{Volume} \Rightarrow 512 : 729$$

Percentage increases of volume  
 $= \frac{729 - 512}{512} = \frac{217}{512} \times 100 = 42.4\%$

**Sol.289.(a)** Area of wall =  $5.6 \times 2.4$   
 Area of 1 tile =  $0.7 \times 0.6 = 0.42$   
 No. of tiles =  $\frac{5.6 \times 2.4}{0.42} = 8 \times 4 = 32$

Required tiles =  $32 + 32 \times \frac{12}{100}$   
 $\Rightarrow 32 + 3.84 = 35.84 \approx 36$

**Sol.290.(d)**  
 Area of square = area of rectangle  
 $484 = 44 \times b \Rightarrow b = 11 \text{ cm}$

**Sol.291.(b)**  
 Area of rectangle wall =  $160 \text{ m}^2$   
 Remaining area =  $160 - \frac{1}{2} \times 8 \times 15$

$\Rightarrow 160 - 60 = 100$   
 Hence, total cost for painting  
 $= 100 \times 11 = 1100$

**Sol.292.(d)** Area of regular hexagon  
 $= \frac{3\sqrt{3}}{2} \times a^2 \Rightarrow \frac{3\sqrt{3}}{2} \times 20^2 = 600\sqrt{3}$   
 $600\sqrt{3} = 1039.2 \text{ cm}^2$   
 $1039.2 \text{ cm}^2 \times \frac{90}{100} \Rightarrow 935.28 \text{ cm}^2$

**Sol.293.(c)**  
 Let side of base square =  $a$  .cm  
 Volume of prism = base of area  $\times$  height  
 $720 \times 1000 \text{ cm}^3 = a^2 \times 200$   
 $a = \sqrt{3600} \Rightarrow 60 \text{ cm}$

**Sol.294.(c)** Volume of a cylinder =  $\pi r^2 h$   
 If radius is doubled  $\rightarrow$  new radius =  $2r$   
 New volume =  $\pi (2r)^2 h = \pi \times 4r^2 h$   
 $= 4 \times (\pi r^2 h)$   
 $\therefore$  Volume becomes four times.

**Sol.295.(a)** Given: External dimensions

$$= 8 \text{ cm} \times 6 \text{ cm} \times 5 \text{ cm}$$

$$\text{Thickness} = 0.5 \text{ cm}$$

Internal dimensions:

$$L = 8 - 2(0.5) = 7 \text{ cm}, B = 6 - 2(0.5) = 5 \text{ cm and } H = 5 - 2(0.5) = 4 \text{ cm}$$

Volume of material (hollow part)

$$= \text{External volume} - \text{Internal volume}$$

$$= (8 \times 6 \times 5) - (7 \times 5 \times 4)$$

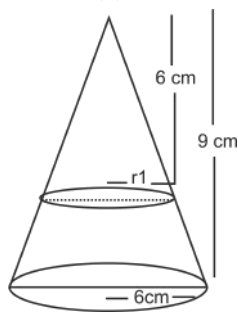
$$= 240 - 140 = 100 \text{ cm}^3$$

**Sol.296.(c)** Lateral Surface Area (L.S.A)

$$= \text{Perimeter of base} \times \text{Height}$$

$$= (3 + 4 + 5) \times 8 = 96 \text{ cm}^2$$

**Sol.297.(b)**



By similarity of cones:

$$\frac{r_1}{R} = \frac{h_1}{H} \Rightarrow \frac{r_1}{6} = \frac{6}{9} \Rightarrow r_1 = 4 \text{ cm}$$

Volume of the frustum

$$= \frac{1}{3} \pi \times (9 - 6) \times (6^2 + 4^2 + 6 \times 4)$$

$$= \frac{1}{3} \pi \times 3 \times 76 = 76\pi \text{ cm}^3$$

**Sol.298.(c)** Volume of cuboid =  $1400 \text{ cm}^3$

$$\Rightarrow L \times B \times H = 1400 \text{ cm}^3$$

Now by checking options one by one, option (c) is satisfied.

**Sol.299.(b)** Area of plot =  $576 \text{ m}^2$

$$\text{Since, Area} = \text{side}^2$$

$$\text{Side} = 24 \text{ m}$$

1 m wide footpath is constructed around it, so side increased by 2 m

$$\text{New side} = 26 \text{ m}$$

$$\text{New area} = 26 \times 26 = 676 \text{ m}^2$$

**Sol.300.(d)** Let length =  $4L$ , Breadth =  $L$

$$A/Q, 2(4L + L) = 200$$

$$\Rightarrow (4L + L) = 100 \Rightarrow L = 20$$

$$\text{Length} = 80 \text{ cm}, \text{Breadth} = 20 \text{ cm}$$

$$\text{Area} = 80 \times 20 = 1600 \text{ cm}^2$$

**Sol.301.(d)** On checking all options one by one, only option (d) satisfy,

$$\text{Dimensions} \rightarrow l = 4 \text{ cm}, b = 14 \text{ cm and}$$

$$h = 24 \text{ cm}$$

$$\text{Sum of the dimension}(l + b + h) = 42 \text{ cm}$$

$$\text{LHS} = l + b + h = 4 + 14 + 24$$

$$= 42 \text{ cm} = \text{RHS}$$

$$\text{Volume of cuboid}(l \times b \times h) = 1344 \text{ cm}^3$$

$$\text{LHS} = l \times b \times h = 4 \times 14 \times 24$$

$$= 1344 \text{ cm}^3 = \text{RHS}$$

**Sol.302.(a)** According to the question:

$$\frac{V_1}{V_2} = \frac{7}{9} = \frac{\pi r_1^2 h_1}{\pi r_2^2 h_2}$$

$$\Rightarrow \frac{7}{9} = \frac{1}{9} \times \frac{h_1}{h_2} \Rightarrow \frac{h_1}{h_2} = \frac{7}{1}$$

So, the ratio of their heights is 7 : 1.

**Sol.303.(b)** Volume of prism (V)

$$= \text{Base Area} \times \text{Height}$$

If the base area is doubled and height remains same,

$$V' = 2 \times \text{Base Area} \times \text{Height} = 2V$$

Thus, the volume of the prism is doubled.

**Sol.304.(b)** According to the question:

$$\frac{1}{3} \times \pi \times r^2 \times 24 = \frac{1}{3} \times \pi \times \left(\frac{r}{2}\right)^2 \times h_2$$

$$h_2 = 96 \text{ cm}$$

**Sol.305.(a)** Area of segment

$$= \text{Area of sector} - \text{Area of triangle}$$

$$= \frac{\theta}{360^\circ} \times \pi r^2 - \frac{1}{2} r^2 \sin \theta$$

$$= \frac{90^\circ}{360^\circ} \times \pi \times 5^2 - \frac{1}{2} \times 5^2 \sin 90^\circ$$

$$= \frac{1}{4} \times 25\pi - \frac{1}{2} \times 25 \times 1$$

$$= \frac{25\pi}{4} - 12.5 \text{ cm}^2.$$

**Sol.306.(c)** Area of a triangle

$$= \frac{1}{2} \times \text{Base} \times \text{height}$$

$$= \frac{1}{2} \times 70 \times 96 = 3360 \text{ m}^2$$

$$\text{Area of rectangle} = 70 \text{ m} \times 35 \text{ m}$$

$$= 2450 \text{ m}^2$$

$$\text{Total area} = 3360 + 2450 = 5810 \text{ m}^2$$

$$\text{So, Required \%} = \frac{3360}{5810} \times 100 = 57.83 \%$$

**Sol.307.(b)**

$$\text{Given: } ab = 48, bc = 54, ca = 72$$

According to the question,

$$\text{Volume of cuboid}(V) = abc$$

$$(abc)^2 = 48 \times 54 \times 72$$

$$abc = \sqrt{6 \times 8 \times 9 \times 6 \times 9 \times 8}$$

$$= 6 \times 8 \times 9 = 432 \text{ cm}^3$$

**Sol.308.(a)** According to the question,

$$\frac{2}{3} \pi r^3 = \pi R^2 h$$

$$\frac{2}{3} \times 9^3 = r^2 \times 6 \Rightarrow r^2 = \frac{2}{3 \times 6} \times 9^3$$

$$r = 9 \text{ cm}$$

**Sol.309.(a)** According to question,

$$\frac{2}{3} \pi r^3 = \pi r^2 h = \frac{1}{3} \pi r^2 \times H$$

$$\pi r^2 3h = \frac{1}{3} \pi r^2 \times H \Rightarrow H = 9h$$

**Sol.310.(d)**

$$\text{Diagonal of cuboid} = \sqrt{l^2 + b^2 + h^2}$$

according to question,

$$26^2 = 9^2 + 15^2 + h^2$$

$$\Rightarrow h^2 = 676 - 81 - 225 = 370$$

$$h = \sqrt{370} \Rightarrow h = 19.2 \text{ cm}$$

**Sol.311.(b)**

$$\text{Slant height (l)} = \sqrt{h^2 + \left(\frac{a}{2}\right)^2}$$

Here, a = side of base

$$= \sqrt{12^2 + \left(\frac{10}{2}\right)^2}$$

$$= \sqrt{144 + 25} = 13 \text{ cm}$$

**Sol.312.(c)**

$$\text{let diagonals of rhombus} = 5x, 6x$$

$$\text{Area of rhombus} = \frac{1}{2} \times d_1 \times d_2$$

$$\Rightarrow \frac{1}{2} \times 5x \times 6x = 135$$

$$\Rightarrow x^2 = 9 \Rightarrow x = 3$$

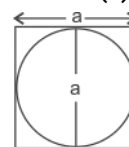
Hence, diagonal are 15, 18

**Sol.313.(b)** the area of the rim

$$= \frac{22}{7} \times (15^2 - 12^2)$$

$$= 81 \times 3.14 = 254.34 \text{ cm}^2$$

**Sol.314.(d)**



$$\text{Let side of the square} = a \Rightarrow \text{Area} = a^2$$

$$\text{Then, radius of the circle} = \frac{a}{2}$$

$$\Rightarrow \text{Area} = \frac{\pi a^2}{4}$$

Percentage of area removed

$$= \frac{\text{Area of circle}}{\text{Area of square}} \times 100 = \frac{\pi}{4} \times 100$$

$$= 78.5\%$$

**Sol.315.(d)** Volume of water displaced =

Volume of sphere

$$\text{Volume of sphere} = \frac{4}{3} \pi r^3$$

$$\text{Volume of sphere} = \frac{4}{3} \pi (7)^3$$

$$= \frac{1372}{3} \pi \text{ cm}^3$$

**Sol.316.(a)** According to the question,

$$\text{Ratio of heights} = \frac{h_{\text{small}}}{h_{\text{full}}} = \frac{6}{30} = \frac{1}{5}$$

The truncated cone and the original cone are similar, their volumes are

The ratio will be equal to the cube of the ratio of their heights:

$$V_{\text{small}} : V_{\text{full}} = \left(\frac{1}{5}\right)^3 = \frac{1}{125}$$

$$\text{volume of frustum} = 125 - 1 = 124$$

$$V_{\text{frustum}} : V_{\text{full}} = 124 : 125$$

**Sol.317.(c)** Number of smaller cones



$$= \frac{\frac{1}{3}\pi R^2 H}{\frac{1}{3}\pi\left(\frac{R}{3}\right)^2 H} = 9$$

**Sol.318.(c) Given:**  $h = r$

$$\text{Required ratio} = \frac{\frac{4}{3}\pi r^3}{\pi r^2 h} = \frac{4}{3}$$

**Sol.319.(b)** Volume = base area  $\times$  height

$$280 = 20 \times \text{height}$$

$$\text{height} = 14 \text{ cm}$$

**Sol.320.(b)** Volume of a cylinder =  $\pi r^2 h$

$$= \frac{22}{7} \times 7 \times 7 \times 10$$

$$= 22 \times 7 \times 10 = 1540 \text{ cm}^3$$

**Sol.321.(b)** Let radius =  $r$  and height =  $h$

$$\Rightarrow \text{Volume of cone} = \frac{1}{3}\pi r^2 h$$

$$\Rightarrow \text{Volume of Cylinder} = \pi r^2 h$$

According to formula:

$$\Rightarrow \text{Volume of Cylinder} = 3 \times \left(\frac{1}{3}\pi r^2 h\right)$$

$$\Rightarrow \text{Volume of Cylinder} = 3 \times \text{volume of cone}$$

So, the volume of the cylinder is **three times** the volume of the cone

**Sol.322.(c)** Volume of box =  $l \times b \times h$

$$= 12 \times 6 \times 4 = 288 \text{ cm}^3$$

**Sol.323.(a)**

$$\text{Area of a triangle} = \frac{1}{2} \times 14 \times 12$$

$$= 84 \text{ cm}^2$$

**Sol.324.(b)**

Curved Surface Area of a cone =  $\pi r l$

$$= 3.14 \times 8 \times 5 = 125.6 \text{ cm}^2$$

**Sol.325.(a)**

Total surface area of hemisphere =  $3\pi r^2$

$$= 3 \times 3.14 \times (6)^2 = 339.12 \text{ cm}^2$$

**Sol.326.(b)** Perimeter of triangle = Sum of all three sides

$$\Rightarrow 30 = 9 + 11 + \text{Third side}$$

$$\Rightarrow \text{Third side} = 30 - 20 = 10 \text{ cm}$$

**Sol.327.(a)** Volume of cone =  $\frac{1}{3}\pi r^2 h$

$$= \frac{1}{3} \times \frac{22}{7} \times 4 \times 4 \times 10$$

$$= \frac{1}{3} \times \frac{22}{7} \times 160$$

$$= \frac{3520}{21} = 167.61 \text{ cm}^3$$

**Sol.328.(c)** According to question,

$$\text{Volume of hemisphere} = V = \frac{2}{3}\pi r^3$$

$$\text{Volume of cylinder} = \pi r^2 (4r) = 4\pi r^3$$

$$\text{Required fraction filled} = \frac{\frac{2}{3}\pi r^3}{4\pi r^3}$$

$$= \frac{2}{3} \times \frac{1}{4} = \frac{2}{12} = \frac{1}{6}$$

**Sol.329.(a)**

TSA of rectangular parallelepiped =  $2(lb + bh + hl)$

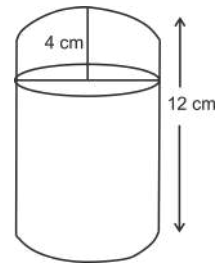
$$= 2(6 \times 5 + 5 \times 4 + 6 \times 4) = 148 \text{ cm}^2$$

**Sol.330.(b)**

Lateral surface area (LSA) of a cone =  $\pi r l$

$$= \pi \times 6 \times 14 = 84\pi \text{ m}^2$$

**Sol.331.(b)**



**Given:**

radius  $r = 4 \text{ cm}$ , total height =  $12 \text{ cm}$

height of the cylinder :  $h = 12 - 4 = 8 \text{ cm}$

Total surface area (TSA) = Curved surface of hemisphere + Lateral surface of cylinder + Base of cylinder

$$= 2\pi r^2 + 2\pi r h + \pi r^2 = \pi r(2r + 2h + r)$$

$$= 3.14 \times 4 \times (8 + 16 + 4) = 351.68 \text{ cm}^2$$

**Sol.332.(c)** Surface area of a brick

$$= 2(10 \times 4 + 4 \times 3 + 10 \times 3) = 164 \text{ cm}^2$$

**Sol.333.(d)** According to the question:

$$\text{Area of triangle} = \frac{1}{2} \times \text{base} \times \text{height}$$

$$175 = \frac{1}{2} \times 25 \times \text{height}$$

$$\text{Height} = 14 \text{ cm}$$

**Sol.334.(c)** Volume of tank = volume of cylinder + volume of hemisphere

$$= \pi r^2 h + \frac{2}{3}\pi r^3 = \pi r^2 \left(h + \frac{2}{3}r\right)$$

$$= 3.14 \times 9 \times \left(6 + \frac{2}{3} \times 3\right) = 226.08 \text{ m}^3$$

**Sol.335.(c)** Volume of wooden block

$$= 16 \times 10 \times 6 = 960 \text{ cm}^3$$

**Sol.336.(a)**

$$\text{Volume of cone} = \frac{1}{3} \times \frac{22}{7} \times 21^2 \times 36$$

$$= 22 \times 3 \times 7 \times 36$$

$$= 16,632 \text{ cm}^3$$

**Sol.337.(a)**

$$\text{Number of small cubes} = \frac{12 \times 6 \times 3}{3 \times 3 \times 3} = 8$$

**Sol.338.(b)** Total surface area of hemisphere ( $3\pi r^2$ ) =  $1,539 \text{ m}^2$

$$\Rightarrow 3\pi r^2 = 1,539$$

$$\Rightarrow r^2 = \frac{1539 \times 7}{3 \times 22} = \frac{10773}{66} = 163.22$$

$$\Rightarrow r = \sqrt{163.22} \text{ m} = 12.77 \text{ m}$$

If the radius is halved:

$$\frac{r}{2} = \frac{12.77}{2} \approx 6.38 \text{ m}$$

Curved surface area =  $2\pi r^2$

$$= 2 \times \frac{22}{7} \times (6.38)^2$$

$$= 2 \times \frac{22}{7} \times 40.70 = 255.82 \text{ m}^2$$

**Sol.339.(a)**

Volume of a cone =  $\frac{1}{3}\pi r^2 h = 84 \pi \text{ cm}^3$

$$\Rightarrow \frac{1}{3}\pi r^2 \times 7 = 84 \pi$$

$$r^2 = \frac{84 \times 3}{7} = 36 \Rightarrow r = 6 \text{ cm}$$

**Sol.340.(a)** Diagonal of the cube =  $\sqrt{75}$

$$\Rightarrow a\sqrt{3} = \sqrt{75} \Rightarrow a = \frac{\sqrt{75}}{\sqrt{3}} = \frac{\sqrt{5 \times 5 \times 3}}{\sqrt{3}}$$

$$= \frac{5\sqrt{3}}{\sqrt{3}} = 5$$

Total surface area of the cube

$$= 6a^2 = 6 \times 25 = 150 \text{ cm}^2$$

**Sol.341.(b) Total surface area of the bottle** = Surface area of the hemisphere + Surface area of the cylinder.

$$= 2\pi r^2 + 2\pi r h$$

$$= 2 \times \frac{22}{7} \times 3.5 \times 3.5 + 2 \times \frac{22}{7} \times 3.5 \times 12$$

$$= 44 \times 1.75 + 528 \times 0.5$$

$$= 77 + 264 = 341 \text{ cm}^2$$

**Sol.342.(a)**  $l^2 = h^2 + r^2$

$$25^2 = r^2 + 20^2$$

$$r^2 = 625 - 400$$

$$r = \sqrt{625 - 400} = \sqrt{225} = 15 \text{ cm}$$

**Sol.343.(d)** Surface area of a hollow hemisphere =  $3\pi R^2 + \pi r^2$

$$= 3 \times \frac{22}{7} \times 64 + \frac{22}{7} \times 36$$

$$= 603.42 + 113.14 = 716.56 \text{ cm}^2$$

**Sol.344.(a)** Slant height

$$= \sqrt{\text{radius}^2 + \text{height}^2} = \sqrt{9^2 + 12^2}$$

$$= \sqrt{81 + 144} = 15 \text{ cm}$$

**Sol.345.(c)**

$$\text{Length of pole} = a\sqrt{3} = 23\sqrt{3} \text{ cm}$$

**Sol.346.(a)** Volume of the cube = Volume of the cuboid

$$a^3 = l \times b \times h \Rightarrow a = \sqrt[3]{12 \times 6 \times 3} = 6 \text{ m}$$

Total surface area of the cube =  $6a^2$

$$= 6 \times 36 = 216 \text{ m}^2$$



**Sol.347.(c)** Radius ( $r_A$ ) of hemisphere A=  $3x$  and radius( $r_B$ ) of hemisphere B =  $x$ 

ATQ

Curved surface area of hemisphere A =

Volume of hemisphere B

$$\Rightarrow 2\pi(r_A^2) = \frac{2}{3}\pi r_B^3$$

$$\Rightarrow 9x^2 = \frac{1}{3}x^3 \Rightarrow x = 9 \times 3 = 27 \text{ cm}$$

**Sol.348.(c)** Circumference of the wheel

$$= 2 \times \frac{22}{7} \times 56 = 44 \times 8 = 352$$

$$\text{Total number of revolutions} = \frac{132000}{352}$$

= 375 revolutions

**Sol.349.(b)**  $l^2 = h^2 + r^2$ 

$$841 = 400 + r^2$$

$$r^2 = 841 - 400 = 441$$

$$\text{Volume of a cone} = \frac{1}{3}\pi r^2 h$$

$$= \frac{1}{3} \times \frac{22}{7} \times 441 \times 20$$

$$= 22 \times 21 \times 20 = 9240 \text{ cm}^3$$

**Sol.350.(a)** Volume of a cylinder =  $\pi r^2 h$ 

$$= \frac{22}{7} \times 49 \times 14 = 2156 \text{ cm}^3$$

**Sol.351.(b)** All angles are right angles**Sol.352.(c)**

$$\text{Area of a rhombus} = \frac{1}{2} \times d_1 \times d_2$$

$$= \frac{1}{2} \times 12 \times 6 = 36 \text{ cm}^2$$

**Sol.353.(a)**

Perimeter of the triangle = sum of all sides

$$= 5 + 9 + 13 = 27 \text{ cm}$$

**Sol.354.(b)** Radius =  $r$ , Height =  $h$ 

$$V = \pi r^2 h$$

$$V_{\text{new}} = \pi r^2 (3h) = 3(\pi r^2 h)$$

$$V_{\text{new}} = 3(V)$$

**Sol.355.(c)** Cylinder's curved surfacearea =  $2\pi r h$ 

$$= 2 \times \frac{22}{7} \times 0.7 \times 5 = 22$$

Plastering cost for the pillar

$$= 22 \times 10 \times 200 = 44,000$$

**Sol.356.(d)** Volume =  $a^3$ new side =  $2a$ 

$$V_{\text{new}} = (2a)^3 = 8a^3 \Rightarrow V_{\text{new}} = 8(V)$$

**Sol.357.(c)** Total surface area of asphere =  $4\pi r^2$ 

Total surface area of a hemisphere

$$= 3\pi r^2$$

$$\text{Ratio} = 4\pi r^2 : 3\pi r^2 = 4 : 3$$

**Sol.358.(c)**Let the width =  $x$  and length =  $3x$ 

According to the question:

Perimeter of the square = Perimeter of the rectangle

$$4a = 2(x + 3x)$$

$$4 \times 16 = 8x \Rightarrow x = 8 \text{ m}, \text{ length} = 24$$

$$\text{Area of the square} = 16 \times 16 = 256 \text{ m}^2$$

$$\text{Area of the rectangle} = 8 \times 24 = 192 \text{ m}^2$$

$$\text{Difference} = 256 - 192 = 64 \text{ m}^2$$

**Sol.359.(c) Perimeter** =  $6 \times 8 = 48 \text{ cm}$ **Sol.360.(b)** The base is a polygon**Sol.361.(b)** Slant height =  $\sqrt{r^2 + h^2}$ 

$$= \sqrt{9^2 + 12^2} = 15 \text{ cm}$$

**Sol.362.(b)** Height =  $3h$ , Radius =  $r$ 

$$V = \frac{1}{3}\pi r^2 h \Rightarrow V_{\text{new}} = \frac{1}{3}\pi r^2 (3h)$$

$$V_{\text{new}} = 3\left(\frac{1}{3}\pi r^2 h\right)$$

$$V_{\text{new}} = 3(V)$$

**Sol.363.(d)** Radius =  $3 \text{ cm}$ Surface area =  $4\pi \times 9 = 36\pi \text{ cm}$ Radius =  $9 \text{ cm}$ Surface area =  $4\pi \times 81 = 324\pi \text{ cm}$ 

The surface area increases 9 times.

**Sol.364.(b)** Internal radius ( $r$ ) =  $10 \text{ cm}$ Thickness =  $0.5 \text{ cm}$ External radius ( $R$ ) =  $10 + 0.5 = 10.5 \text{ cm}$ 

$$V = \frac{4}{3}\pi((10.5)^3 - (10)^3)$$

$$\frac{4}{3} \times \frac{22}{7} \times 157.625 = \frac{13871}{21}$$

$$= 660.52 \text{ cm}^3$$

**Sol.365.(b)** Total surface area =  $3\pi r^2$ 

$$3 \times \frac{22}{7} \times 14 \times 14 = 66 \times 28 = 1848 \text{ cm}^2$$

**Sol.366.(a)**

The area of a circle is always largest.

**Sol.367.(c)** Area of a trapezium

$$= \frac{1}{2}(a + b) \times h$$

$$= \frac{1}{2}(12 + 8) \times 5 = 10 \times 5 = 50 \text{ cm}^2$$

**Sol.368.(a)** Let the other side of the second rectangle be =  $x \text{ cm}$ .

$$18 \times 22 = 12 \times x \Rightarrow x = 33 \text{ cm}$$

**Sol.369.(b) Formula :-** Volume of full cylinder ( $V$ ) =  $\pi r^2 h$ 

Volume of each half-cylinder

$$(V_{\text{half}}) = \frac{1}{2}\pi r^2 h$$

Thus, the volume of each part becomes half of the original volume.

**Sol.370.(c)**Area of trapezium =  $\frac{1}{2} \times (\text{Sum of parallel sides}) \times \text{height}$ 

$$60 = \frac{1}{2} \times (a + b) \times 6 \Rightarrow a + b = 20$$

difference between parallel sides =  $4 \text{ cm}$ So,  $a = 12 \text{ cm}$  and  $b = 8 \text{ cm}$ .**Sol.371.(c)**

$$\text{Area of trapezium} = \frac{1}{2} \times (5 + 7) \times 4$$

$$= 24 \text{ cm}^2$$

Volume of prism = Area of base  $\times$  height

$$= 24 \times 10 = 240 \text{ cm}^3$$

**Sol.372.(a)**CSA of cylinder ( $2\pi rh$ ) =  $264 \text{ cm}^2$ 

$$2 \times \frac{22}{7} \times 3.5 \times h = 264 \Rightarrow h = 12 \text{ cm}$$

**Sol.373.(c)** Big cylinder is melted into 16 small cylinders of same height $\Rightarrow$  Volume ratio = base-area ratio

$$\frac{r_{\text{small}}^2}{r_{\text{big}}^2} = \frac{1}{16}$$

Big radius ( $r_{\text{big}}$ ) =  $8$ 

$$r_{\text{small}} = 8 \times \frac{1}{4} = 2 \text{ cm}$$

**Sol.374.(b)** Length of the side =  $\frac{a}{2}$ Area of square =  $(\text{side length})^2$ 

$$= \left(\frac{a}{2}\right)^2 = \frac{a^2}{4}$$

Hence, the area becomes one-fourth of the original.

**Sol.375.(b)**Area of rectangle =  $25 \times 18 = 450 \text{ m}^2$ Area of triangle =  $\frac{1}{2} \times \text{base} \times \text{Height}$ 

$$= \frac{1}{2} \times 15 \times 10 = 75 \text{ m}^2$$

$$\text{Total Area} = 450 + 75 = 525 \text{ m}^2$$

**Sol.376.(b)** Volume of a solid sphere of

$$\text{radius } 10 \text{ cm} = \frac{4}{3} \times \pi \times 10 \times 10 \times 10$$

$$= \frac{4}{3} \times 1000\pi$$

Volume of a solid sphere of radius  $2 \text{ cm}$ 

$$= \frac{4}{3} \times \pi \times 2 \times 2 \times 2 = \frac{4}{3} \times 8\pi$$

Number of small spheres

$$= \frac{\frac{4}{3} \times 1000\pi}{\frac{4}{3} \times 8\pi} = 125$$

**Sol.377.(a)**  $V = l \times b \times h$ 

$$= 24 \times 14 \times 10 = 3360 \text{ m}^3$$

**Sol.378.(c)** Volume of the tank

$$= 8 \times 6 \times 5 = 240 \text{ m}^3$$

$$1 \text{ m}^3 = 1000 \text{ ltr}$$



$$\therefore 240 \text{ m}^3 = 240 \times 1000 = 240000 \text{ ltr}$$

$$\text{Total cost to fill the tank} = \frac{240000}{1000} \times 50 = 12000$$

**Sol.379.(c)** Volume of cuboid:

$$24 \times 18 \times 6 = 2592 \text{ cm}^3$$

$$\text{Volume of each of 12 cubes} = \frac{2592}{12} = 216 \text{ cm}^3$$

$$\text{Edge of cube} = \sqrt[3]{216} \text{ cm} = 6 \text{ cm}$$

**Sol.380.(c)** Edge of cube = 4cm

5 cubes joined:

Length =  $5 \times 4 = 20$  cm, Breadth = 4 cm, Height = 4 cm

TSA of cuboid =  $2(lb + bh + lh)$

$$= 2(20 \times 4 + 4 \times 4 + 20 \times 4)$$

$$= 2(176) = 352 \text{ cm}^2$$

**Sol.381.(a)** Let diagonals be  $x$  and  $2x$

$$\text{Area of the rhombus} = \left(\frac{1}{2} \times d_1 \times d_2\right)$$

$$= 100 \text{ cm}^2$$

$$\frac{1}{2} \times x \times 2x = 100 \Rightarrow x = 10 \text{ cm}$$

Diagonals:  $x = 10$  cm and  $2x = 20$  cm

**Sol.382.(b)** Volume of a prism = Area of base  $\times$  Height.

Since height is the same, the ratio of volumes = ratio of areas of bases.

$$\therefore \text{Ratio of volumes} = 2 : 5$$

**Sol.383.(a)** Curved surface area of cylinder ( $2\pi rh$ ) =  $220 \text{ cm}^2$

$$2 \times \frac{22}{7} \times r \times 10 = 220 \Rightarrow r = 3.5 \text{ cm}$$

Now,

$$\text{Volume of cylinder} = \pi r^2 h$$

$$= \frac{22}{7} \times (3.5)^2 \times 10 = 385 \text{ cm}^3$$

**Sol.384.(c)**

Let the volume of tank be 100

According to question,

$$\Rightarrow 100 \times \frac{5}{4} \times \frac{7}{8} \times \frac{3}{2} = 164.06$$

So, the increased volume

$$= \frac{164.06 - 100}{100} \times 100 = 64.06\%$$

**Sol.385.(c)** TSA of rectangular tank =  $2(lb + bh + hl)$

$$= 2(8 \times 10 + 10 \times 11 + 11 \times 8) = 556 \text{ m}^2$$

Area of circle =  $\pi r^2$

$$= \frac{22}{7} \times 2.1 \times 2.1 = 13.86 \text{ m}^2$$

Required iron sheet =  $556 - 13.86$

$$= 542.14 \text{ m}^2$$

**Sol.386.(c)** A polynomial cannot have negative exponents, fractional exponents and variables inside roots.

Only option (c) satisfies,

$$\Rightarrow \sqrt{y}, y^{\frac{1}{3}}, \text{ or } y^{-3}$$

These are not allowed in a polynomial.

So, option (c) is NOT a polynomial.

$$\text{Sol.387.(b)} \text{ Side of polygon} = \frac{72}{9} = 8 \text{ cm}$$

Increase per side = 1.5 cm

New side =  $8 + 1.5 = 9.5$  cm

New perimeter =  $9 \times 9.5 = 85.5$  cm.

**Sol.388.(a)** According to the question

Volume of sphere = Volume of cone

$$\Rightarrow \frac{4}{3} \pi r^3 = \frac{1}{3} \pi R^2 h$$

$$\Rightarrow \frac{4}{3} \pi (14)^3 = \frac{1}{3} \pi R^2 (56)$$

$$\Rightarrow R = 14 \text{ cm}$$

**Sol.389.(d)**

$$\text{Surface area of sphere(A)} = 4\pi r^2$$

$$\frac{A_1}{A_2} = \frac{r_1^2}{r_2^2} \Rightarrow \frac{A_1}{A_2} = \frac{4^2}{5^2} = \frac{16}{25}$$

**Sol.390.(b)** Let, number of cone =  $n$

Volume of cylinder =  $n \times$  volume of cone

$$\pi \times 9^2 \times 36 = n \times \frac{1}{3} \pi \times (4.5)^2 \times 8$$

$$36 = n \times \frac{1}{3} \times 0.25 \times 8 \Rightarrow n = 54$$

**Sol.391.(c)**

$$\text{Area of triangle} = \frac{1}{2} \times \text{base} \times \text{height}$$

$$= \frac{1}{2} \times 30 \times 15 = 225 \text{ cm}^2$$

ATQ,

225 = Area of rectangle

$$225 = 25 \times b \Rightarrow b = 9 \text{ cm}$$

**Sol.392.(a)** Given,  $r = 7$  cm and height of cone = 24 cm

$$\Rightarrow \text{Slant height of cone} = \sqrt{7^2 + 24^2} = 25 \text{ cm}$$

$$\text{So, surface area of cone} = \frac{22}{7} \times 7 \times 25$$

$$= 550 \text{ cm}^2$$

Again,

Surface area of cylinder

$$= 2 \times \frac{22}{7} \times 7 \times 15 = 660 \text{ cm}^2$$

$$\text{Base area of cylinder} = \pi \times 7^2 = 22 \times 7 = 154 \text{ cm}^2$$

According to question,

$$\text{Total surface area of candle} = 550 + 660 + 154 = 1364 \text{ cm}^2$$

**Sol.393.(d)** let the side =  $a$

$$\text{Perimeter of equilateral triangle} (3a) = 24$$

$$a = 8$$

$$\text{Area of triangle} = \frac{\sqrt{3}}{4} a^2$$

$$= \frac{\sqrt{3}}{4} \times 8^2 = 16\sqrt{3}$$

**Sol.394.(b)** According to question,

Difference between the perimeter of the park and the path

$$= 4 \times 89 - 4 \times 81$$

$$= 4 \times 8 = 32 \text{ meters}$$

**Sol.395.(a)** Let triangle sides are

respectively,  $15x$ ,  $17x$ ,  $27x$

perimeter of triangle =  $15x + 17x + 27x$

$$= 59x$$

According to question,

$$59x = 2655 \Rightarrow x = 45$$

Smallest side of triangle =  $15 \times 45$

$$= 675 \text{ m}$$

**Sol.396.(b)**

According to question,

$$\frac{4}{3} \pi R^3 = \frac{1}{3} \pi r^2 R$$

$$4R^2 = r^2 \Rightarrow r : R = 2 : 1$$

**Sol.397.(c)** TSA =  $2(lb + bh + hl)$

$$= 2(8 \times 12 + 12 \times 20 + 20 \times 8)$$

$$= 2(496) = 992 \text{ cm}^2$$

**Sol.398.(b)** Area of rectangle =  $30 \times 20 = 600 \text{ m}^2$

$$\text{Area of 4 circles} = 4 \times \frac{22}{7} \times (3.5)^2$$

$$= 154 \text{ m}^2$$

Area left in the grass =  $600 - 154$

$$= 446 \text{ m}^2$$

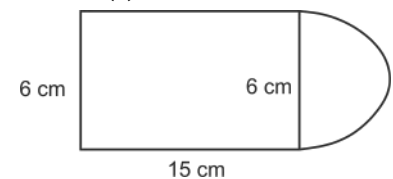
**Sol.399.(b)**

Volume of prism = Base area  $\times$  height

$$= 6 \times \frac{\sqrt{3}}{4} \times 12 \times 12 \times 16 = 3456\sqrt{3}$$

$$= 3456 \times 1.732 \approx 5986 \text{ cm}^3$$

**Sol.400.(a)**



Area of shape = Area of rectangle + area of semicircle

$$= 6 \times 15 + \frac{3.14 \times 3 \times 3}{2} = 90 + 14.13$$

$$= 104.13 \text{ cm}^2$$

**Sol.401.(b)** Area of regular hexagon

$$= 3 \times \frac{\sqrt{3}}{2} \times 36 = 54\sqrt{3}$$

Volume of Prism =  $54\sqrt{3} \times 13$

$$= 1215.9 \text{ cm}^3$$

**Sol.402.(c)**

Area of rectangle =  $20 \times 8 = 160 \text{ cm}^2$

$$\text{Area of semicircle} = \frac{1}{2} \pi (4)^2$$

$$= \frac{1}{2} \times \frac{22}{7} \times 16 \Rightarrow \frac{176}{7} = 25.14 \text{ cm}^2$$

$$\text{Total area} = 160 + 25.14 = 185.14 \text{ cm}^2$$



**Sol.403.(b)**

Volume = Area of base  $\times$  height  
 $= (15)^2 \times 8 = 1800\text{cm}^2$

**Sol.404.(a) A/Q,**

Volume of sphere = Volume of cone

$$\frac{4}{3}\pi r^3 = \frac{1}{3}\pi r^2 h$$

$$h = 4r$$

Curved surface area of sphere =  $4\pi r^2$

Curved surface area of cone

$$= \pi r \sqrt{r^2 + 16r^2}$$

$$\text{Curved surface area of cone} = \pi r^2 \sqrt{17}$$

$$\text{Reqd. ratio} = \frac{4\pi r^2}{\pi r^2 \sqrt{17}} = 4 : \sqrt{17}$$

$$\text{Sol.405.(b)} V_1 = \frac{4}{3}\pi r_1^3 = \frac{4}{3}\pi \times (10)^3$$

ATQ,

$$125 \frac{4}{3}\pi r_2^3 = \frac{4}{3}\pi \times (10)^3$$

$$r_2^3 = \left(\frac{10}{5}\right)^3 \Rightarrow r_2 = 2\text{cm}$$

Reqd. ratio

$$4\pi \times 10^2 : 6 \times 4\pi \times 2^2$$

$$25 : 6$$

**Sol.406.(b) A/Q,**

$$\Rightarrow \frac{2}{3}\pi R^3 = \frac{2}{3}\pi r^3 + \frac{1}{3}\pi r^2 \times r$$

$$\Rightarrow \frac{2}{3}\pi R^3 = \frac{2}{3}\pi r^3 + \frac{1}{3}\pi r^3$$

$$\Rightarrow \frac{2}{3}\pi R^3 = \frac{1}{3}\pi r^3 [2 + 1]$$

$$\Rightarrow \frac{2}{3}\pi R^3 = \pi r^3 \Rightarrow 2R^3 = 3r^3$$

$$\Rightarrow R^3 = 1.5r^3 \Rightarrow \sqrt[3]{1.5} \times r$$

**Sol.407.(a) Area of base =  $6 \times 8 = 48\text{cm}^2$** 

Volume = Area of base  $\times$  height =  $48 \times 14 = 672\text{cm}^3$

**Sol.408.(d) Total area of 2 semicircle =**

$$\pi(14)^2 = 3.14 \times 196 = 615.44\text{cm}^2$$

$$\text{Area of rectangle} = 21 \times 14 = 294\text{cm}^2$$

$$\text{Total area} = 615.44 + 294 = 909.44\text{cm}^2$$

**Sol.409.(a) Lateral surface area of Prism**

= Perimeter of Base  $\times$  height

Pythagorean triplet = (9,40,41)

Perimeter of base =  $9 + 41 + 40 = 90$

Lateral surface area of Prism =  $90 \times 16$

$$= 1440\text{cm}^2$$

**Sol.410.(c) Let the radius of the cylinder and sphere be  $r$  units**

$\Rightarrow$  Height of the cylinder  $h = 2r$  units

Curved Surface area of hemisphere

$$= 2\pi r^2$$

Curved Surface area of the cylinder =  $2\pi rh$

$$= 2\pi r \times 2r = 4\pi r^2$$

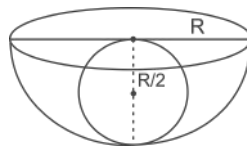
Total surface area of hemisphere mounted on hemisphere is

Curved Surface area of hemisphere +  
 Curved surface area of cylinder + Base  
 area of cylinder =

$$2\pi r^2 + 4\pi r^2 + \pi r^2 = 7\pi r^2.$$

**Sol.411.(d)** Lateral surface area =  
 Perimeter of base  $\times$  height

$$= (8 \times 5) \times 13 = 520\text{cm}^2$$

**Sol.412.(a)**

Now, According to question

$$\frac{2}{3}\pi R^3 : \frac{4}{3}\pi \left(\frac{R}{2}\right)^3$$

$$1 : 2 \times \frac{1}{8} \Rightarrow 4 : 1$$

**Sol.413.(b)**

Let Side of equilateral triangle =  $A$

Side of regular hexagon =  $a$

According to question

$$\Rightarrow 3A = 6a$$

$$\Rightarrow \frac{A}{a} = \frac{6}{3} = \frac{2}{1}$$

Now, Area of equilateral triangle: area of  
 regular hexagon

$$\Rightarrow \frac{\sqrt{3}}{4} \times 2 \times 2 : \frac{3\sqrt{3}}{2} \times 1 \times 1$$

$$\Rightarrow \frac{4\sqrt{3}}{4} : \frac{3\sqrt{3}}{2} \Rightarrow 2 : 3$$

**Sol.414.(a) Given,**

Radius of a sphere whose volume =  
 curved surface area

$$\Rightarrow \frac{4}{3}\pi r^3 = 4\pi r^2$$

$$\Rightarrow \frac{r}{3} = 1 \Rightarrow r = 3 \text{ unit}$$

**Sol.415.(b) According to question,**

Volume of two solid hemisphere =

Volume of sphere + Volume of cone

Let the Cone radius = sphere radius =  $x$

$$\Rightarrow \frac{2}{3}\pi 8R^3 + \frac{2}{3}\pi 27R^3 = \frac{4}{3}\pi x^3 + \frac{1}{3}\pi x^2 \cdot 3x$$

$$\Rightarrow \frac{2}{3}\pi (8R^3 + 27R^3) = \pi x^3 \left(\frac{7}{3}\right)$$

$$\Rightarrow 70\pi R^3 = 7\pi x^3$$

$$\Rightarrow 10R^3 = x^3$$

$$\Rightarrow x = \sqrt[3]{10} \times R$$

**Sol.416.(a) Area of a regular hexagon**

$$A = \frac{3\sqrt{3}}{2} a^2$$

Given,  $a = 10\text{cm}$

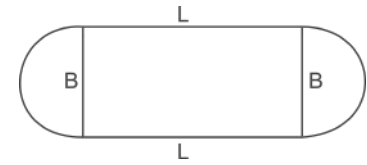
$$A = \frac{3\sqrt{3}}{2} \times 10^2$$

$$A = 150\sqrt{3} \Rightarrow A = 259.5\text{cm}^2$$

$$\text{Plastic cost} = 259.5 \times 4 = 1038$$

After adding the design cost,

$$\text{Total cost per sign} = 1038 + 250 = ₹1288$$

**Sol.417.(b)**

Now, according to question

$$\Rightarrow \pi r^2 = \frac{80}{100} (L \times B)$$

$$\Rightarrow \pi \frac{B^2}{4} = \frac{4}{5} \times L \times B$$

$$\Rightarrow 3.14 \times \frac{B^2}{4} = \frac{4}{5} \times L \times B$$

$$\Rightarrow \frac{3.14 \times 5}{16} = \frac{L}{B}$$

$$\therefore 15.7 : 16 = L : B$$

**Sol.418.(c) Given: Surface area of cube**

$$(6a^2) = 1728\text{cm}^2 \text{ So, } a = 12\sqrt{2}\text{cm}$$

$$\text{Volume of cube } (a^3) = (12\sqrt{2})^3$$

$$= 3456\sqrt{2}\text{cm}^3$$

**Sol.419.(d) Radius =  $\frac{8}{\pi}$ , Volume = 448**

Volume of cylinder ( $\pi r^2 h$ ) = 448

$$\Rightarrow \pi \times \frac{8}{\pi} \times \frac{8}{\pi} \times h = 448$$

$$\Rightarrow h = \frac{448 \times 22}{64 \times 7} \Rightarrow h = 22\text{cm}$$

**Sol.420.(a)**

Ratio  $\rightarrow \text{Sphere}_1 : \text{Sphere}_2$

$$\text{Volume} \rightarrow \frac{4}{3}\pi(4)^3 : \frac{4}{3}\pi(5)^3$$

$$64 : 125$$

**Sol.421.(c) Volume of toy = Volume of  
 hemisphere + volume of cylinder +  
 volume of cone**

$$= \frac{2}{3}\pi r^3 + \pi r^2 h + \frac{1}{3}\pi r^2 h$$

$$= \pi r^2 \left[ \frac{2}{3}r + h + \frac{1}{3}h \right]$$

$$= \frac{22}{7} \times 4.2 \times 4.2 \left[ \frac{2}{3} \times 4.2 + 7 + \frac{7}{3} \right]$$

$$= 22 \times 0.6 \times 4.2 \times \frac{36.4}{3} = 672.672\text{cm}^3$$

**Sol.422.(b)**

$$\text{Area of regular hexagon} = \frac{3\sqrt{3}}{2} a^2$$

According to the question,

$$\frac{3\sqrt{3}}{2} a^2 = 2048\sqrt{3} \Rightarrow a^2 = \frac{4096}{3}$$

$$a = \sqrt{\frac{4096}{3}} = \frac{64}{\sqrt{3}} = \frac{64}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$$

$$= \frac{64\sqrt{3}}{3}\text{cm}^2$$

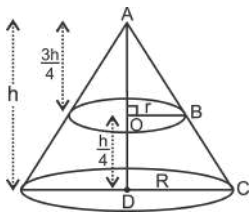
**Sol.423.(d)**

$$\text{CSA of a cone} = \pi Rl = \pi R \sqrt{R^2 + h^2}$$

According to question,

$$\text{Height of small cone} = h - \frac{h}{4} = \frac{3h}{4}$$





$\triangle AOB \sim \triangle ADC$  by AAA rule

$$\Rightarrow \frac{r}{R} = \frac{\frac{3h}{4}}{h} \Rightarrow r = \frac{3R}{4}$$

CSA of a small cone =  $\pi r l_1$

$$= \pi r \sqrt{r^2 + \left(\frac{3h}{4}\right)^2}$$

$$= \pi \times \frac{3R}{4} \times \sqrt{\left(\frac{3R}{4}\right)^2 + \left(\frac{3h}{4}\right)^2}$$

$$= \pi \times \frac{3R}{4} \times \frac{3}{4} \sqrt{R^2 + h^2}$$

Now,

CSA of frustum =  $\pi(R+r)l_2$

$$= \pi \times \left(R + \frac{3R}{4}\right)$$

$$\times \sqrt{\left(R - \frac{3R}{4}\right)^2 + \left(\frac{h}{4}\right)^2}$$

$$= \pi \times \frac{7R}{4} \sqrt{\frac{R^2 + h^2}{16}}$$

$$= \pi \times \frac{7R}{16} \times \sqrt{R^2 + h^2}$$

So,

$$\text{Required ratio} \rightarrow \pi \times \frac{3R}{4} \times \frac{3}{4} \sqrt{R^2 + h^2}$$

$$: \pi \times \frac{7R}{16} \times \sqrt{R^2 + h^2} \text{ or } 9:7$$

**Sol.424.(c)** Volume of cuboid

$$= 5.5 \times 5 \times 3.5 = 96.25 \text{ cm}^3$$

Volume of each gold coins =  $\pi r^2 h$

$$= \frac{22}{7} \times (0.875)^2 \times \frac{2}{10} = 0.48125 \text{ cm}^3$$

$$\text{Hence, no. of gold coins} = \frac{96.25}{0.48125}$$

$$= 200$$

**Sol.425.(a)** Total surface area of the tank

= total surface area of cylinder - area of circular cut

$$= 2\pi r(h+R) - \pi r^2$$

$$= 2\pi \times 5(10+5) - \pi (0.5)^2$$

$$= 2\pi \times 75 - 0.25\pi = 150\pi - \frac{\pi}{4} = \frac{599\pi}{4} \text{ m}^2$$

**Sol.426.(c)** TSA of a cylinder =  $2\pi r(h+r)$

$$= 2 \times \frac{22}{7} \times 3.5(8+3.5)$$

$$= 22 \times 11.5 = 253 \text{ cm}^2$$

According to question,

$$\Rightarrow 5 \text{ cm}^2 = ₹40$$

$$\Rightarrow 253 \text{ cm}^2 = \frac{40}{5} \times 253 = ₹2024$$

**Sol.427.(b)** Given,

$$\Rightarrow 2\pi r = 264 \text{ cm} \Rightarrow r = \frac{264 \times 7}{44} = 42 \text{ cm}$$

Given Bowl is half -full of beverage

$\therefore$  Volume of half -full beverage

$$\text{hemispherical bowl} = \frac{\frac{2}{3}\pi r^3}{2}$$

Now, radius of hemispherical glasses

$$= 3 \text{ cm}$$

According to question ,

$\Rightarrow N \times \text{volume of hemispherical glasses} =$

Volume of half -full beverage

hemispherical bowl

$$\Rightarrow N = \frac{\frac{\frac{2}{3}\pi r^3}{2}}{\frac{\frac{2}{3}\pi r^3}{2}} = \frac{21 \times 42 \times 42}{3 \times 3 \times 3} = 1372$$

**Sol.428.(c)** Given radius = 14 cm

Area of circle =  $\pi r^2$

$$= \frac{22}{7} \times 14 \times 14 = 616 \text{ cm}^2$$

Diagonal of square ( $\sqrt{2}a$ ) = diameter

$$\Rightarrow \sqrt{2}a = 28 \Rightarrow a = 14\sqrt{2} \text{ cm}$$

Area of Square =  $14\sqrt{2} \times 14\sqrt{2} = 14 \times 14$

$$\times 2 = 392 \text{ cm}^2$$

Area of remaining portion =  $616 - 392$

$$= 224 \text{ cm}^2$$

**Sol.429.(c)** Volume of the hemisphere

$$= \frac{2}{3}\pi r^3 \Rightarrow 486\pi = \frac{2}{3}\pi r^3$$

$$\Rightarrow 243 \times 3 = r^3 \Rightarrow r^3 = 729 \Rightarrow r = 9 \text{ cm}$$

According to question,

$$\Rightarrow \text{Radius (r)} = 9 \times 3 = 27 \text{ cm}$$

CSA of a hemisphere =  $2\pi r^2$

$$= 2 \times \pi \times 27 \times 27 = 1458\pi \text{ cm}^2$$

**Sol.430.(c)** Given , internal radius of hemispherical bowl = 15 cm

Diameter and height of cylindrical shaped bottles = 5 cm and 6 cm

Number of bottles required to empty the bowl

$$= \frac{\text{Volume of hemispherical bowl}}{\text{volume of cylindrical shape bottles}} = \frac{\frac{2}{3}\pi r^3}{\pi r^2 h} = \frac{\frac{2}{3} \times 15 \times 15 \times 15}{\frac{5}{2} \times \frac{5}{2} \times 6} = 60$$

**Sol.431.(b)** Volume of cone =  $\frac{1}{3}\pi r^2 h$

$$= \frac{1}{3}\pi \times (7)^2 \times 18 = 294\pi \text{ cm}^3$$

Volume of hollow sphere

$$= \frac{2}{3}\pi(R^3 - r^3) = \frac{2}{3}\pi[R^3 - (20)^3]$$

According to the question,

$$\frac{2}{3}\pi[R^3 - (20)^3] = 294\pi$$

$$R^3 - (20)^3 = 441$$

$$R^3 = 441 + 8000 \Rightarrow R = \sqrt[3]{8441} \text{ cm}^3$$

**Sol.432.(a)** Volume of a spherical item = Volume of a shape

$$\Rightarrow \frac{4}{3}\pi r^3 = \frac{1}{3}\pi r^2 H + \pi r^2 h$$

$$\Rightarrow \frac{4}{3} \times \frac{21}{2} \times \frac{21}{2} \times \frac{21}{2} = \frac{1}{3} \times \frac{21}{2} \times \frac{21}{2}$$

$$\times H + \frac{21}{2} \times \frac{21}{2} \times 7 \Rightarrow \frac{4}{3} \times \frac{21}{2} = \frac{H}{3} + 7$$

$$\Rightarrow 14 = \frac{H}{3} + 7 \Rightarrow H = 21 \text{ cm}$$

Hence, the height of the cone is 21 cm

**Sol.433.(b)**

Each diagonal of hexagon =  $2 \times \text{side}(a)$

$$\text{So, } a = \frac{12}{2} = 6 \text{ cm}$$

$$\text{Now, area of hexagon} = \frac{3\sqrt{3}}{2} a^2$$

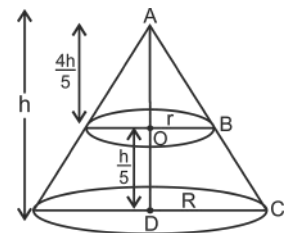
$$= \frac{3\sqrt{3}}{2} \times (6)^2 = 54\sqrt{3} \text{ cm}^2$$

**Sol.434.(d)**

CSA of a cone =  $\pi R l = \pi R \sqrt{R^2 + h^2}$

According to question,

$$\text{Height of small cone} = h - \frac{h}{5} = \frac{4h}{5}$$



$\triangle AOB \sim \triangle ADC$  by AAA rule

$$\Rightarrow \frac{r}{R} = \frac{\frac{4h}{5}}{h}$$

$$\Rightarrow \text{small cone radius (r)} = \frac{4R}{5}$$

TSA of a small cone = CSA of smaller cone + base area of smaller cone

$$\Rightarrow \pi r l_1 + \pi r^2 = \pi \times \frac{4R}{5} \times \left[ \frac{4R}{5} + \right.$$

$$\left. \sqrt{\left(\frac{4R}{5}\right)^2 + \left(\frac{4h}{5}\right)^2} \right] = \pi \times \frac{16R}{25} [R + \sqrt{R^2 + h^2}]$$

Now, CSA of frustum =  $\pi(R+r)l_2$

$$= \pi \times \left(R + \frac{4R}{5}\right) \times \sqrt{\left(R - \frac{4R}{5}\right)^2 + \left(\frac{h}{5}\right)^2}$$

$$= \pi \times \frac{9R}{5} \sqrt{\frac{R^2 + h^2}{25}} = \pi \times \frac{9R}{25} \times \sqrt{R^2 + h^2}$$

$$\text{So, Required ratio} = \pi \times \frac{9R}{25} \times \sqrt{R^2 + h^2}$$

$$: \pi \times \frac{16R}{25} \left[ R + \sqrt{R^2 + h^2} \right]$$

$$= \pi \times \frac{9R}{25} \times L : \pi \times \frac{16R}{25} [R + L]$$

$$= 9L : 16(R+L)$$

**Sol.435.(b)** Given , Radius : Height = 8 : 15



$$\text{Slant height} = \sqrt{r^2 + h^2} = \sqrt{8^2 + 15^2} \\ = \sqrt{289} = 17 \text{ cm}$$

**Sol.436.(d)****Case 1** :- along the length 28 cmCircumference of the circle =  $2\pi r = 28 \text{ cm}$ 

$$\Rightarrow r = \frac{28}{2\pi} = \frac{14}{\pi} \text{ and height} = 16 \text{ cm}$$

$$\text{Volume of cylinder} = \pi r^2 h = \pi \left(\frac{14}{\pi}\right)^2 \times 16 \\ = \frac{3136}{\pi} \text{ cm}^3$$

**Case 2** :- along the breadth 16 cmCircumference of the circle =  $2\pi R = 16 \text{ cm}$ 

$$\Rightarrow 16 = 2\pi R \Rightarrow r = \frac{16}{2\pi} = \frac{8}{\pi} \text{ and}$$

height(h) = 28 cm

$$\text{Volume of cylinder} = \pi R^2 h = \pi \left(\frac{8}{\pi}\right)^2 \times 28 \\ = \frac{1792}{\pi} \text{ cm}^3$$

The ratio of the larger volume to the

$$\text{smaller volume} = \frac{\frac{3136}{\pi}}{\frac{1792}{\pi}} = \frac{3136}{1792} = \frac{7}{4}$$

Hence, Required percentage

$$= \frac{7}{4} \times 100 = 175\%$$

**Sol.437.(a)**1 person occupy the space =  $5 \text{ m}^2$ 12 person occupy the space =  $12 \times 5$ 

$$= 60 \text{ m}^2$$

Then,

$$\Rightarrow \text{Area of the circle} = \pi r^2$$

$$\Rightarrow 60 = \pi r^2 \dots\dots(i)$$

1 person breadth the air =  $20 \text{ m}^3$ 

12 person breadth the air

$$= 12 \times 20 = 240 \text{ m}^3$$

Then,

$$\Rightarrow \text{Volume of the cone} = \frac{1}{3} \pi r^2 h$$

$$\Rightarrow 240 = \frac{1}{3} \times 60 \times h \text{ [from eq. (i)]}$$

$$\Rightarrow h = 12 \text{ m}$$

**Sol.438.(b)** Length of capsule = 15 mm

$$\text{Height of cylindrical part} = 15 - \left(\frac{6}{2}\right)$$

$$= 12 \text{ mm}$$

Now, surface area of capsule = T.S.A of

cylinder + T.S.A of sphere -  $2\pi r^2$ 

$$= (2\pi r^2 + 2\pi r h) + 4\pi r^2 - 2\pi r^2$$

$$= 2\pi r h + 4\pi r^2$$

$$= 2\pi r (h + 2r) = 2 \times \frac{22}{7} \times 3 \times (3 + 12)$$

$$= \frac{12 \times 15 \times 11}{7} \approx 283 \text{ mm}^2$$

Hence, surface area of capsule  $\approx 283 \text{ mm}^2$ **Sol.439.(b)**Total surface area of hemisphere =  $3\pi r^2$ 

According to the question,

$$3\pi r^2 = 7392$$

$$r^2 = 7392 \times \frac{7}{22} \times \frac{1}{3} = 784$$

$$\text{Radius } (r) = \sqrt{784} = 28$$

 $\frac{1}{4}$ th of the hemispherical container is empty

So, total volume of oil container

$$= \frac{3}{4} \times \frac{2}{3} \pi r^3 = \frac{3}{4} \times \frac{2}{3} \times \frac{22}{7} \times (28)^3 \text{ cm}^3$$

Now, oil is to be transferred in small hemispherical bottles

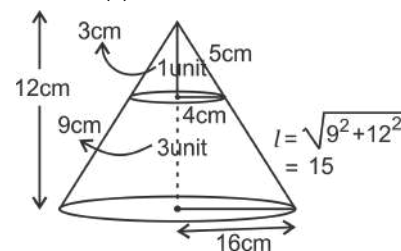
$$\text{Then, volume per bottle} = \frac{2}{3} \pi r^3$$

$$= \frac{2}{3} \times \frac{22}{7} \times (6.3)^3 \text{ cm}^3$$

Hence, required no. of bottles

$$= \frac{\frac{3}{4} \times \frac{2}{3} \times \frac{22}{7} \times (28)^3}{\frac{2}{3} \times \frac{22}{7} \times (6.3)^3} = \frac{\frac{3}{4} \times (28)^3}{(6.3)^3}$$

$$= \frac{3 \times 7 \times 28 \times 28}{6.3 \times 6.3 \times 6.3} = \frac{16464}{250} \approx 66$$

**Sol.440.(a)**Given, radius ( $R$ ) = 16 cm, height = 12 cm and ratio of height of solid = 1 : 3

According to question,

Height of small cone =  $h_1$ 

$$= \frac{1}{4} \times H = \frac{1}{4} \times 12 = 3 \text{ cm}$$

$$\text{Height of frustum} = h_2 = \frac{1}{4} \times H$$

$$= \frac{3}{4} \times 12 = 9 \text{ cm}$$

The section is parallel to the base, so the ratio of the radii will be equal to the ratio of the heights.

$$\text{radius of small cone } (r) = \frac{h_1}{H} \times R$$

$$= \frac{3}{12} \times 16 = 4 \text{ cm}$$

Slant height of small cone ( $l_1$ )

$$= \sqrt{(r_1)^2 + (h_1)^2} = \sqrt{(4)^2 + (3)^2} = 5 \text{ cm}$$

From the question -

$$l_1 : l_2 = 1 : 3 \Rightarrow 5 : l_2 = 1 : 3 \Rightarrow l_2 = 15 \text{ cm}$$

Thus

Slant height of the larger frustum ( $l_2$ )

$$= 15 \text{ cm}$$

$$\text{TSA of small cone} = \pi r_1 l_1 + \pi r_1^2$$

$$= \pi(4 \times 5) + \pi(4)^2 = 20\pi + 16\pi = 36\pi$$

Total surface area of the larger frustum

$$= \pi l_2 (R + r) + \pi (R^2 + r^2)$$

$$= \pi \times 15 \times (20) + \pi (4^2 + 16^2)$$

$$= 300\pi + 272\pi = 572\pi$$

$$\text{Hence, Required ratio} = \frac{36\pi}{572\pi} = 9 : 143$$

**Sol.441.(a)**Radius( $r$ ) = 24 cm, height( $h$ ) = 7 cm

As, (7, 24, 25) are triplet

Then, slant height ( $l$ ) = 25 cm

Area of remaining solid = C.S.A of cylinder + area of base + C.S.A. of cone

$$= 2\pi r h + \pi r^2 + \pi r l$$

$$= \pi r (2h + r + l)$$

$$= \pi \times 24 (2 \times 7 + 24 + 25)$$

$$= \pi \times 24 \times 63 = 1512 \pi \text{ cm}^2$$

**Sol.442.(a)** Let side of one square =  $x \text{ cm}$ Side of another square =  $x + 3$ 

According to question

$$\Rightarrow x^2 + (x + 3)^2 = 225$$

$$\Rightarrow x^2 + x^2 + 9 + 6x = 225$$

$$\Rightarrow 2x^2 + 6x - 216 = 0 \Rightarrow x^2 + 3x - 108 = 0$$

$$\Rightarrow x^2 + 12x - 9x - 108 = 0$$

$$\Rightarrow x(x + 12) - 9(x + 12) = 0$$

$$\Rightarrow (x - 9)(x + 12) = 0$$

$$x = 9, -12$$

 $\therefore$  Sides of two square are = 9 and 12Hence sum of the perimeter of both the square are =  $4 \times 9 + 4 \times 12 = 84 \text{ cm}$ **Sol.443.(b)** CSA of a hemisphere =  $2\pi r^2$ 

$$308 = 2 \times \frac{22}{7} \times r^2$$

$$r^2 = \frac{308 \times 7}{2 \times 22} = 49 \Rightarrow r = 7 \text{ cm}$$

$$\text{Volume of the hemisphere} = \frac{2}{3} \pi r^3$$

$$= \frac{2}{3} \times \frac{22}{7} \times 7 \times 7 \times 7 = 718.66 \text{ cm}^3$$

Now,

$$1 \text{ cm}^3 = 8.7 \text{ gm}$$

$$718.66 \text{ cm}^3 = 718.66 \times 8.7 = 6252.342 \text{ gm} \approx 6252.4 \text{ gm}$$

**Sol.444.(a)** Diagonal of a square who made mid point a square =  $\sqrt{2} \times \text{side}$ 

$$\Rightarrow a = \sqrt{2} \times \text{side} \Rightarrow \text{side} = \frac{a}{\sqrt{2}}$$

$$\text{Area of a square} = \left(\frac{a}{\sqrt{2}}\right)^2 = \frac{a^2}{2}$$

According to question,

$$\text{Remaining area} = a^2 - \frac{a^2}{2}$$

$$= \frac{2a^2 - a^2}{2} = \frac{a^2}{2}$$

**Sol.445.(d)** External radius ( $R$ ) = 2.7 cm

$$\text{Internal radius } (r) = 2.7 - \frac{3}{10} = 2.4 \text{ cm}$$

Now, Volume of pipe =  $\pi(R^2 - r^2)h$ 

$$= \frac{22}{7} \times \{(2.7)^2 - (2.4)^2\} \times 420$$

$$= \frac{22}{7} \times \{7.29 - 5.76\} \times 420$$

$$= 22 \times 1.53 \times 60 = 2019.6 \text{ cm}^3$$

Given,  $1 \text{ cm}^3 = 25 \text{ gram}$

$$\text{So, } 2019.6 \text{ cm}^3 = 25 \times 2019.6$$

$$= 50490 \text{ gm} = 50.49 \text{ kg}$$

**Sol.446.(b)** Volume of solid sphere

$$\left(\frac{4}{3}\pi r^3\right) = \text{Volume of cylindrical pipe}$$

$$(\pi r^2 h) \Rightarrow \frac{4}{3} \times 15 \times 15 \times 15 = (6^2 - 4^2)h$$

$$\Rightarrow h = \frac{\frac{4}{3} \times 15 \times 15 \times 15}{20}$$

$$= 225 \text{ cm or } 2.25 \text{ m}$$

**Sol.447.(c)** According to question,

$$\Rightarrow r + l = h^2 \Rightarrow r + l = 9^2 = 81 \quad \dots (i)$$

$$\Rightarrow l - r = 1 \quad \dots (ii)$$

On solving equation (i) and (ii), we get

$$r = 40 \text{ cm and } l = 41 \text{ cm}$$

$$\text{TSA of cone} = \pi r(r + l)$$

$$= \frac{22}{7} \times 40(41 + 40) = \frac{22 \times 40 \times 81}{7}$$

$$= \frac{71280}{7} = 10182.86 \text{ cm}^2$$

**Sol.448.(b)** TSA of cylinder =  $2\pi r(h + r)$

$$= 2\pi \times 10(30 + 10) = 800\pi \text{ cm}^2$$

According to question,

TSA of larger piece of a cylinder

$$= 800\pi \times \frac{70}{100} = 560\pi \text{ cm}^2$$

$$\Rightarrow 2\pi r(h_1 + r) = 560\pi$$

$$\Rightarrow 2\pi \times 10 \times (h_1 + 10) = 560\pi$$

$$\Rightarrow (h_1 + 10) = 28 \Rightarrow h_1 = 18 \text{ cm}$$

Height( $h_2$ ) of smaller cylinder =  $30 - 18$

$$= 12 \text{ cm}$$

Now,

Volume of sphere = Volume of smaller piece of cylinder

$$\frac{4}{3}\pi R^3 = \pi r^2 h_2 \Rightarrow \frac{4}{3} \times R^3 = 10 \times 10 \times 12$$

$$R^3 = 900 \Rightarrow R = \sqrt[3]{900} \text{ cm}$$

**Sol.449.(a)**

For the largest right circular cone Length

of cone = height of cube = 6 unit

Diameter of cone = side length of cone

= 6 unit

$$\text{Radius of cone} = \frac{6}{2} = 3 \text{ unit}$$

$$\text{Now, volume of cone} = \frac{1}{3}\pi r^2 h$$

$$= \frac{1}{3} \times \pi \times (3)^2 \times 6 = 18\pi \text{ units}^3$$

**Sol.450.(b)** Internal Volume of the large

$$\text{cylindrical vessel } (V_L) = \pi r_1^2 h_1$$

$$= \pi \times 10 \times 10 \times 14 = 1400\pi \text{ cm}^3$$

Internal Volume of small cylindrical

$$\text{vessel } (V_S) = \pi r_2^2 h_2$$

$$= \pi \times 5 \times 5 \times 7 = 175\pi \text{ cm}^3$$

Unfilled volume of big vessel after transferring 6 times =  $1400\pi - 6 \times 175\pi$

$$= 1400\pi - 1050\pi = 350\pi \text{ cm}^3$$

Now,

$$\text{Unfilled volume of big vessel} = 350\pi \text{ cm}^3$$

$$\pi r_3^2 h_3 = 350\pi \text{ cm}^3 \Rightarrow h_3 = \frac{350}{10 \times 10} = 3.5 \text{ cm}$$

$$\text{Sol.451.(b)} \text{ Inner radius}(r) = \frac{12}{2} = 6 \text{ cm}$$

$$\text{Outer radius}(R) = 6 + 2 = 8 \text{ cm}$$

$$\text{Length}(h) = 35 \text{ cm}$$

T.S.A. of cylindrical tube = C.S.A of inner

part + C.S.A of outer part + area of rings

Now,

**C.S.A of inner part + C.S.A of outer part :-**

$$= 2\pi h(R + r) = 2 \times \frac{22}{7} \times 35 \times (8 + 6)$$

$$= 3080 \text{ cm}^2$$

**For area of two ring :-**

$$2 \times \pi(R^2 - r^2) = 2 \times \frac{22}{7} \times [(8)^2 - (6)^2]$$

$$= 2 \times \frac{22}{7} \times [64 - 36] = 176 \text{ cm}^2$$

$$\text{Hence, required area} = 3080 + 176$$

$$= 3256 \text{ cm}^2$$

**Sol.452.(c)**

$$\text{Perimeter of rhombus } (4a) = 160$$

$$\text{Side of rhombus}(a) = 40 \text{ cm}$$

$$\text{Now, } a^2 = \left(\frac{d_1}{2}\right)^2 + \left(\frac{d_2}{2}\right)^2$$

$$(40)^2 = \left(\frac{d_1}{2}\right)^2 + \left(\frac{50}{2}\right)^2$$

$$1600 = \left(\frac{d_1}{2}\right)^2 + 625 \Rightarrow \left(\frac{d_1}{2}\right)^2 = 975$$

$$d_1 = \sqrt{975 \times 4} = \sqrt{3900} = 10\sqrt{39} \text{ cm}$$

$$\text{Area of rhombus} = \frac{1}{2} \times d_1 \times d_2$$

$$= \frac{1}{2} \times 10\sqrt{39} \times 50 = 250\sqrt{39} \text{ cm}^2$$

**Sol.453.(b)** According to question,

$$\Rightarrow \sqrt{2}a_1 : \sqrt{2}a_2 = 5 : 6 \Rightarrow a_1 : a_2 = 5 : 6$$

Now,

$$\Rightarrow a_1^2 : a_2^2 = 25 : 36$$

**Sol.454.(c)**

$$\text{Given, radius of circle}(R) = \frac{(l + r)}{2}$$

According to question,

Total surface area of icecream cone with

lid = area of circle

$$\Rightarrow \pi r^2 + \pi r l = \pi r^2$$

$$\Rightarrow \pi r(r + l) = \pi \times \left(\frac{(l + r)}{2}\right)^2$$

$$\Rightarrow 4r^2 + 4rl = l^2 + r^2 + 2lr$$

$$\Rightarrow 4r^2 - r^2 - l^2 + 4rl - 2lr = 0$$

$$\Rightarrow 3r^2 - l^2 + 2rl = 0 \quad \dots (i)$$

$$\text{Now, } (R) = \frac{(l + r)}{2}$$

$$l = 2R - r \quad \dots (ii)$$

Putting the value of  $l$  in equation (i)

$$3r^2 - (2R - r)^2 + 2r(2R - r) = 0$$

$$3r^2 - 4R^2 - r^2 + 4Rr + 4rR - 2r^2 = 0$$

$$8Rr - 4R^2 = 0$$

$$8Rr = 4R^2$$

$$8r = 4R \Rightarrow R : r = 2 : 1$$

$$\text{Now, } l = 2R - r \Rightarrow l = 2 \times 2 - 1 = 3$$

$$\text{Hence, required ratio} = l : R = 3 : 2$$

**Sol.455.(c)** Volume of hemisphere

$$= \frac{2}{3}\pi r^3 = \left(\frac{2}{3} \times \pi \times 15 \times 15 \times 15\right) \text{ cm}^3$$

Water displaced out =

$$\left(\frac{2}{5} \times \frac{2}{3} \times \pi \times 15 \times 15 \times 15\right) \text{ cm}^3$$

$$\text{Volume of lead shot} = \frac{4}{3}\pi r^3$$

$$= \left(\frac{4}{3} \times \pi \times 0.75 \times 0.75 \times 0.75\right) \text{ cm}^3$$

No of lead shots drop in the hemisphere

$$= \frac{\frac{2}{5} \times \frac{2}{3} \times \pi \times 15 \times 15 \times 15}{\frac{4}{3} \times \pi \times 0.75 \times 0.75 \times 0.75} = 1600$$

**Sol.456.(c)** Volume of the cone =  $\frac{1}{3}\pi r^2 h$

$$= \frac{1}{3} \times \frac{22}{7} \times \frac{13.2}{2} \times \frac{13.2}{2} \times 14$$

$$= 11 \times 4.4 \times 13.2 = 638.88 \text{ cm}^3$$

According to question,

$$\Rightarrow \text{Number of bags} = \frac{638.88}{29.4}$$

$$= 21.73 \approx 22$$

**Sol.457.(b)** For the largest right circular

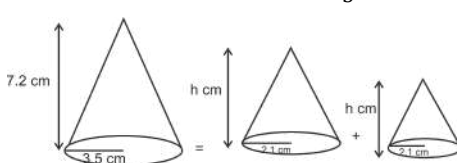
cone that cut out from a cube

Diameter of cone = edge of the cube

$$= 8.4 \text{ cm}$$

$$\text{So, Radius of cone} = \frac{8.4}{2} = 4.2 \text{ cm}$$

**Sol.458.(b)** Volume of cone =  $\frac{1}{3}\pi r^2 h$



$$\frac{1}{3} \times \pi \times 3.5 \times 3.5 \times 7.2$$

$$= 2 \times \left(\frac{1}{3} \times \pi \times 2.1 \times 2.1 \times h\right)$$

$$3.5 \times 3.5 \times 7.2 = 2 \times (2.1 \times 2.1 \times h)$$

$$h = \frac{3.5 \times 3.5 \times 7.2}{2 \times 2.1 \times 2.1} = 10 \text{ cm}$$

**Sol.459.(d)** Side = radius ... (given)

$$\frac{\text{Perimeter of square}}{\text{Perimeter of circle}} = \frac{7}{11}$$

$$\frac{\text{area of square}}{\text{area of circle}} = \frac{\text{side}^2}{\pi r^2} = \frac{1}{\frac{22}{7}} = \frac{7}{22}$$

Hence, required ratio = 7 : 22

**Sol.460.(b)** Area to be painted = external C.S.A. + Internal C.S.A. + area of ring  
 $= 2\pi R^2 + 2\pi r^2 + \pi(R^2 - r^2)$   
 $= 2 \times \frac{22}{7} [(14)^2 + (13.3)^2] + \frac{22}{7} [(14)^2 - (13.3)^2]$   
 $= 2 \times \frac{22}{7} [(196 + 176.89) + \frac{22}{7} [196 - 176.89]]$   
 $= 2 \times \frac{22}{7} \times 372.89 + \frac{22}{7} \times 19.11$   
 $= 44 \times 53.27 + 22 \times 2.73$   
 $= 2343.88 + 60.06 = 2403.94$   
 Cost of painting  
 $= 2403.94 \times 3.75 = 9014.775 \approx 9015$

**Sol.461.(b)**

$$\frac{\text{curved surface area of cylinder}_1}{\text{curved surface area of cylinder}_2}$$

$$= \frac{2\pi RH}{2\pi rh} = \frac{2 \times 5}{3 \times 4} = \frac{5}{6}$$

Required ratio = 5 : 6

**Sol.462.(a)** Outer radius (R) = 9 units  
 Volume of hollow hemispherical steel bowl  
 $= \frac{686}{3} \pi \text{ unit}^3 \Rightarrow \frac{2}{3} \pi r^3 = \frac{686}{3} \pi \text{ unit}^3$   
 $r^3 = 343 \text{ unit}^3$   
 Internal radius (r) = 7 unit

$$\text{Steel used} = \frac{2}{3} \pi (R^3 - r^3)$$

$$= \frac{2}{3} \pi (9^3 - 7^3) = \frac{2}{3} \pi (729 - 343)$$

$$= \frac{2}{3} \pi \times 386 = \frac{772}{3} \pi \text{ unit}^3$$

**Sol.463.(a)** Radius of balloon = 8 feet

$$\text{Volume of spherical balloon} = \frac{4}{3} \pi r^3$$

$$= \frac{4}{3} \times 3.14 \times 512$$

Time taken for balloon to empty

$$= \frac{\text{Volume of spherical balloon}}{\text{Rate of leak}}$$

$$= \frac{\frac{4}{3} \times 3.14 \times 512}{24} = \frac{12.56 \times 64}{9}$$

$$= \frac{803.84}{9} = 89.31 \approx 89 \text{ minutes}$$

**Sol.464.(a)** Volume of hemisphere

$$= N \times \text{volume of cone}$$

$$\frac{2}{3} \pi R^3 = N \times \frac{1}{3} \pi r^2 h$$

$$\frac{2}{3} \times 12^3 = N \times \frac{1}{3} \times 6^2 \times 8$$

$$2 \times 144 \times 12 = N \times 36 \times 8$$

$$N = \frac{144}{12} = 12$$

**Sol.465.(a)** Let the height of the two cone be  $h_1$  and  $h_2$ ,

According to the question,

$$\Rightarrow \frac{\frac{1}{3} \pi r_1^2 h_1}{\frac{1}{3} \pi r_2^2 h_2} = \frac{3}{5} \Rightarrow \frac{4 \times h_1}{25 \times h_2} = \frac{3}{5}$$

$$\Rightarrow \frac{h_1}{h_2} = \frac{75}{20} = \frac{15}{4} \Rightarrow h_1 : h_2 = 15 : 4$$

**Sol.466.(b)** Volume of cylinder =  $\pi r^2 h$   
 $\Rightarrow \frac{22}{7} \times (8)^2 \times 14 = 2816 \text{ cm}^3$

**Sol.467.(b)** Area of sector =  $\frac{1}{2} \times l \times r$   
 $= \frac{1}{2} \times 11 \times 15.75 = 86.625 \text{ cm}^2$

**Sol.468.(c)** Area of isosceles right angle triangle =  $(\frac{\text{hypotenuse}(H)}{2})^2$   
 $\Rightarrow (\frac{H}{2})^2 = 50 \Rightarrow H^2 = 200$   
 $\Rightarrow H = 10\sqrt{2} \text{ unit}$

**Sol.469.(d)**

$$\text{Perimeter of sector} = 2r + \frac{\theta}{360} \times 2\pi r$$

$$\Rightarrow 2 \times 10 + \frac{30}{360} \times 2 \times 3.14 \times 10$$

$$\Rightarrow 20 + \frac{1}{12} \times 62.8$$

$$\Rightarrow 20 + \frac{1}{12} \times 62.8 = 25.2 \text{ (approx)}$$

**Sol.470.(b)** Volume of cone =  $\frac{1}{3} \pi r^2 h$

$$1232 = \frac{1}{3} \times \frac{22}{7} \times r^2 \times 24$$

$$1232 = \frac{176}{7} r^2 \Rightarrow r^2 = \frac{1232 \times 7}{176}$$

$$= r^2 = 49 \Rightarrow r = 7 \text{ cm}$$

$$\text{Slant height} = \sqrt{7^2 + 24^2} = \sqrt{625} = 25 \text{ cm}$$

$$\text{TSA of cone} = \pi r(r + l)$$

$$= \frac{22}{7} \times 7(7 + 25) = 22 \times 32 = 704 \text{ cm}^2$$

**Sol.471.(b)** If a cylinder is formed by rolling the paper along its breadth, breadth becomes perimeter of base and length is equivalent to height.

$$\Rightarrow 2\pi r = 22 \Rightarrow r = \frac{7}{2} = 3.5$$

And  $h = 50$

$$\text{Volume of cylinder} = \pi r^2 h$$

$$= \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \times 50 = 1925 \text{ cm}^3$$

**Sol.472.(d)** Length = 495 cm

Width = 1665 cm

Now, finding HCF of (495 and 1665)

$$\frac{1665}{495} = 3 \text{ (remainder} = 180)$$

$$\frac{495}{180} = 2 \text{ (remainder} = 135)$$

$$\frac{180}{135} = 1 \text{ (remainder} = 45)$$

$$\frac{135}{45} = 3 \text{ (remainder} = 0)$$

$$\text{HCF} (495, 1665) = 45$$

So, dimension of per required tile

$$= 45 \times 45 \text{ cm}^2$$

$$\text{Area of the courtyard} = 495 \times 1665 \text{ cm}^2$$

$$\text{Hence, required tiles} = \frac{495 \times 1665}{45 \times 45}$$

$$= 11 \times 37 = 407$$

**Sol.473.(b)**

$$\text{Area of circular track} = \pi(R_2^2 - R_1^2)$$

$$\Rightarrow \pi[(28)^2 - (25)^2] = \frac{22}{7} [784 - 625]$$

$$= \frac{22}{7} [159]$$

Now, cost of levelling the track

$$= 28 \times \frac{22}{7} [159] = 4 \times 22 \times 159 = ₹13992$$

**Sol.474.(d)** Side of equilateral triangle (a)

$$= \frac{36}{3} = 12 \text{ unit}$$

$$\text{Area of equilateral triangle} = \frac{\sqrt{3}}{4} a^2$$

$$\Rightarrow \frac{\sqrt{3}}{4} \times (12)^2 = 36\sqrt{3} \text{ unit}^2$$

**Sol.475.(a)** Ratio of the sides of two similar triangles =  $\sqrt{25} : \sqrt{144} = 5 : 12$

**Sol.476.(b)** Dimension of largest cube = HCF of (50, 150, 175) = 25

$$\text{Required no} = \frac{\text{volume of cuboid}}{\text{volume of cube}}$$

$$= \frac{50 \times 150 \times 175}{25 \times 25 \times 25} = 84$$

**Sol.477.(d)** lateral surface area of cylinder =  $2\pi r h = 1760$

$$= 2 \times \frac{22}{7} \times r \times 20 = 1760 \Rightarrow r = 14$$

$$\text{volume of cylinder} = \frac{22}{7} \times 14 \times 14 \times 20 = 12320 \text{ cm}^3$$

**Sol.478.(b)** Ratio of curved surface area

$$= \frac{2\pi RH}{2\pi rh} = \frac{2\pi 4 \times 5}{2\pi 5 \times 7} = \frac{4}{7}$$

**Sol.479.(d)** Given height = 8m

Circumference of the base of a conical tent =  $2\pi r = 32\pi \text{ m} \therefore r = 16 \text{ m}$

$$\text{Slant height } l = \sqrt{r^2 + h^2} = \sqrt{25 + 64}$$

$$= \sqrt{320} = 8\sqrt{5} \text{ m}$$

Now, curved surface area ( $\pi r l$ )

$$= \pi \times 16 \times 8\sqrt{5} = 128\sqrt{5} \pi \text{ m}^2$$

**Sol.480.(b)** Length of arc =  $\frac{\theta}{360} \times 2\pi r$

$$= \frac{30}{360} \times 2 \times 3.14 \times 12 = 6.28 \text{ cm}$$

$$\text{Perimeter of the sector} = 6.28 + 12 + 12 = 30.28 \text{ cm}$$

**Sol.481.(b)** Remaining perimeter of pizza

$$= 2\pi r \times \frac{270^\circ}{360^\circ} + 2r$$

$$= 2 \times \frac{22}{7} \times 21 \times \frac{3}{4} + 2 \times 21$$



$$= 99 + 42 = 141 \text{ cm}$$

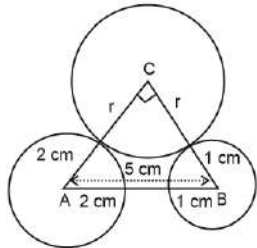
**Sol.482.(b)**

Let the height of conical water tank be  $h$

$$\text{Volume of a cone} = \frac{1}{3} \pi r^2 h$$

$$46200 = \frac{1}{3} \times \frac{22}{7} \times 70 \times 70 \times h$$

$$h = \frac{46200 \times 7 \times 3}{22 \times 70 \times 70} = 9 \text{ m}$$

**Sol.483.(b)**

In  $\triangle ACB$ ,

By using pythagoras theorem,

$$AC^2 + BC^2 = AB^2$$

$$(r + 2)^2 + (r + 1)^2 = 5^2$$

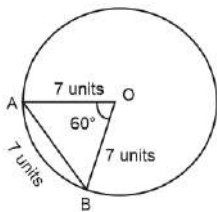
$$r^2 + 4 + 4r + r^2 + 1 + 2r = 25$$

$$2r^2 + 6r - 20 = 0 \Rightarrow r^2 + 3r - 10 = 0$$

$$r^2 + 5r - 2r - 10 = 0 \Rightarrow r(r + 5) - 2(r + 5) = 0$$

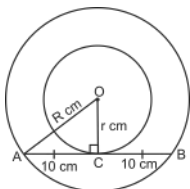
$$(r + 5)(r - 2) = 0$$

Radius can not be negative so  $r = 2 \text{ cm}$

**Sol.484.(c)** According to the question,

Arc length of smaller portion

$$= 2\pi r \times \frac{60^\circ}{360^\circ} = 2\pi \times 7 \times \frac{1}{6} = \frac{7}{3} \pi \text{ units}$$

**Sol.485.(a)**

In the right angled  $\triangle OCA$

$$R^2 = (10)^2 + r^2 \Rightarrow R^2 - r^2 = 100$$

$$\text{Area of annular portion} = \pi(R^2 - r^2)$$

$$= 100\pi \text{ cm}^2$$

**Sol.486.(a)** According to the question,

Radius = 4 unit

Diameter ( $2r$ ) = height ( $h$ ) = 8 unit

$$\text{Volume of cylinder} = \pi r^2 h$$

$$\Rightarrow \pi(4)^2 8 = 128 \pi$$

**Sol.487.(d)** For minute hand

$$\Rightarrow 60 \text{ minute} = 2\pi \Rightarrow 1 \text{ minute} = \frac{2\pi}{60}$$

$$\Rightarrow 25 \text{ minute} = \frac{2\pi}{60} \times 25 = \frac{5\pi}{6}$$

**Sol.488.(d)**

$$\text{Area of sector} = \pi r^2 \times \frac{\text{angle}(\theta)}{360^\circ}$$

$$\Rightarrow 16\pi = \pi \times 8 \times 8 \times \frac{\text{angle}(\theta)}{360^\circ}$$

$$\Rightarrow \frac{360^\circ}{4} = \theta \Rightarrow \text{Angle}(\theta) = 90^\circ$$

**Sol.489.(c)** Circumference of circle =  $2\pi r$ 

$$\Rightarrow 44 = 2 \times \frac{22}{7} \times r \Rightarrow r = 7 \text{ cm}$$

Now,

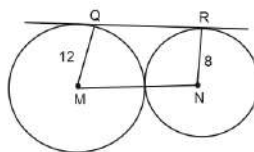
$$\text{Volume of cylinder} = \pi r^2 h$$

$$= \frac{22}{7} \times 7 \times 7 \times 25 = 3850 \text{ cm}^3$$

$$1000 \text{ cm}^3 = 1 \text{ litre}$$

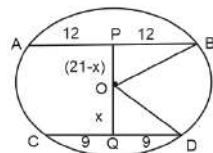
$$1 \text{ cm}^3 = \frac{1}{1000} \text{ litre}$$

$$3850 \text{ cm}^3 = \frac{3850}{1000} = 3.85 \text{ litres}$$

**Sol.490.(b)**

$$\text{Length of QR} = \sqrt{d^2 - (12 - 8)^2}$$

$$= \sqrt{256 - 16} = \sqrt{240} = 4\sqrt{15} \text{ cm}$$

**Sol.491.(a)**

Here,  $OB = OD = \text{radius}$

$$\Rightarrow 12^2 + (21 - x)^2 = 9^2 + x^2$$

$$\Rightarrow 144 + 441 + x^2 - 42x = 81 + x^2$$

$$\Rightarrow 144 + 441 - 81 = 42x$$

$$\Rightarrow 504 = 42x \Rightarrow x = 12$$

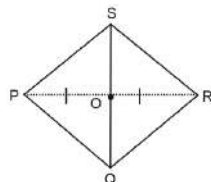
Now in  $\triangle ODQ$

$$OD = \sqrt{81 + 144} = \sqrt{225} = 15$$

So, radius of the circle = 15 cm

**Sol.492.(b)** Diagonal of a cube =  $\sqrt{3}a$ 

$$= \sqrt{3} \times 70 = 70\sqrt{3} \text{ cm}$$

**Sol.493.(b)**

As  $OP = OR$ , So with the help of diagram we can say that,  $\angle SOQ = 180^\circ$

**Sol.494.(c)** Height and diameter of a cylinder are 12 cm and 28 cm

$$\text{Total surface area of cylinder} = 2\pi r(r + h)$$

$$= 2 \times \frac{22}{7} \times 14 (14 + 12) = 88 \times 26$$

$$\text{Curved Surface area of cylinder} = 2\pi rh$$

$$= 2 \times \frac{22}{7} \times 14 \times 12 = 88 \times 12$$

The ratio of the total surface area to the curved surface area =  $88 \times 26 : 88 \times 12 = 13 : 6$

**Sol.495.(d)**

$$\text{Perimeter of the sector} = 2\pi r \times \frac{\theta}{360^\circ} + 2r$$

$$= 2 \times 3.14 \times 14 \times \frac{15^\circ}{360^\circ} + 2 \times 14$$

$$= \frac{3.14 \times 7}{6} + 28 = 3.66 + 28 = 31.66 \text{ cm}$$

**Sol.496.(a)** Let breadth of the rectangular garden =  $x \text{ m}$ 

Length of the rectangular garden

$$= (x + 6) \text{ m}$$

$$\text{Perimeter of rectangular garden } [2(l + b)] = 48 \text{ m}$$

$$\Rightarrow 2(x + 6 + x) = 48 \Rightarrow 2x + 6 = 24$$

$$\Rightarrow 2x = 18 \Rightarrow x = 9 \text{ m}$$

Length = 15 m and breadth = 9 m

Area of rectangular garden =  $l \times b$

$$= 15 \times 9 = 135 \text{ m}^2$$

**Sol.497.(a)**

$$\text{Volume of each cube} = 512 \text{ cm}^3$$

$$\therefore \text{Side of cube} = \sqrt[3]{512} = 8 \text{ cm}$$

Now by joining three cube, length of the cuboid =  $8 + 8 + 8 = 24 \text{ cm}$

Breadth = 8 cm and height = 8 cm

$$\therefore \text{Surface area} = 2(lb + bh + hl)$$

$$= 2(24 \times 8 + 8 \times 8 + 8 \times 24)$$

$$= 2(192 + 64 + 192) = 896 \text{ cm}^2$$

**Sol.498.(a)** Given : radius( $r$ ) = 9 cm and slant height( $l$ ) = 49 cm

Curved surface area of a cone =  $\pi rl$

$$= \frac{22}{7} \times 9 \times 49 = 1386 \text{ cm}^2$$

**Sol.499.(d)**

Area of parallelogram = Base  $\times$  Height

$$\Rightarrow 26 \times 18 = 468 \text{ cm}^2$$

**Sol.500.(d)** Base radius = 10.5 cm

Curved surface area of cylinder ( $2\pi rh$ )

$$= 792 \text{ cm}^2$$

$$h = \frac{792 \times 7 \times 10}{2 \times 22 \times 105} = 12 \text{ cm}$$

$$\text{volume of the cylinder} = \pi r^2 h$$

$$= \frac{22}{7} \times 10.5 \times 10.5 \times 12 = 4158 \text{ cm}^3$$

**Sol.501.(d)** given,  $(a + b) = 9$  and  $ab = 18$   
We know that,

$$(a - b) = \sqrt{(a + b)^2 - 4ab}$$

$$= \sqrt{(9)^2 - 4 \times 18} = \sqrt{81 - 72} = \sqrt{9} = 3$$

then,

$$(a + b) = 9 \dots (i)$$

$$(a - b) = 3 \dots (ii)$$

On solving equation (i) and (ii), we get

$$a = 6 \text{ and } b = 3$$

Now,

$$a^2 - ab + b^2 = 6^2 - 6 \times 3 + 3^2$$



$$= 36 - 18 + 9 = 27$$

**Sol.502.(c)** Area of sector of a circle =  $\pi$

$$r^2 \times \frac{\theta}{360^\circ} \Rightarrow 462 = \frac{22}{7} \times r^2 \times \frac{120^\circ}{360^\circ}$$

$$\Rightarrow r^2 = 462 \times \frac{7}{22} \times 3 = 21 \times 21$$

$$\Rightarrow r = 21 \text{ cm}$$

**Sol.503.(c)**

Total surface area of cube =  $6a^2$

$$\Rightarrow 1152 = 6a^2 \Rightarrow a^2 = 192 \Rightarrow a = 8\sqrt{3} \text{ cm}$$

**Sol.504.(b)**

Surface area of four walls =  $2h(l + b)$

$$= 2 \times 5 (15 + 9) = 10 (24) = 240 \text{ m}^2$$

So,

$$\text{The number of cans} = \frac{240}{40} = 6$$

**Sol.505.(d)** Volume of a right circular

cylindrical box ( $\pi R^2 h$ ) =  $600 \text{ cm}^3$

$$\Rightarrow \frac{22}{7} \times 30 \times 30 \times h = 600$$

$$\Rightarrow h = \frac{600 \times 7}{22 \times 30 \times 30} = \frac{7}{33} = 0.21 \text{ cm}$$

**Sol.506.(d)** According to question ,

$$\Rightarrow \pi R^2 = \pi r_1^2 + \pi r_2^2$$

$$\Rightarrow R^2 = (8^2 + 15^2) \Rightarrow R = 17 \text{ cm}$$

$$\therefore \text{Perimeter of circle} = 2 \times \pi \times 17 = 34\pi \text{ cm}$$

**Sol.507.(b)**

Length : breadth : height =  $3 : 2 : 1$

Breadth (2 unit) = 12 cm

Length (3 unit) = 18 cm

Height (1 unit) = 6 cm

$$\text{Volume} = 18 \times 12 \times 6 = 1296 \text{ cm}^3$$

**Sol.508.(a)** Circumference of base of cylinder = 154 cm

$$2\pi r = 154$$

$$2 \times \frac{22}{7} \times r = 154 \Rightarrow r = \frac{49}{2}$$

Volume of cylindrical vessel =  $\pi r^2 h$

$$= \frac{22}{7} \times \frac{49}{2} \times \frac{49}{2} \times \frac{49}{10}$$

$$= 11 \times 7 \times \frac{49}{2} \times \frac{49}{10} = 9243.5 \text{ cm}^3$$

$$\text{In litres} = 9243.5 \times \frac{1}{1000} = 9.244 \text{ litres}$$

**Sol.509.(d)**  $1 \text{ l} = 1000 \text{ cm}^3$

Volume of water level dropped =  $\pi r^2 h$

$$616 \times 1000 = \frac{22}{7} \times 14 \times 14 \times h$$

$$h = \frac{616000}{22 \times 14 \times 2} = 1000 \text{ cm} = 10 \text{ m}$$

**Sol.510.(b)** Circumference of bigger

$$\text{circle} = 2\pi r_1 + 2\pi r_2$$

$$2\pi R = 2\pi(r_1 + r_2)$$

$$R = 12 + 16 = 28 \text{ cm}$$

$$\text{Radius of bigger circle} = 28 \text{ cm}$$

then diameter = 56 cm

**Sol.511.(c)** Volume of earth taken out

$$= \pi r^2 h = \frac{22}{7} \times 14 \times 14 \times 23$$

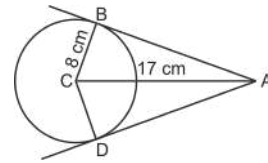
$$= 22 \times 28 \times 23 = 14168 \text{ cm}^3$$

**Sol.512.(b)** T.S.A. of cone = curved surface area + area of base

$$= 550 + \pi r^2 = 550 + \frac{22}{7} \times 7 \times 7$$

$$= 550 + 154 = 704 \text{ cm}^2$$

**Sol.513.(b)**



In  $\triangle ABC$

$\angle ABC = 90^\circ \dots$  (radius  $\perp$  tangent)

hence,

$$AB = \sqrt{AC^2 - BC^2} = \sqrt{17^2 - 8^2}$$

$$= \sqrt{289 - 64} = \sqrt{225} = 15 \text{ cm}$$

$$\text{Area of } \triangle ABC = \frac{1}{2} \times \text{base} \times \text{height}$$

$$= \frac{1}{2} \times AB \times BC = \frac{1}{2} \times 15 \times 8 = 60 \text{ cm}^2$$

Area of quadrilateral ABCD

$$= 2 \times \text{Area of } \triangle ABC = 2 \times 60 = 120 \text{ cm}^2$$

**Sol.514.(d)** Length of sector =  $96\pi$

$$\frac{\theta}{360^\circ} \times 2\pi r = 96\pi \Rightarrow \frac{80^\circ}{360^\circ} \times 2 \times r = 96$$

$$r = \frac{96 \times 36}{16} = 6 \times 36 = 216 \text{ unit}$$

**Sol.515.(c)**

$$\text{Radius of the sphere} = \frac{14}{2} = 7 \text{ cm}$$

$$\text{Volume of the sphere} = \frac{4}{3} \pi r^3$$

$$= \frac{4}{3} \times \frac{22}{7} \times (7)^3 = 1437 \frac{1}{3} \text{ cm}^3$$

**Sol.516.(d)** According to the question,

Let radius and height of cylinder be  $3x$  and  $7x$  respectively,

$$\Rightarrow \frac{\text{CSA of cylinder}}{\text{Volume of cylinder}} = \frac{2\pi rh}{\pi r^2 h}$$

$$\Rightarrow \frac{2}{3} = \frac{2 \times \pi \times 3x \times 7x}{\pi \times 3x \times 3x \times 7x} \Rightarrow x = 1 \text{ unit}$$

So, radius of cylinder =  $3x = 3 \text{ cm}$  and height =  $7x = 7 \text{ cm}$

Now,

Total surface area of cylinder

$$= 2\pi r(r + h) = 2\pi \times 3(3 + 7) = 60\pi \text{ cm}^2$$

**Sol.517.(b)** Ratio of surface area of two

$$\text{spheres} = 4\pi r_1^2 : 4\pi r_2^2 = r_1^2 : r_2^2$$

$$\text{Given: } \frac{r_1^2}{r_2^2} = \frac{25}{36} \Rightarrow \frac{r_1}{r_2} = \frac{5}{6}$$

Now, ratio of their volumes

$$= \frac{4}{3} \pi r_1^3 : \frac{4}{3} \pi r_2^3 = r_1^3 : r_2^3 = \frac{5^3}{6^3} = \frac{125}{216}$$

**Sol.518.(b)** Let the length, breadth and height of cuboid be  $l$ ,  $b$  and  $h$  respectively,

Let sides of cuboid be  $3x$ ,  $4x$  and  $5x$  respectively

Then, Surface area of cuboid

$$= 2(lb + bh + hl)$$

$$= 2(12x^2 + 20x^2 + 15x^2) = 2(47x^2) = 94x^2$$

Now, surface area of cube with side 6 cm

$$= 6 \times (6 \text{ unit})^2 = 216 \text{ unit}^2$$

Now, according to the question,

$$94x^2 = 216 \text{ unit}^2$$

$$x^2 = \frac{216}{94} \Rightarrow x = \sqrt{\frac{216}{94}} \approx \frac{14.69}{9.69}$$

$$= 1.51599$$

Volume of cuboid ( $lbh$ ) =  $3x \times 4x \times 5x$

$$= 60x^3 = 60 \times (1.51599)^3 \approx 210 \text{ unit}^3$$

**Sol.519.(a)** Area of the playground =

$$700 \times \text{CSA of cylinder} = 700 \times 2\pi rh$$

$$= 700 \times 2 \times \frac{22}{7} \times 49 \times 200$$

$$= 4312 \times 10^4 \text{ cm}^2$$

**Sol.520.(d)**

Curved surface area of the cone =  $\pi r l$

$$2200 = \frac{22}{7} \times 28 \times l$$

(slant height)  $l = 25 \text{ cm}$

**Sol.521.(b)** Ratio of surface area and

volume of cube =  $6a^2 : a^3$

$$= 6(3)^2 : (3)^3 = 2 : 1$$

**Sol.522.(b)**

Surface area of sphere =  $4\pi r^2$

$$4\pi r^2 = 2464 \Rightarrow 4 \times \frac{22}{7} \times r^2 = 2464$$

$$r^2 = 2464 \times \frac{7}{22} \times \frac{1}{4} = 196$$

$$r = \sqrt{196} = 14 \text{ cm}$$

$$\text{Now, volume of sphere} = \frac{4}{3} \pi r^3$$

$$= \frac{4}{3} \times \frac{22}{7} \times (14)^3 = 11498 \frac{2}{3} \text{ cm}^3$$

**Sol.523.(b)** By checking options one by one option (b) satisfied

i.e.  $(12 : 16 : 20) = (3 : 4 : 5)$

3, 4 and 5 are the pythagorean triplet so, triangle having side 12 cm, 16 cm and 20 cm is right angled triangle.

$$\text{Sol.524.(d) volume of sphere} = \frac{4}{3} \pi r^3$$

$$= \frac{4}{3} \times \frac{22}{7} \times (r)^3 = 38.808 \text{ cm}^3$$

$$(r)^3 = \frac{38.808 \times 3 \times 7}{22 \times 4}$$

$$= \frac{11 \times 8 \times 0.441 \times 3 \times 7}{11 \times 2 \times 4} = 9.261 \text{ cm}^3$$



$$r = \sqrt[3]{9.261} = \sqrt[3]{\frac{9261}{1000}} = \frac{21}{10} = 2.1 \text{ cm}$$

Hence, diameter of spherical ball ( $2r$ ) = 4.2 cm

**Sol.525.(b)** Length of arc ( $l$ ) =  $\frac{\theta}{360^\circ} 2\pi r$

$$= \frac{45^\circ}{360^\circ} \times 2 \times \pi \times 4$$

$$= \frac{1}{8} \times 2 \times \pi \times 4 = \pi \text{ unit}$$

Now, perimeter of minor sector =  $2r + l$   
 $= 2 \times 4 + \pi = (8 + \pi) \text{ unit}$

**Sol.526.(c)** Curved surface area of hemisphere =  $2\pi r^2 = 2 \times \frac{22}{7} \times \frac{21}{2} \times \frac{21}{2}$   
 $= 3 \times 11 \times 21 = 693 \text{ cm}^2$

**Sol.527.(a)**

Capacity of cylinder =  $8316 \text{ cm}^3$

$$\pi r^2 h = 8316 \Rightarrow \frac{22}{7} \times 21 \times 21 \times h = 8316$$

$$h = 8316 \times \frac{7}{22 \times 21 \times 21} = 6 \text{ cm}$$

**Sol.528.(b)**  $72.9\% = \frac{729}{1000}$

$$\frac{\text{final volume of cube } (a_1^3)}{\text{initial volume of cube } (a^3)} = \frac{729}{1000}$$

$$\frac{a_1}{a} = \frac{9}{10}$$

Hence, required % =  $\frac{1}{10} \times 100 = 10\%$

**Sol.529.(d)**

Total surface area of hemisphere =  $3\pi r^2$   
 $\Rightarrow 3 \times 3.14 \times (6.2)^2 = 362.10 \text{ cm}^2$

**Sol.530.(b)** Required % =  $5 + 5 + \frac{5 \times 5}{100}$   
 $= 10 + 0.25 = 10.25\%$

**Sol.531.(d)** Required bricks

$$= \frac{(10 \times 100) \times 22.5 \times (6 \times 100)}{25 \times 12 \times 9}$$

$$= 40 \times 2.5 \times 50 = 5000$$

**Sol.532.(a)** Weight of the empty box

$$= [(60 \times 40 \times 20) - (58 \times 38 \times 18)] \times 50$$

$$= [48000 - 39672] \times 50 = [8328] \times 50 = 416400 \text{ gm}$$

$$\text{Weight of the empty box} = \frac{416400}{1000}$$

$$= 416.4 \text{ kg}$$

**Sol.533.(d)**

$$\text{Volume of the hemisphere} = \frac{2}{3} \pi r^3$$

Volume of the hemisphere

$$= \frac{2}{3} \times \frac{22}{7} \times 21 \times 21 \times 21$$

$$\text{Volume of the hemisphere} = 19404 \text{ cm}^3$$

**Sol.534.(d)** Difference between TSA and

$$\text{CSA of cylinder} = 2\pi r(h + r) - 2\pi rh$$

$$= 2\pi r[h + r - h] = 2\pi r^2$$

$$\text{Required difference } (2\pi r^2)$$

$$= 2 \times \frac{22}{7} \times 14 \times 14 = 1232 \text{ cm}^2$$

**Sol.535.(c)** To find the maximum possible area of a triangle with a perimeter of 36 units, we assume it's an equilateral triangle.

$$\text{Perimeter} = 3a = 36, a = 12 \text{ units}$$

Area of the equilateral triangle

$$= \frac{\sqrt{3}}{4} \text{side}^2 = \frac{\sqrt{3}}{4} \times 12 \times 12$$

$$= 36\sqrt{3} \text{ square units}$$

**Sol.536.(a)** TSA of the sphere =  $4\pi r^2$

Required cost

$$= 4 \times \pi \times 5 \times 5 \times \frac{80}{\pi} = ₹ 8000$$

**Sol.537.(b)**

3, 4, 5 are triplet of right angle triangle so, Length of the perpendicular

$$= \frac{p \times b}{h} = \frac{3 \times 4}{5} = 2.4 \text{ cm}$$

**Sol.538.(c)** Volume of the cylinder =  $\pi r^2 h$

$$= \frac{22}{7} \times (6x)^2 (7x) = 792$$

$$x^3 = \frac{792 \times 7}{22 \times 36 \times 7} = 1 \text{ so } x = 1$$

So, the radius and height of the cylinder will be 6 and 7

Now, according to the question,

$$\text{CSA of the cylinder} = 2\pi rh$$

$$= 2 \times \frac{22}{7} \times 6 \times 7 = 264 \text{ cm}^2$$

**Sol.539.(c)** Required number of boxes

$$= \frac{2160}{6 \times 10 \times 12} = 3$$

**Sol.540.(c)** Volume of the cone =  $\frac{1}{3} \pi r^2 h$

$$\text{Required ratio} = 6^2 \times 5 : 5^2 \times 3 = 12 : 5$$

**Sol.541.(c)** Volume of sphere =  $\frac{4}{3} \pi r^3$

Ratio - before : after

$$\text{Radius} - 1 : 2$$

$$\text{Volume} - (1)^3 : (2)^3$$

$$= 1 : 8$$

We can clearly see that the volume will be 8 times of the previous volume.

**Sol.542.(b)** CSA of the cylinder =  $2\pi rh$

$$88 = 2 \times \frac{22}{7} \times r \times 14 \Rightarrow r = 1 \text{ cm}$$

$$\text{TSA of cylinder} = 2\pi r(h + r)$$

$$= 2 \times \frac{22}{7} \times 1(14 + 1) = \frac{660}{7} \text{ cm}^2$$

**Sol.543.(c)**

$$\text{Area of the rectangle} = \text{length} \times \text{breadth}$$

$$3600 = 45 \times \text{breadth}$$

$$\text{Breadth} = 80 \text{ m}$$

$$\text{Perimeter of the rectangular field}$$

$$= 2(\text{length} + \text{breadth}) = 2(45 + 80)$$

$$= 250 \text{ m}$$

**Sol.544.(d)**

$$\text{TSA of cuboid} = 2(lb + bh + hl)$$

$$108 = 2(6 \times 3 + 3h + 6h) \Rightarrow 54 = 18 + 9h$$

$$h = \frac{54 - 18}{9} = 4 \text{ m}$$

**Sol.545.(b)** Area of sector =  $\frac{\theta}{360} \pi r^2$

$$= \frac{45}{360} \times 3.14 \times (4)^2 = \frac{1}{8} \times 3.14 \times 16$$

$$= 6.28 \text{ cm}^2$$

**Sol.546.(b)**

$$\text{Volume of right circular cone} = \frac{1}{3} \pi r^2 h$$

According to the question,

Volume of right circular cone

$$= \frac{1}{3} \times \frac{22}{7} \times 10.5 \times 10.5 \times 24 = 2772 \text{ cm}^3$$

**Sol.547.(c)**

$$\text{Effective rate for 2 year} = x + y + \frac{x \times y}{100}$$

Effective rate for 2 year

$$= 12 + 12 + \frac{12 \times 12}{100} = 24 + 1.44 = 25.44$$

Net change in volume

$$= 25.44 + 12 + \frac{25.44 \times 12}{100}$$

$$= 37.44 + 3.0528 = 40.49 \%$$

**Sol.548.(a)** Let radius of cylinder =  $r$

Then according to question,

Height of cylinder =  $12r$

And, radius of each spherical ball =  $\frac{r}{3}$

Volume of cylindrical iron rod =

$x \times \text{volume of each spherical balls}$

$$\pi r^2 h = x \times \frac{4}{3} \pi r^3$$

$$\pi r^2 \times 12r = x \times \frac{4}{3} \pi \left(\frac{r}{3}\right)^3$$

$$12r^3 = x \times \frac{4}{3} \times \frac{r^3}{27}$$

$$x = \frac{12 \times 3 \times 27}{4} = 243$$

Hence, number of spherical ball ( $x$ ) = 243

**Sol.549.(a)**

$$\text{Area of triangle} = \frac{1}{2} \times \text{base} \times \text{height}$$

$$18 = \frac{1}{2} \times 6 \times \text{height} \Rightarrow \text{Height} = 6 \text{ cm}$$

**Sol.550.(b)** Let the sides of the

rectangular box be  $3x$ ,  $5x$  and  $8x$

TSA of cuboid =  $2(lb + bh + hl)$

$$632 = 2(15x^2 + 40x^2 + 24x^2)$$

$$\Rightarrow 316 = 79x^2 \Rightarrow x^2 = 4, x = 2$$



So the sides of the rectangular box will be 6, 10 and 16

$$\text{Volume of cuboid} = lbh \\ = 6 \times 10 \times 16 = 960 \text{ cm}^3$$

**Sol.551.(d)**

Volume of 3 cubes = volume of new cube  
Let the sides of new cubes be  $n$

$$\text{Volume of cube} = a^3 \\ 6^3 + 8^3 + 10^3 = n^3 \\ 216 + 512 + 1000 = n^3 \\ n^3 = 1728, n = 12 \text{ cm} \\ \text{TSA of new cube} = 6a^2 \\ = 6 \times 12 \times 12 = 864 \text{ cm}^2$$

**Sol.552.(a)** CSA of the cylinder =  $2\pi rh$ 

$$352 = 2 \times \frac{22}{7} \times 7 \times h \Rightarrow h = 8 \text{ cm}$$

$$\text{TSA of the cylinder} = 2\pi r(h + r)$$

$$= 2 \times \frac{22}{7} \times 7(8 + 7) = 44 \times 15 = 660 \text{ cm}^2$$

**Sol.553.(c)**  $25\% = \frac{1}{4}$ 

$$\text{TSA of the sphere} = 4\pi r^2 \\ \text{Radius} - 4 : 5$$

$$\text{TSA of the sphere} - (4)^2 : (5)^2 \\ = 16 : 25$$

$$\text{increase \%} = \frac{25 - 16}{16} \times 100 = 56.25\%$$

**Sol.554.(b)** CSA of the cylinder =  $2\pi rh$ 

$$= 2 \times \frac{22}{7} \times 3 \times 14 = 264 \text{ cm}^2$$

**Sol.555.(a)**

$$\text{Total surface area of hemisphere} = 3\pi r^2$$

$$= 3 \times \frac{22}{7} \times (14)^2 = 1848 \text{ cm}^2$$

**Sol.556.(c)** Volume of cylinder =  $\pi r^2 h$ 

$$= \frac{22}{7} \times 2.1 \times 2.1 \times 6.3 = 87.318 \text{ m}^3$$

$$\text{Now, } 1\text{m}^3 = 1000 \text{ litre}$$

$$\text{Hence, capacity of water tank} \\ = 87.318 \times 1000 = 87318 \text{ litre}$$

**Sol.557.(a)** Total surface area of cylinder

$$= 2\pi r(r + h)$$

$$= 2 \times \frac{22}{7} \times 7 \times (7 + 9) = 704 \text{ cm}^2$$

**Sol.558.(d)**

$$\text{Length of the arc} = \frac{\theta}{360} \times 2\pi r$$

$$5.5 = \frac{30}{360} \times 2 \times \frac{22}{7} \times r$$

$$r = \frac{42 \times 5.5}{22} = 10.5 \text{ cm}$$

**Sol.559.(b)** TSA of the cylinder

$$= 2\pi r(h + r) = 2 \times \frac{22}{7} \times 15(27 + 15)$$

$$= 2 \times 22 \times 15 \times 6 = 3960 \text{ cm}^2$$

$$\text{Sol.560.(d)} \text{ Volume of sphere} = \frac{4}{3}\pi r^3$$

Volume of 3 small spheres = volume of 1 big sphere

$$\frac{4}{3}\pi[(3)^3 + (4)^3 + (5)^3] = \frac{4}{3}\pi(R)^3$$

$$27 + 64 + 125 = (R)^3 \Rightarrow 216 = (6)^3 = (R)^3$$

$$R = 6 \text{ cm}$$

**Sol.561.(d)** CSA of the cylinder =  $2\pi rh$ 

$$\text{TSA of the cylinder} = 2\pi r(h + r)$$

$$\text{Required ratio} = 2\pi r(10) : 2\pi r(10 + 30)$$

$$= 10 : 40$$

$$= 1 : 4$$

**Sol.562.(b) Given :** the ratio of cuboid dimensions = 3 : 4 : 5

Let length(l) =  $3x$ , breadth(b) =  $4x$  and height(h) =  $5x$

According to question,

$$\text{Volume of cube} = l \times b \times h$$

$$\Rightarrow 3x \times 4x \times 5x = 1620 \Rightarrow 60x^3 = 1620$$

$$\Rightarrow x^3 = 27 \Rightarrow x = 3 \text{ units}$$

Total surface area of cuboid

$$= 2(lb + bh + hl) = 2(12x^2 + 20x^2 + 15x^2)$$

$$= 2 \times 47x^2 = 2 \times 47 \times 9 = 846 \text{ m}^2$$

**Sol.563.(c)** Total surface area of cylinder

$$= 2\pi r^2 + 2\pi rh$$

$$\Rightarrow 628 = 2 \times \frac{22}{7} \times 5 \times 5 + 2\pi rh$$

$$\Rightarrow 2\pi rh = 628 - \frac{1100}{7}$$

$$\Rightarrow 2\pi rh = 628 - 157.14$$

$$\Rightarrow 2\pi rh = 470.86 \text{ m}^2 \text{ or } 470.8 \text{ m}^2$$

So, curved surface area of a cylinder be  $470.8 \text{ m}^2$

**Sol.564.(a) Given :** diameter of sphere =

28 cm then radius = 14 cm

$$\text{Surface area of sphere} = 4\pi r^2$$

$$\Rightarrow \text{Surface area of sphere}$$

$$= 4 \times \frac{22}{7} \times 14 \times 14 = 2464 \text{ cm}^2$$

Now,

$$\text{Cost of painting the ball}$$

$$= 2464 \times ₹2 = ₹4928$$

**Sol.565.(b)**  $9 \text{ cm} = \frac{9}{100} \text{ m}$ 

$$4 \text{ mm} = \frac{4}{1000} \text{ m}$$

Volume of sphere = volume of cylindrical wire

$$\frac{4}{3}\pi r^3 = \pi r^2 h \Rightarrow \frac{4}{3} \times \frac{9}{100} \times \frac{9}{100} \times \frac{9}{100}$$

$$= \frac{4}{1000} \times \frac{4}{1000} \times h \Rightarrow h = 60.75 \text{ m}$$

**Sol.566.(a)**

Let the height of the triangle =  $3x$

$$\text{And the base is} = 3x \times \frac{2}{3} = 2x$$

$$\text{Area of the triangle} = \frac{1}{2} \times \text{base} \times \text{height}$$

$$363 = \frac{1}{2} \times 3x \times 2x \Rightarrow 121 = x^2, x = 11$$

$$\text{Height of the triangle } (3x) = 33 \text{ cm}$$

**Sol.567.(b)**

$$\text{Total Surface area of cube} = 6a^2$$

Ratio - initial : new

$$\text{Side} - a : 3a$$

$$\text{TSA} - a^2 : 9a^2$$

$$\text{Required \%} = \frac{9a^2 - a^2}{a^2} \times 100 = 800\%$$

**Sol.568.(c)** Total surface area of cuboid

$$= 2(lb + bh + hl)$$

$$2600 = 2(30 \times 20 + 20 \times h + h \times 30)$$

$$\Rightarrow 2600 = 2(600 + 50h)$$

$$\Rightarrow 2600 = 1200 + 100h$$

$$100h = 1400 \Rightarrow \text{height } (h) = 14 \text{ cm}$$

**Sol.569.(c)** According to question,

$$\Rightarrow \frac{\text{Area of } \triangle ABC}{\text{Area of } \triangle PQR} = \frac{AB^2}{PQ^2}$$

$$\Rightarrow \frac{490}{640} = \frac{21^2}{PQ^2} \Rightarrow \frac{49}{64} = \frac{21^2}{PQ^2}$$

Taking square root both side,

$$\Rightarrow \frac{7}{8} = \frac{21}{PQ} \Rightarrow PQ = 24 \text{ cm}$$

**Sol.570.(a) Given:**

$$\text{radius} = \frac{35}{2} \text{ cm and length} = 100 \text{ cm}$$

Surface area of a roller

$$= 2 \times \frac{22}{7} \times \frac{35}{2} \times 100 = 11000 \text{ cm}^2$$

1 revolution of roller = surface area of a roller

$$200 \text{ revolution} = 11000 \times 200$$

$$= 2200000 \text{ cm}^2 = 220 \text{ m}^2$$

$$\text{Hence, Area of the playground} = 220 \text{ m}^2$$

**Sol.571.(c)**

$$\text{Perimeter of rectangle} = 2(l + b)$$

$$70 = 2(l + b) \Rightarrow l + b = 35 \dots (i)$$

$$\text{Diagonal of the rectangle} = \sqrt{l^2 + b^2}$$

$$25 = \sqrt{l^2 + b^2} \Rightarrow l^2 + b^2 = 625 \dots (ii)$$

Now,

$$l + b = 35$$

(on square both sides)

$$(l + b)^2 = 1225 \Rightarrow l^2 + b^2 + 2lb = 1225$$

$$625 + 2lb = 1225 \quad (\text{from e.q. (ii)})$$

$$lb = \frac{1225 - 625}{2} = 300$$

Area of rectangle =  $lb$

Hence, the area of the rectangle is  $300 \text{ cm}^2$

**Sol.572.(a)** Volume of sphere =  $\frac{4}{3}\pi r^3$ 

Let the small number of spheres =  $n$

Volume of big sphere =  $n \times$  volume of small spheres

$$\frac{4}{3}\pi R^3 = n \times \frac{4}{3}\pi r^3$$



$$\Rightarrow (8)^3 = n \times (2)^3 \Rightarrow n = 64$$

**Sol.573.(b)** Surface area of sphere =  $4\pi r^2$

According to the question,

$$4\pi(r + 3.5)^2 - 4\pi(r)^2 = 550$$

$$4\pi[(r + 3.5)^2 - r^2] = 550$$

$$[(r + 3.5)^2 - r^2] = 550 \times \frac{7}{22 \times 4}$$

$$(2r + 3.5) \times 3.5 = 43.75 \Rightarrow 2r + 3.5 = 12.5$$

$$2r = 9 \Rightarrow \text{radius } (r) = 4.5 \text{ cm}$$

**Sol.574.(a)** According to the question,

$$a^2 = \frac{1}{2} \pi r^2 \Rightarrow a^2 = \frac{1}{2} \pi \times (2\sqrt{2})^2 = 4\pi$$

$$a = \sqrt{4\pi} = 2\sqrt{\pi}$$

$$\text{Perimeter of square } (4a) = 8\sqrt{\pi} \text{ cm}$$

**Sol.575.(d)** Volume of cuboid ( $lbh$ )

$$= 36 \text{ cm} \times 22 \text{ cm} \times 12 \text{ cm}$$

$$\text{Volume of each cube} = 6 \times 6 \times 6$$

So, required number of cubes

$$= \frac{36 \times 22 \times 12}{6 \times 6 \times 6} = 44$$

**Sol.576.(b)** Area of isosceles right angle

$$\text{triangle} = \left(\frac{\text{hypotenuse}}{2}\right)^2$$

$$16 = \left(\frac{\text{hypotenuse}}{2}\right)^2 \Rightarrow \text{Hypotenuse} = 8 \text{ m}$$

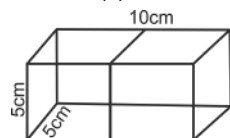
**Sol.577.(b)** Area of sphere =  $4\pi r^2$

$$= 4 \times \frac{22}{7} \times 14 \times 14 = 2464 \text{ cm}^2$$

Cost of painting a ball

$$= 2464 \times 5 = ₹12,320$$

**Sol.578.(a)**



Length of cuboid ( $l$ ) =  $(5 + 5) \text{ cm} = 10 \text{ cm}$

Breadth of cuboid ( $b$ ) =  $5 \text{ cm}$

Height of cuboid ( $h$ ) =  $5 \text{ cm}$

Surface area of cuboid =  $2(lb + bh + hl)$

$$= 2 \times (10 \times 5 + 5 \times 5 + 5 \times 10)$$

$$= 2(50 + 25 + 50) = 250 \text{ cm}^2$$

**Sol.579.(a)** Let breadth of park be  $x \text{ m}$

Length =  $(x + 15) \text{ m}$

Perimeter of rectangular park

$$= 2[(x + 15) + (x)] = 4x + 30$$

According to question,

$$4x + 30 = 110 \Rightarrow x = 20$$

Length of rectangular park

$$= x + 15 = 35 \text{ m}$$

**Sol.580.(d)**

Lateral surface area of cube =  $4a^2$

$$4a^2 = 144 \Rightarrow a = 6 \text{ cm}$$

$$\text{Volume of cube } (a^3) = 6^3 = 216 \text{ cm}^3$$

**Sol.581.(d)** Surface area of cuboid

$$= 2(lb + bh + hl) = 2(5 + 15 + 27) = 94 \text{ cm}^2$$

**Sol.582.(c)** Cylindrical tank is closed

So, total surface area cylindrical tank

$$= 2\pi r(r + h)$$

$$= 2 \times \frac{22}{7} \times 1.5 (1.5 + 3) = 42.42 \text{ m}^2$$

Hence, required metal sheet used

$$= 42.42 \text{ m}^2$$

**Sol.583.(b)**

$$\text{Sides ratio} = \sqrt[3]{1} : \sqrt[3]{125} = 1 : 5$$

$$\text{Surface area of cube} = 6a^2$$

$$\text{Required ratio} = 6 \times (1)^2 : 6 \times (5)^2 = 1 : 25$$

**Sol.584.(d)**

$$\text{Height of the cone} = \sqrt{(10)^2 - (6)^2}$$

$$= \sqrt{100 - 36} = \sqrt{64} = 8 \text{ cm}$$

$$\text{Volume of the cone} = \frac{1}{3} \pi r^2 h$$

$$= \frac{1}{3} \times \pi \times 6 \times 6 \times 8 = 96\pi \text{ cm}^3$$

$$\text{Sol.585.(d) Volume of Sphere} = \frac{4}{3} \pi r^3$$

According to the question,

$$n \times \frac{4}{3} \pi (1)^3 = \frac{4}{3} \pi (9)^3$$

Required number of balls ( $n$ )

$$= \frac{729}{1} = 729$$

**Sol.586.(d)**

As we know, 1 litre =  $1000 \text{ cm}^3$

Then, 10 litres =  $10000 \text{ cm}^3$

Let the height be  $15x$

Breadth of cuboidal water purifier

$$= \frac{1}{5} \times 15x = 3x$$

Length of cuboidal water purifier

$$= \frac{1}{3} (3x + 15x) = 6x$$

Now, volume of cuboidal water purifier

$$= 3x \times 6x \times 15x = 270x^3 \Rightarrow 10000 = 270x^3$$

$$\frac{1000}{27} = x^3 \Rightarrow x = \frac{10}{3}$$

So, the height of cuboidal water purifier

$$= 15 \times \frac{10}{3} = 50 \text{ cm}$$

**Sol.587.(a)**

Total surface area of cube =  $6a^2$

$$\text{Required ratio} = 6 \times (2)^2 : 6 \times (4)^2 = 1 : 4$$

**Sol.588.(c)**

Curved surface area of cone =  $\pi r l$

According to the question,

$$\pi \times 4 \times l = 20\pi$$

$$\text{Slant height } (l) = 5 \text{ cm}$$

$$\text{Height of cone } (h) = \sqrt{l^2 - r^2}$$

$$= \sqrt{(5)^2 - (4)^2} = \sqrt{9} = 3 \text{ cm}$$

$$\text{Now, volume of cone} = \frac{1}{3} \pi r^2 h$$

$$\Rightarrow \frac{1}{3} \times \pi \times (4)^2 \times 3 = 16\pi \text{ cm}^3$$

**Sol.589.(b)** Let the radius of outer circle

be  $R$  and inner circle be  $r$ .

According to the question,

$$2\pi R : 2\pi r = 7 : 6 \Rightarrow R : r = 7 : 6$$

Difference  $(7 - 6) = 1$  unit

1 unit =  $45 \text{ m}$

Hence, radius of inner circle (6 unit)

$$= 45 \times 6 = 270 \text{ m}$$

**Sol.590.(d)**

Let the no. of smaller cube be  $n$

Surface area of larger cube =  $6a_1^2 = 54$

$$a_1^2 = 9 \Rightarrow a_1 = 3 \text{ cm}$$

Surface area of smaller cube

$$= 6a_2^2 = \frac{8}{12} \Rightarrow a_2^2 = \frac{8}{12 \times 6} = \frac{1}{9}$$

$$\Rightarrow a_2 = \frac{1}{3}$$

Volume of larger cube =  $n \times$  volume of smaller cube

$$a_1^3 = n \times a_2^3 \Rightarrow 3^3 = n \times \frac{1}{3^3}$$

$$27 = \frac{n}{27} \Rightarrow n = 729$$

**Sol.591.(d)** Length of cuboid =  $60 \text{ cm}$

$$\text{Breadth of cuboid} = 60 \times \frac{1}{2} = 30 \text{ cm}$$

$$\text{Height of cuboid} = \frac{1}{3} \times 30 = 10 \text{ cm}$$

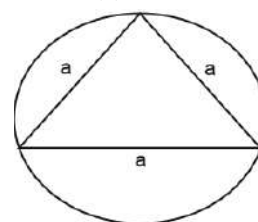
TSA of cardboard box =  $2(lb + bh + hl)$

$$= 2(60 \times 30 + 30 \times 10 + 10 \times 60)$$

$$= 2(1800 + 300 + 600) = 2 \times 2700$$

$$= 5400 \text{ cm}^2$$

**Sol.592.(c)**



$$\text{Circumradius } (R) = \frac{a}{\sqrt{3}}, \text{ Radius } (r) = r$$

$$\text{Given that, } \frac{a}{\sqrt{3}} = r \Rightarrow a = \sqrt{3}r \text{ unit}$$

**Sol.593.(c)**

Area of rhombus = side  $\times$  height

$$70 = \text{side} \times 5 \Rightarrow \text{Side} = \frac{70}{5} = 14 \text{ cm}$$

So, the perimeter of rhombus

$$= 14 \times 4 = 56 \text{ cm}$$

**Sol.594.(c)**

$$\text{Median of equilateral triangle} = \frac{\sqrt{3}}{2} a$$

$$\text{Perimeter of equilateral triangle} = 3a$$

$$\text{Required ratio} = 3a : \frac{\sqrt{3}}{2} \times 3a = 2 : \sqrt{3}$$

**Sol.595.(c)**

$$\text{Inner radius of bowl} = \frac{4}{2} = 2 \text{ cm}$$

$$\text{Outer radius of bowl} = 2 + 0.5 = 2.5 \text{ cm}$$



Required volume of silver used in making

$$\begin{aligned} \text{the bowl} &= \frac{2}{3}\pi(R^3 - r^3) \\ &= \frac{2}{3} \times 3.14 \times [(2.5)^3 - (2)^3] \\ &= \frac{2}{3} \times 3.14 \times 7.625 = \frac{47.885}{3} \\ &= 15.96 \text{ cm}^3 \end{aligned}$$

**Sol.596.(c)**

Length =  $4x$ , Breadth =  $3x$ , Height =  $2x$

Volume = 12288  $\Rightarrow l \times b \times h = 12288$

$$4x \times 3x \times 2x = 12288$$

$$24x^3 = 12288 \Rightarrow x^3 = 512 \Rightarrow x = 8$$

Hence,

$$\text{Length} = 4x = 4 \times 8 = 32 \text{ m}$$

$$\text{Breadth} = 3x = 3 \times 8 = 24 \text{ m}$$

$$\text{Height} = 2x = 2 \times 8 = 16 \text{ m}$$

Total surface area of box =  $2(lb + bh + hl)$

$$= 2(4x \times 3x + 3x \times 2x + 2x \times 4x)$$

$$= 2(12x^2 + 6x^2 + 8x^2) = 2(26x^2)$$

$$= 52 \times 8^2 = 52 \times 64 = 3328 \text{ m}^2$$

Total cost of polishing =  $2 \times 3328$

$$= 6656 \text{ paise or Rs. } 66.56$$

**Sol.597.(c)** Let no. of person be  $x$

Perimeter of brim of hemispherical bowl

$$= 2\pi r = 88 \Rightarrow 2 \times \frac{22}{7} \times r = 88 \Rightarrow \frac{44}{7}r = 88$$

$$r = \frac{88 \times 7}{44} = 14 \text{ cm}$$

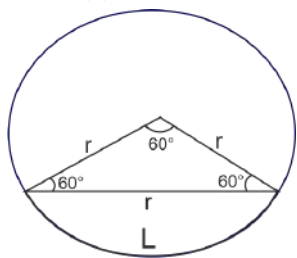
Volume of hemispherical bowl =

$x \times$  Volume of hemispherical glass

$$\frac{2}{3} \times \pi \times 14^3 = x \times \frac{2}{3} \times \pi \times \left(\frac{7}{2}\right)^3$$

$$2744 = \frac{343}{8}x \Rightarrow x = \frac{2744 \times 8}{343} = 64$$

**Sol.598.(b)**



$$\text{Length of the arc}(L) = \frac{\theta}{360^\circ} \times 2\pi r$$

$$= \frac{60^\circ}{360^\circ} \times 2 \times \frac{22}{7} \times 21 = 22 \text{ units}$$

**Sol.599.(a)** Area of parallelogram

= Base( $b$ )  $\times$  Height( $h$ )

According to the question,

$$b = 2h \Rightarrow 338 = 2h \times h$$

$$\Rightarrow 338 = 2h^2 \Rightarrow h^2 = 169 \Rightarrow h = 13 \text{ cm}$$

**Sol.600.(b)** TSA of cone =  $\pi R(L + R)$

$$= \frac{\pi r}{4} \left(4l + \frac{r}{4}\right)$$

$$= \frac{\pi r}{4} \times 4 \left(l + \frac{r}{16}\right) = \pi r \left(l + \frac{r}{16}\right)$$

**Sol.601.(b)** CSA of conical tent =  $\pi r l$

Let the required length of cloth be  $x$  m

$$\text{Slant height of tent}(l) = \sqrt{r^2 + h^2}$$

$$= \sqrt{(21)^2 + (28)^2} = \sqrt{1225} = 35 \text{ m}$$

Area of cloth = CSA of conical tent

$$5 \times x = \frac{22}{7} \times 21 \times 35 \Rightarrow 5x = 2310$$

$$x = \frac{2310}{5} = 462 \text{ m}$$

**Sol.602.(b)**

Volume of cube = 4  $\times$  volume of cuboid

$$a^3 = 4 \times 32 \times 8 \times 4 = 4096 \Rightarrow a = 16$$

Now,  $\frac{\text{TSA of cube}}{\text{TSA of cuboid}}$

$$= \frac{6 \times 16 \times 16}{2(32 \times 8 + 8 \times 4 + 32 \times 4)}$$

$$= \frac{6 \times 16 \times 16}{832} = \frac{24}{13} = 24 : 13$$

**Sol.603.(a)**

$$\text{Radius of copper sphere} = \frac{12}{2} = 6 \text{ cm}$$

$$\text{Radius of wire} = \frac{4}{2} = 2 \text{ mm} = \frac{2}{10}$$

$$= 0.2 \text{ cm}$$

volume of sphere = volume of wire

$$\frac{4}{3} \pi \times 6^3 = \pi \times 0.2^2 \times h$$

$$288 = 0.04h \Rightarrow h = \frac{288}{0.04} = 7200 \text{ cm}$$

**Sol.604.(d)** Length of the largest

possible rod which can be placed in

$$\text{cubical room} = \sqrt{3}a \Rightarrow 42\sqrt{3} = \sqrt{3}a$$

$$\Rightarrow a = 42 \text{ m}$$

Required TSA of sphere

$$= 4 \times \frac{22}{7} \times \left(\frac{42}{2}\right)^2 = 5544 \text{ m}^2$$

**Sol.605.(a)** circumference of circle

$$= 2 \times \frac{22}{7} \times r \Rightarrow 88 = \frac{44}{7}r$$

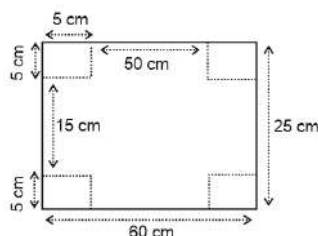
$$r = \frac{88 \times 7}{44} = 14 \text{ cm}$$

$$\text{Area of a minor sector} = \frac{1}{2} \times l r$$

( $l$  = minor arc length)

$$= \frac{1}{2} \times 22 \times 14 = 154 \text{ cm}^2$$

**Sol.606.(c)**



Length of box = 50 cm

Breadth of box = 15 cm

Height of box = 5 cm

$$\text{Volume of box} = l \times b \times h = 50 \times 15 \times 5 = 3750 \text{ cm}^3$$

**Sol.607.(b)**

$$\text{Volume of cylindrical shell} = \pi(R^2 - r^2)h$$

$$= \frac{22}{7} \times (7^2 - 4^2) \times 42$$

$$= 22 \times (49 - 16) \times 6$$

$$= 22 \times 33 \times 6 = 4356 \text{ cm}^3$$

**Sol.608.(d)**

$$\text{Volume of laddoo of radius } 5 \text{ cm} = \frac{4}{3}$$

$$\pi R^3$$

Volume of laddoo of radius 2.5 cm

$$= \frac{4}{3} \pi r^3$$

$$\text{No. of laddoos created} = \frac{\frac{4}{3} \pi R^3}{\frac{4}{3} \pi r^3}$$

$$= \frac{5 \times 5 \times 5}{2.5 \times 2.5 \times 2.5} = 8 \text{ laddoos}$$

**Sol.609.(d)** T.S.A. of spherical vessel

$$= 4\pi r^2 = 4 \times \frac{22}{7} \times 7 \times 7 = 616 \text{ cm}^2$$

Cost per square centimeter

$$= \frac{\text{total cost}}{\text{T.S.A.}} = \frac{21,560}{616} = 35 \text{ Rs.}$$

**Sol.610.(c)**  $\frac{\text{Volume of cone}}{\text{Volume of cylinder}}$

$$= \frac{\frac{1}{3} \pi r^2 h}{\pi R^2 H} = \frac{\frac{1}{3} \times 5 \times 5 \times 6}{2.5 \times 2.5 \times 3} = \frac{8}{3} = 8 : 3$$

**Sol.611.(a)** C.S.A. of cylinder = 2200

$$2\pi r h = 2200$$

$$2 \times \frac{22}{7} \times 7 \times h = 2200$$

$$\Rightarrow h = \frac{2200}{44} = 50 \text{ cm}$$

T.S.A. of cylinder =  $2\pi r(h + r)$

$$= 2 \times \frac{22}{7} \times 7(50 + 7) = 44 \times 57$$

$$= 2508 \text{ cm}^2$$

**Sol.612.(d)** Let  $R$  be external radius and  $r$  be internal radius of a hollow sphere and radius of a cone be  $r_1$

According to question,

Volume of cone = volume of hollow cylinder

$$\frac{1}{3} \pi r_1^2 h = \frac{4}{3} \pi (R^3 - r^3)$$

$$5^2 \times h = 4(5^3 - 3^3)$$

$$25 \times h = 4 \times 98$$

$$h = \frac{4 \times 98}{25} = 15.68 \text{ cm}$$

**Sol.613.(b)**

Surface area of spherical box = 528 cm<sup>2</sup>

$$4\pi r^2 = 528 \text{ cm}^2$$

$$4 \times \frac{22}{7} \times r^2 = 528 \text{ cm}^2$$

$$r^2 = \frac{528 \times 7}{4 \times 22} = 42 \Rightarrow r = \sqrt{42}$$

$$\text{Hence, radius} = \sqrt{42} \text{ cm}$$

**Sol.614.(d)** Area of trapezium =

$$\frac{1}{2} \times \text{sum of parallel sides} \times \text{height}$$

$$\Rightarrow 1188 = \frac{1}{2} \times \text{sum of parallel sides} \times 18$$

$$\text{sum of parallel sides} = 132 \text{ cm}$$

**Sol.615.(b)** Circumference of circular base of cone =  $30\pi$  ft.

$$2\pi r = 30\pi \text{ ft.} \Rightarrow r = 15 \text{ ft.}$$

Height ( $h$ ) of cone = 20 ft.

$$\text{Slant height of cone } (l) = \sqrt{r^2 + h^2}$$

$$\text{Slant height of cone } (l) = \sqrt{225 + 400} = 25 \text{ ft.}$$

$$\text{Area of canvas needed} = \pi \times 15 \times 25 = 375\pi \text{ sq. ft.}$$

**Sol.616.(a)** Volume of cone =  $150\pi$

$$\frac{1}{3} \times \pi \times r^2 \times 18 = 150\pi$$

$$r^2 = 25 \Rightarrow r = 5$$

$$\text{Slant height } (l) = \sqrt{h^2 + r^2} = \sqrt{18^2 + 5^2}$$

$$= \sqrt{324 + 25} = \sqrt{349} = 18.68 \text{ cm}$$

**Sol.617.(d)** If both playgrounds are similar then their dimension must be in proportion.

$$\text{First playground} = \frac{\text{length}}{\text{breadth}} = \frac{224}{160} = \frac{7}{5}$$

By checking option one by one we find that option (d) does not have the same ratio of length and breadth as the first playground.

**Sol.618.(a)** Volume of storage tank

$$= \frac{1}{3} \pi r^2 h = \frac{1}{3} \times \frac{22}{7} \times 7 \times 7 \times 9$$

$$= 21 \times 22 = 462 \text{ cubic ft.}$$

**Sol.619.(b)**

Ratio  $\rightarrow$  length : breadth : height

$$\begin{array}{ccc} 2 & : & 1 \\ & & 2 : 1 \end{array}$$

$$\text{Final} \rightarrow 4 : 2 : 1$$

$$\text{Volume } (8 \text{ m}^3) = 1728 \text{ m}^3$$

$$1 \text{ m}^3 = \frac{1728}{8} = 216 \Rightarrow 1 \text{ unit} = 6$$

$$\text{Length } (4 \text{ m}) = 6 \times 4 = 24 \text{ m}$$

**Sol.620.(b)** Number of candles

$$= \frac{\text{volume of cuboidal solid}}{\text{volume of cylindrical candle}}$$

$$= \frac{70 \times 44 \times 20}{\frac{22}{7} \times 7 \times 7 \times 20} = 20$$

**Sol.621.(a)** Lateral surface area =  $\pi r l$

$$13.2 = \frac{22}{7} \times r \times 4.2$$

$$13.2 = 22 \times r \times 0.6 \Rightarrow r = \frac{13.2}{13.2} = 1 \text{ unit}$$

**Sol.622.(d)** Area of triangle = inradius  $\times$  semiperimeter

$$= 8 \times \frac{125}{2} = 500 \text{ cm}^2$$

**Sol.623.(c)** Volume of metal ball

= volume of (27 identical dice + a ball)

$$\frac{4}{3} \pi R^3 = 27 \times \text{side}^3 + \frac{4}{3} \pi r^3$$

$$\frac{4}{3} \pi R^3 - \frac{4}{3} \pi r^3 = 27 \times \text{side}^3$$

$$\frac{4}{3} \times \frac{22}{7} (27 - 1) = 27 \times \text{side}^3$$

$$\frac{4}{3} \times \frac{22}{7} \times 26 = 27 \times \text{side}^3$$

$$\text{side}^3 = \frac{2288}{21 \times 27} \Rightarrow \text{side}^3 = 4.035 \text{ (approx)}$$

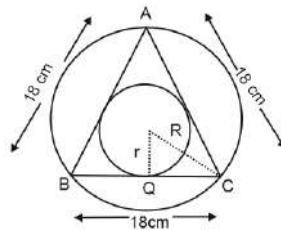
$$\text{Side} = 1.6 \text{ units (approx)}$$

**Sol.624.(c)**  $l : h = 4 : 3$

$$\text{C.S.A. of cone} = 4\sqrt{7}\pi$$

$$\pi r l = 4\sqrt{7}\pi \Rightarrow r \times 4 = 4\sqrt{7} \Rightarrow r = \sqrt{7} \text{ unit}$$

**Sol.625.(d)**



$$\text{Area of equilateral triangle} = \frac{\sqrt{3}}{4} \times a^2$$

$$= \frac{\sqrt{3}}{4} \times 18 \times 18 = 81\sqrt{3} \text{ cm}^2$$

Relation between area of circumcircle, equilateral triangle and incircle

Circumcircle : equilateral triangle :

$$\text{incircle} = 4\pi : 3\sqrt{3} : \pi$$

$$\text{Area of circumcircle} - \text{area of incircle}$$

$$= 4\pi - \pi = 3\pi \Rightarrow 3\sqrt{3} \text{ unit} = 81\sqrt{3} \text{ cm}^2$$

$$1 \text{ unit} = 27 \text{ cm}^2$$

$$3\pi \text{ unit} = \frac{27}{1} \times 3 \times \frac{22}{7} = 254\frac{4}{7} \text{ cm}^2$$

**Sol.626.(d)** Area of circle =  $256\pi \text{ cm}^2$

$$\pi r^2 = 256\pi \Rightarrow r = 16 \text{ cm}$$

$$\text{Diameter} = 2 \times r = 2 \times 16 = 32 \text{ cm}$$

Diameter is the longest chord,

Hence length of longest chord = 32 cm

**Sol.627.(c)** Perimeter of one face of

$$\text{cube } (4a) = 40 \Rightarrow a = 10$$

$$\text{Volume of cube} = a^3 = 10^3 = 1000 \text{ cm}^3$$

$$\text{Sol.628.(c)} \text{ Area of triangle} = \frac{1}{2} \times 15^2 \times$$

$$\sin 30^\circ = \frac{225}{2} \times \frac{1}{2} = \frac{225}{4} = 56.25 \text{ cm}^2$$

$$\text{Sol.629.(a)} \text{ Radius of cone} = \frac{14}{2} = 7 \text{ cm}$$

$$\text{LSA of cone } (\pi r l) = 550$$

$$\Rightarrow \frac{22}{7} \times 7 \times l = 550 \Rightarrow 22l = 550$$

$$\Rightarrow l = \frac{550}{22} = 25$$

Height of the cone

$$= \sqrt{25^2 - 7^2} = \sqrt{576} = 24 \text{ cm}$$

**Sol.630.(c)**

$$\text{Volume of hemisphere} = \frac{2}{3} \pi r^3$$

$$\Rightarrow V = \frac{2}{3} \times \pi \times (9)^3 \Rightarrow V = 486 \pi \text{ cm}^3$$

**Sol.631.(c) Given,**

Height of cylinder = 9 cm

the base area =  $25\pi \text{ cm}^2$

$$\Rightarrow \pi r^2 = 25\pi \text{ cm}^2 \Rightarrow r = 5 \text{ cm}$$

Now,

$$\text{Total Surface Area} = 2\pi r h + 2\pi r^2$$

TSA of cylinder

$$= 2 \times \frac{22}{7} \times 5 \times 9 + 2 \times \frac{22}{7} \times 25$$

$$= 2 \times \frac{22}{7} \times 5 (9 + 5)$$

$$= 2 \times \frac{22}{7} \times 5 \times 14 = 440 \text{ cm}^2$$

**Sol.632.(b)** Let height of the first cone =  $3h$

And height of the second cone =  $h$

Radius of first cone =  $r$

And radius of the second cone =  $2r$

Total volume ( $V$ ) = volume of first cone + volume of second cone

$$\Rightarrow V = \frac{1}{3} \pi r^2 3h + \frac{1}{3} \pi 4r^2 h$$

$$\Rightarrow 100 = \frac{1}{3} \pi (r^2 \cdot 3h + 4r^2 h)$$

$$\Rightarrow 100 = \frac{1}{3} \pi (7r^2 h) \Rightarrow r^2 h = \frac{300}{7\pi}$$

Now,

Difference between the volume of cones

$$= \frac{1}{3} \pi (4r^2 \cdot h - r^2 \cdot 3h)$$

$$\Rightarrow V = \frac{1}{3} \pi (r^2 h) \Rightarrow V = \frac{1}{3} \pi \cdot \frac{300}{7\pi}$$

$$= \frac{100}{7} = 14.28 \approx 14.3 \text{ unit}^3$$

**Sol.633.(a)**

Surface area of cube = 5400 square units

$$\Rightarrow 6a^2 = 5400 \Rightarrow a^2 = 900 \Rightarrow a = 30 \text{ unit}$$

$$\text{Volume of cube} = a^3 = 27000 \text{ Unit}^3$$

Now, New side =  $30 + 5 = 35 \text{ unit}$

New Volume of cube

$$= 35 \times 35 \times 35 = 42875 \text{ Unit}^3$$

$$\text{Increment in volume} = \frac{42875 - 27000}{27000} \times$$

$$100 = \frac{15875}{27000} \times 100 = 58.79 \approx 59 \%$$

**Sol.634.(d)** Let radius =  $r \text{ cm}$

$$\Rightarrow \text{Surface Area of sphere} = 4\pi r^2 \text{ cm}^2$$

New radius =  $r + 2 \text{ cm}$

New surface area of sphere

$$= 4\pi (r + 2)^2 \text{ cm}^2$$

According to question,

$$4\pi (r + 2)^2 - 4\pi r^2 = 704$$



$$\Rightarrow 4 \times \frac{22}{7} (r^2 + 4 + 4r - r^2) = 704$$

$$\Rightarrow 4 + 4r = 56 \Rightarrow 4r = 52 \Rightarrow r = 13 \text{ cm}$$

**Sol.635.(a)**

$$10\% = \frac{1}{10}, 20\% = \frac{1}{5}, 40\% = \frac{2}{5}$$

|         | Original | New |
|---------|----------|-----|
| Length  | 10       | 9   |
| Breadth | 5        | 4   |
| Height  | 5        | 7   |

$$\text{Volume } 250 : 252$$

$$\text{Change in volume} = \frac{252 - 250}{250} \times 100$$

$$= \frac{200}{250} = 0.8\% \text{ increase}$$

$$\text{Sol.636.(b) Radius of cone} = \frac{14}{2} = 7 \text{ m}$$

Volume of cone

$$= \frac{1}{3} \times \frac{22}{7} \times 7^2 \times 12 = 616 \text{ m}^3$$

Required time taken to fill the cone

$$= \frac{616}{2} \times 20 = 6160 \text{ sec}$$

**Sol.637.(b)** Volume of a cuboid =

2 × Volume of a cube

$$12 \times 16 \times 18 = 2 \times \text{Volume of a cube}$$

$$3,456 = 2 \times \text{Volume of a cube}$$

$$\text{Volume of a cube} = 1728$$

$$a^3 = 1728 \Rightarrow a = 12$$

$$\text{TSA of cube} = 6a^2 = 6 \times 12^2 = 864 \text{ cm}^2$$

**Sol.638.(c)** ATQ,

$$h = 4r$$

$$\frac{\text{CSA of cylinder}}{\text{Base area of cylinder}} = \frac{2\pi rh}{\pi r^2} = \frac{2\pi r \times 4r}{\pi r^2} = \frac{8}{1}$$

So, the curved surface area is 8 times its base area.

**Sol.639.(a)** According to the question,

$$\text{Lateral surface area of cylinder} = 2\pi rh$$

$$3862.2 = 2 \times 3.14 \times r \times 15$$

$$3862.2 = 94.2r \Rightarrow r = 41 \text{ cm}$$

$$\text{Volume of cylinder} = \pi r^2 h$$

$$= 3.14 \times 41^2 \times 15 = 79,175.1 \text{ cm}^3$$

$$\text{Sol.640.(a)} L = 2B \text{ and } H = \frac{1}{2}B$$

Volume of cubical shaped cold storage

$$= L \times B \times H$$

$$\Rightarrow 216 = 2B \times B \times \frac{1}{2}B = B^3 \Rightarrow B = 6 \text{ m}$$

$$\text{Then, } L = 2 \times 6 = 12 \text{ m, } H = \frac{1}{2} \times 6 = 3 \text{ m}$$

So, the dimensions are 12 m × 6 m × 3 m

$$\text{Sol.641.(d) Volume of cylinder} = \pi r^2 h$$

$$\text{CSA of cylinder} = 2\pi rh$$

Let  $x$  be multiplied to obtain volume of cylinder

According to the question,

$$2\pi rh \times x = \pi r^2 h \Rightarrow x = \frac{\pi r^2 h}{2\pi rh} = \frac{r}{2}$$

**Sol.642.(a)** Required no of cubes

$$= \frac{\text{volume of cuboid}}{\text{volume of cube}}$$

$$= \frac{500 \times 200 \times 500}{50 \times 50 \times 50} = 400$$

$$\text{Sol.643.(c) Perpendicular} = \sqrt{17^2 - 15^2}$$

$$= \sqrt{289 - 225} = \sqrt{64} = 8 \text{ cm}$$

Area of right angle triangle

$$= \frac{1}{2} \times 8 \times 15 = 60 \text{ cm}^2$$

**Sol.644.(c)**Let length of rectangle =  $l$  meterBreadth of rectangle =  $b$  meter

According to question,

$$\frac{l+6}{b+6} = \frac{4}{3} \Rightarrow 3l - 4b = 6 \text{ ----- (i)}$$

And

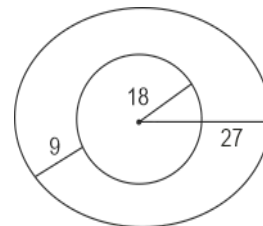
$$\frac{l-6}{b-6} = \frac{2}{1} \Rightarrow l - 2b = -6 \text{ ----- (ii)}$$

On solving equations (i) and (ii) we get,

$$l = 18 \text{ meter, } b = 12 \text{ meter}$$

**Sol.645.(c)** Increment in area of circle

$$= 7 + 7 + \frac{7 \times 7}{100} = 14.49\%$$

**Sol.646.(b)**

Area of the closed figure

$$= \pi(27)^2 - \pi(18)^2$$

$$\Rightarrow \text{Area} = \pi(729 - 324) = 405\pi \text{ cm}^2$$

**Sol.647.(c)** Required area of hemisphere

$$= 2\pi r^2 + 2\pi r^2 \Rightarrow A = 4\pi r^2$$

$$\Rightarrow A = 4 \times \frac{22}{7} \times 21 \times 21 = 5544$$

Hence, cost of painting

$$= 5544 \times 0.05 = 277.2 \text{ Rs.}$$

**Sol.648.(b)**

$$\text{Area of a regular pentagon} = 5 \times \frac{\sqrt{3}}{4} a^2$$

$$\Rightarrow 3920\sqrt{3} = 5 \times \frac{\sqrt{3}}{4} \times a^2$$

$$\Rightarrow a^2 = 3136 \Rightarrow a = 56 \text{ cm}$$

**Sol.649.(a)** Given,Base ( $b$ ) =  $3H$ , Height ( $h$ ) =  $H$ 

$$\text{Area of parallelogram} = 1020 \text{ cm}^2$$

$$\Rightarrow b \times h = 1020 \Rightarrow 3H \times H = 1020$$

$$\Rightarrow H^2 = 340 \Rightarrow H = 2\sqrt{85} \text{ cm}$$

**Sol.650.(a)** According to the question,

Surface area of sphere = Surface area of cube

$$\Rightarrow 4\pi r^2 = 6a^2 \Rightarrow 2\pi r^2 = 3a^2$$

Take square root both side,

$$\Rightarrow \sqrt{2\pi} r = \sqrt{3} a \text{ ....(i)}$$

$$\text{Volume of sphere} = \frac{4}{3}\pi r^3$$

$$\text{Volume of cube} = a^3$$

Required ratio

$$= \frac{\frac{4}{3}\pi r^3}{a^3} = \frac{\frac{4}{3}\pi r^3}{\left(\frac{\sqrt{2\pi} r}{\sqrt{3}}\right)^3} \text{ [from eq. (i)]}$$

$$= \frac{\sqrt{6}}{\sqrt{\pi}}$$

**Sol.651.(d)**Surface area of a sphere = 2464 cm<sup>2</sup>

$$\Rightarrow 4\pi r^2 = 2464 \Rightarrow 4 \times \frac{22}{7} r^2 = 2464$$

$$\Rightarrow r^2 = 196 \Rightarrow r = 14 \text{ cm}$$

Hence,

$$\text{Volume} = \frac{4}{3}\pi r^3$$

$$= \frac{4}{3} \times \frac{22}{7} \times 14 \times 14 \times 14$$

$$= 11498.67 \text{ cm}^3$$

**Sol.652.(b)**

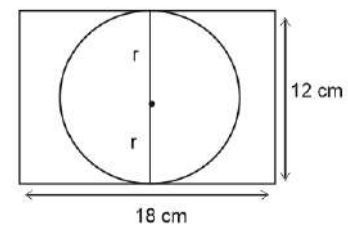
$$\text{As we know, } \left(\frac{D_1}{2}\right)^2 + \left(\frac{D_2}{2}\right)^2 = \text{side}^2$$

$$12^2 + \left(\frac{D_2}{2}\right)^2 = 13^2 \Rightarrow \left(\frac{D_2}{2}\right)^2 =$$

$$= \sqrt{13^2 - 12^2} = 5 \Rightarrow D_2 = 5 \times 2 = 10 \text{ cm}$$

$$\text{Now, Area of rhombus} = \frac{1}{2} \times D_1 \times D_2$$

$$= \frac{1}{2} \times 24 \times 10 = 120 \text{ cm}^2$$

**Sol.653.(d)**

$$\text{Maximum radius (r)} = \frac{12}{2} = 6 \text{ cm}$$

$$\text{Area of circle} = \pi r^2 = \pi \times 6^2 = 36\pi \text{ cm}^2$$

**Sol.654.(c)** LSA of cube =  $4a^2$ 

$$\text{TSA of cube} = 6a^2$$

$$\text{ATQ, } 6a^2 - 4a^2 = 2a^2 = 2 \times 11^2 = 242 \text{ cm}^2$$

$$\text{Sol.655.(d) Volume of cone} = \frac{1}{3}\pi r^2 h$$

$$\text{Volume of cone} \propto r^2 h$$

$$\left(\text{As } \frac{1}{3}\pi \text{ is constant}\right)$$

According to the question,

$$\text{Volume of cone} = \text{Radius}^2 \times \text{Height}$$



$$\begin{aligned}\text{Original} &\rightarrow 100 = 5^2 \times 4 \\ \text{New} &\rightarrow 108 = 6^2 \times 3 \\ \text{\% Change in volume} \\ &= \frac{108 - 100}{100} \times 100 = 8\%\end{aligned}$$

**Sol.656.(b)**

Let the width of rectangular garden be  $x$   
Then, length =  $(x + 8)$  cm  
Perimeter of garden =  $42 \times 2 = 84$  cm  
 $2(x + x + 8) = 84$   
 $2(2x + 8) = 84 \Rightarrow 2x + 8 = 42$   
 $x + 4 = 21 \Rightarrow x = 17$   
So, the area of rectangular garden  
 $= 17 \times 25 = 425 \text{ cm}^2$

**Sol.657.(c)** Volume of cube =  $64 \text{ cm}^3$ 

$$a^3 = 64 \Rightarrow a = 4$$

$$\text{TSA of cube} = 6a^2 = 6 \times 4^2 = 96 \text{ cm}^2$$

**Sol.658.(b)**

Before : After

Ratio of radius = 10 : 13

$$\therefore \text{Ratio of volume} = (10)^3 : (13)^3$$

$$= 1000 : 2197$$

% increase in volume

$$= \frac{2197 - 1000}{1000} \times 100 = 119.7\%$$

**Sol.659.(b)**

$$\text{Diameter} = 21 \text{ cm, Radius} = \frac{21}{2} \text{ cm}$$

$$\text{Volume of hemispherical bowl} = \frac{2}{3} \pi r^3$$

$$= \frac{2}{3} \times \frac{22}{7} \times \frac{21}{2} \times \frac{21}{2} \times \frac{21}{2}$$

$$= 2425.5 \text{ cm}^3$$

**Sol.660.(b)** Before : After

$$\text{Radius} \rightarrow 4^2 : 5^2$$

$$\text{Height} \rightarrow 4 : 5$$

$$\text{Volume} \rightarrow 64 : 125$$

$$\begin{aligned}\text{\% increase in volume} &= \frac{125 - 64}{64} \times 100 \\ &= 95.31\%\end{aligned}$$

**Sol.661.(d)** Volume of cube =  $a^3$ 

$$a^3 = 512 \text{ cubic unit}$$

$$\therefore \text{side of cube (a)} = 8 \text{ unit}$$

$$\begin{aligned}\text{Now, surface area of cube} &= 6 \times a^2 \\ &= 6 \times 8 \times 8 = 384 \text{ square unit}\end{aligned}$$

**Sol.662.(d)**

Let the total number of packed box =  $x$

$\therefore$  volume of big box = volume of small box  $\times x$

$$160 \times 90 \times 60 = 6 \times 5 \times 4 \times x$$

$$\therefore x = 7200$$

Hence no. of packed box = 7200

**Sol.663.(c)**

Let the length =  $6x$ , width =  $3x$ , height =  $2x$   
ATQ,

$$2(LB + BH + HL) = 1800$$

$$2(6x \times 3x + 3x \times 2x + 2x \times 6x) = 1800$$

$$2 \times 36x^2 = 1800 \Rightarrow x^2 = 25 \Rightarrow x = 5$$

$$\begin{aligned}\text{Now, volume of cuboid} &= L \times B \times H = 6x \times \\ &= 3x \times 2x = 36x^3 = 36 \times 5^3 = 4500\end{aligned}$$

**Sol.664.(a)**

Let the no of small cubes be  $n$

ATQ,

Volume of solid cube = Volume of new cube formed

$$48 \times 48 \times 48 = n \times 64$$

$$n = \frac{48 \times 48 \times 48}{64} = 1728$$

**Sol.665.(b)** T.S.A. of cylinder =  $2\pi r(h + r)$ 

$$\frac{18,425}{25} = 2 \times \frac{22}{7} \times r(30 + r)$$

$$\frac{737 \times 7}{11} = 4 \times r(30 + r)$$

$$469 = 4r^2 + 120r$$

$$4r^2 + 120r - 469 = 0$$

$$4r^2 + 134r - 14r - 469 = 0$$

$$2r(2r + 67) - 7(2r + 67) = 0$$

$$(2r + 67)(2r - 7) = 0$$

$$r = -\frac{67}{2} \text{ or } r = \frac{7}{2} \text{ m}$$

Radius cannot be negative

$$\text{therefore, radius (r)} = \frac{7}{2}$$

$$\text{Volume of cylinder} = \pi r^2 h$$

$$= \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \times 30$$

$$\text{Volume of cylinder} = 1155 \text{ m}^3$$

**Sol.666.(c)** According to the question,

T.S.A of hemisphere - C.S.A of hemisphere

$$3\pi r^2 - 2\pi r^2 = \pi r^2 = 4\pi \text{ cm}^2$$

**Sol.667.(a)** According to the question,

C.S.A of cone =  $\pi r l$

$$\text{Required cost} = \frac{22}{7} \times 21 \times 25 \times 1.5$$

$$= 2475 \text{ Rs}$$

**Sol.668.(b)**

Let the number of hemisphere =  $n$

Volume of sphere =  $n \times$  volume of hemisphere

$$\frac{4}{3} \pi \times 12 \times 12 \times 12 = n \times \frac{2}{3} \pi \times 4 \times 4 \times 4$$

$$n = 54$$

**Sol.669.(d)** Given, Perimeter of rhombus

$$= 52 \text{ cm and diagonal (D1)} = 10 \text{ cm}$$

$$\text{Then, side of the rhombus} = 13 \text{ cm}$$

$$\text{Now, } D_1^2 + D_2^2 = 4 \text{ side}^2$$

$$D_2^2 = 676 - 100 \Rightarrow D_2 = 24 \text{ cm}$$

Now, area of rhombus

$$= \frac{1}{2} \times D_1 \times D_2 = \frac{1}{2} \times 10 \times 24 = 120 \text{ cm}^2$$

**Sol.670.(c)** side of square base =  $b$ 

$$\text{slant height} = s$$

$$\text{height of pyramid} = h$$

$$\text{base area of pyramid (b)}^2 = \frac{3}{8} \times 1536$$

$$= 576 \text{ m}^2$$

$$\text{side of square base (b)} = 24 \text{ m}$$

T.S.A. of square based pyramid

$$= (b)^2 + 2 \times b \times s$$

$$1536 = 576 + 2 \times 24 \times s$$

$$\text{Slant height (s)} = \frac{1536 - 576}{48}$$

$$= \frac{960}{48} = 20 \text{ m}$$

$$\text{Height of pyramid (h)} = \sqrt{s^2 - \left(\frac{b}{2}\right)^2}$$

$$= \sqrt{400 - 144} = \sqrt{256} = 16 \text{ m}$$

$$\text{Volume of pyramid} = (b)^2 \times \frac{h}{3}$$

$$= 24 \times 24 \times \frac{16}{3} = 3072 \text{ m}^3$$

**Sol.671.(a)**

$$\text{Volume of sphere} = \frac{4}{3} \times \pi \times (\text{radius})^3$$

when diameter is reduced to half then  
new radius will also reduced to half of  
the former sphere

Therefore,

Volume of new sphere

$$= \frac{4}{3} \times \pi \times \left(\frac{\text{radius}}{2}\right)^3$$

$$= \frac{1}{8} \times \frac{4}{3} \times \pi \times (\text{radius})^3$$

$$= \text{reduced by } \frac{1}{8} \text{ of the former volume}$$

**Sol.672.(c)** According to the question,

$$\text{Volume of cone} = \frac{1}{3} \pi r^2 h$$

Volume of 1st cone : volume of 2nd cone

$$\frac{1}{3} \pi \times 5^2 \times 7 : \frac{1}{3} \pi \times \left(\frac{21}{2}\right)^2 \times 5$$

$$25 \times 7 : \frac{441}{4} \times 5 \Rightarrow 20 : 63$$

**Sol.673.(d)**

$$\text{Diagonal of rectangle} = \sqrt{4^2 + 3^2} = 5$$

According to the question,

Diagonal of rectangle = diameter of hemisphere = 5

$$\text{So, radius} = 2.5 \text{ cm}$$

Now, total surface area of hemisphere

$$= 3\pi \times 2.5 \times 2.5 = \frac{75\pi}{4}$$

**Sol.674.(d)** Let the height and radius of the right circular cone =  $12x$  and  $5x$ 

$$\text{Volume of a cone} = \frac{1}{3} \pi r^2 h$$

$$314 = \frac{1}{3} \times 3.14 \times 25x^2 \times 12x$$

$$x^3 = \frac{314 \times 100 \times 3}{314 \times 25 \times 12} = 1 \Rightarrow x = 1$$

$$\text{Hence, Slant height} = \sqrt{(12x)^2 + (5x)^2}$$



$$= \sqrt{144 + 25} = 13 \text{ cm}$$

**Sol.675.(d)**

Given, diagonal of square =  $24\sqrt{2}$

Then, side of the square = 24 cm

Lateral surface area of square pyramid

$$= 2 \times \text{side} \times \text{slant height}$$

$$= 2 \times 24 \times l = 624 \Rightarrow l = 13 \text{ cm}$$

$$\text{Now, height of pyramid} = \sqrt{13^2 - \left(\frac{24}{2}\right)^2}$$

$$= 5 \text{ cm}$$

So, volume of pyramid

$$= \frac{1}{3} \times \text{side}^2 \times \text{height}$$

$$= \frac{1}{3} \times 24^2 \times 5 = 960 \text{ cm}^3$$

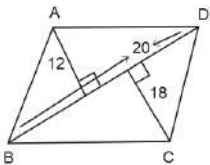
**Sol.676.(d)** According to the question,

Volume of cylinder = Volume of sphere

$$\pi r^2 h = \frac{4}{3} \pi r^3 \Rightarrow 12 \times 12 \times h$$

$$= \frac{4}{3} \times 9 \times 9 \times 9 \Rightarrow h = \frac{4 \times 9 \times 9 \times 9}{3 \times 12 \times 12}$$

$$h = 6.75 \text{ cm}$$

**Sol.677.(d)**

According to the question,

Area of quadrilateral ABCD

$$= \frac{1}{2} \times 20 \times 12 + \frac{1}{2} \times 20 \times 18 = 300 \text{ m}^2$$

**Sol.678.(c)**

Ratio - ball P : ball Q

Diameter - 8 : 2

Radius - 4 : 1

$$\text{Required ratio} = 4^3 : 1^3 = 64 : 1$$

**Sol.679.(c)** Given, Slant height ( $l$ ) = 29 cm

Height ( $h$ ) = 20 cm

$$\text{Radius } (r) = \sqrt{(29)^2 - (20)^2}$$

$$= \sqrt{841 - 400} = \sqrt{441} = 21 \text{ cm}$$

According to question,

$$\frac{\text{T.S.A. of cone}}{\text{Volume of cone}} = \frac{\pi r(l+r)}{\frac{1}{3} \pi r^2 h} = \frac{(l+r)}{\frac{1}{3} r h}$$

$$\frac{\text{T.S.A. of cone}}{\text{Volume of cone}} = \frac{(29+21)}{\frac{1}{3} \times 21 \times 20}$$

$$= \frac{50}{140} = 5 : 14$$

**Sol.680.(b)** Given, Radius ( $R$ ) = 3 cm

Radius of each hemispherical ball after recasting ( $r$ ) = 1.5 cm

$$\text{Number of recasted balls} = \frac{\frac{4}{3} \pi R^3}{\frac{4}{3} \pi r^3}$$

$$= \frac{2 \times (3)^3}{(1.5)^3} = 2 \times \left(\frac{3}{1.5}\right)^3 = 2 \times 8 = 16$$

**Sol.681.(c)** According to the question,

Volume of hollow cylinder =  $\pi(R^2 - r^2)h$

$$= \frac{22}{7} \times (10+7)(10-7) \times 49 = 7854 \text{ cm}^3$$

**Sol.682.(b)**

Volume of sphere = volume of wire

$$\frac{4}{3} \pi r^3 = \pi r^2 h$$

$$\frac{4}{3} \times 3 \times 3 \times 3 = 0.2 \times 0.2 \times h$$

$$h = 900 \text{ cm or 9 m}$$

**Sol.683.(c)** Number of small cubes  $n$

$$= \frac{8 \times 6 \times 6}{0.5 \times 0.5 \times 0.5} = 2304$$

**Sol.684.(c)** Let radius and height of the cone =  $3x$  and  $4x$

$$\text{Slant height } (l) = \sqrt{(3x)^2 + (4x)^2}$$

$$= \sqrt{25x^2} = 5x$$

According to the question,

$$\text{Volume of the cone} = \frac{1}{3} \pi r^2 h$$

$$= \frac{1}{3} \times 3.14 \times 9x^2 \times 4x = 301.44$$

$$= 12x^3 \times 3.14 = 301.44$$

$$= x^3 = \frac{301.44}{12 \times 3.14} = 8 \Rightarrow x = 2$$

$$\text{Slant height } (5x) = 5 \times 2 = 10 \text{ cm}$$

**Sol.685.(c)** According to the question,

$$1 \text{ liter} = 1000 \text{ cm}^3$$

$$50 \text{ liter} = 50,000 \text{ cm}^3$$

$$50 \text{ liter} = 50,000 \times \frac{1}{1000000} = 0.05 \text{ m}^3$$

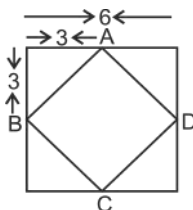
$$\text{Required time} = \frac{120}{0.05}$$

$$= 2400 \text{ min. or 40 hours}$$

**Sol.686.(c)** according to the question,

Area of the square =  $\text{side}^2$

$$36 = \text{side}^2, \text{side} = 6$$



$$\text{Length of AB} = \sqrt{3^2 + 3^2} = \sqrt{18}$$

$$\text{Area of square ABCD} = (\sqrt{18})^2 = 18 \text{ square units}$$

**Sol.687.(a)** Both the median and the altitude are the same in equilateral triangle.

(height) Median of equilateral triangle

$$= \frac{\sqrt{3}}{2} a = \frac{\sqrt{3}}{2} a = 6\sqrt{3} \Rightarrow a = 12 \text{ cm}$$

$$\text{So, area of equilateral triangle} = \frac{\sqrt{3}}{4} a^2$$

$$= \frac{\sqrt{3}}{4} \times 12 \times 12 = 36\sqrt{3} \text{ cm}^2$$

**Sol.688.(b)** Volume of cuboid When areas of three adjacent faces of a cuboid are given =  $\sqrt{pqr} \text{ unit}^3$

**Sol.689.(b)**

Total surface area of cylinder =  $2\pi r(h+r)$

$$2\pi r(h+r) = 64\pi$$

$$2\pi r \times 2r = 64\pi \text{ (given } r = h)$$

$$4\pi r^2 = 64\pi \Rightarrow r = 4, (h = 4)$$

$$\text{Now, } R = 2, H = 8$$

Total surface area of cylinder

$$= 2\pi \times 2(8+2) = 40\pi$$

**Sol.690.(c)** According to the question,

Volume of hollow metal sphere = volume of solid cone

$$\frac{4}{3} \times \pi \times (6^3 - 5^3) = \frac{1}{3} \times \pi \times (5.2)^2 \times h$$

$$h = \frac{4 \times 91}{5.2 \times 5.2} \approx 13.5$$

**Sol.691.(a)** Area of rhombus

$$= \frac{1}{2} \times d_1 \times d_2 = \frac{1}{2} \times 7.5 \times 5.6 = 21 \text{ cm}^2$$

**Sol.692.(a)** According to the question,

Volume of hollow sphere = volume of cone

$$\frac{4}{3} \pi (R^3 - r^3) = \frac{1}{3} \pi r^2 h$$

$$4(3^3 - 2^3) = 4^2 h \Rightarrow 4 \times 19 = 16h$$

$$(\text{height}) h = 4 \frac{3}{4} \text{ cm}$$

**Sol.693.(d)**

$$\text{Total area of ground } (\pi r^2) = 15 \times 5 = 75 \text{ m}^2$$

$$\text{Volume of the cone} = \frac{1}{3} \pi r^2 h$$

According to the question,

$$\frac{1}{3} \pi r^2 h = 30 \times 15 \Rightarrow \frac{1}{3} \times 75 \times h = 30 \times 15$$

$$\text{Height of cone } (h) = 18 \text{ m}$$

**Sol.694.(a)**

Volume of cuboid = Volume of cone

$$11 \times 8 \times 6 = \frac{1}{3} \times \frac{22}{7} \times r^2 \times 14$$

$$r^2 = 36 \Rightarrow r = 6 \text{ cm}$$

**Sol.695.(b)**

Let length, breadth and height of cuboid be  $6x$ ,  $4x$  and  $3x$  respectively

Total surface area of the cuboid

$$= 2(lb + bh + hl)$$

$$2(6x \times 4x + 4x \times 3x + 3x \times 6x) = 972$$

$$108x^2 = 972, x = 3$$



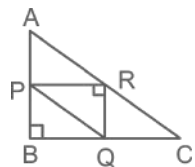
then, length, breadth and height = 18 cm, 12 cm and 9 cm

Diagonal of the cuboid

$$= \sqrt{(18)^2 + (12)^2 + (9)^2}$$

$$= \sqrt{549} = 3\sqrt{61} \text{ cm}$$

**Sol.696.(a)** (5, 12, 13) is a triplate so the triangle will be a right angle triangle.



triangle made by the mid point of the sides ABC triangle will be also right angle triangle

And the area of  $\Delta PRQ$  will be  $\frac{1}{4}$  of the area of  $\Delta ABC$ .

So the area of  $\Delta PRQ = \frac{1}{4} \times \frac{1}{2} \times 5 \times 12$

$$= 7\frac{1}{2} \text{ cm}^2$$

**Sol.697.(d)** C.S.A of cone =  $\pi rl$

T.S.A of cone =  $\pi r(l + r)$

slant height (l) =  $\sqrt{h^2 + r^2}$

$$l = \sqrt{20^2 + 15^2} = 25 \text{ cm}$$

C.S.A of cone =  $\pi \times 15 \times 25 = 375\pi \text{ cm}^2$

T.S.A of cone =  $\pi \times 15(25 + 15) = 600\pi \text{ cm}^2$

**Sol.698.(c)** Surface area of spherical water tank =  $4\pi r^2$

$$= 4 \times \frac{22}{7} \times 3.5 \times 3.5 \Rightarrow 88 \times 5 \times 3.5$$

$$= 44 \times 3.5 = 154 \text{ m}^2$$

Required cost =  $154 \times 13 = 2002 \text{ Rs.}$

**Sol.699.(a)**

Pythagoras triplets :- (20, 21, 29)

Radius of cone = 20 cm

Volume of cone =  $\frac{1}{3} \times \pi r^2 h$

$$= \frac{1}{3} \times \frac{22}{7} \times 20 \times 20 \times 21$$

$$= 400 \times 22 = 8800 \text{ cm}^3$$

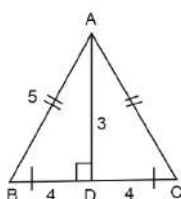
**Sol.700.(d)**

Perimeter of parallelogram =  $2(l + b)$

$$= 48 \text{ cm} \Rightarrow b = 24 - 8 = 16 \text{ cm}$$

Area = base  $\times$  height =  $16 \times 6 = 96 \text{ cm}^2$

**Sol.701.(a)**



Pythagoras triplets : (3, 4, 5)

AD = 3 cm

$$\text{Area of triangle} = \frac{1}{2} \times 8 \times 3 = 12 \text{ cm}^2$$

**Sol.702.(a)** Radius of sphere = 3.5 cm

$$\text{Volume} = \frac{4}{3} \pi r^3$$

$$= \frac{4}{3} \times \frac{22}{7} \times 3.5 \times 3.5 \times 3.5$$

$$= \frac{4}{3} \times 22 \times 0.5 \times 3.5 \times 3.5$$

$$= \frac{44}{3} \times 12.25 = 179.67 \text{ cm}^3$$

**Sol.703.(d)** 0.2 cm refers to 1 km

1 cm refers to 5 km

$1 \text{ cm}^2$  refers to  $25 \text{ km}^2$

$$\text{Now, } 22500 \text{ km}^2 = \frac{1}{25} \times 22500 = 900 \text{ cm}^2$$

**Sol.704.(d)**

$$\text{Total volume} = (15)^3 + (18)^3 + (25)^3$$

Side of newly formed cube

$$= \sqrt[3]{(15)^3 + (18)^3 + (25)^3}$$

$$= \sqrt[3]{3375 + 5832 + 15625}$$

$$= \sqrt[3]{24832} = \sqrt[3]{64 \times 388} = 4\sqrt[3]{388}$$

**Sol.705.(d)**

$$\frac{4}{3} \times \pi \times (9)^3 = x \times \frac{4}{3} \times \pi \times (3)^3$$

$$\Rightarrow x = 27 \text{ laddoos}$$

**Sol.706.(a)**

$$\text{Side of square} = \frac{172}{4} = 43 \text{ cm}$$

$$\text{Area of triangle} = \frac{43 \times 43}{2} = 924.5 \text{ cm}^2$$

**Sol.707.(d)**  $1 \text{ m}^3 = 1000 \text{ litre}$

Volume of cuboid =  $12 \times 9 \times 7.5$

$$= 810 \text{ m}^3 = 810000 \text{ litre}$$

**Sol.708.(a)**

Third side =  $20 - (5 + 6) = 9 \text{ meter}$

Semiperimeter = 10

$$\text{Area} = \sqrt{10 \times (10 - 5)(10 - 6)(10 - 9)}$$

$$= \sqrt{10 \times (5) \times (4) \times (1)} = 10\sqrt{2} \text{ m}^2$$

**Sol.709.(d)** Length of arc =  $\frac{\theta}{360^\circ} \times 2\pi r$

$$= 20.6 - 2 \times 7.4 = 5.8 \text{ cm}$$

$$\text{Area of sector} = \frac{\theta}{360^\circ} \times \pi r^2$$

$$= \left(\frac{\theta}{360^\circ} \times 2\pi r\right) \times \frac{r}{2} = 5.8 \times \frac{7.4}{2}$$

$$= 21.46 \text{ cm}^2$$

**Sol.710.(b)**

Length of arc =  $15.5 - (2 \times 5) = 5.5 \text{ cm}$

$$\text{Area of sector} = \text{length of arc} \times \frac{\text{radius}}{2}$$

$$= 5.5 \times \frac{5}{2} = 13.75 \text{ cm}^2$$

**Sol.711.(a)**

Side of the square =  $\sqrt{30.25} = 5.5 \text{ cm}$

Circumference of circle =  $4 \times 5.5 = 22 \text{ cm}$

$$\text{Radius of circle} = 22 \times \frac{7}{2 \times 22} = 3.5 \text{ cm}$$

$$\text{Area of circle} = \frac{22}{7} \times 3.5 \times 3.5 = 38.5 \text{ cm}^2$$

**Sol.712.(a)** Inner radius of hemispherical bowl = 4 cm

So, outer radius =  $(4 + 1) = 5 \text{ cm}$

Volume of hemispherical bowl

$$= \frac{2}{3} \times \pi \times (5^3 - 4^3) = \frac{122}{3} \pi = 40\frac{2}{3} \pi$$

**Sol.713.(a)** Volume of cylinder =  $\pi r^2 h$

According to the question,

$$\pi \times 5^2 \times 52 = \pi \times r^2 \times 100 \Rightarrow r^2 = 13$$

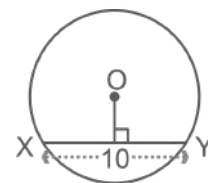
Radius of the wire (r) =  $\sqrt{13}$

**Sol.714.(c)**

Volume of the cylindrical flask =  $\pi r^2 h$

$$= \frac{22}{7} \times 6 \times 6 \times 21 = 2376 \text{ cm}^3$$

**Sol.715.(a)**



Area of the circle =  $\pi r^2$

$$616 = \frac{22}{7} \times r^2 \Rightarrow r = 14 \text{ cm}$$

$$\text{Required distance} = \sqrt{14^2 - 5^2} = \sqrt{171} \text{ cm}$$

